-->

aetna™

(https://www.aetna.com/)

Spinal Surgery: Laminectomy and Fusion

Clinical Policy Bulletins | Medical Clinical Policy Bulletins

Number: 0743

Table Of Contents

Policy History

Last Review
02/18/2026
Effective: 12/14/2007
Next Review: 08/13/2026

Review History

Definitions

Additional Information

Clinical Policy Bulletin Notes

Policy

Scope of Policy

This Clinical Policy Bulletin addresses spinal surgeries.

I. Medical Necessity

Aetna considers the following medically necessary:

A. Cervical laminectomy (and/or an anterior and/or posterior cervical discectomy and fusion) for individuals with herniated discs or other causes of spinal cord or nerve root compression (osteophytic spurring, ligamentous hypertrophy) when *all* of the following criteria are met (see also Additional Notes section):

    1. All other reasonable sources of pain and/or neurological deficit have been ruled out, including but not limited to significant pathology at other spinal level(s) on the advanced imaging radiology report that is/are not part of the surgical request resulting in incomplete surgical planning; *and*
    2. Member has signs or symptoms of neural compression (radiculopathy, neurogenic claudication, myelopathy) associated with the levels being treated; *and*
    3. Advanced imaging studies (CT or MRI) indicate central/lateral recess or foraminal stenosis (graded as moderate, moderate to severe or severe; not mild or mild to moderate), or nerve root or spinal cord compression, at the level corresponding with the clinical findings; *and*
    4. Member has failed at least 6 weeks of conservative therapy * (unless there is an indication for waiver of requirements for conservative management, noted below); *and*
    5. Member's activities of daily living are limited by symptoms of neural compression;

B. Thoracic laminectomy (and/or thoracic discectomy and fusion) for individuals with herniated discs or other causes of thoracic nerve root compression (osteophytic spurring, ligamentous hypertrophy) when *all* of the following criteria are met (see also Additional Notes section):

    1. All other reasonable sources of pain and/or neurological deficit have been ruled out, including but not limited to significant pathology at other spinal level(s) on the advanced imaging radiology report that is/are not part of the surgical request resulting in incomplete surgical planning; *and*

2. Member has signs or symptoms of neural compression (radiculopathy, neurogenic claudication, myelopathy) associated with the levels being treated; *and*
3. Advanced imaging studies (CT or MRI) indicate central/lateral recess or foraminal stenosis (graded as moderate, moderate to severe or severe; not mild or mild to moderate), or nerve root or spinal cord compression, at the level corresponding with the clinical findings; *and*
4. Member has failed at least 6 weeks of conservative therapy * (unless there is an indication for waiver of requirements for conservative management, noted below); an*d*
5. Member's activities of daily living are limited by symptoms of neural compression;

C. Lumbar laminectomy for individuals with a herniated disc when *all* of the following criteria are met (see also Additional Notes section):

1. All other reasonable sources of pain and/or neurological deficit have been ruled out, including but not limited to significant pathology at other spinal level(s) on the advanced imaging radiology report that is/are not part of the surgical request resulting in incomplete surgical planning; *and*
2. Member has signs or symptoms of neural compression (radiculopathy, neurogenic extremity claudication, myelopathy) associated with the levels being treated; *and*
3. Advanced imaging studies (CT or MRI) indicate central/lateral recess or foraminal stenosis (graded as moderate, moderate to severe or severe; not mild or mild to moderate), or nerve root or spinal cord compression, at the level corresponding with the clinical findings; *and*
4. Member has failed at least 6 weeks of conservative therapy * (unless there are indications for waiver of requirements for conservative management, noted below); *and*
5. Member's activities of daily living are limited by symptoms of neural compression;

D. Cervical, thoracic, lumbar, or sacral laminectomy for *any* of the following (see also Additional Notes section):

1. Spinal fracture with displaced fracture fragment(s) causing moderate or worse stenosis or nerve compression confirmed by imaging studies (e.g., CT or MRI) associated with signs/symptoms of nerve compression (or combined with fusion for an unstable fracture – see fusion policies below); *or*
2. Spinal infection confirmed by imaging studies (e.g., CT or MRI); *or*
3. Spinal tumor confirmed by imaging studies (e.g., CT or MRI); *or*
4. Epidural hematomas confirmed by imaging studies (e.g., CT or MRI); *or*
5. Synovial cysts, Tarlov cysts (also known as perineurial cysts and sacral meningeal cysts), or arachnoid cysts causing spinal cord or nerve root compression with unremitting pain, confirmed by imaging studies (e.g., CT or MRI) and with corresponding neurological deficit, where symptoms have failed to respond to six weeks of conservative therapy* (unless there is an indication for waiver of requirements for conservative management, noted below) *or*
6. Spinal stenosis (central, lateral recess or foraminal stenosis) graded as moderate, moderate to severe or severe (not mild or mild to moderate) with unremitting pain, with stenosis confirmed by imaging studies (e.g., CT or MRI) at the level corresponding to neurological findings, where symptoms have failed to respond to six weeks conservative therapy* (unless there is an indication for waiver of requirements for conservative management, noted below); *or*
7. Repair of open spinal dysraphism, or radiographically demonstrated closed spinal dysraphism (including tethered cord) with significant signs or symptoms of lumbosacral spinal dysfunction, or radiographically demonstrated closed spinal dysraphism in asymptomatic young children who are not yet toilet trained or have not yet begun to walk. Surgery

for asymptomatic closed spinal dysraphism in older individuals or clinical tethered cord syndrome without radiographic abnormalities (occult tethered cord syndrome) will be considered upon individual case review. For cases of occult tethered cord syndrome, all other sources of symptoms should be excluded, including consultation notes from appropriate specialists (e.g., a urologist for urologic complaints, a neurologist for neurologic complaints); *or*
8. Other mass lesions confirmed by imaging studies (e.g., CT or MRI), upon individual case review;

E. Lumbar decompression with or without discectomy for rapid progression of neurological impairment (e.g., foot drop, extremity weakness, saddle anesthesia, bladder dysfunction or bowel dysfunction) with central, lateral recess or foraminal stenosis (graded as moderate, moderate to severe or severe; not mild or mild to moderate) confirmed by imaging studies (e.g., CT or MRI) at the levels corresponding to the neurologic findings;
F. Vertebral corpectomy (removal of half* or more of vertebral body, not mere removal of osteophytes and minor decompression) in the treatment of one of the following:

1. For tumors involving one or more vertebrae, *or*
2. Greater than 50 % compression fracture of vertebrae, *or*
3. Retro-pulsed bone fragments, *or*
4. Symptomatic moderate or greater central canal stenosis caused by vertebral body pathology (such as due to fracture, tumor or congenital or acquired deformity of the vertebral body);

G. Cervical spinal fusion for *any* of the following indications (see also Additional Notes section):

1. Cervical kyphosis associated with cord compression; *or*
2. Symptomatic pseudarthrosis (non-union of prior fusion) with radiological (e.g., CT or MRI) demonstration of non-union of prior fusion (lack of bridging bone or abnormal motion at fused segment) after 12 months since fusion

surgery *or* with radiographic evidence of hardware failure (fracture or displacement). **Note:** For cervical pseudoarthrosis not associated with hardware failure or indications for urgent intervention, the member should be nicotine-free (including smoking, use of tobacco products, and nicotine replacement therapy) for at least 6 weeks prior to surgery. For persons with recent (within a year) nicotine use, documentation of nicotine cessation should include a lab report (not surgeon summary) showing blood or urinary nicotine levels of less than or equal to 10 ng/ml (or urinary cotinine levels of less than or equal to 10 ng/ml), drawn within 6 weeks prior to surgery; or

3. Burst fractures, fracture-dislocation associated with mechanical instability, locked facets, or other unstable spinal fractures (see Appendix) confirmed by imaging studies (e.g., CT or MRI); fusion may be combined with a laminectomy; *or*
4. Spinal infection confirmed by imaging studies (e.g., CT or MRI) and/or other studies (e.g., biopsy), which may be combined with a laminectomy; or
5. Spinal tumor, primary or metastatic to spine, confirmed by imaging studies (e.g., CT or MRI), which may be combined with a laminectomy; *or*
6. Atlantoaxial (C1-C2) subluxation (e.g., associated with congenital anomaly, os odontoideum, or rheumatoid arthritis) noted as widening of the atlantodens interval greater than 3 mm confirmed by imaging studies (e.g., CT or MRI); *or*
7. Basilar invagination of the odontoid process into the foramen magnum; *or*
8. Sub-axial (C2-T1) instability confirmed by imaging studies, when *both* of the following are met:

   a. Significant instability (sagittal plane translation of at least 3 mm on flexion and extension views or relative sagittal plane angulation greater than 11 degrees); *and*
   b. Symptomatic unremitting pain that has failed 3 months of conservative management*(unless there is an

indication for waiver of requirements for conservative management, noted below); *or*

9. Adjunct to excision of synovial cysts causing spinal cord or nerve root compression with unremitting pain, confirmed by imaging studies (e.g., CT or MRI) and with corresponding neurological deficit, where symptoms have failed to respond to six weeks of conservative therapy * (unless there is an indication, or progressive neurological deficit, which requires urgent intervention) *or*
10. Clinically significant deformity of the spine (kyphosis, head-drop syndrome, post-laminectomy deformity) that meets any of the following criteria after symptoms/posture has failed to improve with 6 weeks of conservative therapy* and with documentation the member is nicotine free (including smoking, use of tobacco products, and nicotine replacement therapy) for at least 6 weeks prior to surgery:

    a. The deformity prohibits forward gaze; *or*
    b. The deformity is associated with severe neck pain, difficulty ambulating, and interference with activities of daily living; *or*
    c. Documented progression of the deformity;

11. Occipito-cervical instability is seen in Down's syndrome, the mucopolysaccharidoses and rheumatoid arthritis; and occipito-cervical dislocation/instability is seen after high energy trauma. A Power's ratio of greater than 1.0 raises concern for occipito-cervical dislocation, and a basion-dens (BD) or basion-posterior axial interval (BAI) of greater than12 mm suggests occipito-cervical dissociation/instability (combination of BD plus BAI is known as the Harris method and should be measured on CT for best sensitivity). MRI can be used to demonstrate injury to the transverse ligament, alar ligaments, apical ligament and tectorial membrane. Traction should be avoided. Neurological deficits or fatalities are common. Occipito-cervical fusion/stabilization is considered medically necessary in traumatic cases with documented instability

via Harris measurements and ligamentous disruption. When there is signal change in some ligaments only without abnormalities in Powers ratio, BDI, BAI-BDI and X line, conservative management with stabilization may be the appropriate first-line treatment. Occipito-cervical fusion is also considered medically necessary in congenital/acquired cases (e.g., Down's syndrome, the mucopolysaccharidoses, or rheumatoid arthritis) for instability when there is cord compression present at the occipito-cervical junction.

H. Thoracic spinal fusion for *any* of the following indications (see also Additional Notes section):

1. Scoliosis confirmed by imaging studies, with Cobb angle greater than 40 degrees in skeletally immature children and adolescents, or Cobb angle greater than 50 degrees associated with functional impairment in skeletally mature adults, that has failed 3 months of conservative management* (unless there is an indication for urgent intervention, noted below) **Note:** For fusion for scoliosis in children, adolescents and young adults age 25 and younger, see CPB 0398 - Idiopathic Scoliosis (../300_399/0398.html); *or*
2. Thoracic kyphosis resulting in spinal cord compression, or kyphotic curve greater than 75 degrees that is refractory to bracing, that has failed 3 months of conservative management* (unless there is an indication for urgent intervention, noted below); *or*
3. Thoracic pseudarthrosis (defined as absence of bridging bone that connects the vertebrae, or movement of vertebrae at site of prior attempted arthrodesis on dynamic radiographs) after 12 months have elapsed since the time of fusion* (unless there is an indication for urgent intervention, noted below), or if there is pseudarthrosis with additional findings of hardware failure (imaging evidence of fracture/disconnection/dislocation of implants) **Note:** For thoracic pseudoarthrosis not associated with hardware failure or indications for urgent intervention, the member should be nicotine-free (including smoking, use of tobacco

products, and nicotine replacement therapy) for at least 6 weeks prior to surgery. For persons with recent (within a year) nicotine use, documentation of nicotine cessation should include a lab report (not surgeon summary) showing blood or urine nicotine or cotinine levels of less than or equal to 10 ng/ml (or urinary cotinine levels of less than or equal to 10 ng/ml), drawn within 6 weeks prior to surgery; *or*

4. Burst fractures, fracture-dislocation associated with mechanical instability, locked facets, or other unstable spinal fractures (see Appendix) confirmed by imaging studies (e.g., CT or MRI); fusion may be combined with a laminectomy; *or*
5. Spinal infection confirmed by imaging studies (e.g., CT or MRI) and/or other studies (e.g., biopsy), which may be combined with a laminectomy; *or*
6. Spinal tumor, primary or metastatic to spine, confirmed by imaging studies (e.g., CT or MRI), which may be combined with a laminectomy; *or*
7. Spondylolisthesis with segmental instability confirmed by imaging studies (e.g., CT or MRI), when *both* of the following criteria are met:

   a. Significant spondylolisthesis, grades II, III, IV, or V (see Appendix); *and*
   b. Symptomatic unremitting pain that has failed six weeks of conservative management* (unless there is an indication for urgent intervention, noted below); *or*

8. Spinal stenosis where criteria for thoracic decompression in Section I.B. above are met, and *any* of the following is met:

   a. Decompression is performed in an area of segmental instability as manifested by gross movement on flexion-extension radiographs; *or*
   b. Decompression coincides with any degree of anterolisthesis (grades I, II, III, IV or V)); *or*
   c. Decompression creates an iatrogenic instability by the disruption of the posterior elements where facet joint

excision exceeds 50% bilaterally or complete excision of one facet is performed;

I. Lumbar spinal fusion for *any* of the following indications (see also Additional Notes section):

1. Adult scoliosis confirmed by imaging studies, with Cobb angle greater than 50 degrees associated with functional impairment in skeletally mature adults, that has failed 3 months of conservative management* (unless there is an indication for waiver of requirements for conservative management, noted below) **Note:** For fusion for scoliosis in children, adolescents and young adults age 25 and younger, see CPB 0398 - Idiopathic Scoliosis (../300_399/0398.html); *or*
2. Lumbar pseudarthrosis (defined as absence of bridging bone that connects the vertebrae, or movement of vertebrae at site of prior attempted arthrodesis on dynamic radiographs) after 12 months have elapsed since the time of fusion* (unless there is an indication for waiver of requirements for conservative management, noted below), or if there is pseudarthrosis with additional findings of hardware failure (imaging evidence of fracture/disconnection/dislocation of implants) **Note**: For lumbar pseudoarthrosis not associated with hardware failure or indications for urgent intervention, the member should be nicotine-free (including smoking, use of tobacco products, and nicotine replacement therapy) for at least 6 weeks prior to surgery. For persons with recent (within a year) nicotine use (unless there is an indication for waiver of requirements for conservative management, noted below), documentation of nicotine cessation should include a lab report (not surgeon summary) showing blood or urine nicotine levels of less than or equal to 10 ng/ml (or urinary cotinine levels of less than or equal to 10 ng/ml), drawn within 6 weeks prior to surgery; *or*
3. Iatrogenic or degenerative flatback syndrome (defined as members with back pain, stooped posture and limitation of forward gaze), and with significant sagittal imbalance that

has failed 3 months of conservative management* (unless there is an indication for waiver of requirements for conservative management, noted below), when fusion is performed with spinal osteotomy and/or lordotic anterior interbody implants sufficient to restore anterior height and lordosis. Note that sagittal imbalance on standing radiographs of the spine are considered significant where there is:

a. an offset of greater than 5 cm between the sagittal vertebral axis (a plumb line downward from the center of the C7 vertebral body) and the posterior superior aspect of the S1 vertebral body; *or*
b. a pelvic tilt greater than 20 degrees; *or*
c. a lumbar lordosis to pelvic incidence mismatch of greater than or equal to 10 degrees; *or*

4. Spinal fracture causing instability or neural element compromise/compression, spinal dislocation (associated with mechanical instability), locked facets, or fracture with displaced fracture fragment confirmed by advanced imaging studies (CT or MRI) causing neural element compromise, which may be combined with a laminectomy. Excluded are pars defects/pars fractures, subacute or chronic compression fractures, compression fractures without retropulsion that is causing significant canal stenosis, chronic stable odontoid fractures, spinous process fractures, transverse process fractures and stable isolated pedicle fractures; *or*
5. Pars defects (fractures) without significant instability (for spondylolisthesis/instability follow criteria (9) below) and with disc degeneration at the surgical level in members under 18 years of age who have failed at least 6 months of conservative management that includes bracing and at least 6 weeks of formal in-person physical therapy within the past year*. If there is no disc degeneration then pars repair is appropriate; *or*
6. Spinal infection confirmed by imaging studies (e.g., CT or MRI) and/or other studies (e.g., biopsy), which may be

combined with a laminectomy; *or*

7. Spinal tumor confirmed by imaging studies (e.g., CT or MRI), which may be combined with a laminectomy; *or*
8. Spondylolisthesis with segmental instability confirmed by imaging studies (e.g., CT or MRI), with associated symptomatic unremitting low back pain, radiculopathy or neurogenic claudication, when *either* of the following criteria are met:

   a. Radiographic documentation of significant spondylolisthesis, grades II, III, IV, or V (see Appendix) that has failed six weeks of conservative management* (unless there is an indication for waiver of requirements for conservative management, noted below); *or*
   b. Radiographic documentation dynamic instability of at least 4 mm of translation or 10 degrees of angular motion on dynamic imaging that has failed 6 weeks of conservative management* (unless there is an indication for waiver of requirements for conservative management, noted below); *or*

9. Spinal stenosis, where criteria for lumbar decompression in Section I.C. above are met, and *any* of the following is met:

   a. Decompression is performed in an area of segmental instability as manifested by gross movement on flexion-extension radiographs; *or*
   b. Decompression coincides with any degree of anterolisthesis (grades I, II, III, IV or V)); *or*
   c. When performed with initial primary laminectomy/discectomy for nerve root decompression or spinal stenosis and there is documented intraoperative iatrogenic instability; *or*
   d. Revision decompression surgery that will create iatrogenic instability by the disruption of the anterior spinal column or posterior elements where facet joint excision exceed 50 % bilaterally or complete excision of one facet is performed; *or*

J. Spinal surgery in persons with prior spinal surgery when any of the above criteria (I.A. through I.E.) are met;
K. Removal of large anterior cervical osteophytes that are causing dysphagia (with or without anterior fusion) when the osteophytes are deemed to be the cause of the dysphagia by an ENT specialist or a certified speech therapist;
L. Cervical laminoplasty (laminaplasty) for members with at least moderate cervical spinal stenosis at multiple (greater than or equal to 2) levels, where 2 or more vertebral segments are to be decompressed, and the member meets the following criteria:

    1. All other reasonable sources of pain or neurologic deficit/symptoms have been ruled out; *and*
    2. Member has signs or symptoms of spinal cord compression (myelopathy) associated with the levels being treated; *and*
    3. Imaging studies (e.g., CT or MRI) indicate central stenosis (graded as moderate, moderate to severe or severe; not mild or mild to moderate), or spinal cord compression, at the levels corresponding with the clinical findings; *and*
    4. Imaging studies show no more than 3 mm of motion on cervical flexion/extension x-rays and no cervical kyphosis; *and*
    5. Member has failed at least 6 weeks of conservative therapy (unless there is an indication for waiver of requirements for conservative management*, noted below);

M. Exploration of spinal fusion (CPT code 22830) will be separately reimbursed when criteria for fusion for pseudoarthrosis are met (see G.2. and H.3. and I.2) but intra-operative findings reveal solid fusion and no additional procedures are performed. Exploration of spinal fusion (CPT code 22830) is considered incidental to any other procedure in the same anatomic region, and cannot be authorized in combination with other spinal procedures in the same area. Examples include, but are not limited to, hardware removal and revision of fusion. Otherwise, exploration of fusion is considered not medically necessary;

N. Removal of posterior spinal instrumentation is considered medically necessary when *any* of the following criteria is met:

1. Neurologic complication related to the ongoing presence of the instrumentation (e.g., a screw is compressing a nerve root causing radiculopathy);
2. New pain or protruding mass at operative site after significant spinal trauma (e.g., motor vehicle accident);
3. Severe back or neck pain after the fusion has healed related to the spinal instrumentation that has failed conservative management*, including at least 6 weeks of formal in-person physical therapy in the past year;
4. Spinal infection;
5. Symptomatic rod, hook, or screw migration, dislodgment, or breakage;
6. Vascular complications from spinal instrumentation;
7. Further surgery is approved that requires removal of the instrumentation;

Note: While removal of instrumentation necessary for approved fusion surgery is allowable, it is not reimbursable when new instrumentation is inserted at levels including all or part of the previously instrumented segments as it is considered integral to this procedure.

O. Removal of anterior spinal instrumentation is considered medically necessary when *any* of the following criteria is met:

1. Further surgery is approved that requires removal of the instrumentation; *or*
2. Neurologic or vascular complication related to the ongoing presence of the instrumentation; *or*
3. Instrumentation failure/migration that could cause neurologic or vascular complications; *or*
4. Spinal infection; *or*
5. Dysphagia related to esophageal compression by anterior cervical implants (confirmed by ENT evaluation or barium swallow study);

Note: While removal of instrumentation necessary for approved fusion surgery is allowable, it is not reimbursable when new instrumentation is inserted at levels including all or part of the previously instrumented segments as it is considered integral to this procedure.

P. Thoracolumbar osteotomy is considered medically necessary only when there is a significant deformity that meets criteria described elsewhere in this CPB. Laminectomies and TLIFs (which include a laminectomy) cannot be approved for reimbursement separately when combined with osteotomies as these are considered included in the osteotomy procedure;
Q. Excision of a portion of the bony spine (portion of the vertebral body, facets or lamina) for a bony lesion is considered medically necessary when *all* of the following are met:

  1. A bony lesion is documented on formal advanced imaging report (CT or MRI); *and*
  2. Additional radiographic investigation has been completed to define the lesion if appropriate (angiogram, SPECT scan, PET scan); *and*
  3. Bone biopsy has been performed and pathology is available; *and*
  4. The pathology documents a lesion for which the medical literature supports excision and other possible treatments such as radiation have been considered; and
  5. Removal of the entire bone at that level is not indicated due to compression of neural elements or other reasons (corpectomy or laminectomy are not requested and no need for these procedures is documented);

R. Pelvic fixation is considered medically necessary where criteria for lumbar fusion (in I.1.9) are met, for *any* of the following:

  1. With a requested fusion from L2 or above to the sacrum; *or*
  2. A multi-level fusion with non-union or sacral insufficiency fracture; *or*
  3. Fusion of a high grade lumbosacral spondylolisthesis at L5/S1 (Grade 3 or more);

S. Computer-assisted surgical navigation (stereotaxic navigation) is approvable if fusion implants are being placed during surgery on the spinal column when identification of the stereotaxic system being used, registration and review of imaging data sets with verification of accuracy and intraoperative use of the navigation system is documented. Computer-assisted surgical navigation is not approvable during routine decompression surgery. Exceptions may be for decompressions for tumor.

* Conservative measures must be recent (within the past year) and include the following non-surgical measures and medications unless one or more of the requirements for waiver below are met: patient education; active physical therapy (in-person as opposed to home or virtual physical therapy); medications (NSAIDS, acetaminophen, or tricyclic antidepressants), and (where appropriate) identification and management of associated anxiety and depression. The member should participate in physical therapy for the entire required duration of conservative management (six weeks or three months, depending upon the indication for surgery). Physical therapy needs to be confirmed either by the actual PT notes, or by documentation in the member claims history.

The requirement for a trial of conservative measures may be waived in the following situations indicating need for urgent intervention:

A. Spinal cord compression (this does not include nerve root compression);
B. Stenosis causing cauda equina syndrome;
C. Stenosis causing myelopathy (only applicable for cervical and thoracic spine cases, not lumbar);
D. Stenosis causing severe weakness of the muscle(s) innervated by nerves at the requested surgical level(s) as documented by the requesting surgeon (graded 4 minus or less on MRC scale (see Appendix) (Note: 4 minus strength describes muscle activation that is beyond antigravity (3/5) and produces motion against only slight resistance and fails against moderate resistance);

E. Progressive neurological deficit on serial examinations by the same surgeon examiner;
F. Severe stenosis associated with instability (dynamic excursion with flexion/extension or from supine to standing) when fusion is requested (not just decompression only); or
G. And the requirement for a trial of conservative measures may be met if a discharge note from a physical therapist documents the lack of utility of further physical therapy given the member's inability to adapt to a tolerable treatment plan due to exceptional pain.

II. **Additional Notes**

A. For spinal fusion (cervical, lumbar and thoracic):

1. Member should be nicotine-free (including smoking, use of tobacco products, and nicotine replacement therapy) for at least 6 weeks prior to surgery. For persons with recent (within a year) nicotine use (unless there is an indication (e.g., myelopathy, cauda equina syndrome), severe weakness (graded 4 minus or less on MRC scale) or progressive weakness, or an associated infection/tumor and/or fracture), documentation of nicotine cessation should include a lab report (not surgeon summary) showing blood and/or urine nicotine level of less than or equal to 10 ng/ml (or urinary cotinine levels of less than or equal to 10 ng/ml), drawn within 6 weeks prior to surgery;
2. If member is diabetic, their hemoglobin A1C (HbA1c) should be less than 8 percent within 3 months prior to surgery, unless there is an urgent indication for surgery (e.g., myelopathy, cauda equina syndrome), severe weakness (graded 4 minus or less on Medical Research Council [MRC] scale) or progressive weakness, or an associated infection/tumor and/or acute fracture.

B. For elective laminectomy (cervical, lumbar, and thoracic), if member is diabetic, their HbA1c should be less than 8 percent within 3 months prior to surgery, unless there is an indication (e.g., myelopathy, cauda equina syndrome), severe weakness (graded 4 minus or less on MRC scale) or progressive weakness, or an associated infection/tumor and/or acute fracture.

C. Medical records must document that a physical examination, including a neurologic examination, has been performed by or reviewed by the operating surgeon.
D. For purposes of this policy, central stenosis is classified into the following grades: normal or mild stenosis (ligamentum flavum hypertrophy and/or osteophytes and/or or disk bulging without significant narrowing of the central spinal canal); moderate stenosis (central spinal canal is narrowed such that there is minimal spinal fluid visible in the dural sac); severe stenosis (central spinal canal is narrowed and there is only a faint amount of spinal fluid or no fluid in the dural sac; cord morphological change associated with stenosis is a sign of severe stenosis).
For purposes of this policy, foraminal stenosis is classified into the following grades: mild foraminal stenosis (where some perineural fat remains visible around the nerve root); moderate foraminal stenosis (showing minimal perineural fat but no morphological changes); severe foraminal stenosis (showing nerve root morphological change (not just nerve root displacement)).

   For purposes of this policy, any nerve root or spinal cord compression meets criteria regardless of the degree of stenosis on the imaging report. Terms such as mass effect, impingement, abutment, effacement, indentation, flattening or displacement do not substitute for compression.

E. For purposes of this policy, Aetna will consider the official written report of advanced imaging studies (e.g., CT, MRI, myelogram). If the operating surgeon disagrees with the official written report, the surgeon should document that disagreement. The surgeon should discuss the disagreement with the provider who did the official interpretation, and there should also be a written addendum to the official report indicating agreement or disagreement with the operating surgeon. Advanced imaging studies are required prior to any spinal surgery. The imaging should be performed within the past year, or after the onset of the current constellation of symptoms or any relevant surgical procedures, whichever is sooner.

F. Paraspinal muscle flap closure or tissue transfer (by plastic or other surgeon (including the operating surgeon)) at the time of spinal surgeries is considered not medically necessary as wound closure is included in the surgery and not separately reimbursable and there is no literature evidence that particular methods of wound closure produce improved outcomes.

III. **Experimental, Investigational, or Unproven**

The following procedures are considered experimental, investigational, or unproven for all indications not listed above because of insufficient evidence of their effectiveness:

- Avulsion of spinal nerves;
- Cervical, thoracic and lumbar laminectomy and/or fusion for indications not listed as medically necessary;
- Cervical laminoplasty for treatment of cervical kyphosis, mechanical axial neck pain (cervical strain) without myelopathy, and for all other indications not listed as medically necessary;
- Certain fusion procedures: for inter-laminiar lumbar instrumented fusion (ILIF) and Coflex-F implant for lumbar fusion, see CPB 0016 - Back Pain: Invasive Procedures (../1_99/0016.html). See also CPB 0772 - Axial Lumbar Interbody Fusion (AxiaLIF) (0772.html);
- Computer-assisted surgical navigation for laminectomies. **Note**: Robotic assistance is considered integral to the primary procedure and not separately reimbursed;
- Lumbar spinal fusion for degenerative disc disease and all other indications not listed as medically necessary;
- Occipito-cervical fusion/stabilization not listed as medically necessary in Section G;
- Removal of the C1 tubercle or transverse process or cervical laminectomy for the treatment of internal jugular vein stenosis;

- Surgery for Bertolotti's syndrome (fusion or resection of the lumbosacral articulation).

IV. **Related CMS Coverage Guidance**

This Clinical Policy Bulletin (CPB) supplements but does not replace, modify, or supersede existing Medicare Regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs). The supplemental medical necessity criteria in this CPB further define those indications for services that are proven safe and effective where those indications are not fully established in applicable NCDs and LCDs. These supplemental medical necessity criteria are based upon evidence-based guidelines and clinical studies in the peer-reviewed published medical literature. The background section of this CPB includes an explanation of the rationale that supports adoption of the medical necessity criteria and a summary of evidence that was considered during the development of the CPB; the reference section includes a list of the sources of such evidence. While there is a possible risk of reduced or delayed care with any coverage criteria, Aetna believes that the benefits of these criteria – ensuring patients receive services that are appropriate, safe, and effective – substantially outweigh any clinical harms.

This CPB is being used supplement certain Medicare LCDs and NCDs on spinal surgery (LCD L37848, Lumbar Fusion; LCD L38033, Cervical Disc Arthroplasty; LCD L39773, Cervical Fusion; LCD L39762, Cervical Fusion; LCD L39741, Cervical Fusion; LCD L39788, Cervical Fusion; LCD L39788, Cervical Fusion; NCD 150.10, Lumbar Artificial Disc Replacement) to require radiologic documentation of at least moderate stenosis or other radiologic evidence of neural compression. This criterion will reduce the risk of having surgery to decompress nerves that are not compressed, and thereby increase the likelihood of success of surgery. This CPB is also used to define appropriate multi-modal conservative management, as some patients may respond to conservative management, potentially delaying or avoiding the need for surgery and its inherent risks (see, e.g., Bono, et al., 2011; Brox, et al., 2003; Brox, et al., 2006; Brox, et al., 2010; Corp, et al., 2021; Delitto, et al., 2015;

Engquist, et al., 2015; Evans, et al., 2023; Fairbank, et al., 2005; Froholdt, et al., 2012; Kuiper, et al., 2009; Liang, et al., 2019; Lown Institute, 2024; Minetama, et al., 2018; Minetama, et al., 2022; Nikolaidis, et al., 2010; Plener, et al., 2023; Walker, et al., 2008; Zaina, et al., 2016). We believe the clinical benefits of avoiding unnecessary and potentially harmful care outweighs any potential risks in delayed or reduced access to care.

Code of Federal Regulations (CFR): 42 CFR 417; 42 CFR 422; 42 CFR 423. *Internet-Only Manual (IOM) Citations:* CMS IOM Publication 100-02, Medicare Benefit Policy Manual; CMS IOM Publication 100-03 Medicare National Coverage Determination Manual. *Medicare Coverage Determinations:* Centers for Medicare & Medicaid Services (CMS), Medicare Coverage Database [Internet]. Baltimore, MD: CMS; updated periodically. Available at: Medicare Coverage Center (https://www.cms.gov/medicare/coverage/center?redirect=/center/coverage.asp). Accessed November 7, 2023.

V. **Related Policies**

- CPB 0016 - Back Pain: Invasive Procedures (../1_99/0016.html)
- CPB 0181 - Evoked Potential Studies (../100_199/0181.html) - for use of evoked potentials in spinal surgery
- CPB 0398 - Idiopathic Scoliosis (../300_399/0398.html)
- CPB 0411 - Bone and Tendon Graft Substitutes and Adjuncts (../400_499/0411.html) - for use of mesenchymal stem cell therapy for spinal fusion
- CPB 0591 - Intervertebral Disc Prostheses (../500_599/0591.html) - for hybrid lumbar/cervical fusion with artificial disc replacement for the management of back and neck pain/spinal disorders
- CPB 0772 - Axial Lumbar Interbody Fusion (AxiaLIF) (0772.html)
- CPB 0931 - Chiari Malformation Decompression Surgery (../900_999/0931.html) - for laminectomy as treatment for Chiari malformation

CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
Stereotactic Computer - assisted surgical navigational procedures:	
CPT codes covered if selection criteria are met:	
61783	Stereotactic computer-assisted (navigational) procedure; spinal (List separately in addition to code for primary procedure)
Stereotactic Computer - assisted surgical navigation :	
CPT codes covered if selection criteria are met:	
22840	Posterior non-segmental instrumentation (eg, Harrington rod technique, pedicle fixation across 1 interspace, atlantoaxial transarticular screw fixation, sublaminar wiring at C1, facet screw fixation) (List separately in addition to code for primary procedure)
22842	Posterior segmental instrumentation (eg, pedicle fixation, dual rods with multiple hooks and sublaminar wires); 3 to 6 vertebral segments (List separately in addition to code for primary procedure)
22844	13 or more vertebral segments (List separately in addition to code for primary procedure)
22845	Anterior instrumentation; 2 to 3 vertebral segments (List separately in addition to code for primary procedure)
22846	4 to 7 vertebral segments (List separately in addition to code for primary procedure)
22847	8 or more vertebral segments (List separately in addition to code for primary procedure)
22853	Insertion of interbody biomechanical device(s) (eg, synthetic cage, mesh) with integral anterior instrumentation for device anchoring (eg, screws, flanges), when performed, to intervertebral disc space in conjunction with interbody arthrodesis, each interspace (List separately in addition to code for primary procedure)

Code	Code Description
22854	Insertion of intervertebral biomechanical device(s) (eg, synthetic cage, mesh) with integral anterior instrumentation for device anchoring (eg, screws, flanges), when performed, to vertebral corpectomy(ies) (vertebral body resection, partial or complete) defect, in conjunction with interbody arthrodesis, each contiguous defect (List separately in addition to code for primary procedure)
22859	Insertion of intervertebral biomechanical device(s) (eg, synthetic cage, mesh, methylmethacrylate) to intervertebral disc space or vertebral body defect without interbody arthrodesis, each contiguous defect (List separately in addition to code for primary procedure)
Cervical laminectomy (and/or an anterior cervical discectomy, corpectomy, and cervical fusion) for herniated disc:	
CPT codes covered if selection criteria are met:	
22548	Arthrodesis, anterior transoral or extraoral technique, clivus-C1-C2 (atlas-axis), with or without excision of odontoid process
22551	Arthrodesis, anterior interbody, including disc space preparation,
22552	Arthrodesis, anterior interbody, including disc space preparation, discectomy, osteophytectomy and decompression of spinal cord and/or nerve roots; cervical below C2; each additional interspace
22554	Arthrodesis, anterior interbody technique; including minimal discectomy to prepare interspace (other than for decompression); cervical below c2
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
63040	Laminotomy (hemilaminectomy), with decompression of nerve root(s), including partial facetectomy, foraminotomy and/or excision of herniated intervertebral disc, reexploration, single interspace; cervical
+ 63043	each additional cervical interspace (List separately in addition to code for primary procedure)

Code	Code Description
63075	Discectomy, anterior, with decompression of spinal cord and/or nerve root(s), including osteophytectomy; cervical, single interspace
+63076	Cervical, each additional interspace (List separately in addition to code for primary procedure)
63081	Vertebral corpectomy (vertebral body resection), partial or complete, anterior approach with decompression of spinal cord and/or nerve root(s); cervical, single segment
+63082	cervical, each additional segment (List separately in addition to code for primary procedure)
Other CPT codes related to the CPB:	
+ 63035	Laminotomy (hemilaminectomy), with decompression of nerve root(s), including partial facetectomy, foraminotomy and/or excision of herniated intervertebral disc; each additional interspace, cervical or lumbar (List separately in addition to code for primary procedure)
83036	Hemoglobin; glycosylated (A1C)
83037	glycosylated (A1C) by device cleared by FDA for home use
99406	Smoking and tobacco use cessation counseling visit; intermediate, greater than 3 minutes up to 10 minutes
99407	intensive, greater than 10 minutes
ICD-10 codes covered if selection criteria are met:	
C41.2	Malignant neoplasm of vertebral column
D16.6	Benign neoplasm of vertebral column
M25.78	Osteophyte, vertebrae [of spine causing spinal cord or nerve root compression, confirmed by imaging studies]
M46.31	Infection of intervertebral disc (pyogenic), occipito-atlanto-axial region
M46.32	Infection of intervertebral disc (pyogenic), cervical region
M46.33	Infection of intervertebral disc (pyogenic), cervicothoracic region
M50.00 - M50.03	Cervical disc disorder with myelopathy

Code	Code Description
M50.20 - M50.23	Other cervical disc displacement
M53.2X1 - M53.2X3	Spinal instabilities, cervical region
M53.81	Other specified dorsopathies, occipito-atlanto-axial region
M53.82	Other specified dorsopathies, cervical region
M53.83	Other specified dorsopathies, cervicothoracic region
M54.12 - M54.13	Radiculopathy, cervical and cervicothoracic region [cervical nerve root compression]
M62.81	Muscle weakness (generalized)
S12.01XA - S12.01XS	Stable burst fracture of first cervical vertebra
S12.02XA - S12.02XS	Unstable burst fracture of first cervical vertebra
S13.100A - S13.29XS	Subluxation and dislocation of cervical vertebrae

Code	Code Description
S12.000A, S12.000B, S12.001A, S12.001B, S12.01XA, S12.01XB, S12.02XA, S12.02XB, S12.030A, S12.030B, S12.031A, S12.031B, S12.040A, S12.040B, S12.041A, S12.041B, S12.090A, S12.090B, S12.091A, S12.091B, S12.100A, S12.100B, S12.101A, S12.101B, S12.130A, S12.130B, S12.131A, S12.131B, S12.14XA, S12.14XB, S12.150A, S12.150B, S12.151A, S12.151B, S12.190A, S12.190B, S12.191A,	Cervical vertebral fracture, initial encounter [Acute fracture]

Code	Code Description
S12.191B, S12.200A, S12.200B, S12.201A, S12.201B, S12.230A, S12.230B, S12.231A, S12.231B, S12.24XA, S12.24XB, S12.250A, S12.250B, S12.251A, S12.251B, S12.290A, S12.290B, S12.291A, S12.291B, S12.300A, S12.300B, S12.301A, S12.301B, S12.330A, S12.330B, S12.331A, S12.331B, S12.34XA, S12.34XB, S12.350A, S12.350B, S12.351A, S12.351B, S12.390A, S12.390B, S12.391A, S12.391B,	

Code	Code Description
S12.400A, S12.400B, S12.401A, S12.401B, S12.430A, S12.430B, S12.431A, S12.431B, S12.44XA, S12.44XB, S12.450A, S12.450B, S12.451A, S12.451B, S12.490A, S12.490B, S12.491A, S12.491B, S12.500A, S12.500B, S12.501A, S12.501B, S12.530A, S12.530B, S12.531A, S12.531B, S12.54XA, S12.54XB, S12.550A, S12.550B, S12.551A, S12.551B, S12.590A, S12.590B, S12.591A, S12.591B, S12.600A,	

Code	Code Description
S12.600B, S12.601A, S12.601B, S12.630A, S12.630B, S12.631A, S12.631B, S12.64XA, S12.64XB, S12.650A, S12.650B, S12.651A, S12.651B, S12.690A, S12.690B, S12.691A, S12.691B	
ICD-10 codes not covered if selection criteria are met:	
F17.200 - F17.299	Nicotine dependence
Z72.0	Tobacco use
Cervical laminoplasty:	
CPT codes covered if selection criteria are met:	
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
63050	Laminoplasty, cervical, with decompression of the spinal cord, 2 or more vertebral segments
63051	Laminoplasty, cervical, with decompression of the spinal cord, 2 or more vertebral segments; with reconstruction of the posterior bony elements (including the application of bridging bone graft and non-segmental fixation devices [eg, wire, suture, mini-plates], when performed)
Other CPT codes related to the CPB:	
72125 - 72127	Computed tomography, cervical spine

Code	Code Description
72141 - 72142	Magnetic resonance (eg, proton) imaging, spinal canal and contents, cervical
ICD-10 codes covered if selection criteria are met:	
M47.011 - M47.019	Anterior spinal artery compression syndromes
M48.00 - M48.03, M99.20 - M99.21, M99.30 - M99.31, M99.40 - M99.41, M99.50 - M99.51, M99.60 - M99.61, M99.70 - M99.71	Spinal stenosis of cervical region
M50.00 - M50.03	Cervical disc disorder with myelopathy
M54.12	Radiculopathy, cervical region
ICD-10 codes not covered for indications listed in the CPB (not all-inclusive):	
M40.00 - M40.03, M40.10 - M40.13, M40.202 - M40.203, M40.292 - M40.293	Cervical kyphosis

Code	Code Description
M46.41 - M46.43, M50.80 - M50.83, M50.90 - M50.93	Other and unspecified cervical disc disorders
M47.11 - M47.13	Cervical spondylosis with myelopathy
M47.21 - M47.23, M47.811 - M47.813, M47.891 - M47.893	Cervical spondylosis without myelopathy
M50.10 - M50.13, M54.11, M54.13	Cervical disc disorder with radiculopathy
M50.20 - M50.23	Other cervical disc displacement
M50.30 - M50.33	Other cervical disc degeneration
M53.0	Cervicocranial syndrome
M53.1	Cervicobrachial syndrome
M53.81 - M53.83	Other specified dorsopathies
M54.2	Cervicalgia
M54.81	Occipital neuralgia
M54.89 - M54.9	Other and unspecified dorsalgia
M62.40 - M62.49, M62.830 - M62.838	Contracture of muscle

Code	Code Description
M96.1	Postlaminectomy syndrome, not elsewhere classified
M99.10 - M99.11	Subluxation complex, cervical
Q76.412 - Q76.413	Congenital cervical kyphosis
S13.100A - S13.29XS	Subluxation and dislocation of cervical vertebra
S13.4XXA – S13.4XXS, S13.8XXA – S13.8XXS, S16.1XXA – S16.1XXS	Sprain and strain of ligaments of cervical spine
Thoracic laminectomy (and/or thoracic discectomy and fusion):	
CPT codes covered if selection criteria are met:	
22222	Osteotomy of spine, including discectomy, anterior approach, single vertebral segment; thoracic
22532	Arthrodesis, lateral extracavitary technique, including minimal discectomy to prepare interspace (other than for decompression); thoracic
+22534	each additional vertebral segment (List separately in addition to code for primary procedure)
22556	Arthrodesis, anterior interbody technique, including minimal discectomy to prepare interspace (other than for decompression); thoracic
+22585	each additional interspace (List separately in addition to code for primary procedure)
22800	Arthrodesis, posterior, for spinal deformity, with or without cast; up to 6 vertebral segments
22802	7 to 12 vertebral segments
22804	13 or more vertebral segments
22808	Arthrodesis, anterior, for spinal deformity, with or without cast; 2 to 3 vertebral segments
22810	4 to 7 vertebral segments

Code	Code Description
22812	8 or more vertebral segments
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
63003	Laminectomy with exploration and/or decompression of spinal cord and/or cauda equina, without facetectomy, foraminotomy or discectomy (eg, spinal stenosis), 1 or 2 vertebral segments; thoracic
63016	more than 2 vertebral segments; thoracic
63046	Laminectomy, facetectomy and foraminotomy (unilateral or bilateral with decompression of spinal cord, cauda equina and/or nerve root[s], [eg, spinal or lateral recess stenosis]), single vertebral segment; thoracic
+63048	each additional segment, cervical, thoracic, or lumbar (List separately in addition to code for primary procedure)
63077	Discectomy, anterior, with decompression of spinal cord and/or nerve root(s), including osteophytectomy; thoracic, single interspace
+63078	each additional interspace (List separately in addition to code for primary procedure)
Other CPT codes related to the CPB:	
83036	Hemoglobin; glycosylated (A1C)
83037	glycosylated (A1C) by device cleared by FDA for home use
ICD-10 codes covered if selection criteria are met:	
C41.2	Malignant neoplasm of vertebral column
D16.6	Benign neoplasm of vertebral column
G54.3	Thoracic root disorders, not elsewhere classified [nerve root compression]
M24.28	Disorder of ligament vertebrae [spinal ligament hypertrophy]
M46.33	Infection of intervertebral disc (pyogenic), cervicothoracic region
M46.34	Infection of intervertebral disc (pyogenic), thoracic region
M46.35	Infection of intervertebral disc (pyogenic), thoracolumbar region

Code	Code Description
M47.14 - M47.15	Others pondylosis with myelopathy thoracic or thoracolumar region
M48.00 - M48.08	Spinal stenosis
M48.9	Spondylopathy, unspecified
M51.04 - M51.05	Intervertebral disc disorders with myelopathy, thoracic or thoracolumbar region
M51.24 - M51.25	Other intervertebral disc displacement, thoracic or thoracolumbar region
M96.1	Postlaminectomy syndrome, not elsewhere classified [thoracic region]
S14.2XXA – S14.2XXS, S24.2XXA – S24.2XXS, S34.21XA – S34.22XS, S34.4XXA – S34.4XXS	Injury to multiple sites of nerve roots and spinal plexus
S22.000A – S22.089S	Thoracic fracture [Acute fracture]

Code	Code Description
S23.101A – S23.101S, S23.111A – S23.111S, S23.121A – S23.121S, S23.123A – S23.123S, S23.131A – S23.131S, S23.133A – S23.133S, S23.141A – S23.141S, S23.143A – S23.143S, S23.151A – S23.151S, S23.153A – S23.153S, S23.161A – S23.161S, S23.163A – S23.163S, S23.171A – S23.171S	Dislocation of thoracic vertebra
Lumbar laminectomy for herniated disc:	
CPT codes covered if selection criteria are met:	
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
63030	Laminotomy (hemilaminectomy), with decompression of nerve root(s), including partial facetectomy, foraminotomy and/or excision of herniated intervertebral disc; 1 interspace, lumbar

Code	Code Description
63042	Laminotomy (hemilaminectomy), with decompression of nerve root(s), including partial facetectomy, foraminotomy and/or excision of herniated intervertebral disc, reexploration, single interspace; lumbar
+ 63044	each additional lumbar interspace (List separately in addition to code for primary procedure)
63047	Laminectomy, facetectomy and foraminotomy (unilateral or bilateral with decompression of spinal cord, cauda equina and/or nerve root[s], [eg, spinal or lateral recess stenosis]), single vertebral segment; lumbar
63056	Transpedicular approach with decompression of spinal cord, equina and /or nerve root(s) (eg, herniated intervertebral disc), single segment; lumbar (including transfacet, or lateral extraforaminal approach) (eg, far lateral herniated intervertebral disc)
Other CPT codes related to the CPB:	
+ 63035	Laminotomy (hemilaminectomy), with decompression of nerve root(s), including partial facetectomy, foraminotomy and/or excision of herniated intervertebral disc, including open and endoscopically-assisted approaches; each additional interspace, cervical or lumbar (List separately in addition to code for primary procedure)
+ 63048	Laminectomy, facetectomy and foraminotomy (unilateral or bilateral with decompression of spinal cord, cauda equina and/or nerve root[s], [eg, spinal or lateral recess stenosis]), single vertebral segment; each additional segment, cervical, thoracic, or lumbar (List separately in addition to code for primary procedure)
+ 63057	Transpedicular approach with decompression of spinal cord, equina and/or nerve(s) (eg, herniated intervertebral disc), single segment; each additional segment, thoracic or lumbar (List separately in addition to code for primary procedure)
83036	Hemoglobin; glycosylated (A1C)
83037	glycosylated (A1C) by device cleared by FDA for home use

Code	Code Description
ICD-10 codes covered if selection criteria are met:	
C41.2	Malignant neoplasm of vertebral column
D16.6	Benign neoplasm of vertebral column
G83.4	Cauda equina syndrome
M46.35	Infection of intervertebral disc (pyogenic), thoracolumbar region
M46.36	Infection of intervertebral disc (pyogenic), lumbar region
M46.37	Infection of intervertebral disc (pyogenic), lumbosacral region
M51.06	Intervertebral disc disorders with myelopathy, lumbar region
M51.26 - M51.27	Other intervertebral disc displacement, lumbar and lumbosacral region
M62.81	Muscle weakness (generalized)

Code	Code Description
S32.000A, S32.000B, S32.001A, S32.001B, S32.002A, S32.002B, S32.008A, S32.008B, S32.009A, S32.009B, S32.010A, S32.010B, S32.011A, S32.011B, S32.012A, S32.012B, S32.018A, S32.018B, S32.019A, S32.019B, S32.020A, S32.020B, S32.021A, S32.021B, S32.022A, S32.022B, S32.028A, S32.028B, S32.029A, S32.029B, S32.030A, S32.030B, S32.031A, S32.031B, S32.032A, S32.032B, S32.038A,	Lumbar vertebral fracture [Acute fracture]

Code	Code Description
S32.038B, S32.039A, S32.039B, S32.040A, S32.040B, S32.041A, S32.041B, S32.042A, S32.042B, S32.048A, S32.048B, S32.049A, S32.049B, S32.050A, S32.050B, S32.051A, S32.051B, S32.052A, S32.052B, S32.058A, S32.058B, S32.059A, S32.059B	
Cervical, thoracic, lumbar or sacral laminectomy other than for herniated disc:	
CPT codes covered if selection criteria are met:	
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
63001	Laminectomy with exploration and/or decompression of spinal cord and/or cauda equina, without facetectomy, foraminotomy or discectomy (eg, spinal stenosis), 1 or 2 vertebral segments; cervical
63003	thoracic
63005	lumbar, except for spondylolisthesis
63011	sacral

Code	Code Description
63012	Laminectomy with removal of abnormal facets and/or pars inter-articularis with decompression of cauda equina and nerve roots for spondylolisthesis, lumbar (Gill type procedure)
63015	Laminectomy with exploration and/or decompression of spinal cord and/or cauda equina, without facetectomy, foraminotomy or discectomy (eg, spinal stenosis), more than 2 vertebral segments; cervical
63016	thoracic
63017	lumbar
63030	1 interspace, lumbar
+ 63035	each additional interspace, cervical or lumbar (List separately in addition to code for primary procedure)
63040	Laminotomy (hemilaminectomy), with decompression of nerve root(s), including partial facetectomy, foraminotomy and/or excision of herniated intervertebral disc, reexploration, single interspace; cervical
63042	lumbar
+ 63043	each additional cervical interspace (List separately in addition to code for primary procedure)
+ 63044	each additional lumbar interspace (List separately in addition to code for primary procedure)
63045	Laminectomy, facetectomy and foraminotomy (unilateral or bilateral with decompression of spinal cord, cauda equina and/or nerve root[s], [eg, spinal or lateral recess stenosis]), single vertebral segment; cervical
63046	thoracic
63047	lumbar
+ 63048	each additional segment, cervical, thoracic, or lumbar (List separately in addition to code for primary procedure)

Code	Code Description
63052	Laminectomy, facetectomy, or foraminotomy (unilateral or bilateral with decompression of spinal cord, cauda equina and/or nerve root[s] [eg, spinal or lateral recess stenosis]), during posterior interbody arthrodesis, lumbar; single vertebral segment (List separately in addition to code for primary procedure)
63053	Laminectomy, facetectomy, or foraminotomy (unilateral or bilateral with decompression of spinal cord, cauda equina and/or nerve root[s] [eg, spinal or lateral recess stenosis]), during posterior interbody arthrodesis, lumbar; each additional segment (List separately in addition to code for primary procedure)
63055	Transpedicular approach with decompression of spinal cord, equina and /or nerve root(s) (eg, herniated intervertebral disc), single segment; thoracic
63056	lumbar (including transfacet, or lateral extraforaminal approach) (eg, far lateral herniated intervertebral disc)
+ 63057	each additional segment, thoracic or lumbar (List separately in addition to code for primary procedure)
63200	Laminectomy, with release of tethered spinal cord, lumbar
63265	Laminectomy for excision or evacuation of intraspinal lesion other than neoplasm, extradural; cervical
63266	thoracic
63267	lumbar
63268	sacral
63273	Laminectomy for excision of intraspinal lesion other than neoplasm, intradural; sacral
63278	Laminectomy for biopsy/excision of intraspinal neoplasm; extradural, sacral
63283	intradural, sacral
Other CPT codes related to the CPB:	
83036	Hemoglobin; glycosylated (A1C)
83037	glycosylated (A1C) by device cleared by FDA for home use

Code	Code Description
ICD-10 codes covered if selection criteria are met:	
C41.2	Malignant neoplasm of vertebral column, excluding sacrum and coccyx
C70.1	Malignant neoplasm of spinal meninges
C72.0 - C72.1	Malignant neoplasm of spinal cord and cauda equina
C79.31 - C79.32	Secondary malignant neoplasm of brain and spinal cord
C79.49	Secondary malignant neoplasm of other parts of nervous system
C79.51 - C79.52	Secondary malignant neoplasm of bone and bone marrow
D16.6	Benign neoplasm of vertebral column, excluding sacrum and coccyx
D32.1	Benign neoplasm of spinal meninges
D33.4	Benign neoplasm of spinal cord
D42.0 - D42.9	Neoplasm of uncertain behavior of meninges
D43.0 - D43.2, D43.4	Neoplasm of uncertain behavior of brain and spinal cord
D48.0	Neoplasm of uncertain behavior of bone and articular cartilage
G06.1	Intraspinal abscess and granuloma
I62.00 - I62.03	Nontraumatic subdural hemorrhage
I62.1	Nontaumatice extradural hemorrhage
M46.20 - M46.39	Osteomyelitis of vertebra and infection of intervertebral disc (pyogenic) [spinal]
M48.00 - M48.08	Spinal stenosis
M71.38	Other bursal cyst, other site [of spine causing spinal cord or nerve root compression, confirmed by imaging studies (e.g., CT or MRI) and with corresponding neurological deficit]
M86.18, M86.28, M86.68	Other acute, subacute and other chonic osteomyelitis, other site [spinal]

Code	Code Description
S14.0XXA – S14.159S	Injury of spinal cord at neck level [causing spinal cord or nerve root compression, confirmed by imaging studies (e.g., CT or MRI) and with corresponding neurological deficit]
S24.0XXA – S24.159S	Injury of spinal cord at thorax level [causing spinal cord or nerve root compression, confirmed by imaging studies (e.g., CT or MRI) and with corresponding neurological deficit]
S31.000A - S31.000S	Unspecified open wound of lower back and pelvis without penetration into retroperitoneum
S32.000A – S32.000S	Fracture of lumbar vertebra
S33.100A – S33.141S	Subluxation and dislocation of lumbar vertebra
S34.101A – S34.139S	Other and unspecified injury of lumbar and sacral spinal cord
ICD-10 codes not covered if selection criteria are met:	
I82.C11 - I82.C29	Embolism and thrombosis of internal jugular vein [Internal jugular vein stenosis]
Cervical, lumbar, sacral or thoracic laminectomy for Tarlov cysts:	
CPT codes covered if selection criteria are met:	
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
63273	Laminectomy for excision of intraspinal lesion other than neoplasm, intradural; sacral
63295	Osteoplastic reconstruction of dorsal spinal elements, following primary intraspinal procedure (List separately in addition to code for primary procedure)
ICD-10 codes covered if selection criteria are met:	
G96.191	Perineural cyst
Lumbar decompression:	
CPT codes covered if selection criteria are met:	
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)

Code	Code Description
62330	Decompression, percutaneous, with partial removal of the ligamentum flavum, including laminotomy for access, epidurography, and imaging guidance (ie, CT or fluoroscopy), bilateral; one interspace, lumbar
62331	additional interspace(s), lumbar (List separately in addition to code for primary procedure)
63005	Laminectomy with exploration and/or decompression of spinal cord and/or cauda equina, without facetectomy, foraminotomy or discectomy (eg, spinal stenosis), 1 or 2 vertebral segments; lumbar, except for spondylolisthesis
63012	Laminectomy with removal of abnormal facets and/or pars inter-articularis with decompression of cauda equina and nerve roots for spondylolisthesis, lumbar (Gill type procedure)
63017	Laminectomy with exploration and/or decompression of spinal cord and/or cauda equina, without facetectomy, foraminotomy or discectomy (eg, spinal stenosis), more than 2 vertebral segments; lumbar
63030	Laminotomy (hemilaminectomy), with decompression of nerve root(s), including partial facetectomy, foraminotomy and/or excision of herniated intervertebral disc; 1 interspace, lumbar
Other CPT codes related to the CPB:	
+ 63035	Laminotomy (hemilaminectomy), with decompression of nerve root(s), including partial facetectomy, foraminotomy and/or excision of herniated intervertebral disc; each additional interspace, cervical or lumbar (List separately in addition to code for primary procedure)
ICD-10 codes covered if selection criteria are met:	
G83.4	Cauda equina syndrome
M21.371 - M21.379	Foot drop
M48.061 - M48.062	Spinal stenosis, lumbar region

Code	Code Description
Vertebral Corpectomy:	
CPT codes covered if selection criteria are met:	
22818	Kyphectomy, circumferential exposure of spine and resection of vertebral segment(s) (including body and posterior elements); single or 2 segments
22819	3 or more segments
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
63081 - 63091	Vertebral corpectomy (vertebral body resection), partial or complete
63101 - 63103	Vertebral corpectomy (vertebral body resection), partial or complete, lateral extracavitary approach with decompression of spinal cord and/or nerve root(s) (eg, for tumor or retropulsed bone fragments)
ICD-10 codes covered if selection criteria are met:	
C41.2	Malignant neoplasm of vertebral column, excluding sacrum and coccyx
C79.51 - C79.52	Secondary malignant neoplasm of bone and bone marrow
D16.6	Benign neoplasm of vertebral column, excluding sacrum and coccyx
M48.00 - M48.08	Spinal stenosis
M48.50xA - M48.58xS	Collapsed vertebra, not elsewhere classified

Code	Code Description
S22.000A - S22.000S, S22.010A - S22.010S, S22.020A - S22.020S, S22.030A - S22.030S, S22.040A - S22.040S, S22.050A - S22.050S, S22.060A - S22.060S, S22.070A - S22.070S, S22.080A - S22.080S, S32.000A - S32.000S, S32.010A - S32.010S, S32.020A - S32.020S, S32.030A - S32.030S, S32.040A - S32.040S, S32.050A - S32.050S	Wedge compression fracture of thoracic vertebra

Code	Code Description
S12.01xA - S12.01xS, S12.02xA - S12.02xS, S22.001A - S22.001S, S22.002A - S22.002S, S22.011A - S22.011S, S22.012A - S22.012S, S22.021A - S22.021S, S22.022A - S22.022S, S22.031A - S22.031S, S22.032A - S22.032S, S22.041A - S22.041S, S22.042A - S22.042S, S22.051A - S22.051S, S22.052A - S22.052S, S22.061A - S22.061S, S22.062A - S22.062S, S22.071A - S22.071S, S22.072A - S22.072S, S22.081A -	Burst fracture

Code	Code Description
S22.081S, S22.082A - S22.082S, S32.001A - S32.001S, S32.002A - S32.002S, S32.011A - S32.011S, S32.012A - S32.012S, S32.021A - S32.021S, S32.022A - S32.022S, S32.031A - S32.031S, S32.032A - S32.032S, S32.041A - S32.041S, S32.042A - S32.042S, S32.051A - S32.051S, S32.052A - S32.052S	
Cervical spinal fusion:	
CPT codes covered if selection criteria are met:	
22548	Arthrodesis, anterior transoral or extraoral technique, clivus-C1-C2 (atlas-axis), with or without excision of odontoid process
22551	Arthrodesis, anterior interbody, including disc space preparation, discectomy, osteophytectomy and decompression of spinal cord and/or nerve roots; cervical below C2
+22552	cervical below C2, each additional interspace (List separately in addition to code for separate procedure)

Code	Code Description
22554	Arthrodesis, anterior interbody technique, including minimal discectomy to prepare interspace (other than for decompression); cervical below C2
+22585	each additional interspace (List separately in addition to code for primary procedure)
22590	Arthrodesis, posterior technique, craniocervical (occiput-C2)
22595	Arthrodesis, posterior technique, atlas-axis (C1-C2)
22600	Arthrodesis, posterior or posterolateral technique, single level; cervical below C2 segment
+22614	each additional vertebral segment (List separately in addition to code for primary procedure)
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
Other CPT codes related to the CPB:	
22830	Exploration of spinal fusion
80323	Alkaloids, not otherwise specified [Blood or Urinary Nicotine]
83036	Hemoglobin; glycosylated (A1C)
83037	glycosylated (A1C) by device cleared by FDA for home use
ICD-10 codes covered if selection criteria are met:	
C41.2	Malignant neoplasm of vertebral column [cervical]
C72.0 - C72.1	Malignant neoplasm of spinal cord and cauda equina [cervical]
C79.49	Secondary malignant neoplasm of other parts of nervous system [cervical spinal cord]
C79.51 - C79.52	Secondary malignant neoplasm of bone and bone marrow [cervical spine]
D16.6	Benign neoplasm of vertebral column [cervical]
D33.4	Benign neoplasm of spinal cord [cervical]
D43.4	Neoplasm of uncertain behavior of spinal cord [cervical]
D48.0	Neoplasm of uncertain behavior of bone and articular cartilage [cervical spine]
D49.2	Neoplasm of unspecified behavior of bone, soft tissue, and skin

Code	Code Description
D49.7	Neoplasm of unspecified behavior of endocrine glands and other parts of nervous system [cervical spinal cord]
E76.01 - E76.03	Mucopolysaccharidosis, type I
E76.1	Mucopolysaccharidosis, type II
E76.210 - E76.29	Other mucopolysaccharidoses
E76.3	Mucopolysaccharidosis, unspecified
G06.1	Intraspinal abscess and granuloma [cervical]
G54.2	Cervical root disorders, not elsewhere classified [nerve root compression]
G95.20 - G95.29	Other and unspecified cord compression [cervical cord compression]
M05.00 - M05.9	Rheumatoid arthritis with rheumatoid factor
M06.00 - M06.9	Other rheumatoid arthritis
M40.03, M40.12 - M40.13, M40.202 - M40.203, M40.292 - M40.293	Kyphosis [cervical]
M40.40, M40.50	Lordosis [cervical]

Code	Code Description
M41.02 - M41.03, M41.112 - M41.113, M41.122 - M41.123, M41.22 - M41.23, M41.41 - M41.43, M41.52 - M41.53, M41.82 - M41.83	Scoliosis [cervical]
M43.11 - M43.13	Spondylolisthesis, occipito-atlanto-axial, cervical and cervicothoracic region [subaxial spondylolisthesis]
M43.3	Recurrent atlantoaxial dislocation with myelopathy
M43.4	Other recurrent atlantoaxial dislocation
M46.31	Infection of intervertebral disc (pyogenic), occipito-atlanto-axial region
M46.32	Infection of intervertebral disc (pyogenic), cervical region
M46.33	Infection of intervertebral disc (pyogenic), cervicothoracic region
M47.011 - M47.013, M47.021 - M47.022, M47.11 - M47.13, M47.21 - M47.23, M47.811 - M47.813, M47.891 - M47.893	Spondylosis [cervical cord compression]

Code	Code Description
M48.01 - M48.03	Spinal stenosis, [cervical region]
M48.31	Traumatic spondylopathy, occipito-atlanto-axial region
M50.00 - M50.93	Cervical disc disorders [cervical cord compression]
M53.2X1	Spinal instabilities, osccipito-atlanto-axial region
M54.11 - M54.13	Radiculopathy [cervical nerve root compression]
M62.81	Muscle weakness (generalized)
M66.18	Rupture of synovium, other site [synovial cyst]
M71.38	Other bursal cyst, other site [synovial cyst]
M96.0	Pseudarthrosis after fusion or arthrodesis [cervical]
M96.1	Postlaminectomy syndrome, not elsewhere classified [cervical]
Q67.5	Congenital deformity of spine [cervical]
Q75.9	Congenital malformation of skull and face bones, unspecified [basilar invagination]
Q76.2	Congenital spondylolisthesis [cervical]
Q76.3	Congenital scoliosis due to congenital bony malformation [cervical]
Q76.49	Other congenital malformations of spine, not associated with scoliosis [cervical]
Q90.0 - Q90.9	Down syndrome
S12.000A – S12.9XXS	Fracture of cervical vertebra and other parts of neck

Code	Code Description
S12.000A, S12.000B, S12.001A, S12.001B, S12.01XA, S12.01XB, S12.02XA, S12.02XB, S12.030A, S12.030B, S12.031A, S12.031B, S12.040A, S12.040B, S12.041A, S12.041B, S12.090A, S12.090B, S12.091A, S12.091B, S12.100A, S12.100B, S12.101A, S12.101B, S12.130A, S12.130B, S12.131A, S12.131B, S12.14XA, S12.14XB, S12.150A, S12.150B, S12.151A, S12.151B, S12.190A, S12.190B, S12.191A,	Cervical vertebral fracture, initial encounter [Acute fracture]

Code	Code Description
S12.191B, S12.200A, S12.200B, S12.201A, S12.201B, S12.230A, S12.230B, S12.231A, S12.231B, S12.24XA, S12.24XB, S12.250A, S12.250B, S12.251A, S12.251B, S12.290A, S12.290B, S12.291A, S12.291B, S12.300A, S12.300B, S12.301A, S12.301B, S12.330A, S12.330B, S12.331A, S12.331B, S12.34XA, S12.34XB, S12.350A, S12.350B, S12.351A, S12.351B, S12.390A, S12.390B, S12.391A, S12.391B,	

Code	Code Description
S12.400A, S12.400B, S12.401A, S12.401B, S12.430A, S12.430B, S12.431A, S12.431B, S12.44XA, S12.44XB, S12.450A, S12.450B, S12.451A, S12.451B, S12.490A, S12.490B, S12.491A, S12.491B, S12.500A, S12.500B, S12.501A, S12.501B, S12.530A, S12.530B, S12.531A, S12.531B, S12.54XA, S12.54XB, S12.550A, S12.550B, S12.551A, S12.551B, S12.590A, S12.590B, S12.591A, S12.591B, S12.600A,	

Code	Code Description
S12.600B, S12.601A, S12.601B, S12.630A, S12.630B, S12.631A, S12.631B, S12.64XA, S12.64XB, S12.650A, S12.650B, S12.651A, S12.651B, S12.690A, S12.690B, S12.691A, S12.691B	
S13.0XXA – S13.0XXS	Dislocation and sprain of joints and ligaments at neck level
S13.4XXA - S13.4XXS	Sprain of ligaments of cervical spine
S14.0XXA – S14.9XXS	Injury of nerves and spinal cord at neck level
T84.418A - T84.418S	Breakdown (mechanical) of other internal orthopedic devices, implants and grafts [hardware failure]
T84.428A - T84.428S	Displacement of other internal orthopedic devices, implants and grafts [hardware failure]
T84.498A - T84.498S	Other mechanical complication of other internal orthopedic devices, implants and grafts [hardware failure]
Z98.1	Arthrodesis status [non-union of prior fusion] [cervical]
ICD-10 codes not covered for indications listed in the CPB (not all inclusive):	
F17.200 - F17.299	Nicotine dependence
Z72.0	Tobacco use

Code	Code Description
Thoracic spinal fusion:	
CPT codes covered if selection criteria are met:	
22532	Arthrodesis, lateral extracavitary technique, including minimal discectomy to prepare interspace (other than for decompression); thoracic
+22534	thoracic or lumbar, each additional vertebral segment (List separately in addition to code for primary procedure)
22556	Arthrodesis, anterior interbody technique, including minimal discectomy to prepare interspace (other than for decompression); thoracic
22610	Arthrodesis, posterior or posterolateral technique, single level; thoracic (with lateral transverse technique, when performed)
+22614	each additional vertebral segment (List separately in addition to code for primary procedure)
22800	Arthrodesis, posterior, for spinal deformity, with or without cast; up to 6 vertebral segments
22802	7 to 12 vertebral segments
22804	13 or more vertebral segments
22808	Arthrodesis, anterior, for spinal deformity, with or without cast; 2 to 3 vertebral segments
22810	4 to 7 vertebral segments
22812	8 or more vertebral segments
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
Other CPT codes related to the CPB:	
22830	Exploration of spinal fusion
80323	Alkaloids, not otherwise specified [Blood or Urinary Nicotine]
83036	Hemoglobin; glycosylated (A1C)
83037	glycosylated (A1C) by device cleared by FDA for home use
97110 - 97546	Therapeutic procedures

Code	Code Description
ICD-10 codes covered if selection criteria are met:	
C41.2	Malignant neoplasm of vertebral column, excluding sacrum and coccyx
C72.0	Malignant neoplasm of spinal cord
C79.31 - C79.32	Secondary malignant neoplasm of brain and spinal cord [thoracic spinal cord]
C79.51 - C79.52	Secondary malignant neoplasm of bone and bone marrow [spine, spinal (column)]
D16.6	Benign neoplasm of vertebral column
D33.4	Benign neoplasm of spinal cord
D43.0 - D43.2, D43.4	Neoplasm of uncertain behavior of brain and spinal cord
D48.0	Neoplasm of uncertain behavior of bone and articular cartilage [spine, spinal (column)]
D49.2	Neoplasms of unspecified nature of bone, soft tissue, and skin [spine, spinal (column)]
D49.7	Neoplasm of unspecified behavior of endocrine glands and other parts of nervous system [thoracic spinal cord]
G06.1	Intraspinal abscess
M40.00 - M40.299, M96.2 - M96.3	Kyphosis (acquired)
M41.112 - M41.9	Kyphosis and scoliosis
M43.13	Spondylolisthesis, cervicothoracic region
M43.14	Spondylolisthesis, thoracic region
M43.15	Spondylolisthesis, thoracolumbar region
M46.33	Infection of intervertebral disc (pyogenic), cervicothoracic region
M46.34	Infection of intervertebral disc (pyogenic), thoracic region
M46.35	Infection of intervertebral disc (pyogenic), thoracolumbar region
M47.14	Other spondylosis with myelopathy, thoracic region

Code	Code Description
M48.04 - M48.05	Spinal stenosis thoracic region
M51.04 - M51.05	Intervertebral disc disorders with myelopathy, thoracic region
M51.24 - M51.25	Other intervertebral disc displacement, thoracic region
M53.2X1- M53.2X9	Spinal instabilities
M53.84	Other specified dorsopathies, thoracic region [locked facets]
M54.14 - M54.15	Radiculopathy, thoracic region
M96.0	Pseudarthrosis after fusion or arthrodesis
S12.000A – S12.691S, S22.000A – S22.089S, S32.000A – S32.2XXS	Fracture of spinal column, [open, closed, non-union, with or without spinal cord injury]
S21.201A – S21.259S, S21.401A – S21.95XS	Open wound of back wall of thorax without or with penetration into thoracic cavity [allowed when billed with S23.1xx codes only]

Code	Code Description
S23.101A – S23.101S, S23.111A – S23.111S, S23.121A – S23.121S, S23.123A – S23.123S, S23.131A – S23.131S, S23.133A – S23.133S, S23.141A – S23.141S, S23.143A – S23.143S, S23.151A – S23.151S, S23.153A – S23.153S, S23.161A – S23.161S, S23.163A – S23.163S, S23.171A – S23.171S, S23.29XA – S23.29XS	Dislocation of thoracic vertebra [open dislocation must be billed with S21.2xx or S21.4xx codes]
S24.101A – S24.159S	Other and unspecified injury of thoracic spinal cord
T84.418A - T84.418S	Breakdown (mechanical) of other internal orthopedic devices, implants and grafts [hardware failure]
T84.428A - T84.428S	Displacement of other internal orthopedic devices, implants and grafts [hardware failure]
T84.498A - T84.498S	Other mechanical complication of other internal orthopedic devices, implants and grafts [hardware failure]

Code	Code Description
Z98.1	Arthrodesis status
ICD-10 codes not covered for indications listed in the CPB (not all inclusive):	
F17.200 - F17.299	Nicotine dependence
Z72.0	Tobacco use
Lumbar spinal fusion:	
CPT codes covered if selection criteria are met:	
22533	Arthrodesis, lateral extracavitary technique, including minimal discectomy to prepare interspace (other than for decompression); lumbar
22558	Arthrodesis, anterior interbody technique, including minimal discectomy to prepare interspace (other than for decompression); lumbar
22612	Arthrodesis, posterior or posterolateral technique, single level; lumbar (with or without lateral transverse technique)
+ 22614	each additional vertebral segment (List separately in addition to code for primary procedure)
22630	Arthrodesis, posterior interbody technique, including laminectomy and/or discectomy to prepare interspace (other than for decompression), single interspace; lumbar
22632	each additional interspace (list separately in addition to code for primary procedure)
22633	Arthrodesis, combined posterior or posterolateral technique with posterior interbody technique, including laminectomy and/or discectomy sufficient to prepare interspace (other than for decompression), single interspace; lumbar
22634	each additional interspace and segment (List separately in addition to code for primary procedure)
22800	Arthrodesis, posterior, for spinal deformity, with or without cast; up to 6 vertebral segments
22802	7 to 12 vertebral segments
22804	13 or more vertebral segments
22808	Arthrodesis, anterior, for spinal deformity, with or without cast; 2 to 3 vertebral segments

Code	Code Description
22810	4 to 7 vertebral segments
22812	8 or more vertebral segments
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
Other CPT codes related to the CPB:	
20225	Biopsy, bone, trocar, or needle; deep (eg, vertebral body, femur)
20250	Biopsy, vertebral body, open; thoracic
20251	Biopsy, vertebral body, open; lumbar or cervical
22207	Osteotomy of spine, posterior or posterolateral approach, 3 columns, 1 vertebral segment (eg, pedicle/vertebral body subtraction); lumbar
+ 22208	each additional vertebral segment (List separately in addition to code for primary procedure)
22214	Osteotomy of spine, posterior or posterolateral approach, 1 vertebral segment; lumbar
+ 22216	each additional vertebral segment (List separately in addition to primary procedure)
22224	Osteotomy of spine, including discectomy, anterior approach, single vertebral segment; lumbar
22830	Exploration of spinal fusion
22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
72125	Computed tomography, cervical spine; without contrast material
72126	with contrast material
72127	without contrast material, followed by contrast material(s) and further sections
72128	Computed tomography, thoracic spine; without contrast material
72129	with contrast material
72130	without contrast material, followed by contrast material(s) and further sections

Code	Code Description
72131	Computed tomography, lumbar spine; without contrast material
72132	with contrast material
72133	without contrast material, followed by contrast material(s) and further sections
72141	Magnetic resonance (eg, proton) imaging, spinal canal and contents, cervical; without contrast material
72142	with contrast material(s)
72146	Magnetic resonance (eg, proton) imaging, spinal canal and contents, thoracic; without contrast material
72147	with contrast material(s)
72148	Magnetic resonance (eg, proton) imaging, spinal canal and contents, lumbar; without contrast material
72149	with contrast material(s)
72156	Magnetic resonance (eg, proton) imaging, spinal canal and contents, without contrast material, followed by contrast material(s) and further sequences; cervical
72157	thoracic
72158	lumbar
72159	Magnetic resonance angiography, spinal canal and contents, with or without contrast material(s)
75705	Angiography, spinal, selective, radiological supervision and interpretation
78811	Positron emission tomography (PET) imaging; limited area (eg, chest, head/neck)
78812	skull base to mid-thigh
78813	whole body
78814	Positron emission tomography (PET) with concurrently acquired computed tomography (CT) for attenuation correction and anatomical localization imaging; limited area (eg, chest, head/neck)
78815	skull base to mid-thigh
78816	whole body

Code	Code Description
78830	Radiopharmaceutical localization of tumor, inflammatory process or distribution of radiopharmaceutical agent(s) (includes vascular flow and blood pool imaging, when performed); tomographic (SPECT) with concurrently acquired computed tomography (CT) transmission scan for anatomical review, localization and determination/detection of ṁ pathology, single area (eg, head, neck, chest, pelvis) or acquisition, single day imaging
78831	Radiopharmaceutical localization of tumor, inflammatory process or distribution of radiopharmaceutical agent(s) (includes vascular flow and blood pool imaging, when performed); tomographic (SPECT), minimum 2 areas (eg, pelvis and knees, chest and abdomen) or separate acquisitions (eg, lung ventilation and perfusion), single day imaging, or single area or acquisition over 2 or more days
78832	Radiopharmaceutical localization of tumor, inflammatory process or distribution of radiopharmaceutical agent(s) (includes vascular flow and blood pool imaging, when performed); tomographic (SPECT) with concurrently acquired computed tomography (CT) transmission scan for anatomical review, localization and determination/detection of pathology, minimum 2 areas (eg, pelvis and knees, chest and abdomen) or separate acquisitions (eg, lung ventilation and perfusion), single day imaging, or single area or acquisition over 2 or more days
78835	Radiopharmaceutical quantification measurement(s) single area (List separately in addition to code for primary procedure)
80323	Alkaloids, not otherwise specified [Blood or Urinary Nicotine]
83036	Hemoglobin; glycosylated (A1C)
83037	glycosylated (A1C) by device cleared by FDA for home use
97110 - 97546	Therapeutic procedures
99406	Smoking and tobacco use cessation counseling visit; intermediate, greater than 3 minutes up to 10 minutes
99407	intensive, greater than 10 minutes
Other HCPCS codes related to the CPB:	
L0625 – L0651	Lumbar-sacral orthosis [Lumbar bracing]

Code	Code Description
ICD-10 codes covered if selection criteria are met:	
C41.2	Malignant neoplasm of vertebral column, excluding sacrum and coccyx
C70.1	Malignant neoplasm of spinal meninges
C79.31 - C79.32	Secondary malignant neoplasm of brain and spinal cord
C79.49	Secondary malignant neoplasm of other parts of nervous system
C79.51 - C79.52	Secondary malignant neoplasm of bone and bone marrow
D16.6	Benign neoplasm of vertebral column
D16.8	Benign neoplasm of pelvic bones, sacrum and coccyx
D32.1	Benign neoplasm of spinal meninges
D33.4	Benign neoplasm of spinal cord
D42.0 - D42.9	Neoplasm of uncertain behavior of meninges
D43.0 - D43.2, D43.4	Neoplasm of uncertain behavior of brain and spinal cord
D48.0	Neoplasm of uncertain behavior of bone and articular cartilage
G06.1	Intraspinal abscess and granuloma
G83.4	Cauda equina syndrome
M40.35 - M40.37	Flatback syndrome, thoracolumbar, lumbar, and lumbosacral region
M40.50 - M40.57	Lordosis, unspecified
M41.00 - M41.35 M41.80 - M41.9	Scoliosis
M43.00 - M43.09	Spondylolysis
M43.15	Spondylolisthesis, thoracolumbar region
M43.16	Spondylolisthesis, lumbar region

Code	Code Description
M43.17	Spondylolisthesis, lumbosacral region
M46.20	Osteomyelitis of vertebra, site unspecified
M46.30	Infection of intervertebral disc (pyogenic), site unspecified
M46.35	Infection of intervertebral disc (pyogenic), thoracolumbar region
M46.36	Infection of intervertebral disc (pyogenic), lumbar region
M46.37	Infection of intervertebral disc (pyogenic), lumbosacral region
M48.061 - M48.07	Spinal stenosis, lumbar and lumbosacral region
M48.50xA - M48.58xS M80.08xA - M80.08xS M84.48xA - M84.48xS M84.58xA - M84.58xS M84.68xA - M84.68xS	Pathologic fracture of vertebrae
M51.06	Intervertebral disc disorders with myelopathy, lumbar region
M53.2X1 - M53.2X9	Spinal instabilities
M62.81	Muscle weakness (generalized)
M86.18	Other acute osteomyelitis, other site [spinal]
M86.28	Subacute osteomyelitis, other site [spinal]
M86.68	Other chronic osteomyelitis, other site [spinal]
M89.8X0	Other specified disorders of bone, multiple sites [Spinal bony lesion]
M89.8X8	Other specified disorders of bone, other site [Spinal bony lesion]
M96.0	Pseudoarthrosis after fusion or arthrodesis
M96.5	Postradiation scoliosis
Q76.2	Congenital spondylolisthesis

Code	Code Description
S31.000A - S31.000S	Unspecified open wound of lower back and pelvis without penetration into retroperitoneum
S32.000A - S32.059S	Fracture of lumbar vertebra
S32.10xA - S32.19xS	Fracture of sacrum
S33.100A - S33.141S	Subluxation and dislocation of lumbar vertebra
S34.101A - S34.129S	Other and unspecified injury of lumbar spinal cord
T84.418A - T84.418S	Breakdown (mechanical) of other internal orthopedic devices, implants and grafts [hardware failure]
T84.428A - T84.428S	Displacement of other internal orthopedic devices, implants and grafts [hardware failure]
T84.498A - T84.498S	Other mechanical complication of other internal orthopedic devices, implants and grafts [hardware failure]
T84.81xA - T84.89xS	Other specified complications of internal orthopedic prosthetic devices, implants and grafts [hardware failure]
Z98.1	Arthrodesis status
Numerous options	Nonunion of fracture [Codes not listed due to expanded specificity]
ICD-10 codes not covered for indications listed in the CPB (not all inclusive):	
F17.200 - F17.299	Nicotine dependence
M51.34 - M51.379	Other thoracic, thoracolumbar and lumbosacral intervertebral disc degeneration
Q76.49	Other congenital malformations of spine, not associated with scoliosis [Bertolotti’s syndrome]
Z72.0	Tobacco use

Code	Code Description
Cervical osteophytes causing dysphagia:	
CPT codes covered if selection criteria are met:	
22110	Partial excision of vertebral body, for intrinsic bony lesion, without decompression of spinal cord or nerve root(s), single vertebral segment; cervical
+22116	each additional vertebral segment (List separately in addition to code for primary procedure)
+22848	Pelvic fixation (attachment of caudal end of instrumentation to pelvic bony structures) other than sacrum (List separately in addition to code for primary procedure)
ICD-10 codes covered if selection criteria are met:	
M25.78	Osteophyte, vertebrae
R13.10 – R13.19	Dysphagia
Removal of posterior spinal instrumentation:	
CPT codes covered if selection criteria are met:	
22850	Removal of posterior nonsegmental instrumentation (eg, Harrington rod)
22852	Removal of posterior segmental instrumentation
Other CPT codes related to the CPB:	
97110 - 97546	Therapeutic procedures
ICD-10 codes covered if selection criteria are met:	
G89.11	Acute pain due to trauma
G89.18	Other acute postprocedural pain
M53.80 – M53.9	Other specified dorsopathies
M54.10 – M54.18	Radiculopathies
M54.2	Cervicalgia
M54.30 – M54.32	Sciatica
M54.40 – M54.42	Lumbago with sciatica

Code	Code Description
M54.50 – M54.59	Low back pain
M54.6	Pain in thoracic spine
M54.9	Dorsalgia, unspecified [backache]
M62.81	Muscle weakness (generalized)
M79.12	Myalgia of auxiliary muscles, head and neck
M79.18	Myalgia, other site
M79.2	Neuralgia and neuritis, unspecified
T84.296A - T84.296S	Other mechanical complication of internal fixation device of vertebrae
T84.63XA – T84.63XS	Infection and inflammatory reaction due to internal fixation device of spine
Removal of anterior spinal instrumentation:	
CPT codes covered if selection criteria are met:	
22855	Removal of anterior instrumentation
ICD-10 codes covered if selection criteria are met:	
T84.296A - T84.296S	Other mechanical complication of internal fixation device of vertebrae
T84.63XA – T84.63XS	Infection and inflammatory reaction due to internal fixation device of spine
Thoracolumbar osteotomy:	
CPT codes covered if selection criteria are met:	
22212	Osteotomy of spine, posterior or posterolateral approach, 1 vertebral segment; thoracic
22216	each additional vertebral segment (List separately in addition to primary procedure)
22226	Osteotomy of spine, including discectomy, anterior approach, single vertebral segment; each additional vertebral segment (List separately in addition to code for primary procedure)
Other CPT codes related to the CPB:	
22612	Arthrodesis, posterior or posterolateral technique, single interspace; lumbar (with lateral transverse technique, when performed)

Code	Code Description
22614	each additional interspace (List separately in addition to code for primary procedure)
22630	Arthrodesis, posterior interbody technique, including laminectomy and/or discectomy to prepare interspace (other than for decompression), single interspace, lumbar
22632	each additional interspace (List separately in addition to code for primary procedure)
22633	Arthrodesis, combined posterior or posterolateral technique with posterior interbody technique including laminectomy and/or discectomy sufficient to prepare interspace (other than for decompression), single interspace, lumbar
22634	each additional interspace (List separately in addition to code for primary procedure)
63003	Laminectomy with exploration and/or decompression of spinal cord and/or cauda equina, without facetectomy, foraminotomy or discectomy (eg, spinal stenosis), 1 or 2 vertebral segments; thoracic
63005	lumbar, except for spondylolisthesis
63012	Laminectomy with removal of abnormal facets and/or pars inter-articularis with decompression of cauda equina and nerve roots for spondylolisthesis, lumbar (Gill type procedure)
63016	Laminectomy with exploration and/or decompression of spinal cord and/or cauda equina, without facetectomy, foraminotomy or discectomy (eg, spinal stenosis), more than 2 vertebral segments; thoracic
63017	lumbar
63046	Laminectomy, facetectomy and foraminotomy (unilateral or bilateral with decompression of spinal cord, cauda equina and/or nerve root[s], [eg, spinal or lateral recess stenosis]), single vertebral segment; thoracic
63047	lumbar
63048	each additional vertebral segment, cervical, thoracic, or lumbar (List separately in addition to code for primary procedure)

Code	Code Description
63052	Laminectomy, facetectomy, or foraminotomy (unilateral or bilateral with decompression of spinal cord, cauda equina and/or nerve root[s] [eg, spinal or lateral recess stenosis]), during posterior interbody arthrodesis, lumbar; single vertebral segment (List separately in addition to code for primary procedure)
63053	each additional vertebral segment (List separately in addition to code for primary procedure)
Computer-assisted surgical navigation and Robotic assistance:	
CPT codes not covered for indications listed in the CPB:	
61783	Stereotactic computer-assisted (navigational) procedure; spinal (List separately in addition to code for primary procedure)
HCPCS codes not covered for indications listed in the CPB:	
S2900	Surgical techniques requiring use of robotic surgical system (list separately in addition to code for primary procedure)
Transection or avulsion of spinal nerve:	
CPT codes covered if selection criteria are met:	
64772	Transection or avulsion of other spinal nerve, extradural
ICD-10 codes covered if selection criteria are met:	
C41.2	Malignant neoplasm of vertebral column
C41.4	Malignant neoplasm of pelvic bones, sacrum and coccyx
C47.9	Malignant neoplasm of peripheral nerves and autonomic nervous system, unspecified
C70.1	Malignant neoplasm of spinal meninges
C72.0	Malignant neoplasm of spinal cord
C79.49	Secondary malignant neoplasm of other parts of nervous system
C79.51 – C79.52	Secondary malignant neoplasm of bone and bone marrow
D16.6	Benign neoplasm of vertebral column
D16.8	Benign neoplasm of pelvic bones, sacrum and coccyx
D32.1	Benign neoplasm of spinal meninges
D33.4	Benign neoplasm of spinal cord

Code	Code Description
D42.1	Neoplasm of uncertain behavior of spinal meninges
D43.4	Neoplasm of uncertain behavior of spinal cord
Paraspinal muscle flap closure and tissue transfer:	
CPT codes not covered for indications listed in the CPB:	
14000	Adjacent tissue transfer or rearrangement, trunk; defect 10 sq cm or less
14001	defect 10.1 sq cm to 30.0 sq cm
14301	Adjacent tissue transfer or rearrangement, any area; defect 30.1 sq cm to 60.0 sq cm
14302	each additional 30.0 sq cm, or part thereof (List separately in addition to code for primary procedure)
15734	Muscle, myocutaneous, or fasciocutaneous flap; trunk

Code	Code Description
Other CPT codes related to the CPB:	
0656T, 0657T, 0790T, 22210, 22212, 22214, 22216, 22220, 22222, 22224, 22226, 22510, 22511, 22512, 22513, 22514, 22515, 22532, 22533, 22534, 22548, 22551, 22552, 22554, 22556, 22558, 22585, 22590, 22595, 22600, 22610, 22612, 22614, 22630, 22632, 22633, 22634, 22800, 22802, 22804, 22808, 22810, 22812, 22818, 22819, 22830, 22836, 22837, 22838, 22840, 22841, 22842, 22843, 22844, 22845, 22846, 22847, 22848, 22849, 22850, 22852, 22853, 22854, 22855, 22856, 22857, 22858, 22859, 22860, 22861, 22862, 22865,	Spinal surgeries

Code	Code Description
27278, 27279, 27280, 61343, 63001, 63003, 63005, 63011, 63012, 63015, 63016, 63017, 63020, 63030, 63035, 63040, 63042, 63043, 63044, 63045, 63046, 63047, 63048, 63050, 63051, 63052, 63053, 63055, 63056, 63057, 63064, 63066, 63075, 63076, 63077, 63078, 63081, 63082, 63085, 63086, 63090, 63091, 63185, 63190, 63200, 63265, 63266, 63267	
Other HCPCS codes related to the CPB:	
C1821	Interspinous process distraction device (implantable)

Background

The lifetime incidence of low back pain (LBP) in the general population is reported to be 60 % to 90 % with annual incidence of 5 %. According to the National Center for Health Statistics (Patel, 2007), each year, 14.3 % of new patient visits to primary care physicians are for LBP, and nearly 13 million physician visits are related to complaints of chronic LBP. The causes of LBP are numerous. For individuals with acute LBP, the precise etiology can be identified in only about 15 % of cases (Lehrich et al, 2007).

The initial evaluation of patients with LBP involves ruling out potentially serious conditions such as infection, malignancy, spinal fracture, or a rapidly progressing neurologic deficit suggestive of the cauda equina syndrome, bowel or bladder dysfunction, or weakness, which suggest the need for early diagnostic testing. Patients without these conditions are initially managed with conservative therapy. The most common pathological causes of LBP are attributed to herniated lumbar discs (lumbar disc prolapse, slipped disc), lumbar stenosis and lumbar spondylolisthesis (Lehrich and Sheon, 2007).

Spondylolisthesis refers to the forward slippage of one vertebral body with respect to the one beneath it. This most commonly occurs at the lumbosacral junction with L5 slipping over S1, but it can occur at higher levels as well. It is classified based on etiology into 5 types: dysplastic, defect in pars inter-articularis, degenerative, traumatic, and pathologic. The most common grading system for spondylolisthesis is the Meyerding grading system for severity of slippage, which categorizes severity based upon measurements on lateral X-ray of the distance from the posterior edge of the superior vertebral body to the posterior edge of the adjacent inferior vertebral body. The distance is then reported as a percentage of the total superior vertebral body length (see Appendix).

Guidelines for the approach to the initial evaluation of LBP have been issued by the Agency for Healthcare Research and Quality (1994) and similar conclusions were reached in systematic reviews (Jarvik et al, 2002; Chou et al, 2007; NICE, 2009). For adults less than 50 years of age with no signs or symptoms of systemic disease, symptomatic therapy without imaging is appropriate. For patients 50 years of age and older or those whose findings suggest systemic disease, plain radiography and simple laboratory tests can almost completely rule out underlying systemic diseases. Advanced imaging should be reserved for patients who are considering surgery or those in whom systemic disease is strongly suspected. Conservative care without immediate imaging is also considered appropriate for patients with radiculopathy, as long as symptoms are not bilateral or associated with urinary retention. Magnetic resonance imaging (MRI) should be performed if the latter symptoms are present or if patients do not improve with conservative therapy for 4 to 6 weeks. Ninety percent of acute attacks of sciatica will resolve with conservative management within 4 to 6 weeks; only 5 % remain disabled longer than 3 months (Gibson and Waddell, 2007; Lehrich and Sheon, 2007; AHCPR 1994).

Conservative Management for LBP Includes

- Avoidance of activities that aggravate pain
- Chiropractic manipulation in the first 4 weeks if there is no radiculopathy
- Cognitive support and reassurance that recovery is expected
- Education regarding spine biomechanics
- Exercise program
- Heat/cold modalities for home use
- Limited bed rest with gradual return to normal activities
- Low impact exercise as tolerated (e.g., stationary bike, swimming, walking)
- Pharmacotherapy (e.g., non-narcotic analgesics, NSAIDs (as second-line choices), avoid muscle relaxants, or only use during the first week, avoid narcotics)
- Physical therapy

In the American Pain Society/American College of Physicians Clinical Practice Guideline on "Nonpharmacologic Therapies for Acute and Chronic Low Back Pain," Chou and Huffman (2007) reached the following conclusions: "Therapies with good evidence of moderate efficacy for chronic or subacute low back pain are cognitive-behavioral therapy, exercise, spinal manipulation, and interdisciplinary rehabilitation. For acute low back pain, the only therapy with good evidence of efficacy is superficial heat."

According to a draft technology assessment prepared for the Agency for Healthcare Research and Quality (AHRQ) by the Duke Evidence-based Practice Center on spinal fusion for treatment of degenerative disease affecting the lumbar spine (AHRQ, 2006), conservative treatments are generally performed routinely before any surgery is considered in axial back pain. These include medical management (such as NSAIDs, etc.), pain management, injections, physical therapy, exercise and various forms of cognitive rehabilitation. Such conservative treatments are seldom applied in a comprehensive, well-organized rehabilitation program, although some such programs do exist. Conservative treatments are usually tried for at least 6 to 12 months before surgery for any form of lumbar fusion is considered. Several reviews of these therapies noted that there is no evidence about the effectiveness of any of these therapies for low back or radicular pain beyond about 6 weeks. In addition, the assessment stated that almost all lumbar spine surgery, including lumbar fusion, is performed to reduce the subjective individual symptoms of radiculopathy; thus, patient education to inform patients of their treatment options is considered critical. The other indications for lumbar fusion focus on improvement in axial lumbar pain (i.e., near the midline and not involving nerve roots or leg pain). These indications include lumbar instability, such as degenerative lumbar scoliosis, spondylolisthesis for axial pain alone, and for less common problems, such as discitis, lumbar flat back syndrome, neoplastic bone invasion and collapse, and chronic fractures, such as osteoporotic fractures which develop into burst fractures over time. The assessment concluded that, "The evidence for lumbar spinal fusion does not conclusively demonstrate short-term or long-term benefits compared with non-surgical treatment, especially when considering patients over 65 years of age, for degenerative disc disease; for spondylolisthesis, considerable uncertainty exists due to lack of data, particularly for older patients."

The National Institute for Clinical Excellence's (NICE, 2009) guidance on early management of people with non-specific LBP stated that it is important to help people with persistent non-specific LBP self-manage their condition. The guidance stated that one of the following treatment options should be offered to the patient: (i) an exercise program, (ii) a course of manual therapy (i.e., spinal manipulation, spinal mobilization, massage), (iii) a course of acupuncture, and (iv) pharmacological therapy. Referral to a combined physical and psychological treatment program may be appropriate for individuals who have received at least one less intensive treatment and have high disability and/or significant psychological distress. The guidance stated "

[t]here is evidence that manual therapy, exercise and acupuncture individually are cost-effective management options compared with usual care for persistent non-specific low back pain. The cost implications of treating people who do not respond to initial therapy and so receive multiple back care interventions are substantial. It is unclear whether there is added health gain for this subgroup from either multiple or sequential use of therapies." In addition, the guidance stated that imaging is not necessary for the management of non-specific LBP. An MRI is appropriate only for people who have failed conservative care, including a combined physical and psychological treatment program, and are considering a referral for an opinion on spinal fusion.

The American Pain Society Clinical Practice Guideline *Interventional Therapies, Surgery, and Interdisciplinary Rehabilitation for Low Back Pain* (Chou et al, 2009) stated "[r]ates of certain interventional and surgical procedures for back pain are rising. However, it is unclear if methods for identifying specific anatomic sources of back pain are accurate, and effectiveness of some interventional therapies and surgery remains uncertain or controversial." Included in the guideline are the following recommendations.

The APS guideline stated that, in patients with chronic non-radicular LBP, provocative discography is not recommended as a procedure for diagnosing LBP (strong recommendation, moderate-quality evidence) (Chou et al, 2009).

In patients with non-radicular LBP who do not respond to usual, non-interdisciplinary interventions, the APS guideline recommended that clinicians consider intensive interdisciplinary rehabilitation with a cognitive/behavioral emphasis (strong recommendation, high-quality evidence) (Chou et al, 2009).

In patients with non-radicular LBP, common degenerative spinal changes, and persistent and disabling symptoms, the APS guideline recommended that clinicians discuss risks and benefits of surgery as an option (weak recommendation, moderate-quality evidence) (Chou et al, 2009).

The guideline recommended that shared decision-making regarding surgery for non-specific LBP include a specific discussion about intensive interdisciplinary rehabilitation as a similarly effective option, the small to moderate average benefit from surgery versus non-interdisciplinary non-surgical therapy, and the fact that the majority of such patients who undergo surgery do not experience an optimal outcome (defined as minimum or no pain, discontinuation of or occasional pain medication use, and return of high-level function) (Chou et al, 2009).

The APS guideline explained that for persistent non-radicular LBP with common degenerative changes (e.g., degenerative disc disease), fusion surgery is superior to non-surgical therapy without interdisciplinary rehabilitation in 1 trial, but no more effective than intensive interdisciplinary rehabilitation in 3 trials (Chou et al, 2009). Compared with non-interdisciplinary, non-surgical therapy, average benefits are small for function (5-10 points on a 100-point scale) and moderate for improvement in pain (10-20 points on a 100-point scale). Furthermore, more than half of the patients who undergo surgery do not experience an "excellent" or "good" outcome (i.e., no more than sporadic pain, slight restriction of function, and occasional analgesics). Although operative deaths are uncommon, early complications occur in approximately 18 % of patients who undergo fusion surgery in randomized trials. Instrumented fusion is associated with enhanced fusion rates compared with non-instrumented fusion, but insufficient evidence exists to determine whether instrumented fusion improves clinical outcomes, and additional costs are substantial. In addition, there is insufficient evidence to recommend a specific fusion method (anterior, posterolateral, or circumferential), though more technically difficult procedures may be associated with higher rates of complications.

In patients with persistent and disabling radiculopathy due to herniated lumbar disc or persistent and disabling leg pain due to spinal stenosis, the APS guideline recommended that clinicians discuss risks and benefits of surgery as an option (strong recommendation, high-quality evidence) (Chou et al, 2009). It is recommended that shared decision-making regarding surgery include a specific discussion about moderate average benefits, which appear to decrease over time in patients who undergo surgery.

The APS guideline explained that for persistent and disabling radiculopathy due to herniated lumbar disc, standard open discectomy and microdiscectomy are associated with moderate short-term (through 6 to 12 weeks) benefits compared to non-surgical therapy, though differences in outcomes in some trials are diminished or no longer present after 1 to 2 years (Chou et al, 2009). In addition, patients tend to improve substantially either with or without discectomy, and continued non-surgical therapy in patients who have had symptoms for at least 6 weeks does not appear to increase risk for cauda equina syndrome or paralysis.

If conservative management fails to relieve symptoms of radiculopathy and there is strong evidence of dysfunction of a specific nerve root confirmed at the corresponding level by findings demonstrated by CT or MRI, further evaluation and more invasive treatment, including spine surgery, may be proposed as a treatment option. The primary rationale of any form of surgery for disc prolapse is to provide decompression of the affected nerve root to relieve the individual's symptoms. It involves the removal of all or part of the lamina of a lumbar vertebra. The addition of fusion with or without instrumentation is considered when there are concerns about instability. Open discectomy, performed with or without the use of an operating microscope, is the most common surgical technique applied, but there are now a number of other less invasive surgical approaches. The surgical treatment of sciatica with discectomy is reportedly ineffective in a sizable percentage of patients, and re-herniation occurs after 5 % to 15 % of such procedures. Thus, it would be ideal to define the optimal type of treatment for the specific types of prolapse (Carragee et al, 2003).

Different fusion procedures, including anterior lumbar interbody fusion, posterolateral fusion, posterior lumbar interbody fusion and transforaminal lumbar interbody fusion, and anterior-posterior combined fusion, do not vary significantly in pain or disability outcomes, although there are qualitative differences in complications related to the surgical approach. Prior to the 1980's both anterior and posterior non-instrumented lumbar fusions were commonly performed, using primarily bone graft. As pedicle screws became more widely used, it was noted that the rate of fusion increased from 65 % with bone graft alone to nearly 95 % with the instrumentation to provide internal support for the bone graft. The increased stiffness from the insertion of screws and rods has been hypothesized to lead to increased degeneration at spine segments adjacent to the fusion.

Anterior spine procedures, through either the peritoneum or retroperitoneum, require no posterior muscle and ligamentous dissection and result in less post-operative axial back pain. This approach is generally recommended for the treatment of axial LBP in young individuals. The usual criteria for consideration of an anterior lumbar fusion (or anterior lumbar arthroplasty) include a young person (i.e., age 20 to 40 years), who on MRI scan has either one or two dark discs, a concordant discogram indicating the axial pain is likely arising from the degenerated joints, and failure of previous conservative measures to improve the back pain over a period of time, with a minimum of 6 month conservative treatment. However, according to AHRQ (2006), the discogram remains highly controversial, and recent reports suggest that relying on the MRI findings of a dark disc and limiting the discogram to just those levels may improve the definition of a "positive discorgram". The AHRQ assessment stated, "However, the high rate of false positives with normal disc spaces is problematic, as well as the high rate of prevalence of dark disc syndrome." As patients age into their 40’s and 50’s the disc and facet degenerative processes slowly worsen, and it is less likely to find patients with isolated arthritis, thus, anterior fusion is less often recommended for older patients. Posterior fusion may be preferable for older individuals in order to stabilize facet joint disease. However, the posterior approach involves significant muscle dissection, resulting in severe back pain in the post-operative period, and is avoided by some surgeons.

The natural history of sciatica is favorable, with resolution of leg pain within 8 weeks from onset in the majority of patients (Peul et al, 2007). Dutch guidelines on the diagnosis and treatment of the lumbo-sacral radicular syndrome (Stam, 1996) recommended the option of lumbar-disk surgery in patients who have sciatica if symptoms do not improve after 6 weeks of conservative treatment. To determine the optimal timing of surgery, investigators (Peul et al, 2007) randomly assigned patients (n = 283) who had had severe sciatica for 6 to 12 weeks to early surgery or to prolonged conservative treatment with surgery if needed. The primary outcomes were the score on the Roland Disability Questionnaire, the score on the visual analog scale for leg pain, and the patient's report of perceived recovery during the first year after randomization. Repeated-measures analysis according to the intention-to-treat principle was used to estimate the outcome curves for both groups. Of 141 patients assigned to undergo early surgery, 125 (89 %) underwent microdiscectomy after a mean of 2.2 weeks. Of 142 patients designated for conservative treatment, 55 (39 %) were treated surgically after a mean of 18.7 weeks. There was no significant overall difference in disability scores during the first year ($p = 0.13$). Relief of leg pain was faster for patients assigned to early surgery ($p < 0.001$). Patients assigned to early surgery also reported a faster rate of perceived recovery (hazard ratio, 1.97; 95 % confidence interval [CI]: 1.72 to 2.22; $p < 0.001$). In both groups, however, the probability of perceived recovery after 1 year of follow-up was 95 %. The investigators concluded that the 1-year outcomes were similar for patients assigned to early surgery and those assigned to conservative treatment with eventual surgery if needed, but the rates of pain relief and of perceived recovery were faster for those assigned to early surgery.

A Cochrane systematic review (2007) on surgical interventions for lumbar disc prolapse identified 40 randomized controlled trials and 2 quasi-randomized trials on the surgical management of lumbar disc prolapse. However, the authors identified only 4 studies (Weber, 1983; Greenfield, 2003; Butterman, 2004; Weinstein, 2006) that compared discectomy with conservative management. The authors stated that these studies contain major design weaknesses, particularly on the issues of sample size, randomization, blinding, and duration of follow-up. Furthermore, outcome measures in clinical studies of LBP have not been standardized making it difficult to compare the results of clinical studies of similar treatment.

The first study (Weber, 1983) compared the results of surgical versus conservative treatment for lumbar disc herniation confirmed by radiculography ($n = 126$) with 10 years of follow-up observation. The author reported a significantly better result in the surgically treated group at the 1-year follow-up examination; however, after 4 years the difference was no longer statistically significant. Only minor changes took place during the last 6 years of observation. The trial was not blinded and 26 % of the conservative group crossed-over to surgery.

In a prospective, randomized study, Buttermann (2004), evaluated the efficacy of epidural steroid injection versus discectomy in the treatment of patients with a large, symptomatic lumbar herniated nucleus pulposus (n = 100). The discectomy patients had the most rapid decrease in symptoms, with 92 % to 98 % of the patients reporting that the treatment had been successful over the various follow-up periods. Of the 50 patients who had undergone epidural steroid injection, 42 % to 56 % reported the treatment had been effective. Those who did not obtain relief from the injection had a subsequent discectomy (27 of 50 patients). The epidural steroid injection trial group did not appear to have any adverse outcomes as a result of their delay in receiving surgery. The author concluded that discectomy was more effective in reducing symptoms and disability associated with a large herniated lumbar disc than epidural steroid injection; however, the epidural steroid injection was found to be effective for the follow-up period of 3 years by nearly 50 % of the patients who had not had improvement with 6 or more weeks of non-invasive care.

The Spine Patient Outcomes Research Trial (SPORT) was designed to compare the effectiveness of surgical and non-surgical treatment among participants with confirmed diagnoses of intervertebral disk herniation, spinal stenosis, and degenerative spondylolisthesis. The SPORT included 13 multi-disciplinary spine centers across the United States. To assess the efficacy of standard open discectomy versus non-operative treatment individualized to the patient for lumbar intervertebral disk herniation, the SPORT observational cohort (Weinstein et al, 2006) conducted a randomized clinical trial (n = 501) with image-confirmed lumbar intervertebral disk herniation and persistent signs and symptoms of radiculopathy for at least 6 weeks. The authors reported limited adherence to the assigned treatment: 50 % of patients assigned to surgery received surgery within 3 months of enrollment, while 30 % of those assigned to non-operative treatment received surgery in the same period. Intent-to-treat analyses demonstrated substantial improvements for all primary and secondary outcomes in both treatment groups. Between-group differences in improvements were consistently in favor of surgery for all periods but were small and not statistically significant for the primary outcomes. The authors reported that both surgical and non-operative treatment groups improved substantially over a 2-year period.

However, the large numbers of patients who crossed over between assigned groups precluded any conclusions about the comparative effectiveness of operative therapy versus usual care.

The fourth study (Greenfield, 2003), available only as an abstract, compared microdiscectomy with a low-tech physical therapy regime and educational approach in patients with LBP and sciatica with a small or moderate disc prolapse. At 12 and 18 months there were statistically significant differences in pain and disability favoring the surgical group; however, by 24 months there was no difference between the 2 groups.

The Cochrane systematic review (2007) concluded: (i) most lumbar disc prolapses resolve naturally with conservative management and the passage of time; (ii) there is considerable evidence that surgical discectomy provides effective clinical relief for carefully selected patients with sciatica due to lumbar disc prolapse that fails to resolve with conservative management. It provides faster relief from the acute attack of sciatica, although any positive or negative effects on the long-term natural history of the underlying disc disease are unclear. There is still a lack of scientific evidence on the optimal timing of surgery. The amount of cross-over in these trials makes it likely that the intent-to-treat analysis underestimates the true effect of surgery; but the resulting confounding also makes it impossible to draw any firm conclusions about the efficacy of surgery.

In a randomized controlled study, Brox et al (2006) compared the effectiveness of lumbar fusion with posterior transpedicular screws and cognitive intervention and exercises on 60 patients aged 25 to 60 years with LBP lasting longer than 1 year after previous surgery for disc herniation. Cognitive intervention consisted of a lecture intended to give the patient an understanding that ordinary physical activity would not harm the disc and a recommendation to use the back and bend it. This was reinforced by 3 daily physical exercise sessions for 3 weeks. The primary outcome measure was the Oswestry Disability Index (ODI). The success rate was 50 % in the fusion group and 48 % in the cognitive intervention/exercise group. The authors concluded that for patients with chronic LBP after previous surgery for disc herniation, lumbar fusion failed to show any benefit over cognitive intervention and exercise.

The American Association of Neurological Surgeons/Congress of Neurological Surgeons (AANS/CNS) Guideline's for the Performance of Fusion Procedures for Degenerative Disease of the Lumbar Spine (Resnick, 2005), is a series of guidelines that deal with the methodology of guideline formation, the assessment of outcomes following lumbar fusion, recommendations that involve the diagnostic modalities helpful for the pre- and post-operative evaluation of patients considered candidates for or treated with lumbar fusion, followed by recommendations dealing with specific patient populations. Finally, several surgical adjuncts, including pedicle screws, intra-operative monitoring, and bone graft substitutes are discussed, and recommendations are made for their use.

In their review of the literature, the AANS/CNS committee found that several authors published their experience in the surgical management of patients with stenosis and spondylolisthesis treated with decompression with or without fusion. These results are variable and all studies involved non-validated outcome measures. Many of the published reviews presented flawed results due to poorly defined outcome measures, inadequate numbers of patients, and comparison of dissimilar treatment groups. As a result, most of the published studies on lumbar fusion were not included in their review. However, the committee stated that, "The aforementioned results do not detract from the importance of this document; rather, we can now clearly see the need for the neurosurgical community to design and complete prospective randomized controlled studies to answer the many lingering clinical questions with rigorous scientific power." The guidelines concluded that "The precise definition of instability or kyphosis has varied among researchers and requires further study."

Investigators from the SPORT trial (Weinstein et al, 2007) compared surgical versus non-surgical treatment for lumbar degenerative spondylolisthesis. Candidates who had at least 12 weeks of symptoms and image-confirmed degenerative spondylolisthesis were offered enrollment in a randomized cohort (n = 304) or an observational cohort (n = 303). Eighty-six percent of patients had grade 1 slippage and 14 % had grade 2. However, all patients had neurogenic claudication or radicular leg pain with associated neurologic signs, spinal stenosis shown on cross-sectional imaging, and degenerative spondylolisthesis. Treatment was standard decompressive laminectomy (with or without fusion) or usual non-surgical care. The primary outcome measures were the 36-Item Short-Form General Health Survey (SF - 36) bodily pain and physical function scores (100-point scales, with higher scores indicating less severe symptoms) and the modified ODI (100-point scale, with lower scores indicating less severe symptoms) at 6 weeks, 3 months, 6 months, 1 year, and 2 years. The investigators reported high 1 year cross-over rates in the randomized cohort (approximately 40 % in each direction) but moderate in the observational cohort (17 % cross-over to surgery and 3 % cross-over to non-surgical care). The intention-to-treat analysis for the randomized cohort showed no statistically significant effects for the primary outcomes. The as-treated analysis for both cohorts combined showed a significant advantage for surgery at 3 months that increased at 1 year and diminished only slightly at 2 years. The treatment effects at 2 years were 18.1 for bodily pain (95 % CI: 14.5 to 21.7), 18.3 for physical function (95 % CI: 14.6 to 21.9), and -16.7 for the ODI (95 % CI: -19.5 to -13.9). The investigators concluded that patients with degenerative spondylolisthesis and spinal stenosis treated surgically showed substantially greater improvement in pain and function during a period of 2 years than patients treated non-surgically. However, the investigators stated, "Often patients fear they will get worse without surgery, but the patients receiving nonsurgical treatment, on average, showed moderate improvement in all outcomes." No conclusion is drawn regarding selection criteria for percentage of vertebral slippage in individuals with spondylolisthesis considered for fusion.

Vokshoor (2004) stated that before surgery is considered for adult patients with degenerative spondylolisthesis, minimal neurologic signs, or mechanical back pain alone, conservative measures should be exhausted, and a thorough evaluation of social and psychological factors should be undertaken. Indications for surgical intervention (fusion) include:

- Any high-grade slip (greater than 50 %)
- Iatrogenic spondylolisthesis

- Neurologic signs – Radiculopathy (unresponsive to conservative measures), myelopathy, neurogenic claudication
- Traumatic spondylolisthesis
- Type 1 and type 2 slips, with evidence of instability and progression of listhesis
- Type 3 (degenerative) listhesis with gross instability and incapacitating pain.

This is consistent with Wheeless (2008) who stated that for spondylolisthesis, posterior spine fusion should be limited to those patients who do not respond to conservative measures and whose slips are greater than 50 %.

Matsudaira and colleagues (2005) compared outcomes following decompression laminectomy combined with posterolateral fusion and pedicle screw instrumentation (n = 19) versus a laminoplasty technique without fusion (n = 18) in patients with grade I lumbar degenerative spondylolisthesis and reported no significant difference in the degree of clinical improvement between the 2 groups at the 2 year follow-up.

Randomized controlled trials have shown results of fusion to be equivalent to those of structured exercise and cognitive intervention. In a retrospective study on lumbar fusion outcomes among Washington State compensated workers with chronic back pain (n = 1,950), Maghout et al (2006) reported that fusions with cages increased from 3.6 % in 1996 to 58.1 % in 2001. Overall disability rate at 2 years after fusion was 63.9 %, the re-operation rate was 22.1 %, and the rate for other complications was 11.8 %. The use of cages or instrumentation was associated with an increased complication risk compared with bone-only fusions without improving disability or reoperation rates. Legal, work-related, and psychologic factors predicted worse disability. Discography and multi-level fusions predicted greater reoperation risk. The authors concluded that the use of intervertebral fusion devices rose rapidly after their introduction in 1996 and that this increased use was associated with an increased complication risk without improving disability or reoperation rates.

In a systematic review of randomized trials comparing lumbar fusion surgery to non-surgical treatment of chronic back pain associated with lumbar disc degeneration, Mirza et al (2007) compared outcomes in 4 trials that focused on non-specific chronic back. One study suggested greater improvement in back-specific disability for fusion compared to unstructured nonoperative care at 2 years, but the trial did not report data according to intent-to-treat principles. Three trials suggested no substantial difference in disability scores at 1-year and 2-years when fusion was compared to a 3-week cognitive-behavior treatment addressing fears about back injury. However, 2 of these trials were under-powered to identify clinically important differences. The third trial had high rates of cross-over (greater than 20 % for each treatment) and loss to follow-up (20 %); it is unclear how these affected results. The authors concluded that surgery may not be more efficacious than structured cognitive-behavior therapy, however, methodological limitations of the randomized trials prevent firm conclusions.

According to the American College of Physicians/American Pain Society Clinical Practice Guideline, *Diagnosis and Treatment of Low Back Pain* (2007), studies on LBP show large variations in practice patterns on diagnostic tests and treatments, although costs of care can differ substantially, patients seem to experience similar outcomes. The guideline makes the following recommendations:

1. Recommendation 1: A focused history and physical should be conducted to determine whether the back pain is: (i) non-specific; (ii) associated with radiculopathy or spinal stenosis; or (iii) due to another specific spinal cause. The history should include an assessment of psychosocial risk factors, which predict risk of chronic disabling back pain (strong recommendation, moderate-quality evidence).
2. Recommendation 2: For patients with non-specific LBP, imaging or other diagnostic tests should not be routinely obtained (strong recommendation, moderate-quality evidence).
3. Recommendation 3: For patients with LBP when severe or progressive neurologic deficits are present or when serious underlying conditions are suspected on the basis of history and physical examination, diagnostic imaging and testing should be obtained (strong recommendation, moderate-quality evidence).

4. Recommendation 4: For patients with persistent LBP and signs or symptoms of radiculopathy or spinal stenosis who are also considered candidates for surgery or epidural steroid injection (for suspected radiculopathy), MRI (preferred) or CT should be performed (strong recommendation, moderate-quality evidence).
5. Recommendation 5: Patients with LBP should be advised to remain active, and information about effective self-care options, including evidence-based information on the expected course of LBP, should be provided (strong recommendation, moderate-quality evidence).
6. Recommendation 6: For patients with LBP, the use of medications with proven benefits should be considered in conjunction with back care information and self-care. Severity of baseline pain, functional deficits, potential benefits, risks, and relative lack of long-term efficacy and safety data should be considered before initiating therapy (strong recommendation, moderate-quality evidence). First-line medication options for most patients are acetaminophen or non-steroidal anti-inflammatory drugs.
7. Recommendation 7: For patients with LBP who do not improve with self-care options, non-pharmacologic therapy with proven benefits should be considered. For acute LBP, spinal manipulation may be considered. For chronic or subacute LBP, intensive inter-disciplinary rehabilitation, exercise therapy, acupuncture, massage therapy, spinal manipulation, yoga, cognitive-behavioral therapy, or progressive relaxation may be considered (weak recommendation, moderate-quality evidence).

The Washington State Health Technology Assessment Program commissioned the ECRI Institute, an independent, non-profit health services research agency, to conduct an assessment of lumbar fusion and discography in patients with chronic uncomplicated degenerative disc disease (DDD) associated with chronic LBP. In a draft assessment (2007), the ECRI Institute stated that they did not find sufficient evidence that lumbar fusion surgery is more effective to a clinically meaningful degree than non-surgical treatments for any of the following patient populations, comparisons and outcomes:

- Meta-analysis of post-operative changes in Oswestry disability scores from 2 moderate quality randomized controlled trials

(RCTs) (n = 413) revealed no clinically meaningful difference between fusion and intensive exercise/rehabilitation plus cognitive behavioral therapy (CBT) in patients without prior back surgery, although the difference slightly favored fusion. Strength of evidence: Weak.

- The evidence was insufficient to determine whether lumbar fusion provides a greater improvement in back pain (1 moderate-quality RCT, n = 64) or quality of life (no acceptable evidence) compared to intensive exercise/rehabilitation plus CBT in patients without prior back surgery.
- The evidence from 1 moderate quality RCT (n = 60) was insufficient to determine the relative benefits of lumbar fusion compared to intensive exercise/rehabilitation in patients with prior back surgery.
- The evidence from 1 moderate quality RCT (n = 294) was insufficient to determine the relative benefits of lumbar fusion compared to conventional physical therapy in patients with or without prior back surgery.

The ECRI Institute assessed the rates of adverse events (peri-operative, long-term events, and reoperations) for lumbar fusion surgery and non-surgical treatments and reported the following:

- Categories of adverse events most frequently reported in fusion studies include reoperation (0 - 46 %), infection (0 - 9 %), various device-related complication (0 - 17.8 %), neurologic complications (0.7 - 26 %), thrombosis (0 - 4 %), bleeding/vascular complications (0 - 13 %), and dural injury (0.5 - 29 %)
- Lumbar fusion leads to significantly higher rates of early adverse events compared to non-intensive physical therapy or intensive exercise/rehabilitation plus CBT.
- Lumbar fusion leads to significantly higher rates of late adverse events at 2-year follow-up compared to non-intensive physical therapy or intensive exercise/rehabilitation plus CBT.
- None of the 4 RCTs comparing fusion to non-intensive physical therapy or intensive exercise/rehabilitation plus CBT reported any adverse events occurring in patients who only received non-operative care. Most of the reported adverse events could not have occurred in patients who did not undergo surgery.

The ECRI assessment stated that there is insufficient evidence to determine what patient characteristics are associated with differences in the benefits and adverse events of lumbar fusion surgery.

A number of recently published clinical studies demonstrating benefits of physical therapy (PT) for persons with spinal stenosis (see, e.g., Minetama et al, 2022; Ammendolia et al, 2022; Schnieder et al, 2019; Minetama et al, 2018; Jacobi et al, 2021; Chow et al, 2019; Ammendolia et al, 2018; and Marchand et al, 2019). A number of studies demonstrated the benefits of supervised PT over self-directed exercise (see, e.g., Comer et al, 2013; Minetama et al, 2019; and Minetama et al, 2021).

Contraindications

The ECRI assessment identified one guideline citing absolute contraindications for lumbar fusion and 3 guidelines reporting relative contraindications.

The Washington State Department of Labor and Industries (2004) cited the following as an absolute contraindication for lumbar fusion:

- Initial laminectomy/discectomy related to unilateral compression of a lumbar nerve root

The Washington State Department of Labor and Industries (2004) cited the following as relative contraindications for lumbar fusion:

- Current evidence of a factitious disorder
- Current smoking
- Greater than 12 months of disability (time-loss compensation benefits) prior to consideration of fusion
- High degrees of somatization on clinical or psychological evaluation
- Multiple level degenerative disease of the lumbar spine
- No evidence of functional recovery (return to work) for at least 6 months following the most recent spine surgery
- Presence of a personality disorder or major psychiatric illness

- Psychosocial factors that are correlated with poor outcome, such as history of drug or alcohol abuse
- Severe physical deconditioning.

The American Association of Neurological Surgeons reviewed evidence on lumbar fusion for the treatment of disc herniation and radiculopathy, and concluded: "There is insufficient evidence to recommend a treatment guideline." However, they did comment that lumbar spinal fusion is not recommended as a routine treatment following primary disc excision in patients with a herniated lumbar disc causing radiculopathy, though it may be of use for patients with herniated discs and evidence of preoperative lumbar spinal disability or deformity, for patients with significant chronic axial LBP and radiculopathy due to disc herniation, or for patients with recurrent lumbar disc herniation.

For patients with low back complaints in general, the American College of Occupational and Environmental Medicine (2005) stated that patients with co-morbidities including cardiac or respiratory disease, diabetes, or mental illness, are poor candidates for back surgery in general.

A cervical laminectomy (may be combined with an anterior approach) is sometimes performed when acute cervical disc herniation causes cord compression or in cervical disc herniations refractory to conservative measures. Studies have shown that an anterior discectomy with fusion is the recommended procedure for central or anterolateral soft disc herniation, while a posterior laminotomy-foraminotomy may be considered when technical limitations for anterior access exist (e.g., short thick neck) or when the individual has had prior surgery at the same level (Windsor, 2006).

Discectomy alone is regarded as a technique that most frequently results in spontaneous fusion (70 % to 80 %). Additional fusion techniques include the use of bone grafts (autograft, allograft or artificial) with or without cages and/or the use of an anterior plate. A Cochrane systematic review (2004) reported the results of 14 studies (n = 939) that evaluated three comparisons of different fusion techniques for cervical degenerative disc disease and concluded that discectomy alone has a shorter operation time, hospital stay, and post-operative absence from work than discectomy with fusion with no statistical difference for pain relief and rate of fusion. The authors concluded that more conservative techniques (discectomy alone, autograft) perform as well or better than allograft, artificial bone, and additional instrumentation; however, the low quality of the trials reviewed prohibited extensive conclusions and more studies with better methodology and reporting are needed.

To identify whether there is an advantage to instrumented or non-instrumented spinal fusion over decompression alone for patients with degenerative lumbar spondylolisthesis on the surgical management of degenerative lumbar spondylolisthesis, Martin et al (2007) reported the results of 13 studies in a systematic review; however, the studies were generally of low methodologic quality. Abstracted outcomes included clinical outcome, reoperation rate, and solid fusion status. Analyses were separated into: (i) fusion versus decompression alone, and (ii) instrumented fusion versus non-instrumented fusion. A satisfactory clinical outcome was significantly more likely with fusion than with decompression alone; however, the clinical benefit favoring fusion decreased when analysis was limited to studies where the majority of the patients reported neurologic symptoms (e.g., intermittent claudication and/or leg pain). The use of adjunctive instrumentation significantly increased the probability of attaining solid fusion, but no significant improvement in clinical outcome was reported. There was a non-significant trend toward lower repeat operations with fusion compared with both decompression alone and instrumented fusion. The authors concluded that spinal fusion may lead to a better clinical outcome than decompression alone. No conclusion about the clinical benefit of instrumenting a spinal fusion could be made; however, there is moderate evidence that the use of instrumentation improves the chance of achieving solid fusion.

Tsutsumimoto and colleagues (2008) retrospectively examined the surgical outcomes of un-instrumented posterolateral lumbar fusion for a minimum of 8 years (average of 9.5 years), by comparing cases exhibiting union with those exhibiting non-union. Un-instrumented posterolateral lumbar fusion was performed for the treatment of lumbar canal stenosis (LCS) with degenerative spondylolisthesis. The study included 42 patients, and the follow-up rate was 82.4 %. The mean age of the patients was 64.1 years (range of 46 to 77 years). Eight patients exhibited fusion at the L3-L4 level and 34 patients, at the L4-L5 level. The fusion status was assessed using plain radiographs. The clinical outcomes were evaluated using the Japanese Orthopedic Association (JOA) scores. Non-union was noted in 26 % (11/42) of the patients. There were no statistically significant differences between the groups exhibiting union and non-union with respect to age, sex, pre-operative JOA score, or pre-operative lumbar instability. The union group achieved better operative results than the non-union group at the 5-year and final follow-up (p = 0.006 and 0.008, respectively), although there was no significant difference in the percent recovery at 1 and 3 - year follow-up (p = 0.515 and 0.506, respectively). A stepwise regression analysis revealed that the best combination of predictors for recovery at the time of final follow-up included the fusion status and the presence of co-morbid disease.

A BlueCross BlueShield Association Technology Evaluation Center (BCBSA, 2007) assessment on the artificial lumbar disc commented on the evidence for spinal fusion for degenerative disc disease. The assessment stated that "The effectiveness of fusion for chronic degenerative disc disease is not well established. There are few clinical trials and results are inconsistent." One of the reasons the assessment concluded that the artificial disc is unproven is that the clinical studies compared it to spinal fusion, which is itself an unproven treatment for degenerative disc disease. The assessment stated that "Surgical arthrodesis, or fusion, is considered the current standard surgical treatment for degenerative disc disease which is not responsive to other treatments. Elimination of motion across the disc space and reduction of loads on disc tissues theoretically result in pain relief. Evidence supporting the efficacy of fusion is relatively sparse."

National survey data showed a rapid increase in the use of spinal fusion (i.e., annual numbers of procedures increased by 77 % from 1996 to 2001) as a result of new technological advances (e.g., bone morphogenetic protein), financial incentives, and controversial expansion of indications (e.g., discogenic back pain without evidence of sciatica), and a high rate of re-operations. These increases were not associated with reports of clarified indications or improved efficacy of various fusion techniques for various indications (Deyo et al, 2004, 2005). The review of spinal fusion surgery by Deyo et al (2004) stated that, "Fundamental problems plague the study of spinal fusion, including the lack of definitive methods to confirm a solid fusion, a weak association between solid fusion and pain relief, and the placebo effect of surgery for pain relief." They further stated that, "Evidence-based practice for degenerative spine disorders might reserve the use of spinal fusions for spondylolisthesis and only rare cases of disk herniation or spinal stenosis without spondylolisthesis," and that "More evidence from clinical trials should be required for degenerative disk disease to be an accepted indication." Regarding the use of "emerging spinal implants," such as artificial discs, the review states that, "If ongoing trials suggest results equivalent to those of spinal fusion, it may be faint praise, given the paucity of evidence that spinal fusion is safe and effective for common indications."

A review of the literature by Turner et al (1992) found no randomized trials of fusion. Combining many studies of fusion performed for many different clinical indications, the authors found an average of 68 % of patients reported a satisfactory outcome. A 1999 Cochrane review (Gibson et al) concluded that at that time there was no acceptable evidence of any form of fusion for degenerative lumbar spondylosis, back pain, or "instability." The authors could find no randomized clinical trials comparing fusion to a nonsurgical alternative, only trials which compared surgical techniques of fusion to each other.

Two published clinical trials comparing lumbar fusion to a non-surgical alternative treatment for patients with chronic back pain due to degenerative disc disease have been published since the Cochrane review. Fritzell et al (2001) conducted a multi-center randomized controlled trial comparing 3 techniques of lumbar fusion to non-surgical treatment. Enrollment criteria included patients (n = 294) with chronic pain, severe disability, pain attributed to degenerative disc disease, and no neurologic compromise due to herniated disc, spondylolisthesis, or spinal stenosis. There was no specified non-surgical treatment, but it was described as commonly used physical therapies. In terms of patients' overall assessment, 63 % of patients receiving fusion reported being better or much better, compared to 29 % of control patients. Critics of the study have pointed to the modest effect of surgery (up to 30 % mean score change), and the fact that control patients may not have received optimal nonsurgical treatment (Deyo et al, 2004).

The other randomized trial, by Brox et al (2003), assigned a specific cognitive and exercise regimen to the non-surgical patients. Enrollment criteria for this study were roughly similar to the other clinical trial, and outcomes were assessed at 1 year. In this study, patients receiving fusion reported improvements ranging from 36 to 49 % on pain and disability scales, but patients in the control arm also reported similar improvements in these scores, resulting in differences which were not statistically significant for most outcomes. Although this trial was much smaller (n = 64) than the study by Fritzell et al (2001), the point estimates of effect for each arm are very similar to each other, and confidence intervals sufficiently narrow to rule out a large clinical benefit of surgery. The authors believed that the difference in results between the 2 studies was caused by the specific intervention used in the non-surgical group, which produced improvements similar to the surgical fusion group.

The relative sparseness of controlled clinical trial data regarding the effectiveness of fusion for degenerative disc disease makes the validity of it as a valid comparator to total disc replacement uncertain. It can not be ruled out that some of the improvements associated with lumbar fusion are due to natural history, placebo effects, or co-interventions such as rehabilitation and exercise programs. It would be difficult to compare retrospective studies of fusion with case series results of artificial disc because of the very restrictive selection criteria for the artificial disc. Complicating the evaluation of fusion is the variety of techniques and devices used to perform the procedure. Pedicle screws and intervertebral fusion cages are 2 types of devices implanted during some procedures. Clinical trial results comparing use of these devices have not produced consistent results (BCBSA, 2007).

Common complications of fusion reported by Deyo et al (2004) include instrument failure (7 %), complications at the bone donor site (11 %), neural injuries (3 %), and failure to achieve a solid fusion or pseudarthrosis (15 %). In addition, fusion is thought to cause increased rate of disc degeneration in spinal segments adjacent to the fusion.

The Washington State Health Technology Assessment Program commissioned Spectrum Research Inc., an independent, clinical research organization, to conduct a systematic review of the evidence on the safety, efficacy, and cost-effectiveness of artificial disc replacement

(Dettori et al, 2008). One of the questions posed to the reviewers was whether there is evidence of the efficacy/effectiveness of artificial disc replacement compared with comparative therapies, including spinal fusion. The assessment reviewed the effectiveness of artificial disc replacement compared with spinal fusion in patients with degenerative disc disease and concluded that there is moderate evidence that the efficacy/effectiveness of lumbar artificial disc replacement is comparable with anterior lumbar interbody fusion or circumferential fusion up to 2 years following surgery; however, the authors stated that a non-inferiority trial requires that the reference treatment have an established efficacy or that it is in widespread use. For the lumbar spine, the authors noted that the efficacy of lumbar fusion for degenerative disc disease remains uncertain, especially when it is compared with non-operative care; thus, this limits the ability to fully answer the efficacy/effectiveness question of artificial disc replacement.

Four randomized controlled trials comparing lumbar fusion to non-surgical treatments found that nearly 15 % (58/399) of patients receiving lumbar fusion experienced complications. The most frequent complications reported included re-operation (with rates ranging from 0 % - 46.1 %), infection (0 % - 9 %), device-related complications (0 % - 17.8 %), neurologic complications (0.7 % - 25.8 %), thrombosis (0 % - 4 %), bleeding/vascular complications (0 % - 12.8 %), and dural injury (0.5 % - 29 %). In another study, a 12 % 2-year incidence rate of major complications following lumbar spinal fusion was reported, with a re-operation rate of 14.6 % for that population. Other complications reported in the literature include the potential for adjacent segment degeneration (development of disc degeneration, hypertrophic facets, dynamic instability, and/or spinal stenosis in adjacent levels), pseudoarthrosis, bone graft donor site pain and infection, instrumentation prominence or failure, neural injuries, and failure to relieve pain (Dettori et al, 2008).

Failed back surgery syndrome (FBSS), a condition in which there is failure to improve satisfactorily after back surgery, is characterized by intractable pain and various degrees of functional disability after lumbar spine surgery. A review of the literature on failed degenerative lumbar spine surgery (Diwan et al, 2003) estimated that 10 % to 15 % of patients who have undergone a spinal decompression procedure and 15 % to 20 % of patients who have had a spinal fusion procedure for degenerative disease of the lumbar spine undergo revision lumbar surgery within 3 to 5 years due to significant back and leg symptoms. However, most studies do not give the time to reoperation from the initial surgery. AHRQ (2006) reviewed studies that reported the incidence of adjacent segment disease requiring reoperation following lumbar or lumbo-sacral fusion and reported that annualized reoperation rates ranged from 0 % to 3.7 % and 1.7 % to 3.4 % for non-fusion lumbar surgery based on the over-all re-opeartion rates of the studies and the average time to follow-up.

The major causes for reoperation include fibrosis and adhesions, spinal instability, recurrent herniated disk, and inadequate decompression (Skaf et al, 2005). Over time, recurrent lumbar stenosis may occur at the same level (due to persistent or even enhanced motion at that level) or at adjacent levels due to the natural course of disease progression. It is hypothesized that fusion at one level increases stress on joints at adjacent levels during ordinary spine motion, hence leading to accelerated degenerative joint disease at these adjacent levels, as compared to the natural history of disease progression. Whether an instrumented fusion may increase adjacent segment disease is another controversial point, but without much evidence (AHRQ, 2006).

The etiology of FBSS can be poor patient selection, incorrect diagnosis, sub-optimal selection of surgery, poor technique, failure to achieve surgical goals, and/or recurrent pathology. Successful intervention in this difficult patient population requires a detailed history to accurately identify symptoms, rule out extra-spinal causes, identify a specific spinal etiology, and assess the psychological state of the patient. Only after these factors have been assessed can further treatment be planned (Guyer et al, 2006).

Relevant outcome studies are rarely diagnosis specific, and high level research studies comparing surgical and non-surgical approaches to FBSS studies have not been published to date. Surgical strategies focus on decompressing neural impingement or fusing unstable or putatively painful intervertebral discs. Non-surgical interventions range from nerve root specific blocks for pain relief to multi-disciplinary rehabilitation programs geared toward improving function (Hazard, 2006).

Herron (1994) reported the results of recurrent disc herniation treated by repeat laminectomy and discectomy. Fifty recurrences were treated in 46 patients, an average of 7 years and 1 month after the previous laminectomy. Thirty-four patients were treated for 37 recurrences at the same level, with 3 undergoing a third laminectomy and discectomy. Twelve patients were treated for 13 recurrences at a different level. Four patients underwent a third laminectomy and discectomy for recurrent disc herniation. Forty-one patients had follow-up of at least 1 year and average follow-up was 4 years and 6 months. There were 28 good (69 %), 10 fair (24 %), and 3 poor (7 %) results. Patients with pending litigation or work-related injuries (5 good, 5 fair, and 3 poor) did less well overall than those without these issues (23 good, 5 fair, and 0 poor). Heron stated that, "Fusion is not routinely required in patients undergoing repeat laminectomy and discectomy for recurrent disc herniation. In the absence of objective evidence of spinal instability, recurrent disc herniation may be adequately treated by repeat lumbar laminectomy and discectomy alone".

Fritsch et al (1996) conducted a retrospective review of 182 revisions on FBSS from the years 1965 to 1990 and analyzed the reasons for failure of primary discectomy, the outcome of the revisions, and factors that influenced those outcomes. The reported re-intervention rates after lumbar discectomy ranged from 5 % to 33 % depending on the type of surgical procedure. The authors' former investigations reported a revision rate of 10.8 % in evaluating 1,500 lumbar discectomies. A total of 182 revisions were performed on 136 patients; 44 patients (34 %) were revised multiple times. Recurrent or un-influenced sciatic pain and neurologic deficiency or lumbar instability led to re-intervention. Recurrent lumbar disc herniation was mainly found at the first re-intervention. In multiple revision patients the rate of epidural fibrosis and instability increased to greater than 60 %. In 80 % of the patients the results were satisfactory in short-term evaluation, decreasing to 22 % in long-term follow up (2 to 27 years). Laminectomy performed in primary surgery could be detected as the only factor leading to a higher rate of revisions. The authors noted a trend toward poor results after recurrent disc surgery due to the development of epidural fibrosis and instability. In severe discotomy syndrome, a spinal fusion appeared to be more successful than multiple fibrinolyses.

Phillips and Cunningham (2002) conducted a review of the peer-reviewed publications that investigated etiologies and treatments for neurogenic pain in patients who have undergone previous spinal surgery. The authors recommended that in the absence of profound or progressive neurologic deficits, most patients with chronic back and leg pain who have undergone previous spinal surgery be treated non-operatively, however, additional decompressive surgical intervention may be justified in patients with well-defined, discrete pathology amenable to surgical correction who have been refractory to conservative care.

During a 2-year period, Duggal et al (2004) treated patients diagnosed with FBSS with anterior lumbar interbody fusion. Clinical and radiological outcomes were recorded in a prospective, non-randomized, longitudinal manner. Neurological, pain, and functional outcomes were measured pre-operatively and 12 months after surgery. Operative data, peri-operative complications, and radiological and clinical outcomes were recorded. Thirty-three patients with a pre-operative diagnosis of FBSS, with degenerative disc disease (n = 17), post-surgical spondylolisthesis (n = 13), or pseudarthrosis (n = 3), underwent anterior lumbar interbody fusion. Back pain, leg pain, and functional status improved significantly, by 76 % ($p < 0.01$), 80 % ($p < 0.01$), and 67 % ($p < 0.01$), respectively. The authors found anterior lumbar interbody fusion to be a safe and effective procedure for the treatment of FBSS for selected patients.

Skaf et al (2005) prospectively studied 50 patients with FBSS. The underlying pathology was identified and all the patients were treated surgically. Redo surgery was targeted at correcting the underlying pathology: removal of recurrent or residual disk, release of adhesions with neural decompression, and fusion with or without instrumentation. The post-surgical outcome was studied using the ODQ. The average pre-operative ODQ mean score was 80.8; the average post-operative ODQ mean score was 36.6 at 1 month and 24.2 at 1 year. Best scores were obtained at 3 months of follow-up in most cases. Successful outcome (greater than 50 % pain relief) was achieved in 92 % of the patients at 1 year. The authors concluded that successful management of patients with FBSS could be achieved with proper patient selection, correct pre-operative diagnosis, and adequate surgical procedure targeting the underlying pathology.

Fu et al (2005) evaluated the long-term outcomes of repeat surgery for recurrent lumbar disc herniation and compared the results of disc excision with and without posterolateral fusion in a retrospective study. The sample included 41 patients who underwent disc excision with or without posterolateral fusion, with an average follow-up of 88.7 months (range of 60 to 134 months). Clinical symptoms were assessed based on the Japanese Orthopedic Association Back Scores. All medical and surgical records were examined and analyzed, including pain-free interval, intra-operative blood loss, length of surgery, and post-surgery hospital stay. Clinical outcome was excellent or good in 80.5 % of patients, including 78.3 % of patients undergoing a discectomy alone, and 83.3 % of patients with posterolateral fusion. The recovery rate was 82.2 %, and the difference between the fusion and non-fusion groups was insignificant (p = 0.799). The difference in the post-operative back pain score was also insignificant (p = 0.461). These 2 groups were not different in terms of age, pain-free interval, and follow-up duration. Intra-operative blood loss, length of surgery, and length of hospitalization were significantly less in patients undergoing discectomy alone than in patients with fusion. The authors concluded that repeat surgery for recurrent sciatica is effective in cases of true recurrent disc herniation.

Papadopoulos et al (2006) retrospectively reviewed a total of 27 patients who had undergone revision discectomies for recurrent lumbar disc herniations to assess their clinical outcomes. Patients were compared with a control group of 30 matched patients who had undergone only a primary discectomy. The spine module of the MODEMS outcome instrument was used to evaluate patients' satisfaction, pain and functional ability following discectomy, as well as quality of life. Patients were also asked whether they were improved or worsened with surgery. Those undergoing revision surgery were asked whether the improvement following the second surgery was more or less than the improvement following the first surgery. Improvement following the repeat discectomy was not statistically different from the improvement that occurred in patients who underwent just the primary operation. Differences in residual numbness/tingling in the leg and/or the foot as well as in frequency of back and/or buttock pain were identified. The authors concluded that revision discectomy is as efficacious as primary discectomy in selected patients.

An assessment of spinal fusion by the Andalusian Agency for Health Technology Assessment (AETSA) (Martinez Ferez et al, 2009) concluded that the available scientific evidence about spinal fusion is scarce and is based on low or moderate quality studies. The assessment found no clear evidence that fusion combined with spinal decompression provides some benefits in degenerative lumbar stenosis. The assessment found weak evidence in favor of spinal fusion in contrast with decompression for degenerative spondylolisthesis, but it is based on studies of low methodological quality. The report found that, for degenerative discopathies, spinal fusion does not provide clinical benefits compared to structured and intense conservative treatments with cognitive-behavioral therapy; on the contrary, it seems to provide benefits compared to less intensive treatments which are commonly used in practice. For degenerative discopathies with radicular compression, spinal fusion does not seem to provide a clear clinical benefit compared to decompression alone. The report found that total replacement of the degenerated discs by artificial discs such as Charité and Pro-Disc L shows results at least as good or better than those obtained by spinal fusion, which is considered the standard treatment in these cases. The report concluded that clinical trials of higher quality are necessary in order to get clear results about the real benefit which fusion provides for the treatment of the spinal degenerative pathologies which have been included in this report.

Anterior spinal fusion with instrumentation has been used for many years in the treatment of thoracolumbar and lumbar curves in adolescent idiopathic scoliosis (AIS). Tis et al (2010) reported the intermediate radiographical and pulmonary function testing (PFT) data from patients who underwent open instrumented anterior spinal fusion (OASF) using modern, rigid instrumentation for the treatment of primary thoracic (Lenke 1) adolescent idiopathic scoliosis (AIS). Of 101 potential patients who underwent OASF with a minimum 5-year follow-up, 85 (85 %) were studied. Standing radiographs were analyzed before surgery and at first standing erect, 2-year, and 5-year follow-up. Data on PFT were collected before surgery and at 5 years after surgery. Complete 5-year follow-up was obtained in 85 patients. Five years after surgery, the mean coronal correction was 26 degrees (51 %; $p < 0.05$) and the thoracolumbar/lumbar curve improved 16 degrees (51 %). There was a 9-degree ($p < 0.001$) increase in kyphosis, and there were 9 patients (11 %) in whom the C7 plumb line translated greater than 2 cm. There was a

6.7 % decrease in predicted forced expiratory volume in one second over the 5-year period, from 75.5 % +/- 13 % before surgery to 68.8 % +/- 2 % at 5-year follow-up (p = 0.007); however, there was no significant change in forced vital capacity. There were 3 significant adverse events: 1 implant breakage requiring re-operation and 2 cases of progression of the main thoracic curve requiring re-operation. The authors concluded that OASF is a reproducible and safe method to treat thoracic AIS. It provides good coronal and sagittal correction of the main thoracic and compensatory thoracolumbar/lumbar curves that is maintained with intermediate term follow-up. In skeletally immature children, this technique can cause an increase in kyphosis beyond normal values, and less correction of kyphosis should be considered during instrumentation. As with any procedure that employs a thoracotomy, pulmonary function is mildly decreased at final follow-up.

In a retrospective review, Kelly and colleagues (2010) evaluated a group of patients based on Scoliosis Research Society (SRS)-30 and Oswestry data as well as radiographical and MRI findings; and reported the results of long-term follow-up of anterior spinal fusion with instrumentation for thoracolumbar and lumbar curves in AIS. Eighteen patients were available for review. Average follow-up for this study was 16.97 years. Based on SRS-30 and the Oswestry Disability Index data, most patients had good function scores and acceptable pain levels. Radiographs demonstrated no progression of the thoracolumbar or thoracic curves. Implant failure was identified in 2 patients. Radiographical changes of early degenerative disc disease were identified in most patients but had no correlation with SRS or Oswestry data. These degenerative changes were evident on both radiographs and MRI. The authors concluded that the anterior approach in the treatment of thoracolumbar and lumbar curves in AIS offers good long-term functional outcomes for patients. Despite expected degenerative changes, patients scored well on the SRS and Oswestry tests, and were able to pursue careers and family activities.

Lehman and Lenke (2007) reviewed the case of a 44-year-old woman who underwent long-segment fusion and an artificial disc replacement (ADR). There have been many reported advantages and disadvantages of stopping the fusion at L5, with the theoretical benefits being preserved motion, shorter operative time, allowing the remaining disc to compensate for curve correction cephalad in the lumbar spine, and a decreased likelihood for the development of a pseudarthrosis at that distal level. As the issue of the fate of the L5 to S1 motion segment continues to be debated, these investigators presented the case of a medium-segment thoraco-lumbar fusion carried down to the L4 stable vertebra, an intervening healthy L4 to L5 disc space, with the placement of an artificial disc arthroplasty at the L5 to S1 level for a degenerative and discographically positive pain generator. At 2-year follow-up, her L5 to S1 artificial disc replacement level shows 11 degrees range of motion (ROM) and consolidated fusion from T12 to L4 with complete resolution of her axial back pain. Her T12 to L4 construct is stable, and the L4 to L5 level is unaffected at the latest follow-up. Her clinical outcome has been excellent with her return to a very active lifestyle. The authors concluded that ADR below a long-segment fusion is a viable alternative to performing fusion to additional motion segments.

In an in-vitro human cadaveric biomechanical study, Erkan et al (2009) compared the biomechanical properties of a 2-level Maverick disc replacement at L4 to L5, L5 to S1, and a hybrid model consisting of an L4 to L5 Maverick disc replacement with an L5 to S1 anterior lumbar interbody fusion using multi-directional flexibility test. A total of 6 fresh human cadaveric lumbar specimens (L4 to S1) were subjected to unconstrained load in axial torsion (AT), lateral bending (LB), flexion (F), extension (E), and flexion-extension (FE) using multi-directional flexibility test. Four surgical treatments – intact, 1-level Maverick at L5 to S1, 2-level Maverick between L4 and S1, and the hybrid model (anterior fusion at L5 to S1 and Maverick at L4 to L5) were tested in sequential order. The ROM of each treatment was calculated. The Maverick disc replacement slightly reduced intact motion in AT and LB at both levels. The total FE motion was similar to the intact motion. However, the E motion is significantly increased (approximately 50 % higher) and F motion is significantly decreased (30 % to 50 % lower). The anterior fusion using a cage and anterior plate significantly reduced spinal motion compared with the condition ($p < 0.05$). No significant differences were found between 2-level Maverick disc prosthesis and the hybrid model in terms of all motion types at L4 to L5 level ($p > 0.05$). The authors concluded that the Maverick disc preserved total motion but altered the motion pattern of the intact condition. This result is similar to unconstrained devices such as Charité. The motion at L4 to L5 of the hybrid model is similar to that of 2-level Maverick disc replacement. The fusion procedure using an anterior plate significantly reduced intact motion. The authors concluded that clinical studies are recommended to validate the effectiveness of the hybrid model.

On behalf of the American Pain Society, Chou et al (2009) systematically evaluated benefits and harms of surgery for non-radicular back pain with common degenerative changes, radiculopathy with herniated lumbar disc, and symptomatic spinal stenosis. For non-radicular LBP with common degenerative changes, these researchers found fair evidence that fusion is no better than intensive rehabilitation with a cognitive-behavioral emphasis for improvement in pain or function, but slightly to moderately superior to standard (non-intensive) non-surgical therapy. Less than half of patients experience optimal outcomes (defined as no more than sporadic pain, slight restriction of function, and occasional analgesics) following fusion. Clinical benefits of instrumented versus non-instrumented fusion are unclear. For radiculopathy with herniated lumbar disc, these investigators found good evidence that standard open discectomy and microdiscectomy are moderately superior to non-surgical therapy for improvement in pain and function through 2 to 3 months. For symptomatic spinal stenosis with or without degenerative spondylolisthesis, they found good evidence that decompressive surgery is moderately superior to non-surgical therapy through 1 to 2 years. For both conditions, patients on average experience improvement either with or without surgery, and benefits associated with surgery decrease with long-term follow-up in some trials. Although there is fair evidence that ADR is similarly effective compared to fusion for single level degenerative disc disease and that an inter-spinous spacer device is superior to non-surgical therapy for 1- or 2-level spinal stenosis with symptoms relieved with forward flexion, insufficient evidence exists to judge long-term benefits or harms. The authors concluded that surgery for radiculopathy with herniated lumbar disc and symptomatic spinal stenosis is associated with short-term benefits compared to non-surgical therapy, although benefits diminish with long-term follow-up in some trials. For non-radicular back pain with common degenerative changes, fusion is no more effective than intensive rehabilitation, but associated with small-to-moderate benefits compared to standard non-surgical therapy. This review did not mention the hybrid use of lumbar fusion and ADR for the management of LBP/spinal disorders.

Brox et al (2010) compared the long-term effectiveness of surgical and non-surgical treatment in patients with chronic LBP. The study was conducted at 4 university hospitals in Norway. The limitations on study enrollment ensured that patients with more significant symptoms and findings were not included in the protocol. All participants had LBP for at least 1 year, moderate disability, and evidence of disk degeneration at L4-L5 or L5-S1; those with symptomatic spinal stenosis were excluded from study participation. Similarly, patients with disk herniation or lateral recess stenosis plus signs of radiculopathy were excluded, as were those with generalized disk degeneration, ongoing serious somatic or psychiatric disease, or "reluctance" (term not defined) to undergo one of the study treatments. Participants were randomized to receive instrumented transpedicular fusion or non-surgical therapy. The non-surgical therapy was very intensive and included initial education, support, and physical training sessions that lasted an average of 25 hours per week over 3 weeks. There were 4 to 7 participants assigned to this training at a time, and they stayed in a hotel for patients during the 3 weeks. Specialists in physical medicine and rehabilitation guided the program, and participants also met with a peer who had previously completed the non-surgical program. At the end of the 3 weeks, participants were prescribed a home exercise program. The primary study outcome was the Oswestry disability index, which measures both pain and disability. Researchers also followed participants' ratings of treatment effectiveness, quality of life, and effects of the interventions on medication use and time missed from work. The study focused on these results measured at 4 years after randomization, and results were adjusted to account for sex, age, previous surgery for disk herniation, and baseline pain and disability scores. Of 234 eligible patients, 124 were enrolled in the trials. Baseline data were similar for the 2 groups. The mean age of participants was 42 years, and 72 % were women. The average duration of LBP was 9 years, and the mean severity of back pain was 64 on a scale of 0 to 100, with 100 being the most severe pain. Both treatment groups professed stronger beliefs in surgical versus non-surgical treatment of chronic LBP at baseline. In the surgical group, the rates of undergoing surgery were 88 % at 1 year and 91 % at 4 years. The respective rates of surgery in the non-surgical group were 5 % and 24 %. Study follow-up was excellent, with rates of 92 % and 86 % in the surgical and non-surgical groups at 4 years. Beyond comparing surgical and non-surgical treatment for chronic LBP, the study also gave some insight into the use of healthcare and other resources by these patients. Only a slight majority of patients saw a physician for back pain in the year before study follow-up at year 4. Less than 25 % received physical therapy. However, the rate of repeat surgery after the initial study surgery was 25 % over 4 years. This high repeat surgery rate was recorded despite the fact that no major adverse events related to surgery occurred through year 1 of the study.

Participants who received surgery were more than twice as likely to receive a disability pension, regardless of their randomized group. However, it would be wrong to infer that surgery itself promoted a higher rate of disability. These patients had surgery in response to more severe symptoms, and were therefore more likely to receive a disability pension in the first place. Moreover, applications for disability pension from patients who had received surgery could have received more favorable reviews. There were no differences between randomized groups in the outcomes of pain and disability in either intent-to-treat or as-treated analyses at 4 years. The mean Oswestry disability index score declined in both groups from an approximate mean of 44 at baseline to 28 at 4 years. Among secondary outcomes, the only difference between treatment groups was a reduction in fear and avoidance of physical activity, favoring the non-surgical group. Measurements of general function improved by approximately 40 % in both groups, and life satisfaction also improved. The number of participants returning to work improved with both treatments to a similar degree, and the proportions of participants rating their treatment as successful at 1 year were 61 % and 65 % in the surgical and non-surgical cohorts, respectively. Use of pain medication was higher among participants who received surgery, but any difference between treatment groups was not significant on intent-to-treat analysis.

Lumbar Fusion for Degenerative Disease

Yavin and colleagues (2017) noted that due to uncertain evidence, lumbar fusion for degenerative indications is associated with the greatest measured practice variation of any surgical procedure. These investigators summarized the current evidence on the comparative safety and efficacy of lumbar fusion, decompression-alone, or non-operative care for degenerative indications. They carried out a systematic review using PubMed, Medline, Embase, and the Cochrane Central Register of Controlled Trials (up to June 30, 2016). Comparative studies reporting validated measures of safety or efficacy were included. Treatment effects were calculated through DerSimonian and Laird random effects models. The literature search yielded 65 studies (19 RCTs, 16 prospective cohort studies, 15 retrospective cohort studies, and 15 registries) enrolling a total of 302,620 patients. Disability, pain, and patient satisfaction following fusion, decompression-alone, or non-operative care were dependent on surgical indications and study methodology. Relative to decompression-alone, the risk of re-operation following fusion was increased for spinal stenosis (relative risk [RR] 1.17, 95 % CI: 1.06 to 1.28) and decreased for spondylolisthesis (RR 0.75, 95 % CI: 0.68 to 0.83). Among patients with spinal stenosis, complications were more frequent following fusion (RR 1.87, 95 % CI: 1.18 to 2.96). Mortality was not significantly associated with any treatment modality. The authors concluded that positive clinical change was greatest in patients undergoing fusion for spondylolisthesis while complications and the risk of re-operation limited the benefit of fusion for spinal stenosis. The relative safety and efficacy of fusion for chronic LBP suggested careful patient selection is needed.

Kreiner et al (2020) noted that the North American Spine Society's (NASS) Evidence Based Clinical Guideline for the Diagnosis and Treatment of Low Back Pain features evidence-based recommendations for diagnosing and treating adult patients with non-specific low back pain (LBP). The guideline is intended to reflect contemporary treatment concepts for non-specific LBP as reflected in the highest quality clinical literature available on this subject as of February 2016. The objective of the guideline is to provide an evidence-based tool to aid spine specialists when making clinical decisions for adult patients with non-specific LBP. This article provided a brief summary of the evidence-based recommendations for diagnosing and treating patients with this condition. This guideline is the product of the Low Back Pain Work Group of NASS' Evidence-Based Clinical Guideline Development Committee. The methods used to develop this guideline were detailed in the complete guideline and technical report available on the NASS website. In brief, a multi-disciplinary work group of spine care specialists convened to identify clinical questions to address in the guideline. The literature search strategy was developed in consultation with medical librarians.

Upon completion of the systematic literature search, evidence relevant to the clinical questions posed in the guideline was reviewed. Work group members utilized NASS evidentiary table templates to summarize study conclusions, identify study strengths and weaknesses, and assign levels of evidence. Work group members participated in webcasts and in-person recommendation meetings to update and formulate evidence-based recommendations and incorporate expert opinion when necessary. The draft guideline was submitted to an internal and external peer-review process and eventually approved by the NASS Board of Directors. A total of 82 clinical questions were addressed, and the answers were summarized in this article. The respective recommendations were graded according to the levels of evidence of the supporting literature. The evidence-based clinical guideline has been created using techniques of evidence-based medicine and best available evidence to aid practitioners in the diagnosis and treatment of adult patients with nonspecific low back pain. The entire guideline document, including the evidentiary tables, literature search parameters, literature attrition flow-chart, suggestions for future research, and all of the references, is available electronically on the NASS website at: Clinical Guidelines (https://www.spine.org/ResearchClinicalCare/QualityImprovement/ClinicalGuidelines.aspx).

Key Findings/Recommendations Regarding Fusion (https://www.spine.org/Portals/0/assets/downloads/ResearchClinicalCare/Guidelines/LowBackPain.pdf)

- Surgical Question 1. In patients with low back pain, does surgical treatment vs medical/interventional treatment alone decrease the duration of pain, decrease the intensity of pain, increase the functional outcomes of treatment and improve the return-to-work rate?

  A systematic review of the literature yielded no studies to adequately address this question.

- Surgical Question 3. In patients undergoing fusion surgery for low back pain, which fusion technique results in the best outcomes for the following: decreased duration of pain, decreased intensity of pain, increased functional outcomes of treatment and improved return-to-work rate?

There is insufficient evidence to make a recommendation for or against a particular fusion technique for the treatment of low back pain. Grade of Recommendation: I

- Surgical Question 4. In patients undergoing fusion surgery for low back pain, are clinical outcomes, including duration of pain, intensity of pain, functional outcomes and return-to-work status, different for multi-level fusions vs single level fusions?

  A systematic review of the literature yielded no studies to adequately address this question.

- Surgical Question 5. In patients undergoing fusion surgery for low back pain, does radiographic evidence of fusion correlate with decreased duration of pain, decreased intensity of pain, increased functional outcomes of treatment and improved return-to-work rate?

  There is insufficient evidence to make a recommendation regarding whether radiographic evidence of fusion correlates with better clinical outcomes in patients with low back pain. Grade of Recommendation: I

- Surgical Question 8. In patients undergoing fusion surgery for low back pain, does the use of minimally invasive techniques decrease the duration of pain, decrease the intensity of pain, increase the functional outcomes of treatment and improve the return-to-work rate compared to open fusion techniques?

  A systematic review of the literature yielded no studies to adequately address this question.

- Surgical Question 11. In patients with low back pain, does fusion treatment decrease the duration of pain, decrease the intensity of pain, increase the functional outcomes of treatment and improve the return-to-work rate compared to treatment with:

  - discectomy;
  - discectomy plus rhizotomy; and

- decompression alone

A systematic review of the literature yielded no studies to adequately address this question.

Laminoplasty

Laminoplasty (laminaplasty) may be indicated in patients with myelopathy and multiple-level cervical spondylosis, such as in congenital cervical stenosis. When cervical spinal stenosis is severe, various symptoms may develop which include pain, weakness in arms and/or legs and unsteadiness in the gait (myelopathy).

For mild conditions conservative treatment may be sufficient. When symptoms are severe or progressive then a surgical treatment may be necessary. Surgical goals include a decompression of all compressed levels of the spine and stabilization with solid fusion. Surgical techniques are very dependent upon the specific problems of each patient. Anterior and posterior surgical approaches can be applied. In certain cases a decompression laminoplasty without fusion may be employed.

Laminoplasty involves opening and fusion of the cervical spinal canal. During laminoplasty the laminae are split and then held apart by bone struts, sutures or other techniques, in order to enlarge the spinal canal diameter. This procedure is usually performed on the cervical spine and may be used in an effort to lessen the chance of deformity that can develop when a facetectomy or laminectomy is performed alone.

In a retrospective study, Sakaura and colleagues (2005) compared the long-term outcomes after laminoplasty and anterior spinal fusion (ASF) for patients with cervical myelopathy secondary to disc herniation. The authors concluded that because the 2 procedures provided the same neurological improvement, the risks of bone graft complication with ASF must be weighed against the risks of chronic neck pain associated with laminoplasty for determining the best technique. For these investigators, laminoplasty is the procedure of choice for cervical myelopathy due to disc herniation except for patients with single-level disc herniation without developmental canal stenosis, who are considered to be good surgical candidates for ASF.

Ohnari et al (2006) noted that cervical laminoplasty is a good strategy for cervical myelopathy, but some post-operative patients complain of obstinate axial symptoms after surgery (i.e., nuchal pain, neck stiffness, and shoulder pain). It was reported that these symptoms proved to be more serious than has been believed and should be considered in the evaluation of the outcome of cervical spinal surgery. However, axial symptoms are sometimes recognized before surgery, or also after corpectomy. These investigators examined the difference in axial symptoms before and after laminoplasty and discussed the characteristics of these symptoms as a surgical complication. They conducted a questionnaire survey and reviewed the medical records of respondents. A total of 180 patients who underwent a spinous process-splitting laminoplasty for cervical myelopathy caused by degenerative disease in the authors' institution from 1993 until 2002 were included in the study, and were followed-up for 2 years or longer after surgery. Major outcome measures were self-report measures and functional measures. The questionnaire elicited information as follows: the location and characteristics of pre- and post-operative symptoms, frequency and duration of post-operative symptoms, and impairment in activities of everyday living, analgesic use, and the duration of use of cervical orthosis after surgery. The researchers divided axial symptoms into 4 characteristics based on previous reports: (i) pain, (ii) heaviness, (iii) stiffness, and (iv) other. An illustration of the upper back on which respondents could mark each characteristic was used to acquire information about the location of axial symptoms. The following information was gathered from medical records and statistically analyzed: whether post-operative axial symptoms were related or not, age, sex, neurological findings, the period of cervical orthosis, surgery time, blood loss, with or without reconstruction surgery of the semi-spinalis cervicis muscle, and pre-operative axial symptoms. For all of the 51 respondents, the average time since surgery was 4.1 years at the time of investigation; 42 patients complained of post-operative axial symptoms; 26 patients stated the duration of symptoms after surgery to be more than 2 years. The surgical outcome of this group, however, did not differ from that of the 2-year-or-less group. Axial symptoms, which accounted for 13.3 % of all answers about post-operative impairment of everyday living, were similar to hand numbness. Of respondents with post-operative axial symptoms, 52.2 % stated the frequency of affliction to be "all day long", but 34.8 % replied "rarely" to frequency of use of analgesics. Axial symptoms in the nuchal region increased from 45.2 % to 48.6 % after surgery. Stiffness was the most common characteristic before and following surgery, but pain significantly increased from 24.6 % before surgery to 38.4 % after surgery. It was speculated that the principal manifestation of axial symptoms might be pain and that the nuchal region might be the predominant region for axial symptoms. There was no significant difference in age, blood loss, operative time, sex, duration of use of cervical orthosis, reconstructive surgery, and pre-operative symptoms between 2 groups – those who complained of axial symptoms after surgery, and those who did not. The authors concluded that axial symptoms were not usually so severe as to require analgesic use and did not worsen the Japanese Orthopedic Association score after surgery; symptoms were, however, considered to continuously affect everyday life as much as hand numbness. Regarding their features, the authors speculated that the main characteristics of axial symptoms might be pain and that the nuchal region might be the predominant region for axial symptoms. These findings are consistent with the hypothesis that laminoplasty is not, as such, an effective treatment for axial neck pain and that axial symptoms may in fact be worsened by the procedure.

The Work Loss Data Institute's clinical practice guideline on "Neck and upper back (acute & chronic)" (2011) stated that a relative contraindication to laminoplasty is pre-operative neck pain as disruption of the musculature can aggravate axial pain.

In a meta-analysis, Sun and associates (2015) compared the clinical outcomes of anterior approaches (anterior cervical corpectomy with fusion, cervical discectomy with fusion) and posterior approaches (laminectomy, laminoplasty) in multilevel cervical spondylotic myelopathy (MCSM) patients. PubMed, Embase, Scopus, and the Cochrane library were searched for literatures up to March 27, 2015 without language restriction. The reference lists of selected articles were also screened. Heterogeneity was identified using Q test and I2 statistic. A fixed effect model was used for homogeneous data and a random effects model for heterogeneous data. Weighted mean difference (WMD) or odds ratio (OR) with 95 % confidence intervals (CIs) were calculated. Subgroup analysis was conducted according to the cause of MCSM. A total of 17 articles were selected. Higher post-Japanese Orthopedic Association

(JOA, $p = 0.002$) and shorter length of stay ($p = 0.004$) were found in anterior approaches group compared with posterior approaches. Moreover, operation time was shorter ($p < 0.00001$) and neurological recovery rate was higher ($p = 0.005$) in ossification of posterior longitudinal ligament patients who underwent posterior approaches. Complication rate of posterior approaches was lower in spinal stenosis subgroup ($p < 0.0001$). The authors concluded that MCSM patients who underwent anterior approaches showed superior post-JOA and shorten length of stay. However, the outcomes such as operation time and complication rate are associated with the cause of MCSM. Therefore, the favorable surgical strategy for MCSM still needs more studies.

Lee and colleagues (2015) stated that posterior cervical surgery (expansive laminoplasty (EL) or laminectomy followed by fusion (LF)) is usually performed in patients with MCSM (greater than or equal to 3). However, the superiority of either of these techniques is still open to debate. These investigators compared clinical outcomes and post-operative kyphosis in patients undergoing EL versus LF by performing a meta-analysis. Included in the meta-analysis were all studies of EL versus LF in adults with multi-level CSM in MEDLINE (PubMed), EMBASE, and the Cochrane library. A random-effects model was applied to pool data using the MD for continuous outcomes, such as the JOA grade, the cervical curvature index (CCI), and the visual analog scale (VAS) score for neck pain. A total of 7 studies comprising 302 and 290 patients treated with EL and LF, respectively, were included in the final analyses. Both treatment groups showed slight cervical lordosis and moderate neck pain in the baseline state. Both groups were similarly improved in JOA grade (MD 0.09, 95 % CI: -0.37 to 0.54, $p = 0.07$) and neck pain VAS score (MD -0.33, 95 % CI: -1.50 to 0.84, $p = 0.58$). Both groups evenly lost cervical lordosis. In the LF group, lordosis seemed to be preserved in long-term follow-up studies, although the difference between the 2 treatment groups was not statistically significant. The authors concluded that both EL and LF lead to clinical improvement and loss of lordosis evenly. There is no evidence to support EL over LF in the treatment of multi-level CSM. Any superiority between EL and LF remains in question, although the LF group showed favorable long-term results.

In a systematic review and meta-analysis, Huang and colleagues (2016) compared the effectiveness between anterior corpectomy (CORP) and posterior laminoplasty (LAMP) for the treatment of multi-level cervical myelopathy. These investigators searched Medline, Embase, PubMed, OVID, Web of Science and the Cochrane Central Register of Controlled Trials databases for all relevant articles that compared the 2 operations for the treatment of multi-level cervical myelopathy. Exclusion criteria were non-controlled studies, combined anterior and posterior surgery, follow-up of less than 1 year and patients with tumors, trauma, soft disc herniation or previous surgery. The following outcome measures were extracted: Japanese orthopedic association (JOA) score, neurological recovery rate, surgical complications, re-operation rate, operation time and blood loss. A total of 7 high quality studies were included in the meta-analysis. There was no significant difference in pre-operative JOA score [$p > 0.05$, WMD 0.31 (-0.16, 0.79)] and complication rate [$p > 0.05$, OR 1.26 (0.82, 1.94)] between the 2 groups. Significant less re-operation rate [$p < 0.05$, OR 8.16 (3.10, 21.51)], operation time [$p < 0.05$, WMD 67.94 (50.69, 85.20)] and blood loss [$p < 0.05$, WMD 170.06 (80.05, 260.08)] were found in posterior LAMP group. Whereas, patients in anterior CORP group obtained a better post-operative JOA score [$p < 0.05$, WMD 2.02 (1.61, 2.43)] and neurological recovery rate [$p < 0.05$, WMD 7.22 (0.36, 14.08)] than that in posterior LAMP group. The authors concluded that anterior CORP has a higher post-operative JOA score and neurological recovery rate compared with posterior LAMP. However, significant higher re-operation rate, operation time and blood loss should be taken into consideration when anterior CORP is used. They stated that high-quality randomized controlled trials (RCTs) with long-term follow-up and large sample size are needed.

Chen and colleagues (2016) compared the safety and effectiveness of anterior corpectomy and fusion (ACF) with laminoplasty for the treatment of patients diagnosed with cervical ossification of the posterior longitudinal ligament (OPLL). These investigators searched electronic databases for relevant studies that compared the use of ACF with laminoplasty for the treatment of patients with OPLL. Data extraction and quality assessment were conducted, and statistical software was used for data analysis. The random effects model was used if there was heterogeneity between studies; otherwise, the fixed effects model was used. A total of 10 non-RCTs involving 819 patients were included. Post-operative JOA score (p = 0.02, 95 % CI: 0.30 to 2.81) was better in the ACF group than in the laminoplasty group. The recovery rate was superior in the ACF group for patients with an occupying ratio of OPLL of greater than or equal to 60 % (p < 0.00001, 95 % CI: 21.27 to 34.44) and for patients with kyphotic alignment (p < 0.00001, 95 % CI: 16.49 to 27.17). Data analysis also showed that the ACF group was associated with a higher incidence of complications (p = 0.02, 95 % CI: 1.08 to 2.59) and re-operations (p = 0.002, 95 % CI: 1.83 to 14.79), longer operation time (p = 0.01, 95 % CI: 17.72 to 160.75), and more blood loss (p = 0.0004, 95 % CI: 42.22 to 148.45). The authors concluded that for patients with an occupying ratio greater than or equal to 60 % or with kyphotic cervical alignment, ACF appeared to be the preferable treatment method. Nevertheless, laminoplasty appeared to be safe and effective for patients with an occupying ratio less than 60 % or with adequate cervical lordosis. However, it must be emphasized that a surgical strategy should be made based on the individual patient. They stated that further RCTs comparing the use of ACF with laminoplasty for the treatment of OPLL should be performed to make a more convincing conclusion.

In a systematic review and meta-analysis, Qin and colleagues (2018) compared the clinical efficacy, post-operative complication and surgical trauma between anterior cervical corpectomy (ACCF) and fusion LAMP for the treatment of oppressive myelopathy owing to cervical OPLL. An comprehensive search of literature was implemented in 3 electronic databases (Embase, PubMed, and the Cochrane library); RCTs and non-RCTs published since January 1990 to July 2017 that compared ACCF versus LAMP for the treatment of cervical oppressive myelopathy owing to OPLL were acquired. Exclusion criteria were non-human studies, non-controlled studies, combined anterior and posterior operative approach, the other anterior or posterior approaches involving cervical discectomy and fusion and laminectomy with (or without) instrumented fusion, revision surgeries, and cervical myelopathy caused by cervical spondylotic myelopathy. The quality of the included articles was evaluated according to GRADE. The main outcome measures included: pre-operative and post-operative Japanese Orthopedic Association (JOA) score; neuro-functional recovery rate; complication rate; re-operation rate; pre-operative and post-operative C2 to C7 Cobb angle; operation time and intra-operative blood loss; and subgroup analysis was performed according to the mean pre-operative canal occupying ratio (Subgroup A: the mean pre-operative canal occupying ratio less than 60 %, and Subgroup B: the mean pre-operative canal occupying ratio greater than or equal to 60 %). A total of 10 studies containing 735 patients were included in this meta-analysis. And all of the selected studies were non-RCTs with relatively low quality as assessed by GRADE. The results revealed that there was no obvious statistical difference in pre-operative JOA score between the ACCF and LAMP groups in both subgroups. Also, in subgroup A (the mean pre-operative canal occupying ratio of less than 60 %), no obvious statistical difference was observed in the post-operative JOA score and neuro-functional recovery rate between the ACCF and LAMP groups. But, in subgroup B (the mean pre-operative canal occupying ratio of greater than or equal to 60 %), the ACCF group illustrated obviously higher post-operative JOA score and neuro-functional recovery rate than the LAMP group ($p < 0.01$, WMD 1.89 [1.50, 2.28] and $p < 0.01$, WMD 24.40 [20.10, 28.70], respectively). Moreover, the incidence of both complication and reoperation was markedly higher in the ACCF group compared with LAMP group ($p < 0.05$, OR 1.76 [1.05, 2.97] and $p < 0.05$, OR 4.63 [1.86, 11.52], respectively). In addition, the pre-operative cervical C2 to C7 Cobb angle was obviously larger in the LAMP group compared with ACCF group ($p < 0.05$, WMD - 5.77 [- 9.70, - 1.84]). But no statistically obvious difference was detected in the post-operative cervical C2 to C7 Cobb angle between the 2 groups. Furthermore, the ACCF group showed significantly more operation time as well as blood loss compared with LAMP group ($p < 0.01$, WMD 111.43 [40.32,182.54], and $p < 0.01$, WMD 111.32 [61.22, 161.42], respectively). The authors concluded that when the pre-operative canal occupying ratio was less than 60 %, no palpable difference was tested in post-operative JOA score and neuro-functional recovery rate. But, when the pre-operative canal occupying ratio was greater than or equal to 60 %, ACCF was associated with better post-operative JOA score and the recovery rate of neurological function compared with LAMP. Synchronously, ACCF in the cure for cervical myelopathy owing to OPLL led to more surgical trauma and more incidence of complication and re-operation. On the other hand, LAMP had gone a diminished post-operative C2 to C7 Cobb angle, that might be a cause of relatively higher incidence of post-operative late neuro-functional deterioration. In summary, when the pre-operative canal occupying ratio was less than 60 %, LAMP appeared to be safe and effective. However, when the pre-operative canal occupying ratio was greater than or equal to 60 %, the authors preferred to choose ACCF while complications could be controlled by careful manipulation and advanced surgical techniques.

Tamai and colleagues (2017) examined the characteristics of C3-C4 level cervical spondylotic myelopathy (CSM) in elderly patients (C3-C4CSM) (main analysis) and validated the post-operative outcomes of anterior cervical discectomy and fusion (ACDF) and of LAMP (subgroup analysis). The main analysis included 180 patients with CSM, divided into 2 groups (C3-C4CSM group, n=46; conventional CSM group, n=134) according to the findings of the pre-operative physical examination and magnetic resonance imaging (MRI). The subgroup analysis included 46 patients with C3-C4CSM, divided into 2 groups (ACDF group, n=21; LAMP group, n=25) according to surgical technique. Pre-operative demographics and post-operative outcomes were compared. The age at surgery was higher, disease duration was shorter, and pre-operative JOA score was lower in the C3-C4CSM group than in the conventional CSM group. Although the C3-C4 range of motion (ROM) was significantly higher, that of other levels was significantly lower in the C3-C4CSM group. The antero-posterior diameter for levels C3-C7 was significantly larger in the C3-C4CSM group. In the subgroup analysis using the repeated-measures analysis of variance, the post-operative JOA scores, and VAS of neck pain were significantly better in the ACDF group. The authors concluded that higher age, shorter disease duration, and worse JOA scores appeared to be characteristic of C3-C4CSM. In the management of C3-C4CSM, ACDF provided better surgical outcomes than did LAMP; hypermobility at the C3-C4 level, a radiological characteristic of C3-C4CSM, may be one of key factors affecting surgical outcome. The chance to diagnose C3-C4CSM is increasing with the increasing healthy life expectancy. To enable effective resolution of symptoms, C3-C4CSM must be distinguished from conventional CSM.

Xu and colleagues (2017) noted that ACDF and LAMP are used for the treatment of multi-level cervical myelopathy. Despite their widespread applications certain differences are noted between the ACDF and LAMP procedures. These researchers carried out a meta-analysis to compare the clinical outcomes, complications, and surgical trauma between ACDF and LAMP for the treatment of multi-level cervical myelopathy. Medline, Embase, Google Scholar, and Cochrane databases were used for the search of relevant studies until September 2016. The studies aimed to compare the ACDF and LAMP procedures for the treatment of multi-level cervical myelopathy. Title and abstract screening was carried out concomitantly, whereas full text screening was carried out independently. A random effect model was used for heterogeneous data. The data that did not follow heterogeneous pattern were pooled by a fixed effect model in order to examine the MD for continuous outcomes and the OR for dichotomous outcomes, respectively. A total of 6 articles out of 1,351 citations (379 participants) were eligible. Significant differences were noted between the 2 groups in the cobb angle of C2 to C7 (MD = 4.00, 95 % CI: 0.83 to 7.17; p = 0.01) and with regard to the incidence of associated complications (OR = 3.61, 95 % CI: 1.72 to 7.59; p = 0.0007). However, no apparent differences were noted in the variables blood loss (MD = -24.16, 95 % CI: -174.47 to 126.15; p = 0.75), operation time ((MD = 32.81, 95 % CI: -26.76 to 92.38; p = 0.28), recovery rate of JOA score (MD = 4.00, 95 % CI: 0.83 to 7.17; p = 0.01) and incidence of associated complications (OR = 3.61, 95 % CI: 1.72 to 7.59). The authors concluded that the present meta-analysis demonstrated that the rate of complications was lower in the laminoplasty. However, the cobb angle of C2 to C7 was decreased in the ACDF group at the final follow-up period compared with the baseline. The outcomes of the variables blood loss, operation time, ROM and recovery rate of JOA score, were similar in the 2 groups.

Jiang and colleagues (2017) compared the radiological and clinical outcomes of 2 treatments for multi-level cervical degenerative disease: ACDF versus plate-only open-door laminoplasty (laminoplasty). Patients were randomized on a 1:1 randomization schedule with 17 patients in the ACDF group and 17 patients in the laminoplasty group. Clinical outcomes were assessed by a VAS, JOA scores, operative time, blood loss, rates of complications, drainage volume, discharge days after surgery, and complications. The cervical spine curvature index (CI) and ROM were assessed with radiographs. The mean VAS score, the mean JOA score, and the rate of complications did not differ significantly between groups. The laminoplasty group had greater blood loss, a longer operative time, more drainage volume, and a longer hospital stay than the ACDF group. There were no significant differences in the CI and ROM between the 2 groups at baseline and at each follow-up time-point; ROM in both groups decreased significantly after surgery. The authors concluded that both ACDF and laminoplasty were safe and effective treatments for multi-level cervical degenerative disease; ACDF caused fewer traumas than laminoplasty.

In a multi-center, international, prospective cohort study, Fehlings and colleagues (2017) compared outcomes of cervical laminoplasty (LP) and cervical laminectomy and fusion (LF). A total of 266 surgically treated symptomatic degenerative cervical myelopathy (DCM) patients undergoing cervical decompression using LP (n = 100) or LF (n = 166) were included. The outcome measures were the modified JOA score (mJOA), Nurick grade, Neck Disability Index (NDI), Short-Form 36v2 (SF36v2), length of hospital stay, length of stay in the intensive care unit (ICU), treatment complications, and re-operations. Differences in outcomes between the LP and LF groups were analyzed by analysis of variance and analysis of covariance. The dependent variable in all analyses was the change score between baseline and 24-month follow-up, and the independent variable was surgical procedure (LP or LF). In the analysis of co-variance, outcomes were compared between cohorts while adjusting for gender, age, smoking, number of operative levels, duration of symptoms, geographic region, and baseline scores. There were no differences in age, gender, smoking status, number of operated levels, and baseline Nurick, NDI, and SF36v2 scores between the LP and LF groups. Pre-operative mJOA was lower in the LP compared with the LF group (11.52 ± 2.77 and 12.30 ± 2.85, respectively, p = 0.0297). Patients in both groups showed significant improvements in mJOA, Nurick grade, NDI, and SF36v2 physical and mental health component scores 24 months after surgery (p < 0.0001). At 24 months, mJOA scores improved by 3.49 (95 % CI: 2.84 to 4.13) in the LP group compared with 2.39 (95 % CI: 1.91 to 2.86) in the LF group (p = 0.0069). Nurick grades improved by 1.57 (95 % CI: 1.23 to 1.90) in the LP group and 1.18 (95 % CI: 0.92 to 1.44) in the LF group (p = 0.0770). There were no differences between the groups with respect to NDI and SF36v2 outcomes. After adjustment for pre-operative characteristics, surgical factors and geographic region, the differences in mJOA between surgical groups were no longer significant. The rate of treatment-related complications in the LF group was 28.31 % compared with 21.00 % in the LP group (p = 0.1079). The authors concluded that both LP and LF were effective at improving clinical disease severity, functional status, and quality of life (QOL) in patients with DCM . In an unadjusted analysis, patients treated with LP achieved greater improvements on the mJOA at 24-month follow-up than those who received LF; however, these differences were insignificant following adjustment for relevant confounders.

Humadi and associates (2017) noted that in the late 1990s, spinal surgeons experimented by using maxilla-facial fixation plates as an alternative to sutures, anchors, and local spinous process autografts to provide a more rigid and lasting fixation for laminoplasty. This eventually led to the advent of laminoplasty mini-plates, which are currently used. In a systematic review and meta-analysis, these investigators compared laminoplasty techniques with plate and without plate with regard to functional outcome results. Qualitative and quantitative analyses were performed to evaluate the currently available studies in an attempt to justify the use of a plate in laminoplasty. The principal finding of this study was that there was no statistically significant difference in clinical outcome between the 2 different techniques of laminoplasty. The authors concluded that there is insufficient evidence in the literature to support one technique over the other, and hence, there is no evidence to support change in practice (using or not using the plate in laminoplasty); a RCT will give a better comparison between the 2 groups.

Phan and co-workers (2017) noted that surgical approaches for MCSM include posterior cervical surgery via laminectomy and fusion (LF) or expansive laminoplasty (EL). The relative benefits and risks of either approach in terms of clinical outcomes and complications are not well established. These investigators performed a systematic review and meta-analysis to address this topic. Electronic searches were performed using 6 databases from their inception to January 2016, identifying all relevant RCTs and non-RCTs comparing LF vs EL for multi-level cervical myelopathy. Data was extracted and analyzed according to pre-defined end-points. From 10 included studies, there were 335 patients who underwent LF compared to 320 patients who underwent EL. There was no significant difference found post-operatively between LF and EL groups in terms of post-operative JOA ($p = 0.39$), VAS neck pain ($p = 0.93$), post-operative CCI ($p = 0.32$) and Nurich grade ($p = 0.42$). The total complication rate was higher for LF compared to EL (26.4 versus 15.4 %, RR 1.77, 95 % CI: 1.10 to 2.85, $I2 = 34$ %, $p = 0.02$). Re-operation rate was found to be similar between LF and EL groups (p =

0.52). A significantly higher pooled rate of nerve palsies was found in the LF group compared to EL (9.9 versus 3.7 %, RR 2.76, p = 0.03). No significant difference was found in terms of operative time and intra-operative blood loss. The authors concluded that from the available low-quality evidence, LF and EL approaches for MCSM demonstrated similar clinical improvement and loss of lordosis. However, a higher complication rate was found in LF group, including significantly higher nerve palsy complications. This requires further validation and investigation in larger sample-size prospective and randomized studies.

Lau and colleagues (2017) stated that cervical curvature is an important factor when deciding between laminoplasty and laminectomy with posterior spinal fusion (LPSF) for CSM. These researchers compared outcomes following laminoplasty and LPSF in patients with matched post-operative cervical lordosis. Adults undergoing laminoplasty or LPSF for cervical CSM from 2011 to 2014 were identified. Matched cohorts were obtained by excluding LPSF patients with post-operative cervical Cobb angles outside the range of laminoplasty patients. Clinical outcomes and radiographic results were compared. A subgroup analysis of patients with and without pre-operative pain was performed, and the effects of cervical curvature on pain outcomes were examined. A total of 145 patients were included: 101 who underwent laminoplasty and 44 who underwent LPSF. Pre-operative Nurick scale score, pain incidence, and VAS neck pain scores were similar between the 2 groups. Patients who underwent LPSF had significantly less pre-operative cervical lordosis (5.8° versus 10.9°, p = 0.018). Pre-operative and post-operative C2 to C7 sagittal vertical axis (SVA) and T-1 slope were similar between the 2 groups. Laminoplasty cases were associated with less blood loss (196.6 versus 325.0 ml, p < 0.001) and trended toward shorter hospital stays (3.5 versus 4.3 days, p = 0.054). The peri-operative complication rate was 8.3 %; there was no significant difference between the groups. LPSF was associated with a higher long-term complication rate (11.6 % versus 2.2 %, p = 0.036), with pseudarthrosis accounting for 3 of 5 complications in the LPSF group. Follow-up cervical Cobb angle was similar between the groups (8.8° versus 7.1°, p = 0.454). At final follow-up, LPSF had a significantly lower mean Nurick score (0.9 versus 1.4, p = 0.014). Among patients with pre-operative neck pain, pain incidence (36.4 % versus 31.3 %, p = 0.629) and VAS neck pain (2.1 versus 1.8, p = 0.731) were similar between the groups. Similarly, in patients without pre-operative pain, there was no significant difference in pain incidence (19.4 % versus 18.2 %, p = 0.926) and VAS neck pain (1.0 versus 1.1, p = 0.908). For laminoplasty, there was a significant trend for lower pain incidence (p = 0.010) and VAS neck pain (p = 0.004) with greater cervical lordosis, especially when greater than 20° (p = 0.011 and p = 0.018). Mean follow-up was 17.3 months. The authors concluded that for patients with CSM, LPSF was associated with slightly greater blood loss and a higher long-term complication rate, but offered greater neurological improvement than laminoplasty. In cohorts of matched follow-up cervical sagittal alignment, pain outcomes were similar between laminoplasty and LPSF patients. However, among laminoplasty patients, greater cervical lordosis was associated with better pain outcomes, especially for lordosis greater than 20°. Cervical curvature (lordosis) should be considered as an important factor in pain outcomes following posterior decompression for MCSM.

Qian and colleagues (2018) noted that the efficacy of laminoplasty in patients with cervical kyphosis is controversial. These investigators examined the impact of the initial pathogenesis on the clinical outcomes of laminoplasty in patients with cervical kyphosis. A total of 137 patients with CSM or OPLL underwent laminoplasty from April 2013 to May 2015. Subjects were divided into the following 4 groups: lordosis with CSM (LC), kyphosis with CSM (KC), lordosis with OPLL (LO), and kyphosis with OPLL (KO). The clinical outcome measures included the VAS and mJOA scores, ROM, and the cervical global angle (CGA). The mean VAS and mJOA scores improved significantly in all groups after surgery. The changes in VAS and mJOA scores were significantly smaller, and the JOA recovery rate was significantly lower, in the KC group than in the LC and KO groups. The mean change in the CGA was greatest in the KC group (greater than 8° towards kyphosis). The pre-operative ROM was negatively correlated with the change in CGA and the JOA recovery rate in the KO and KC groups. The authors concluded that laminoplasty is suitable for patients with cervical lordosis and those with mild cervical kyphosis and OPLL, but is not recommended for patients with kyphosis and CSM, particularly those with a large ROM pre-operatively. This study had several drawbacks. First, it was a retrospective study. Second, it was limited to a single institution. Third, a relatively small number of patients were involved, and the follow-up duration was short. These researchers plan to conduct a further study involving a larger number of patients and a longer follow-up.

Lee, et al. (2016) assessed postoperative cervical lordosis, clinical outcome, and progression of ossification of the posterior longitudinal ligament (OPLL) in patients with cervical spondylotic myelopathy (CSM) by the OPLL. The posterior approach is usually used for multilevel (≥3) CSM and is decided based on cervical lordosis and instability. OPLL, 1 cause of CSM, makes decreased neck motion and is progressed by neck motion. In OPLL patients, it may be asked whether motion-preserving surgery is still helpful. The investigators reviewed 57 patients of CSM by OPLL who underwent 3 posterior surgeries, laminoplasty, laminectomy alone (LA), and laminectomy with fusion (LF), and followed up minimum 24 months. Postoperative cervical sagittal balance was measured using by the C2-C7 sagittal vertical axis (SVA), cervical curvature index, and C2-C7 Cobb angle. The clinical outcome was analyzed by the neck disability index and the visual analog scale for axial pain. OPLL progression was measured by length and depth growth. A linear mixed model was used to evaluate the differences between each time point and baseline score. Cervical lordosis, C2-C7 Cobb angle, and cervical curvature index decreased gradually in all patients. SVA was maintained in the LF group only and increased in the others (P=0.01). Clinical outcomes, neck disability index, and visual analogue scale were evenly improved in all groups. In patients showing SVA≥40 mm at baseline, neck pain increased in the laminoplasty group but was stationary in the LF group. Progression of OPLL was observed more frequent in the LA group than in the LF group. The investigators concluded that posterior surgeries resulted in clinical improvements although with loss of cervical lordosis in CSM with OPLL patients. OPLL may worsen more frequently after LA. LF and laminoplasty are preferable techniques in this condition, with the former better for patients with high baseline SVA distances.

Intradiscal Lumbar Interbody Fusion and Posterior Stabilization for the Treatment of Lumbar Degenerative Disc Disease

Fiori and colleagues (2020) reported the preliminary results of a novel full percutaneous interbody fusion technique for the treatment of DDD resistant to conservative treatment with posterior stabilization with rods and screws and trans-foraminal placement of an 8-mm-width intradiscal cage. A total of 79 patients with lumbar spine DDD resistant to medical therapy and/or spondylolisthesis up to grade 2 were treated. These researchers carried out pre-operative X-rays, computed tomography (CT) and MRI. The outcomes were assessed using the VAS score and the ODI at a 1-, 6- and 12-month follow-up and also included X-rays to examine the correct bone fusion and the absence of complications. Mean operation time was 130 mins, and mean post-operative time until hospital discharge was 2 days. Post-operative values for VAS scores and ODI improved significantly compared to pre-operative data: Mean pre-procedural VAS was 7.49 ± 0.69 and decreased at 12-month follow-up to 1.31 ± 0.72, and mean pre-procedural ODI was 29.94 ± 1.67 and decreased at 12-month follow-up to 12.75 ± 1.44. No poor results were reported, and no post-procedural sequelae were observed. The authors concluded that these preliminary findings demonstrated a safe and feasible full percutaneous alternative procedure and represented a minimally invasive management of DDD with LBP resistant to medical therapy with or without lumbar spondylolisthesis up to grade 2. These preliminary findings need to be validated by well-designed studies.

Laminectomy with Instrumented Fusion Versus Laminoplasty for the Treatment of Multilevel Cervical Spondylotic Myelopathy

Lin and colleagues (2019) noted that posterior laminectomy with instrumented fusion and laminoplasty are widely used for the treatment of multi-level cervical spondylotic myelopathy (MCSM). There is great controversy over the preferred surgical method. In a systematic review and meta-analysis, these investigators evaluated the safety and clinical outcomes between laminectomy with instrumented fusion and laminoplasty for the treatment of MCSM. Related studies that compared the effectiveness of laminectomy with instrumented fusion and laminoplasty for the treatment of MCSM were acquired by a comprehensive search in PubMed, Embase, the Cochrane library, CNKI, VIP, and WanFang up to April 2018. Included studies were evaluated according to eligibility criteria. The main end-points included: pre-operative and post-operative JOA scores, pre-operative and post-operative visual analog scale (VAS), pre-operative and post-operative cervical ROM, pre-operative and post-operative cervical curvature index (CCI), overall complication rate, C5 nerve palsy rate, axial symptoms rate, operation time and blood loss. A total of 15 studies were included in this meta-analysis. All of the selected studies were of high quality as indicated by the Newcastle-Ottawa scale (NOS). Among 1,131 patients, 555 underwent laminectomy with instrumented fusion and 576 underwent laminoplasty. The results of this meta-analysis indicated no significant difference in pre-operative and post-operative JOA scores, pre-operative and post-operative VAS, pre-operative and post-operative CCI, pre-operative ROM and axial symptoms rate. However, compared with laminoplasty, laminectomy with instrumented fusion exhibited a higher overall complication rate [RR=1.99, 95 % CI: 1.24 to 3.21), p < 0.05], a higher C5 palsy rate [RR=2.22, 95 % CI: 1.30 to 3.80, p< 0.05], a decreased post-operative ROM [standard mean difference (SMD)=-1.51, 95 % CI: -2.14 to -0.88), p< 0.05], a longer operation time [SMD=0.51, 95 % CI: 0.12 to 0.90, p< 0.05] and increased blood loss [SMD=0.47, 95 % CI: 0.30 to 0.65, p< 0.05]. The authors concluded that these findings suggested that both posterior laminectomy with instrumented fusion and laminoplasty were effective for MCSM. However, laminoplasty appeared to allow for a greater ROM, lower overall complication and C5 palsy rates, shorter operation time and lower blood loss. Moreover, these researchers stated that future well-designed, RCTS are needed to confirm these findings.

The authors stated that this review had several drawbacks. First, only 1 of the included studies was a RCT. Second, there was variability in choosing the indicators to evaluate clinical outcomes between the included studies, indicating a lack of standard outcome measurements. Third, the length of follow-up varied between studies, and this was important for surgical outcome evaluations. Finally, clinical heterogeneity might be caused by the various indications for operations.

In a meta-analysis, Yuan and associates (2019) evaluated the safety and efficacy between laminectomy and fusion (LF) versus laminoplasty (LP) for the treatment of MCSM. These investigators searched electronic databases using PubMed, Medline, Embase, Cochrane Controlled Trial Register, and Google Scholar for relevant studies that compared the clinical effectiveness of LF and LP for the treatment of patients with MCSM. The following outcome measures were extracted: the JOA scores, CCI, VAS, Nurich grade, re-operation rate, complications, rate of nerve palsies. Newcastle Ottawa Quality Assessment Scale (NOQAS) was used to evaluate the quality of each study. Data analysis was conducted with RevMan 5.3. A total of 14 studies were included in this meta-analysis. No significant difference was observed in terms of post-operative JOA score (p=0.29), VAS-neck pain (p =0.64), CCI (p=0.24), Nurich grade (p =0.16) and re-operation rate (p =0.21) between LF and LP groups. Compared with LP group, the total complication rate (OR 2.60, 95 % CI: 1.85 to 3.64, I=26 %, p < 0.00001) and rate of nerve palsies (OR 3.18, 95 % CI: 1.66 to 6.11, I=47 %, p=0.0005) was higher in the LF group. The authors concluded that the findings of this meta-analysis demonstrated that surgical treatments of MCSM were similar in terms of most clinical outcomes using LF and LP. However, LP was found to be superior than LF in terms of nerve palsy complications; and these researchers stated that this requires further validation and investigation in high-quality RCTs with large sample size and long-term follow-up.

The authors stated that this meta-analysis had several drawbacks. First, in most the studies selected were not RCT, while it did not influence the credibility of the results. Second, laminoplasty had different techniques, such as open door and French door and these differences were not considered. Third, the current research has not been registered and there may be some small bias, but these investigators still followed the steps of system evaluation strictly. Finally, clinical heterogeneity might be caused by the various indications for surgery and the surgical technologies used at the different treatment centers.

Laminectomy Versus Laminoplasty for the Treatment of Spinal Cord Tumors

Sun and colleagues (2019) stated that laminectomy (LAMT) and laminoplasty (LAMP) have been wildly applied on patients with spinal cord tumors (SCTs). However, the clinical efficacy of LAMT versus LAMP remains controversial. In a systematic review and meta-analysis, these researchers examined the safety and efficacy of LAMT compared with LAMP in the treatment of SCTs. They searched several English and Chinese databases (PubMed, Embase, The Cochrane Library, CBM, CNKI and WanFang) to identify relevant RCTs or observational studies (OSs). The quality of included studies was assessed by the Cochrane Collaboration's tool and the NOS. The pooled analysis was conducted by RevMan 5.3 software. The outcome measures included the primary and secondary outcomes. Subgroups analysis was performed to examine the impact of study type, age, type of tumor, tumor size, surgical levels, follow-up time, surgical methods (whether with fusion) on the outcome measures. A total of 16 studies of 1,096 patients with SCTs were included in this meta-analysis. The results showed that statistically significant difference between LAMT and LAMP groups was found in terms of effective recovery rate (ERR) ($p = 0.003$), blood loss ($p < 0.00001$), hospital stays ($p = 0.006$), spinal deformity ($p = 0.01$), cerebrospinal fluid (CSF) leak ($p < 0.00001$). However, there was no significant difference in total resection rate of tumor ($p = 0.21$) and operation time ($p = 0.14$). In subgroup analysis, the results indicated that age, type of tumor, follow-up time, surgical levels and methods were the influence factors for spinal deformity incidence. The authors concluded that LAMP might be a safer and more effective surgical method in the treatment of SCTs. In addition, the advantage of fusion in preventing the occurrence of spinal deformity should not to be ignored. However, due to the lack of high quality RCT studies and adequate data, the safety and validity of LAMP was undermined.

Bilateral Laminotomy Versus Total Laminectomy for the Treatment of Lumbar Spinal Stenosis

Pietrantonio and colleagues (2019) noted that lumbar spinal stenosis (LSS) is the most common spinal disease in the geriatric population, and is characterized by a compression of the lumbo-sacral neural roots from a narrowing of the lumbar spinal canal. LSS can result in symptomatic compression of the neural elements, requiring surgical treatment if conservative management fails. Different surgical techniques with or without fusion are current therapeutic options. These investigators reported the long-term clinical outcomes of patients who underwent bilateral laminotomy compared with total laminectomy for LSS. They retrospectively reviewed all the patients treated surgically by the senior author for LSS with total laminectomy and bilateral laminotomy with a minimum of 10 years of follow-up. Patients were divided into 2 treatment groups (total laminectomy, group 1; and bilateral laminotomy, group 2) according to the type of surgical decompression. Clinical outcomes measures included the VAS, the 36-Item Short-Form Health Survey (SF-36) scores, and the ODI. In addition, surgical parameters, re-operation rate, and complications were evaluated in both groups. A total of 214 patients met the inclusion and exclusion criteria (105 and 109 patients in groups 1 and 2, respectively). The mean age at surgery was 69.5 years (range of 58 to 77). Comparing pre- and post-operative values, both groups showed improvement in ODI and SF-36 scores; at final follow-up, a slightly better improvement was noted in the laminotomy group (mean ODI value of22.8, mean SF-36 value of 70.2), considering the worse pre-operative scores in this group (mean ODI value of 70, mean SF-36 value of 38.4) with respect to the laminectomy group (mean ODI of 68.7 versus mean SF-36 value of 36.3), but there were no statistically significant differences between the 2 groups. Significantly, in group 2 there was a lower incidence of re-operations (15.2 % versus 3.7 %, $p = 0.0075$). The authors concluded that bilateral laminotomy allowed adequate and safe decompression of the spinal canal in patients with LSS; this technique ensured a significant improvement in patients' symptoms, disability, and QOL. Clinical outcomes were similar in both groups, but a lower incidence of complications and iatrogenic instability has been shown in the long-term in the bilateral laminotomy group.

Laminectomy with Instrumented Fixation in the Treatment of Adjacent Segmental Disease Following Anterior Cervical Corpectomy and Fusion (ACCF) Surgery

In a retrospective, observational study, Yang and colleagues (2019) examined the clinical efficacy of laminectomy with instrumented fixation in treatment of adjacent segmental diseases following anterior cervical corpectomy and fusion (ACCF) surgery. Between January 2008 and December 2015, a total of 48 patients who underwent laminectomy with instrumented fixation to treat adjacent segmental diseases following ACCF surgery, were enrolled into this study. Subjects were followed-up for at least 2 years. Pain assessment was determined by VAS score and Neck Disability Index (NDI) score; neurological impairment was evaluated by JOA score; and radiographic parameters were also compared. All comparisons were determined by paired t-test with appropriate Bonferronni correction; VAS score pre-operatively and at last follow-up was 5.28 ± 2.35 versus 1.90 ± 1.06 ($p < 0.001$); JOA score pre-operatively and at last follow-up was 8.2 ± 3.6 versus 14.5 ± 1.1 ($p < 0.001$); NDI score pre-operatively and at last follow-up was 30.5 ± 12.2 versus 10.6 ± 5.8 ($p < 0.001$). Moreover, the losses of cervical lordosis and C2 to C7 ROM after laminectomy were significant (both $p < 0.005$), but not sagittal vertical axis distance. Post-operative complications were few or mild. The authors concluded that safety and clinical effectiveness can be guaranteed when the patients undergo laminectomy with instrumented fixation to treat adjacent segmental diseases following ACCF surgery. Moreover, these researchers stated that a randomized clinical trial with a large sample is needed to validate these findings.

The authors stated that this study had several drawbacks. First, the retrospective character of the study design was bound to cause some selection bias. Second, this study was only an observational study; a comparative study would be better. Third, the small sample size (n = 48).

Laminectomy for Tarlov Cysts (Perineurial Cysts and Sacral Meningeal Cysts)

Seo and colleagues (2014) noted that Tarlov cysts (TCs – also known as perineurial cysts and sacral meningeal cysts) are lesions of the nerve root that are often observed in the sacral area. There is debate regarding whether symptomatic TCs should be treated surgically. These researchers presented the findings of 3 patients with symptomatic TCs who were treated surgically, and introduced sacral re-capping laminectomy. Patients complained of low back pain (LBP) and hypesthesia on lower extremities (LEs). These investigators operated with sacral re-capping technique for all 3 patients. The outcome measure was baseline visual analog scale (VAS) score and post-operative follow-up magnetic resonance images (MRIs). All patients were completely relieved of symptoms following operation. The authors concluded that although not sufficient to address controversies, the findings of this small case series introduced successful use of a particular surgical technique to treat sacral TC, with resolution of most symptoms and no sequelae.

Del Castillo-Calcaneo and associates (2017) noted that TCs are focal dilations of arachnoid and dura mater of the spinal posterior nerve root sheath that appear as cystic lesions of the nerve roots typically in the lower spine, especially in the sacrum, which can cause radicular symptoms when they increase in size and compress the nerve roots. Different open procedures have been described to treat TCs, but no minimally invasive procedures have been described to effectively address this pathology. These investigators reported the findings of a 29-year old woman who presented with right LE pain and weakness. A MRI scan demonstrated a lumbo-sacral TC that protruded through the right L5 to S1 foramina. Through a small laminotomy, cyst drainage followed by neck ligation using a Scanlan modified technique through tubular retractors was performed. The patient recovered full motor function within the first days post-operatively and showed no signs of relapse at 6-month follow-up. The authors concluded that minimally invasive spine surgery through tubular retractors could be safely performed for successful excision and ligation of TC using a Scanlan modified technique.

Haouas and co-workers (2019) stated that TC is a local dilation of the sub-arachnoid space formed within the nerve root and filled with cerebrospinal fluid (CSF). There is no consensus on the best treatment of symptomatic sacral TCs. Many methods have been used to treat these symptomatic lesions, with variable results. These investigators reported a case-series study including 20 patients undergoing surgery for sacral TCs. The outcomes were satisfactory; 80 % of patients improved without neurological worsening in the post-operative period. The surgical technique (sacral laminectomy + cyst puncture + establishment of dural sheath) described for the first time in this study appeared to have been effective in the 20 cases reported in this study.

Chen and colleagues (2019) noted that although laminae are not viewed as essential structures for spinal integrity; however, in the sacrum the anatomical weakness and gravity makes it a vulnerable area for CSF accumulation and expansion. The congenital or post-operative defects of sacral laminae, such as in patients with spina bifida, make this area more susceptible to forming progressive dural ectasia, pseudo-meningocele, or expansile TC. Furthermore, adhesions between the dura and surrounding soft tissue after laminectomy can cause some local symptoms, which are difficult to relieve. These investigators proposed that sacral laminoplasty with titanium mesh could provide a rigid support and barrier to resolve these sacral lesions and local symptoms. From January 2016 to December 2017, patients with progressive CSF-containing lesions in the sacral area and defective sacral laminae were included in the study. After repair of the lesion, these researchers carried out sacral laminoplasty with titanium mesh in each patient. Subsequently, the soft tissue and skin were closed primarily. A total of 6 patients were included; 4 with repaired myelomeningocele had progressive dural ectasia; 1 patient with lipomyelo-meningocele previously underwent detethering surgery and developed post-operative pseudo-meningocele; and 1 patient had a symptomatic TC; 4 of the 6 patients presented with

LBP and local tenderness. During follow-up (ranging of 13 to 37 months), all 6 patients experienced no recurrence of dural ectasia or pseudo-meningocele and were symptoms-free. The authors concluded that sacral laminoplasty with titanium mesh was a safe and effective procedure for treating progressive sacral dural ectasia and refractory pseudo-meningocele, preventing CSF leakage as well as relieving local symptoms that may occur years after previous surgery for spina bifida.

Furthermore, an UpToDate review on "Closed spinal dysraphism: Clinical manifestations, diagnosis, and management" (Khoury, 2020) states that "Although no clear consensus exists, the main indication for neurosurgery is new onset or progression of neurologic symptoms related to the CSD or tethered cord syndrome. Early neurosurgical intervention also is warranted for severe neonatal symptoms such as bowel obstruction. Additional indications for neurosurgical intervention include cases where the spinal cord is internally exposed, such as with intra-sacral meningocele, to decrease the risk of infection and meningitis, and patients who need vertebral stabilization or pain relief. In contrast, severely disabled patients with static deficits related to CSD are unlikely to benefit from surgery. One series reported that such patients did not improve even when operated in infancy".

Minimally Invasive Decompression with Posterior Elements Preservation Versus Laminectomy and Fusion for Lumbar Degenerative Spondylolisthesis

Ricciardi and colleagues (2020) stated that chronic LBP can be due to many different causes, including degenerative spondylolisthesis (DS). For patients who do not respond to conservative management, surgery remains the most effective treatment. Open laminectomy alone and laminectomy and fusion (LF) for DS have been widely examined, however, no meta-analyses have compared minimally invasive decompression with posterior elements preservation (MID) techniques and LF. Minimally invasive techniques might provide specific advantages that were not recognized in previous studies that pooled different decompression strategies together. These researchers carried out a systematic review and meta-analysis, according to the PRISMA statement, of comparative studies reporting surgical, clinical and radiological outcomes of MID and LF for DS. A total of 3,202 papers were screened and 7 were finally included in the meta-analysis. MID is associated with a shorter surgical duration and hospitalization stay, and a lower intra-operative blood loss and residual LBP; however, the residual disability grade was lower in the LF group. Complication rates were similar between the 2 groups. The rate of adjacent segment degeneration was lower in the MID group, whereas data on radiological outcomes were heterogeneous and not suitable for data-pooling. The authors concluded that this meta-analysis suggested that MID might be considered as an effective alternative to LF for DS. Moreover, these researchers stated that further clinical studies are needed to confirm these findings, better-examine radiological outcomes, and identify patient subgroups that may benefit the most from specific techniques.

Cervical Laminoplasty (Laminaplasty)

Laminoplasty (laminaplasty) may be indicated in patients with myelopathy and multiple-level cervical spondylosis, such as in congenital cervical stenosis. When cervical spinal stenosis is severe, various symptoms may develop which include pain, weakness in arms and/or legs and unsteadiness in the gait (myelopathy).

For mild conditions conservative treatment may be sufficient. When symptoms are severe or progressive then a surgical treatment may be necessary. Surgical goals include a decompression of all compressed levels of the spine and stabilization with solid fusion. Surgical techniques are very dependent upon the specific problems of each patient. Anterior and posterior surgical approaches can be applied. In certain cases a decompression laminoplasty without fusion may be employed.

Laminoplasty involves opening and fusion of the cervical spinal canal. During laminoplasty the laminae are split and then held apart by bone struts, sutures or other techniques, in order to enlarge the spinal canal diameter. This procedure is usually performed on the cervical spine and may be used in an effort to lessen the chance of deformity that can develop when a facetectomy or laminectomy is performed alone.

In a retrospective study, Sakaura and colleagues (2005) compared the long-term outcomes after laminoplasty and anterior spinal fusion (ASF) for patients with cervical myelopathy secondary to disc herniation. The authors concluded that because the 2 procedures provided the same neurological improvement, the risks of bone graft complication with ASF must be weighed against the risks of chronic neck pain associated with laminoplasty for determining the best technique. For these investigators, laminoplasty is the procedure of choice for cervical myelopathy due to disc herniation except for patients with single-level disc herniation without developmental canal stenosis, who are considered to be good surgical candidates for ASF.

Ohnari et al (2006) noted that cervical laminoplasty is a good strategy for cervical myelopathy, but some post-operative patients complain of obstinate axial symptoms after surgery (i.e., nuchal pain, neck stiffness, and shoulder pain). It was reported that these symptoms proved to be more serious than has been believed and should be considered in the evaluation of the outcome of cervical spinal surgery. However, axial symptoms are sometimes recognized before surgery, or also after corpectomy. These investigators examined the difference in axial symptoms before and after laminoplasty and discussed the characteristics of these symptoms as a surgical complication. They conducted a questionnaire survey and reviewed the medical records of respondents. A total of 180 patients who underwent a spinous process-splitting laminoplasty for cervical myelopathy caused by degenerative disease in the authors' institution from 1993 until 2002 were included in the study, and were followed-up for 2 years or longer after surgery. Major outcome measures were self-report measures and functional measures. The questionnaire elicited information as follows: the location and characteristics of pre- and post-operative symptoms, frequency and duration of post-operative symptoms, and impairment in activities of everyday living, analgesic use, and the duration of use of cervical orthosis after surgery. The researchers divided axial symptoms into 4 characteristics based on previous reports: pain, heaviness, stiffness, and other pain, heaviness, stiffness, and other.

An illustration of the upper back on which respondents could mark each characteristic was used to acquire information about the location of axial symptoms. The following information was gathered from medical records and statistically analyzed: whether post-operative axial symptoms were related or not, age, sex, neurological findings, the period of cervical orthosis, surgery time, blood loss, with or without reconstruction surgery of the semi-spinalis cervicis muscle, and pre-operative axial symptoms. For all of the 51 respondents, the average time since surgery was 4.1 years at the time of investigation; 42 patients complained of post-operative axial symptoms; 26 patients stated the duration of symptoms after surgery to be more than 2 years. The surgical outcome of this group, however, did not differ from that of the 2-year-or-less group. Axial symptoms, which accounted for 13.3 % of all answers about post-operative impairment of everyday living, were similar to hand numbness. Of respondents with post-operative axial symptoms, 52.2 % stated the frequency of affliction to be "all day long", but 34.8 % replied "rarely" to frequency of use of analgesics. Axial symptoms in the nuchal region increased from 45.2 % to 48.6 % after surgery. Stiffness was the most common characteristic before and following surgery, but pain significantly increased from 24.6 % before surgery to 38.4 % after surgery. It was speculated that the principal manifestation of axial symptoms might be pain and that the nuchal region might be the predominant region for axial symptoms. There was no significant difference in age, blood loss, operative time, sex, duration of use of cervical orthosis, reconstructive surgery, and pre-operative symptoms between 2 groups -- those who complained of axial symptoms after surgery, and those who did not. The authors concluded that axial symptoms were not usually so severe as to require analgesic use and did not worsen the Japanese Orthopedic Association score after surgery; symptoms were, however, considered to continuously affect everyday life as much as hand numbness. Regarding their features, the authors speculated that the main characteristics of axial symptoms might be pain and that the nuchal region might be the predominant region for axial symptoms. These findings are consistent with the hypothesis that laminoplasty is not, as such, an effective treatment for axial neck pain and that axial symptoms may in fact be worsened by the procedure.

The Work Loss Data Institute's clinical practice guideline on "Neck and upper back (acute & chronic)" (2011) stated that a relative contraindication to laminoplasty is pre-operative neck pain as disruption of the musculature can aggravate axial pain.

In a meta-analysis, Sun and associates (2015) compared the clinical outcomes of anterior approaches (anterior cervical corpectomy with fusion, cervical discectomy with fusion) and posterior approaches (laminectomy, laminoplasty) in multilevel cervical spondylotic myelopathy (MCSM) patients. PubMed, Embase, Scopus, and the Cochrane library were searched for literatures up to March 27, 2015 without language restriction. The reference lists of selected articles were also screened. Heterogeneity was identified using Q test and I2 statistic. A fixed effect model was used for homogeneous data and a random effects model for heterogeneous data. Weighted mean difference (WMD) or odds ratio (OR) with 95 % confidence intervals (CIs) were calculated. Subgroup analysis was conducted according to the cause of MCSM. A total of 17 articles were selected. Higher post-Japanese Orthopedic Association (JOA, p = 0.002) and shorter length of stay (p = 0.004) were found in anterior approaches group compared with posterior approaches. Moreover, operation time was shorter (p < 0.00001) and neurological recovery rate was higher (p = 0.005) in ossification of posterior longitudinal ligament patients who underwent posterior approaches. Complication rate of posterior approaches was lower in spinal stenosis subgroup (p < 0.0001). The authors concluded that MCSM patients who underwent anterior approaches showed superior post-JOA and shorten length of stay. However, the outcomes such as operation time and complication rate are associated with the cause of MCSM. Therefore, the favorable surgical strategy for MCSM still needs more studies.

Lee and colleagues (2015) stated that posterior cervical surgery (expansive laminoplasty (EL) or laminectomy followed by fusion (LF)) is usually performed in patients with MCSM (greater than or equal to 3). However, the superiority of either of these techniques is still open to debate. These investigators compared clinical outcomes and post-operative kyphosis in patients undergoing EL versus LF by performing a meta-analysis. Included in the meta-analysis were all studies of EL versus LF in adults with multi-level CSM in MEDLINE (PubMed), EMBASE, and the Cochrane library. A random-effects model was applied to pool data using the MD for continuous outcomes, such as the JOA grade, the cervical curvature index (CCI), and the visual analog scale (VAS) score for neck pain. A total of 7 studies comprising 302 and 290 patients treated with EL and LF, respectively, were included in the final analyses. Both treatment groups showed slight cervical lordosis and moderate neck pain in the baseline state. Both groups were similarly improved in JOA grade (MD 0.09, 95 % CI: -0.37 to 0.54, p = 0.07) and neck pain VAS score (MD -0.33, 95 % CI: -1.50 to 0.84, p = 0.58). Both groups evenly lost cervical lordosis. In the LF group, lordosis seemed to be preserved in long-term follow-up studies, although the difference between the 2 treatment groups was not statistically significant. The authors concluded that both EL and LF lead to clinical improvement and loss of lordosis evenly. There is no evidence to support EL over LF in the treatment of multi-level CSM. Any superiority between EL and LF remains in question, although the LF group showed favorable long-term results.

Anterior Corpectomy Versus Posterior Laminoplasty for Multi-Level Cervical Myelopathy

In a systematic review and meta-analysis, Huang and colleagues (2016) compared the effectiveness between anterior corpectomy (CORP) and posterior laminoplasty (LAMP) for the treatment of multi-level cervical myelopathy. These investigators searched Medline, Embase, PubMed, OVID, Web of Science and the Cochrane Central Register of Controlled Trials databases for all relevant articles that compared the 2 operations for the treatment of multi-level cervical myelopathy. Exclusion criteria were non-controlled studies, combined anterior and posterior surgery, follow-up of less than 1 year and patients with tumors, trauma, soft disc herniation or previous surgery. The following outcome measures were extracted: Japanese orthopedic association (JOA) score, neurological recovery rate, surgical complications, re-operation rate, operation time and blood loss. A total of 7 high quality studies were included in the meta-analysis. There was no significant difference in pre-operative JOA score [$p > 0.05$, WMD 0.31 (-0.16, 0.79)] and complication rate [$p > 0.05$, OR 1.26 (0.82, 1.94)] between the 2 groups. Significant less re-operation rate [$p < 0.05$, OR 8.16 (3.10, 21.51)], operation time [$p < 0.05$, WMD 67.94 (50.69, 85.20)] and blood loss [$p < 0.05$, WMD 170.06 (80.05, 260.08)] were found in posterior LAMP group. Whereas, patients in anterior CORP group obtained a better post-operative JOA score [$p < 0.05$, WMD 2.02 (1.61, 2.43)] and neurological recovery rate [$p < 0.05$, WMD 7.22 (0.36, 14.08)] than that in posterior LAMP group. The authors concluded that anterior CORP has a higher post-operative JOA score and neurological recovery rate compared with posterior LAMP. However, significant higher re-operation rate, operation time and blood loss should be taken into consideration when anterior CORP is used. They stated that high-quality randomized controlled trials (RCTs) with long-term follow-up and large sample size are needed.

Anterior Corpectomy and Fusion Versus Laminoplasty for Cervical Ossification of Posterior Longitudinal Ligament

Chen and colleagues (2016) compared the safety and effectiveness of anterior corpectomy and fusion (ACF) with laminoplasty for the treatment of patients diagnosed with cervical ossification of the posterior longitudinal ligament (OPLL). These investigators searched electronic databases for relevant studies that compared the use of ACF with laminoplasty for the treatment of patients with OPLL. Data extraction and quality assessment were conducted, and statistical software was used for data analysis. The random effects model was used if there was heterogeneity between studies; otherwise, the fixed effects model was used. A total of 10 non-RCTs involving 819 patients were included. Post-operative JOA score (p = 0.02, 95 % CI: 0.30 to 2.81) was better in the ACF group than in the laminoplasty group. The recovery rate was superior in the ACF group for patients with an occupying ratio of OPLL of greater than or equal to 60 % (p < 0.00001, 95 % CI: 21.27 to 34.44) and for patients with kyphotic alignment (p < 0.00001, 95 % CI: 16.49 to 27.17). Data analysis also showed that the ACF group was associated with a higher incidence of complications (p = 0.02, 95 % CI: 1.08 to 2.59) and re-operations (p = 0.002, 95 % CI: 1.83 to 14.79), longer operation time (p = 0.01, 95 % CI: 17.72 to 160.75), and more blood loss (p = 0.0004, 95 % CI: 42.22 to 148.45). The authors concluded that for patients with an occupying ratio greater than or equal to 60 % or with kyphotic cervical alignment, ACF appeared to be the preferable treatment method. Nevertheless, laminoplasty appeared to be safe and effective for patients with an occupying ratio less than 60 % or with adequate cervical lordosis. However, it must be emphasized that a surgical strategy should be made based on the individual patient. They stated that further RCTs comparing the use of ACF with laminoplasty for the treatment of OPLL should be performed to make a more convincing conclusion.

In a systematic review and meta-analysis, Qin and colleagues (2018) compared the clinical efficacy, post-operative complication and surgical trauma between anterior cervical corpectomy (ACCF) and fusion LAMP for the treatment of oppressive myelopathy owing to cervical OPLL. An comprehensive search of literature was implemented in 3 electronic databases (Embase, PubMed, and the Cochrane library); RCTs and non-RCTs published since January 1990 to July 2017 that compared ACCF versus LAMP for the treatment of cervical oppressive myelopathy owing to OPLL were acquired. Exclusion criteria were non-human studies, non-controlled studies, combined anterior and posterior operative approach, the other anterior or posterior approaches involving cervical discectomy and fusion and laminectomy with (or without) instrumented fusion, revision surgeries, and cervical myelopathy caused by cervical spondylotic myelopathy. The quality of the included articles was evaluated according to GRADE. The main outcome measures included: pre-operative and post-operative Japanese Orthopedic Association (JOA) score; neuro-functional recovery rate; complication rate; re-operation rate; pre-operative and post-operative C2 to C7 Cobb angle; operation time and intra-operative blood loss; and subgroup analysis was performed according to the mean pre-operative canal occupying ratio (Subgroup A: the mean pre-operative canal occupying ratio less than 60 %, and Subgroup B: the mean pre-operative canal occupying ratio greater than or equal to 60 %). A total of 10 studies containing 735 patients were included in this meta-analysis. And all of the selected studies were non-RCTs with relatively low quality as assessed by GRADE. The results revealed that there was no obvious statistical difference in pre-operative JOA score between the ACCF and LAMP groups in both subgroups. Also, in subgroup A (the mean pre-operative canal occupying ratio of less than 60 %), no obvious statistical difference was observed in the post-operative JOA score and neuro-functional recovery rate between the ACCF and LAMP groups. But, in subgroup B (the mean pre-operative canal occupying ratio of greater than or equal to 60 %), the ACCF group illustrated obviously higher post-operative JOA score and neuro-functional recovery rate than the LAMP group ($p < 0.01$, WMD 1.89 [1.50, 2.28] and $p < 0.01$, WMD 24.40 [20.10, 28.70], respectively). Moreover, the incidence of both complication and reoperation was markedly higher in the ACCF group compared with LAMP group ($p < 0.05$, OR 1.76 [1.05, 2.97] and $p < 0.05$, OR 4.63 [1.86, 11.52], respectively). In addition, the pre-operative cervical C2 to C7 Cobb angle was obviously larger in the

LAMP group compared with ACCF group ($p < 0.05$, WMD - 5.77 [- 9.70, - 1.84]). But no statistically obvious difference was detected in the post-operative cervical C2 to C7 Cobb angle between the 2 groups. Furthermore, the ACCF group showed significantly more operation time as well as blood loss compared with LAMP group ($p < 0.01$, WMD 111.43 [40.32,182.54], and $p < 0.01$, WMD 111.32 [61.22, 161.42], respectively). The authors concluded that when the pre-operative canal occupying ratio was less than 60 %, no palpable difference was tested in post-operative JOA score and neuro-functional recovery rate. But, when the pre-operative canal occupying ratio was greater than or equal to 60 %, ACCF was associated with better post-operative JOA score and the recovery rate of neurological function compared with LAMP. Synchronously, ACCF in the cure for cervical myelopathy owing to OPLL led to more surgical trauma and more incidence of complication and re-operation. On the other hand, LAMP had gone a diminished post-operative C2 to C7 Cobb angle, that might be a cause of relatively higher incidence of post-operative late neuro-functional deterioration. In summary, when the pre-operative canal occupying ratio was less than 60 %, LAMP appeared to be safe and effective. However, when the pre-operative canal occupying ratio was greater than or equal to 60 %, the authors preferred to choose ACCF while complications could be controlled by careful manipulation and advanced surgical techniques.

Anterior Cervical Discectomy and Fusion Versus Posterior Laminoplasty

Tamai and colleagues (2017) examined the characteristics of C3-C4 level cervical spondylotic myelopathy (CSM) in elderly patients (C3-C4CSM) (main analysis) and validated the post-operative outcomes of anterior cervical discectomy and fusion (ACDF) and of LAMP (subgroup analysis). The main analysis included 180 patients with CSM, divided into 2 groups (C3-C4CSM group, n=46; conventional CSM group, n=134) according to the findings of the pre-operative physical examination and magnetic resonance imaging (MRI). The subgroup analysis included 46 patients with C3-C4CSM, divided into 2 groups (ACDF group, n=21; LAMP group, n=25) according to surgical technique. Pre-operative demographics and post-operative outcomes were compared. The age at surgery was higher, disease duration was shorter, and pre-operative JOA score was lower in the C3-C4CSM group than in the conventional CSM group. Although the C3-C4 range of motion (ROM) was significantly higher, that of other levels was significantly lower in the C3-C4CSM group. The antero-posterior diameter for levels C3-C7 was significantly larger in the C3-C4CSM group. In the subgroup analysis using the repeated-measures analysis of variance, the post-operative JOA scores, and VAS of neck pain were significantly better in the ACDF group. The authors concluded that higher age, shorter disease duration, and worse JOA scores appeared to be characteristic of C3-C4CSM. In the management of C3-C4CSM, ACDF provided better surgical outcomes than did LAMP; hypermobility at the C3-C4 level, a radiological characteristic of C3-C4CSM, may be one of key factors affecting surgical outcome. The chance to diagnose C3-C4CSM is increasing with the increasing healthy life expectancy. To enable effective resolution of symptoms, C3-C4CSM must be distinguished from conventional CSM.

Xu and colleagues (2017) noted that ACDF and LAMP are used for the treatment of multi-level cervical myelopathy. Despite their widespread applications certain differences are noted between the ACDF and LAMP procedures. These researchers carried out a meta-analysis to compare the clinical outcomes, complications, and surgical trauma between ACDF and LAMP for the treatment of multi-level cervical myelopathy. Medline, Embase, Google Scholar, and Cochrane databases were used for the search of relevant studies until September 2016. The studies aimed to compare the ACDF and LAMP procedures for the treatment of multi-level cervical myelopathy. Title and abstract screening was carried out concomitantly, whereas full text screening was carried out independently. A random effect model was used for heterogeneous data. The data that did not follow heterogeneous pattern were pooled by a fixed effect model in order to examine the MD for continuous outcomes and the OR for dichotomous outcomes, respectively. A total of 6 articles out of 1,351 citations (379 participants) were eligible. Significant differences were noted between the 2 groups in the cobb angle of C2 to C7 (MD = 4.00, 95 % CI: 0.83 to 7.17; p = 0.01) and with regard to the incidence of associated complications (OR = 3.61, 95 % CI: 1.72 to 7.59; p = 0.0007). However, no apparent differences were noted in the variables blood loss (MD = -24.16, 95 % CI: -174.47 to 126.15; p = 0.75), operation time ((MD = 32.81, 95 % CI: -26.76 to 92.38; p = 0.28), recovery rate of JOA score (MD = 4.00, 95 % CI: 0.83 to 7.17; p = 0.01) and incidence of associated complications (OR = 3.61, 95 % CI: 1.72 to 7.59). The authors concluded that the present meta-analysis demonstrated that the rate of complications was lower in the laminoplasty. However, the cobb angle of C2 to C7 was decreased in the ACDF group at the final follow-up period compared with the baseline. The outcomes of the variables blood loss, operation time, ROM and recovery rate of JOA score, were similar in the 2 groups.

Anterior Cervical Discectomy and Fusion Versus Plate-Only Open-Door Laminoplasty

Jiang and colleagues (2017) compared the radiological and clinical outcomes of 2 treatments for multi-level cervical degenerative disease: ACDF versus plate-only open-door laminoplasty (laminoplasty). Patients were randomized on a 1:1 randomization schedule with 17 patients in the ACDF group and 17 patients in the laminoplasty group. Clinical outcomes were assessed by a VAS, JOA scores, operative time, blood loss, rates of complications, drainage volume, discharge days after surgery, and complications. The cervical spine curvature index (CI) and ROM were assessed with radiographs. The mean VAS score, the mean JOA score, and the rate of complications did not differ significantly between groups. The laminoplasty group had greater blood loss, a longer operative time, more drainage volume, and a longer hospital stay than the ACDF group. There were no significant differences in the CI and ROM between the 2 groups at baseline and at each follow-up time-point; ROM in both groups decreased significantly after surgery. The authors concluded that both ACDF and laminoplasty were safe and effective treatments for multi-level cervical degenerative disease; ACDF caused fewer traumas than laminoplasty.

Laminectomy and Fusion Versus Laminoplasty for Degenerative Cervical Myelopathy

In a multi-center, international, prospective cohort study, Fehlings and colleagues (2017) compared outcomes of cervical laminoplasty (LP) and cervical laminectomy and fusion (LF). A total of 266 surgically treated symptomatic degenerative cervical myelopathy (DCM) patients undergoing cervical decompression using LP (n = 100) or LF (n = 166) were included. The outcome measures were the modified JOA score (mJOA), Nurick grade, Neck Disability Index (NDI), Short-Form 36v2

(SF36v2), length of hospital stay, length of stay in the intensive care unit (ICU), treatment complications, and re-operations. Differences in outcomes between the LP and LF groups were analyzed by analysis of variance and analysis of covariance. The dependent variable in all analyses was the change score between baseline and 24-month follow-up, and the independent variable was surgical procedure (LP or LF). In the analysis of co-variance, outcomes were compared between cohorts while adjusting for gender, age, smoking, number of operative levels, duration of symptoms, geographic region, and baseline scores. There were no differences in age, gender, smoking status, number of operated levels, and baseline Nurick, NDI, and SF36v2 scores between the LP and LF groups. Pre-operative mJOA was lower in the LP compared with the LF group (11.52 ± 2.77 and 12.30 ± 2.85, respectively, $p = 0.0297$). Patients in both groups showed significant improvements in mJOA, Nurick grade, NDI, and SF36v2 physical and mental health component scores 24 months after surgery ($p < 0.0001$). At 24 months, mJOA scores improved by 3.49 (95 % CI: 2.84 to 4.13) in the LP group compared with 2.39 (95 % CI: 1.91 to 2.86) in the LF group ($p = 0.0069$). Nurick grades improved by 1.57 (95 % CI: 1.23 to 1.90) in the LP group and 1.18 (95 % CI: 0.92 to 1.44) in the LF group ($p = 0.0770$). There were no differences between the groups with respect to NDI and SF36v2 outcomes. After adjustment for pre-operative characteristics, surgical factors and geographic region, the differences in mJOA between surgical groups were no longer significant. The rate of treatment-related complications in the LF group was 28.31 % compared with 21.00 % in the LP group ($p = 0.1079$). The authors concluded that both LP and LF were effective at improving clinical disease severity, functional status, and QOL in patients with DCM. In an unadjusted analysis, patients treated with LP achieved greater improvements on the mJOA at 24-month follow-up than those who received LF; however, these differences were insignificant following adjustment for relevant confounders.

Cervical Laminoplasty Using Mini-Plates

Humadi and associates (2017) noted that in the late 1990s, spinal surgeons experimented by using maxilla-facial fixation plates as an alternative to sutures, anchors, and local spinous process autografts to provide a more rigid and lasting fixation for laminoplasty. This eventually led to the advent of laminoplasty mini-plates, which are currently used. In a systematic review and meta-analysis, these investigators compared laminoplasty techniques with plate and without plate with regard to functional outcome results. Qualitative and quantitative analyses were performed to evaluate the currently available studies in an attempt to justify the use of a plate in laminoplasty. The principal finding of this study was that there was no statistically significant difference in clinical outcome between the 2 different techniques of laminoplasty. The authors concluded that there is insufficient evidence in the literature to support one technique over the other, and hence, there is no evidence to support change in practice (using or not using the plate in laminoplasty); a RCT will give a better comparison between the 2 groups.

Laminectomy and Fusion Versus Laminoplasty for Multi-Level Cervical Myelopathy

Phan and co-workers (2017) noted that surgical approaches for MCSM include posterior cervical surgery via laminectomy and fusion (LF) or expansive laminoplasty (EL). The relative benefits and risks of either approach in terms of clinical outcomes and complications are not well established. These investigators performed a systematic review and meta-analysis to address this topic. Electronic searches were performed using 6 databases from their inception to January 2016, identifying all relevant RCTs and non-RCTs comparing LF vs EL for multi-level cervical myelopathy. Data was extracted and analyzed according to pre-defined end-points. From 10 included studies, there were 335 patients who underwent LF compared to 320 patients who underwent EL. There was no significant difference found post-operatively between LF and EL groups in terms of post-operative JOA ($p = 0.39$), VAS neck pain ($p = 0.93$), post-operative CCI ($p = 0.32$) and Nurich grade ($p = 0.42$). The total complication rate was higher for LF compared to EL (26.4 versus 15.4 %, RR 1.77, 95 % CI: 1.10 to 2.85, $I2 = 34$ %, $p = 0.02$). Re-operation rate was found to be similar between LF and EL groups ($p = 0.52$). A significantly higher pooled rate of nerve palsies was found in the LF group compared to EL (9.9 versus 3.7 %, RR 2.76, $p = 0.03$). No significant difference was found in terms of operative time and intra-operative blood loss. The authors concluded that from the available low-quality evidence, LF and EL approaches for MCSM demonstrated similar clinical improvement and loss of lordosis. However, a higher complication rate was found in LF group, including significantly higher nerve palsy complications. This requires further validation and investigation in larger sample-size prospective and randomized studies.

Lau and colleagues (2017) stated that cervical curvature is an important factor when deciding between laminoplasty and laminectomy with posterior spinal fusion (LPSF) for CSM. These researchers compared outcomes following laminoplasty and LPSF in patients with matched post-operative cervical lordosis. Adults undergoing laminoplasty or LPSF for cervical CSM from 2011 to 2014 were identified. Matched cohorts were obtained by excluding LPSF patients with post-operative cervical Cobb angles outside the range of laminoplasty patients. Clinical outcomes and radiographic results were compared. A subgroup analysis of patients with and without pre-operative pain was performed, and the effects of cervical curvature on pain outcomes were examined. A total of 145 patients were included: 101 who underwent laminoplasty and 44 who underwent LPSF. Pre-operative Nurick scale score, pain incidence, and VAS neck pain scores were similar between the 2 groups. Patients who underwent LPSF had significantly less pre-operative cervical lordosis (5.8° versus 10.9°, $p = 0.018$). Pre-operative and post-operative C2 to C7 sagittal vertical axis (SVA) and T-1 slope were similar between the 2 groups. Laminoplasty cases were associated with less blood loss (196.6 versus 325.0 ml, $p < 0.001$) and trended toward shorter hospital stays (3.5 versus 4.3 days, $p = 0.054$). The peri-operative complication rate was 8.3 %; there was no significant difference between the groups. LPSF was associated with a higher long-term complication rate (11.6 % versus 2.2 %, $p = 0.036$), with pseudarthrosis accounting for 3 of 5 complications in the LPSF group. Follow-up cervical Cobb angle was similar between the groups (8.8° versus 7.1°, $p = 0.454$). At final follow-up, LPSF had a significantly lower mean Nurick score (0.9 versus 1.4, $p = 0.014$). Among patients with pre-operative neck pain, pain incidence (36.4 % versus 31.3 %, $p = 0.629$) and VAS neck pain (2.1 versus 1.8, $p = 0.731$) were similar between the groups. Similarly, in patients without pre-operative pain, there was no significant difference in pain incidence (19.4 % versus 18.2 %, $p = 0.926$) and VAS neck pain (1.0 versus 1.1, $p = 0.908$). For laminoplasty, there was a significant trend for lower pain incidence ($p = 0.010$) and VAS neck pain ($p = 0.004$) with greater cervical lordosis, especially when greater than 20° ($p = 0.011$ and $p = 0.018$). Mean follow-up was 17.3 months. The authors concluded that for patients with CSM, LPSF was associated with slightly greater blood loss and a higher long-term complication rate, but offered greater neurological improvement than laminoplasty. In cohorts of matched follow-up cervical sagittal alignment, pain outcomes were similar between laminoplasty and LPSF patients. However, among laminoplasty patients, greater cervical lordosis was associated with better pain outcomes, especially for lordosis greater than 20°. Cervical curvature (lordosis) should be considered as an important factor in pain outcomes following posterior decompression for MCSM.

Cervical Kyphosis with Cervical Spondylotic Myelopathy

Qian and colleagues (2018) noted that the efficacy of laminoplasty in patients with cervical kyphosis is controversial. These investigators examined the impact of the initial pathogenesis on the clinical outcomes of laminoplasty in patients with cervical kyphosis. A total of 137 patients with CSM or OPLL underwent laminoplasty from April 2013 to May 2015. Subjects were divided into the following 4 groups: lordosis with CSM (LC), kyphosis with CSM (KC), lordosis with OPLL (LO), and kyphosis with OPLL (KO). The clinical outcome measures included the VAS and mJOA scores, ROM, and the cervical global angle (CGA). The mean VAS and mJOA scores improved significantly in all groups after surgery. The changes in VAS and mJOA scores were significantly smaller, and the JOA recovery rate was significantly lower, in the KC group than in the LC and KO groups. The mean change in the CGA was greatest in the KC group (greater than 8° towards kyphosis). The pre-operative ROM was negatively correlated with the change in CGA and the JOA recovery rate in the KO and KC groups. The authors concluded that laminoplasty is suitable for patients with cervical lordosis and those with mild cervical kyphosis and OPLL, but is not recommended for patients with kyphosis and CSM, particularly those with a large ROM pre-operatively. This study had several drawbacks. First, it was a retrospective study. Second, it was limited to a single institution. Third, a relatively small number of patients were involved, and the follow-up duration was short. These researchers plan to conduct a further study involving a larger number of patients and a longer follow-up.

Laminoplasty Versus Laminectomy

Lee et al (2016) assessed postoperative cervical lordosis, clinical outcome, and progression of ossification of the posterior longitudinal ligament (OPLL) in patients with cervical spondylotic myelopathy (CSM) by the OPLL. The posterior approach is usually used for multilevel (≥3) CSM and is decided based on cervical lordosis and instability. OPLL, 1 cause of CSM, makes decreased neck motion and is progressed by neck motion. In OPLL patients, it may be asked whether motion-preserving surgery is still helpful. The investigators reviewed 57 patients of CSM by OPLL who underwent 3 posterior surgeries, laminoplasty, laminectomy alone (LA), and laminectomy with fusion (LF), and followed up minimum 24 months. Postoperative cervical sagittal balance was measured using by the C2-C7 sagittal vertical axis (SVA), cervical curvature index, and C2-C7 Cobb angle. The clinical outcome was analyzed by the neck disability index and the visual analog scale for axial pain. OPLL progression was measured by length and depth growth. A linear mixed model was used to evaluate the differences between each time point and baseline score. Cervical lordosis, C2-C7 Cobb angle, and cervical curvature index decreased gradually in all patients. SVA was maintained in the LF group only and increased in the others (P=0.01). Clinical outcomes, neck disability index, and visual analogue scale were evenly improved in all groups. In patients showing SVA≥40 mm at baseline, neck pain increased in the laminoplasty group but was stationary in the LF group. Progression of OPLL was observed more frequent in the LA group than in the LF group. The investigators concluded that posterior surgeries resulted in clinical improvements although with loss of cervical lordosis in CSM with OPLL patients. OPLL may worsen more frequently after LA. LF and laminoplasty are preferable techniques in this condition, with the former better for patients with high baseline SVA distances.

Comparison of Laminectomy and Fusion Versus Laminoplasty

In a meta-analysis, Yuan and colleagues (2019) evaluated the safety and efficacy between laminectomy and fusion (LF) versus LAMP for the treatment of MCSM. These investigators searched electronic databases using PubMed, Medline, Embase, Cochrane Controlled Trial Register, and Google Scholar for relevant studies that compared the clinical effectiveness of LF and LAMP for the treatment of patients with MCSM.

The following outcome measures were extracted: the JOA scores, CCI, VAS, Nurich grade, re-operation rate, complications, rate of nerve palsies. Newcastle Ottawa Quality Assessment Scale (NOQAS) was used to evaluate the quality of each study. Data analysis was conducted with RevMan 5.3. A total of 14 studies were included in this meta-analysis. No significant difference was observed in terms of post-operative JOA score (p=0.29), VAS neck pain (p = 0.64), CCI (p=0.24), Nurich grade (p=0.16) and re-operation rate (p = 0.21) between LF and LAMP groups. Compared with LAMP group, the total complication rate (OR 2.60, 95 % CI: 1.85 to 3.64, I=26 %, p < 0.00001) and rate of nerve palsies (OR 3.18, 95 % CI: 1.66 to 6.11, I=47 %, p=0.0005) was higher in the LF group. The authors concluded that this meta-analysis showed that surgical treatments of MCSM were similar in terms of most clinical outcomes using LF and LAMP. However, LAMP was found to be superior than LF in terms of nerve palsy complications. This requires further validation and investigation in prospective, larger sample-size, randomized trials.

Laminoplasty for Cervical Ossification of the Posterior Longitudinal Ligament

Xu and co-workers (2021) noted that considerable controversy exists over surgical procedures for ossification of the posterior longitudinal ligament (OPLL). In a meta-analysis, these researchers compared the clinical outcome of anterior decompression and fusion (ADF) with LAMP in treatment of cervical myelopathy due to OPLL. PubMed, Embase and the Cochrane Register of Controlled Trials database were searched to identify potential clinical studies that compared ADF with LAMP for cervical myelopathy owing to OPLL. These investigators also manually searched the reference lists of articles and reviews for possible relevant studies. A total of 13 studies with 1,120 patients were included in this analysis. Subgroup analyses were performed by the canal occupying ratio of OPLL. Overall, the mean pre-operative JOA score was similar between 2 groups. Compared with LAMP group, ADF group was higher at the mean post-operative JOA scores and mean recovery rate, re-operation rate, and longer at mean operation time. There were not significant differences in mean blood loss and complication rate between 2 groups. In subgroup analysis, ADF had a higher mean post-operative JOA score and recovery rate than LAMP in cases of OPLL with occupying ratios of greater than or equal to 50 %, while those difference were not found in cases of OPLL with occupying ratios of less than 50 %. The authors concluded that ADF attained better neurological improvement compared with LAMP in treatment of cervical myelopathy due to OPLL, especially in cases of OPLL with occupying ratios of greater than or equal to 50 %. Complication rate was similar between 2 groups, but ADF could increase the risk of re-operation.

Yoshii and colleagues (2020) stated that the optimal surgical procedure for the treatment of cervical OPLL remains controversial because there are few comprehensive studies investigating the surgical methods. These researchers conducted a systematic review and meta-analysis to examine the evidence and compare the surgical outcomes of ADF and LAMP, which are representative procedures for cervical OPLL. They carried out a literature search using PubMed, Embase, and the Cochrane Library to identify comparative studies of ADF and LAMP for cervical OPLL. The language was restricted to English, and the year of publication was from January 1980 to December 2018. These investigators extracted outcomes from the studies, such as pre-operative and post-operative JOA score, cervical alignment, surgical complications and re-operation rate; and a meta-analysis was performed for these surgical outcomes. A total of 12 studies were obtained, including 1 prospective cohort study and 11 retrospective cohort studies. In the meta-analysis, neurological recovery rate in JOA score was greater in ADF than in LAMP, especially in patients with a large canal occupying ratio (greater than or equal to 60 %) and pre-operative kyphotic alignment; ADF also exhibited more favorable results in post-operative cervical alignment. In contrast, operating time and intra-operative blood loss were greater in ADF. Surgical complications were more frequently observed in ADF, leading to higher rates of re-operation. The authors concluded that this systematic review and meta-analysis showed both the merits and shortcomings of ADF and LAMP; ADF resulted in more favorable neurological recovery compared to LAMP, especially for patients with massive OPLL and kyphotic alignment. Post-operative cervical lordosis was also better preserved in ADF. However, ADF was associated with greater surgical invasion and higher incidences of surgical complications.

Laminoplasty for Cervical Spondylotic Amyotrophy

Luo and colleagues (2019) noted that cervical spondylotic amyotrophy (CSA) is characterized by upper limb muscle weakness and atrophy, without sensory deficits. The pathophysiology of CSA has been attributed to selective injury to the ventral nerve root and/or anterior horn of the spinal cord. These investigators discussed the history of CSA and described the epidemiology, etiology, pathophysiology, classification, clinical features, radiological and electrophysiological assessment, diagnosis, differential diagnosis, natural history and treatment of CSA. They carried out a comprehensive search of PubMed, Embase, Cochrane library and Web of Science databases from their inception to April 3, 2018. Clinically, CSA is classified into 3 types: a proximal-type (involving the scapular muscles, deltoid and biceps), a distal-type (involving the triceps and muscles of the forearm and hand), and a diffuse-type (involving features of both the distal- and proximal-type). Diagnosis requires documentation of muscle atrophy, without significant sensory deficits, supported by careful neurological, radiological and neurophysiological assessments, with amyotrophic lateral sclerosis (ALS), Parsonage-Turner syndrome, rotator cuff tear and Hirayama disease being the principle differential diagnoses. Conservative management of CSA includes cervical traction, neck immobilization and physical therapy, with vitamin B12 or E administration being useful in some patients. Surgical treatment, including anterior decompression and fusion or laminoplasty, with or without foraminotomy, is indicated after conservative treatment failure. Factors associated with a poor outcome include the distal-type CSA, long symptom duration, older age and greater pre-operative muscle weakness. The authors concluded that although the disease process of CSA is self-limited, treatment remains challenging, leaving scope for future studies.

Smoking Cessation and Spinal Fusion Outcomes

In a systematic review and meta-analysis of published observational cohort studies, Pearson et al (2016) examined the increased risk smokers have of experiencing a delayed and/or non-union in fractures, spinal fusion, osteotomy, arthrodesis or established non-unions. These investigators employed Medline, Embase, Allied and Complementary Medicine Database (AMED) as well as Web of Science Core Collection from 1966 to 2015 to gather data. Observational cohort studies that reported adult smokers and non-smokers with delayed and/or non-union or time to union of the fracture, spinal fusion, osteotomy, arthrodesis or established non-union were eligible. Total of 2 authors screened titles, abstracts and full papers. Data were extracted by 1 author and checked independently by a second. The relative risk ratios (RRs) of smoking versus non-smoking and the mean difference (MD) in time to union patients developing a delayed and/or non-union were calculated. The search identified 3,013 articles; of which, 40 studies were included. The meta-analysis of 7,516 procedures revealed that smoking was linked to an increased risk of delayed and/or non-union. When considered collectively, smokers have 2.2 (1.9 to 2.6) times the risk of experiencing delayed and/or non-union. In all the subgroups, the increased risk was always 1.6 times or higher that of non-smokers. In the patients where union did occur, it was a longer process in the smokers. The data from 923 procedures were included and revealed an increase in time to union of 27.7 days (14.2 to 41.3). The authors concluded that the findings of this study showed that smokers took 27.7 days (14.2 to 41.3) longer for union to occur for fractures, osteotomy, arthrodesis and established non-union. Smokers had double the risk of non-union 2.2 (1.9 to 2.6) for fractures, osteotomy, arthrodesis and established non-union. Smokers should be encouraged to abstain from smoking to improve the outcome of these orthopedic treatments.

Dengler et al (2017) noted that 3 recently published prospective studies have shown that minimally invasive sacroiliac joint fusion (SIJF) using triangular titanium implants produced better outcomes than conservative management for patients with pain originating from the SIJ. Due to limitations in individual trial sample size, analyses of predictors of treatment outcome were not performed. In a pooled patient-level analysis of 2 multi-center randomized controlled trials (RCTs) and 1 prospective, single-arm, multi-center trial, these researchers identified predictors of outcome of conservative and minimally invasive surgical management of pain originating from the SIJ. They pooled individual patient data from the 3 studies and used random effects models with multi-variate regression analysis to identify predictors for treatment outcome separately for conservative and minimally invasive surgical treatment. Outcome measures included VAS, ODI, and EuroQOL-5D (EQ-5D). These investigators included 423 patients assigned to either non-surgical management (NSM, n = 97) or SIJF (n = 326) between 2013 and 2015. The reduction in SIJ pain was 37.9 points larger [95 % confidence interval (95 % CI): 32.5 to 43.4, p < 0.0001] in the SIJF group than in the NSM group. Similarly, the improvement in ODI was 18.3 points larger (9 5% CI: 14.3 to 22.4), p < 0.0001). In NSM, these investigators found no predictors of outcome. In SIJF, a reduced improvement in outcome was predicted by smoking (p = 0.030), opioid use (p = 0.017), lower patient age (p = 0.008), and lower duration of SIJ pain (p = 0.028). The authors concluded that these findings supported the view that SIJF resulted in better treatment outcome than conservative management of SIJ pain and that a higher margin of improvement could be predicted in non-smokers, non-opioid users, and patients of increased age and with longer pain duration. Level of Evidence = 1.

Phan et al (2018) stated that anterior lumbar interbody fusion (ALIF) is a surgical technique indicated for the treatment of several lumbar pathologies. Smoking has been suggested as a possible cause of reduced fusion rates after ALIF, although the literature regarding the impact of smoking status on lumbar spine surgery is not well established. In a retrospective study, these researchers examined the impact of peri-operative smoking status on the rates of peri-operative complications, fusion, and adverse clinical outcomes in patients undergoing ALIF surgery. They carried out an analysis on a prospectively maintained database of 137 patients, all of whom underwent ALIF surgery by the same primary spine surgeon. Smoking status was defined by the presence of active smoking in the 2 weeks before the procedure. Outcome measures included fusion rates, surgical complications, SF-12, and ODI. Patients were separated into non-smokers (n = 114) and smokers (n = 23). Univariate analysis demonstrated that the percentage of patients with successful fusion differed significantly between the groups (69.6 % versus 85.1 %, p = 0.006). Pseudarthrosis rates were shown to be significantly associated with peri-operative smoking. Results for other post-operative complications and clinical outcomes were similar for both groups. On multi-variate analysis, the rate of failed fusion was significantly greater for smokers than non-smokers (odds ratio [OR] 37.10, p = 0.002). The authors concluded that the rate of successful fusion after ALIF surgery was found to be significantly lower for smokers compared with non-smokers. No significant association was found between smoking status and other peri-operative complications or adverse clinical outcomes.

Zhuang et al (2020) noted that smoking cessation represents an opportunity to reduce both short- and long-term effects of smoking on complications after lumbar fusion and smoking-related morbidity and mortality. However, the cost-effectiveness of smoking-cessation interventions before lumbar fusion is not fully known. These investigators created a decision-analytic Markov model to examine the cost-effectiveness of 5 smoking-cessation strategies (behavioral counseling, nicotine replacement therapy [NRT], bupropion or varenicline monotherapy, and a combined intervention) before single-level, instrumented lumbar postero-lateral fusion (PLF) from the health payer perspective. Probabilities, costs, and utilities were obtained from published sources. These researchers calculated the costs and quality-adjusted life years (QALYs) associated with each strategy over multiple time horizons and accounted for uncertainty with probabilistic sensitivity analyses (PSAs) consisting of 10,000 second-order Monte Carlo simulations. Every smoking-cessation intervention was more effective and less costly than usual care at the lifetime horizon. In the short-term, behavioral counseling, NRT, varenicline monotherapy, and the combined intervention were also cost-saving, while bupropion monotherapy was more effective but more costly than usual care. The mean lifetime cost savings for behavioral counseling, NRT, bupropion monotherapy, varenicline monotherapy, and the combined intervention were $3,291 (standard deviation [SD], $868), $2,571 (SD, $479), $2,851 (SD, $830), $6,767 (SD, $1,604), and $34,923 (SD, $4,248), respectively. The minimum efficacy threshold (relative risk [RR] for smoking cessation) for lifetime cost savings varied from 1.01 (behavioral counseling) to 1.15 (varenicline monotherapy). A PSA revealed that the combined smoking-cessation intervention was always more effective and less costly than usual care. The authors concluded that even brief smoking-cessation interventions yielded large short-term and long-term cost savings. Smoking-cessation interventions before PLF could both reduce costs and improve patient outcomes as health payers/systems shift toward value-based reimbursement (e.g., bundled payments) or population health models. Level of Evidence = II.

Debono et al (2021) stated that enhanced recovery after surgery (ERAS) evidence-based protocols for peri-operative care have led to improvements in outcomes in numerous surgical areas, through multi-modal optimization of patient pathway, reduction of complications, improved patient experience and reduction in the length of stay. ERAS represent a relatively new paradigm in spine surgery. In a multi-disciplinary consensus review, these investigators summarized the literature and proposed recommendations for the peri-operative care of patients undergoing lumbar fusion surgery with an ERAS program. Under the impetus of the ERAS Society, a multi-disciplinary guideline development group was constituted by bringing together international experts involved in the practice of ERAS and spine surgery. This group identified 22 ERAS items for lumbar fusion. They performed a systematic search in the English language in Medline, Embase, and Cochrane Central Register of Controlled Trials. Systematic reviews, RCTs, and cohort studies were included, and the evidence was graded according to the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) system. Consensus recommendation was reached by the group after a critical appraisal of the literature. A total of 256 articles were included to develop the consensus statements for 22 ERAS items; 1 ERAS item (pre-habilitation) was excluded from the final summary due to very poor quality and conflicting evidence in lumbar spinal fusion. From these remaining 21 ERAS items, 28 recommendations were included. All recommendations on ERAS protocol items were based on the best available evidence. These included 9 pre-operative, 11 intra-operative, and 6 post-operative recommendations. They spanned topics from pre-operative patient education and nutritional evaluation, intra-operative anesthetic and surgical techniques, and post-operative multi-modal analgesic strategies. The level of evidence for the use of each recommendation was presented. The ERAS Society recommend a combined smoking cessation therapy at a minimum of 4 weeks before surgery. Quality of Evidence = Moderate; recommendation grade = Strong.

In a retrospective review of prospectively collected registry data, Gatot et al (2022) examined the effect of smoking on 2 years post-operative functional outcomes, satisfaction, and radiologic fusion in non-diabetic patients undergoing minimally invasive transforaminal lumbar interbody fusion (TLIF) for degenerative spine conditions. Prospectively collected registry data of non-diabetic patients who underwent primary single-level minimally invasive TLIF in a single institution was reviewed. Patients were stratified based on smoking history. All patients were examined pre-operatively and post-operatively using the Numerical Pain Rating Scale for back pain and leg pain, ODI, SF-36 Physical and Mental Component Scores. Satisfaction was evaluated using the North American Spine Society (NASS) questionnaire . Radiographic fusion rates were compared. A total of 187 patients were included, of which 162 were non-smokers, and 25 had a positive smoking history. In this multi-variate analysis, smoking history was insignificant in predicting for minimal clinically important difference attainment rates in Physical Component Score and fusion grading outcomes. However, in terms of satisfaction score, positive smoking history remained a significant predictor (OR = 4.7, 95 % confidence interval [CI]: 1.10 to 20.09, p = 0.036). The authors concluded that non-diabetic patients with a positive smoking history had lower satisfaction scores but comparable functional outcomes and radiologic fusion 2 years after single-level TLIF. These investigators stated that thorough pre-operative counseling and smoking cessation advice may help to improve patient satisfaction following minimally invasive spine surgery. Level of Evidence = III.

Khalid et al (2022) stated that while there are several reports on the impact of smoking tobacco on spinal fusion outcomes, there is minimal literature on the influence of modern smoking cessation therapies on such outcomes. These researchers examined the outcomes of single-level lumbar fusion surgery in active smokers and in smokers undergoing recent cessation therapy. MARINER30, an all-payer claims database, was used to identify patients undergoing single-level lumbar fusions between 2010 and 2019. The primary outcomes were the rates of any complication, symptomatic pseudarthrosis, need for revision surgery, and all-cause re-admission within 30 and 90 days. The exact matched population analyzed in this study contained 31,935 patients undergoing single-level lumbar fusion with 10,645 (33 %) in each of the following groups: (i) active smokers; (ii) patients on smoking cessation therapy; and (iii) those without any smoking history. Patients undergoing smoking cessation therapy have reduced odds of developing any complication following surgery (OR 0.86, 95 % CI: 0.80 to 0.93) when compared with actively smoking patients. Non-smokers and patients on cessation therapy had a significantly lower rate of any complication compared with the smoking group (9.5 % versus 17 % versus 19 %, respectively). The authors concluded that when compared with active smoking, pre-operative smoking cessation therapy within 90 days of surgery decreased the likelihood of all-cause post-operative complications. However, there were no between-group differences in the likelihood of pseudarthrosis, revision surgery, or re-admission within 90 days.

In a systemic review and meta-analysis, Nunna et al (2022) examined the effect of tobacco smoking on risk of nonunion following spinal fusion. These investigators carried out a systematic search of Medline, Embase, Cochrane Central Register of Controlled Trials, and the Cochrane Database of Systematic Reviews from inception to December 31, 2020. Cohort studies directly comparing smokers with non-smokers that provided the number of non-unions and fused segments were included. Following data extraction, the risk of bias was assessed using the Quality in Prognosis Studies Tool, and the strength of evidence for non-union was examined using the GRADE system. All data analysis was carried out in Review Manager 5, and a random effects model was used. A total of 20 studies assessing 3,009 participants, which included 1,117 (37 %) smokers, met inclusion criteria. Pooled analysis found that smoking was associated with increased risk of non-union compared to not smoking 1 year or less following spine surgery (RR 1.91, 95 % CI: 1.56 to 2.35). Smoking was significantly associated with increased non-union in those receiving either allograft (RR 1.39, 95 % CI: 1.12 to 1.73) or autograft (RR 2.04, 95 % CI: 1.54 to 2.72). Both multi-level and single-level fusions carried increased risk of non-union in smokers (RR 2.30, 95 % CI: 1.64 to 3.23; RR 1.79, 95 % CI 1.12 to 2.86, respectively). The authors concluded that smoking status carried a global risk of non-union for spinal fusion procedures regardless of follow-up time, location, number of segments fused, or grafting material. Moreover, these investigators stated that further comparative studies with robust methodology are needed to establish treatment guidelines tailored to smokers.

Connor et al (2022) noted that tobacco use is associated with complications after surgical procedures, including poor wound healing, surgical site infections, and cardiovascular events. In a retrospective, database study, these researchers used the Nationwide Readmissions Database (NRD) to determine if tobacco use is associated with increased 30- and 90-day re-admission among patients undergoing surgery for degenerative spine disorders. Patients who underwent elective spine surgery were identified in the NRD from 2010 to 2014. The study population included patients with degenerative spine disorders treated with discectomy, fusion, or decompression. Descriptive and multi-variate logistic regression analyses were carried out to identify patient and hospital factors associated with 30- and 90-day re-admission, with significance set at p value of < 0.001. Within 30 days, 4.8 % of patients were re-admitted at a median time of 9 days. The most common reasons for 30-day re-admission were post-operative infection (12.5 %), septicemia (3.5 %), and post-operative pain (3.0 %). Within 90 days, 7.3 % were re-admitted at a median time of 18 days. The most common reasons for 90-day re-admission were post-operative infection (9.6 %), septicemia (3.5 %), and pneumonia (2.3 %). After adjustment for patient and hospital characteristics, tobacco use was independently associated with re-admission at 90 days (OR 1.05, 95 % CI: 1.03 to 1.07, p < 0.0001; but not 30 days (OR 1.02, 95 % CI: 1.00 to 1.05, p = 0.045). The authors concluded that tobacco use was associated with re-admission within 90 days after cervical and thoracolumbar spine surgery for degenerative disease. Tobacco use is a known risk factor for adverse health events and therefore should be considered when selecting patients for spine surgery.

Carboxyhemoglobin Levels and Smoking Cessation

Schimmel et al (2018) stated that non-invasive screening of carboxyhemoglobin saturation (SpCO) in the emergency department (ED) to detect occult exposure is increasingly common. The SpCO threshold to consider exposure in smokers is up to 9 %. The literature supporting this cut-off is inadequate, and the impact of active smoking on SpCO saturation remains unclear. In a prospective, cohort, pilot study, these researchers characterized baseline SpCO in a cohort of smokers outdoors; and secondary objectives were to examine the impact of active smoking on SpCO and to compare SpCO between smokers and non-smokers. This trial was carried out in 2 outdoor urban public areas in the U.S., in a convenience sample of adult smokers. SpCO saturations were assessed non-invasively before, during, and 2 mins after cigarette smoking with pulse CO-oximetry. Analyses included descriptive statistics, correlations, and a generalized estimating equation model. A total of 85 smokers had mean baseline SpCO of 2.7 % (SD 2.6) and peak of 3.1 %

(SD 2.9), while 15 controls had SpCO 1.3 % (SD 1.3). This was a significant difference. Time since last cigarette was associated with baseline SpCO, and active smoking increased mean SpCO. There was correlation among individual smokers' SpCO levels before, during, and 2 mins after smoking, indicating smokers tended to maintain their baseline SpCO level. The authors concluded that this study was the 1st to measure SpCO during active smoking in an uncontrolled environment. It suggested that 80 % of smokers had SpCO 5 % or less, but potentially lent support for the current 9 % as a threshold, depending on clinical context.

The authors stated that this study had several drawbacks. First, this pilot study enrolled a cohort of smokers in an outdoor environment. Volunteers were disproportionally male and young and may not be representative of all smokers. Although it was the authors; objective to enroll typical smokers, participants may have represented an overall healthier group given they were ambulatory and otherwise well, in contrast to those encountered in a home, institutional, or healthcare setting. Second, this study monitored a patient smoking a single cigarette. Patients who serially smoke multiple cigarettes may reach an SpCO level higher than those found in this cohort; self-reported smoking habits may have been inaccurate. Third, the sample size may have been under-powered to detect associations between some of the variables. These investigators were unable to meaningfully compare males and females, given females were significantly under-represented in the convenience sample. Fourth, although the study design intended to capture pre-smoking data and an extended observation period after smoking, logistically, it was often not feasible to identify smokers in public before initiation of smoking, and frequently they would not remain for prolonged observation. However, once identified, all smokers allowed a during and post-cigarette measurement at 2 mins.

Plastic Surgeons-Associated Paraspinal Muscle Flap Closure at the Time of Spinal Surgeries

Devulapalli et al (2017) noted that spinal resections could result in defects requiring soft-tissue reconstruction. In a retrospective study, these investigators examined their institutional experience with reconstruction of spinal defects and identified risk factors predictive of wound complications, focusing on timing of reconstruction with ablative surgery. They reviewed patients who underwent spinal resection and required soft-tissue reconstruction from 2002 to 2014. Logistic regression was carried out to identify risk factors for complications. Of 289 reconstructions carried out in 259 patients, 64 cases (22.1 %) had major wound complications requiring re-operation. Lumbo-sacral defects were the most common location (43.6 %) and para-spinous muscle flaps were the preferred reconstructive method used for all defect regions. A total of 224 reconstructions (77.5 %) were carried out immediately at the time of spinal surgery, and 65 (22.5 %) were carried out in delayed fashion as a result of wound complications from previous spinal surgery. Patients undergoing immediate reconstruction had significantly lower rates of instrumentation removal (0.9 % versus 4.6 %; $p = 0.043$), unplanned re-operations (0.5 versus 1.3; $p < 0.001$), and mortality (0.9 % versus 9.2 %; $p < 0.001$) compared with those undergoing delayed reconstruction. On logistic regression analysis, presence of instrumentation (OR, 3.2; $p = 0.012$), requirement for a free flap (OR, 9.0; $p = 0.016$), and spinal cord exposure (OR, 2.6; $p = 0.036$) were associated with increased odds of a major wound complication. The authors concluded that they have identified several factors (e.g., presence of instrumentation, prior radiation therapy, and certain wound locations) that appeared to portend poor outcomes. These investigators noted that the use of a multi-disciplinary approach by neurosurgical and plastic surgery teams to provide immediate soft-tissue reconstruction with loco-regional muscle flaps at the time of ablative spinal surgery may play a role in improving wound-related outcomes. Moreover, these researchers stated that future, ongoing studies will use control groups to aid in determining the optimal reconstructive algorithm in these complex cases. Level of Evidence = III.

The authors stated that this review had several drawbacks. First, a key element missing from the study design and to make more sound recommendations regarding the indication for and value of reconstruction was a control group of patients undergoing ablative spinal surgery and managed without any soft-tissue reconstruction. These researchers hoped to further investigate this information in the future to add to the literature, given that a controlled trial randomizing patients to no flap, immediate flap, and delayed flap groups appeared ethically and practically not feasible. Second, reconstructions were carried out by multiple surgeons over 10 years, with significant variation in techniques;

thus, these were unable to be controlled for in the analysis. Third, the size of the wound defect was not consistently recorded, which may act as a significant confounder to predicting success of reconstruction. Fourth, these researchers included many causes for spinal resection into their review to add power to their analysis for outcomes of reconstruction; however, this may have created significant variation in their patient population.

In a retrospective, single-center study, Weissler et al (2019) examined wound complications following plastic surgeon closure of the primary spinal surgery in a large patient population. This study entailed spinal surgeries closed by a single plastic surgeon at a large academic hospital. Descriptive statistics were used and outcomes in this sample were compared with previously published outcomes using 2-sample z tests. A total of 928 surgeries were reviewed, of which 782 were included – 715 operations were for degenerative conditions of the spine, 22 for trauma, 30 for neoplasms, and 14 for congenital conditions. A total of 421 were lumbo-sacral procedures (53.8 %) and 361 (46.2 %) cervical; and 14 patients (1.8 %) required re-admission with 30 days. This compared favorably to a pooled analysis of 488,049 patients, in which the 30-day re-admission rate was found to be 5.5 % ($z = 4.5$, $p < 0.0001$). A total of 7 patients (0.89 %) had wound infection and 3 (0.38 %) wound dehiscence post-operatively, compared with a study of 22,430 patients in the American College of Surgeons National Surgical Quality Improvement Program (ACS-NSQIP) database who had an infection incidence of 2.2 % ($z = 2.5$, $p = 0.0132$) and 0.3 % dehiscence rate ($z = 0.4$, $p = 0.6889$). The combined incidence of wound complications in the present sample, 1.27 %, was less than the combined incidence of wound complications in the population of 22,430 patients ($z = 2.2$, $p = 0.029$). The authors concluded that 30-day re-admissions and wound complications were intensely scrutinized quality metrics that may result in reduced reimbursements and other penalties for hospitals. Plastic surgeon closure of index spinal cases decreased these adverse outcomes. Moreover, these researchers stated that further investigations must be carried out to examine if the increased cost of plastic surgeon involvement in these cases would be off-set by the savings represented by fewer re-admissions and complications. Moreover, these investigators stated that having shown proof-of-concept (POC), they planned to expand data collection to other plastic surgeons closing spine surgeries to increase the sample size and more robustly determine the effects of specialty involvement. These investigators stated that a larger sample of patients with a greater variety of surgeons will allow for much-needed cost-benefit analysis of this practice as well.

The authors stated that this study had several drawbacks; the most obvious of which was its retrospective design. It relied on adequate reporting of complications; thus, any bias within the study cohort could not be ascertained. The inclusion of spine surgery closures by a single plastic surgeon may be a source of bias; however, the patients in the present sample were likely at greater, not less, risk for complications than a general spine patient population given that they were preselected for plastic surgeon involvement, showing that plastic surgeon closures decrease adverse outcomes compared to general spine patients even in a risky population. Because each case was included on an independent basis, rather than on a per-patient basis, certain patients may have been counted as multiple complications over the study period. This, again, would tend to bias the number of adverse outcomes observed upwards, as high-risk patients would contribute disproportionately to observed complications. These researchers mainly reviewed peri-operative, 30-day outcomes -- it remained to be determined whether plastic surgeon closure of these index cases has any effect on long-term hardware failure, fusion success, or re-operation. A final drawback of this study related to its reliance on previously published research: there was no way to know how many operations in the NSQIP or pooled cohorts may have been closed by plastic surgeons. The possible comparisons with published data were limited by availability -- there was no study with a population similar to the present one to which length of stay (LOS) or seroma/hematoma rates could be compared.

Inglesby et al (2020) noted that para-spinous muscle flap closure offers an innovative option in difficult-to-manage post-spinal surgery wounds. Available studies were mixed in terms of success and complication rates of these flap procedures, with most sources citing a wound complication rate of 20 %. In a retrospective, case-series study, these investigators examined the effectiveness and complication profile of the use of para-spinous muscle flaps for closure of complex spinal wounds. This study examined the hospital course of 58 patients undergoing para-spinous flap closure following spinal surgery between the years 2014 and 2018.

Information gathered includes: demographics, surgery indication, location, and length of incision on the spine, nutrition labs, previous spinal surgeries, pre-operative wound class, operative times, hospital LOS, and complication rates including re-operation, wound infection, and other post-operative complications. Of the 58 patients undergoing spinal muscle flap closure, 51 (87.93 %) had undergone previous spinal surgery with an average of 2.12 previous surgeries in these patients. Mean albumin and pre-albumin were 2.62 and 13.75, respectively. A total of 4/58 (6.90 %) developed a wound infection or experienced a continuation of their chronic osteomyelitis. Of the 57 patients who had spinal instrumentation, 3 (5.26 %) had spinal implants removed at the time of surgery, and 2 (3.51 %) had it removed or replaced later for mechanical complications. No patients had instrumentation removed for chronic infections, 1 patient (1.72 %) experienced re-operation for wound-related complications. These rates were lower than most complication rates in the current literature. The authors concluded that the plastic and reconstructive para-spinous muscle flap has promising results as a closure option for complex spinal wounds following neurosurgical cases. Moreover, these researchers stated that further investigations with larger number of participants are needed to determine the applicability of these findings to the general population. Level of Evidence = IV.

The authors stated that this study had 2 main drawbacks. First ,the retrospective nature of the chart review limited their ability to draw causative conclusions. Second, this was a relatively small cohort of patients (n = 58) who underwent surgery with a single plastic surgeon and neurosurgeon, which limited generalization. The small sample size also limited meaningful statistical analysis; however, the relatively low post-operative wound complications compared to routine spinal wound closures should promote further investigation on the use of muscle flaps for improving surgical outcomes in high-risk patients undergoing complex spine procedures.

Wright et al (2020) noted that advances in surgical technology and adjuvant therapies along with an aging and increasingly morbid U.S. population have resulted in an increase in complex spine surgery. With this increase comes an elevated risk of complications, including those related to the surgical wound, with some studies reporting wound complication incidences approaching 45 %. In a retrospective, single-center study, these researchers hypothesized that immediate muscle flap closure would improve outcomes in high-risk patients. A total of 301 consecutive index cases of spinal wound closure using local muscle flaps performed by the senior author at a single center between 2006 and 2018 were reviewed. The primary outcome was major wound complication (re-operation and/or re-admission because of surgical-site infection, late infection, dehiscence, seroma, or hematoma). Logistic regression analysis was carried out to identify predictors of this endpoint. Major wound complications occurred in 6.6 % of patients (re-operation, 3.6 %; re-admission, 3.0 %), with a 6.0 % infection rate, and 5 cases requiring instrumentation removal because of infection. Risk factors identified included radiotherapy (OR, 5.9; $p = 0.004$), age of 65 years or older (OR, 2.8; $p = 0.046$), and prior spine surgery (OR, 4.3; $p = 0.027$). The incidence of major wound complication increased dramatically with each additional risk factor. Mean drain dwell duration was 21.1 ± 10.0 days and not associated with major wound complications, including infection (OR, 1.04; $p = 0.112$). The authors concluded that immediate local muscle flap closure following complex spine surgery on high-risk patients was associated with an acceptable rate of wound complications and, as these data demonstrated, was safe and effective. These investigators stated that consideration should be given to immediate muscle flap closure in appropriately selected patients. Level of Evidence = III.

The authors stated that drawbacks of this trial included the fact that it was performed at a single center and involved the work of a single senior surgeon. Other drawbacks entailed the retrospective design and the use of historical references from the literature rather than internal controls. Lastly, the regression analysis was limited in terms of power because of the small number of major complications (20 cases of major wound complication), which prevented the inclusion of more than 3 co-variates in the model.

Garg and Mehta (2020) noted that their personal experience with incisional vacuum-assisted closure (VAC) dressing -- especially in morbidly obese patients who typically have serosanguinous discharge in the early post-operative period due to fat necrosis -- has been favorable. In patients with extensive scarring due to prior surgeries -- which most often, in the authors' setup were pediatric spinal deformity patients with implanted growing rods -- assistance from a plastic surgeon may be sought to apply tissue expanders. The authors believed that this article raised 2 important questions. Q1 -- can we identify the pre-operative risk factors in patients undergoing spine surgery that places them at an increased risk of post-operative wound complications? Q2 -- can we demonstrate clear-cut benefits in terms of healthcare costs, of liaison between spine surgeons and plastic surgeons in such selected patients? These researchers hoped the readers of North American Spine Society Journal (NASSJ) are stimulated to undertake studies which can answer these questions.

Zhong et al (2021) noted that plastic surgeons conduct layered musculo-cutaneous flap closures in high-risk spine patients such as revision posterior spinal fusion and complex deformity correction surgeries. However, few studies have evaluated outcomes of revision fusion carried out with plastic surgical closures, especially in non-deformity thoraco-lumbar spinal surgery. In a retrospective, cohort analysis, these researchers compared outcomes of plastic versus spine surgeon wound closure in revision 1 to 4 level thoraco-lumbar fusions. Patient charts were reviewed for demographics and peri-operative outcomes. Patients were divided into 2 cohorts: wound closures carried out by spine surgeons and those closed by plastic surgeons. Outcomes were analyzed before and after propensity score match for prior levels fused, iliac fixation, and levels fused at index surgery. Significance was set at $p < 0.05$. A total of 357 (87.3 %) spine surgeon (SS) and 52 (12.7 %) plastic surgeon (PS) closures were identified. PS group had significantly higher number of levels fused at index (PS 2.7 ± 1.0 versus SS 1.8 ± 0.9, $p < 0.001$) and at prior surgeries (PS 1.8 ± 1.2 versus SS 1.0 ± 0.9, $p < 0.001$), and rate of iliac instrumentation (PS 17.3 % versus SS 2.8 %, $p < 0.001$). Plastics closure was an independent risk factor for hospital LOS of greater than 5 days (OR 2.3) and post-operative seroma formation (OR 7.8). After propensity score match, PS had higher rates of seromas (PS 36.5 % versus SS 3.8 %, $p < 0.001$). There were no differences between PS and SS groups in surgical outcomes, peri-operative complication, surgical site infection (SSI), seroma requiring aspiration, or return to operating room at all time-points until follow-up ($p > 0.05$ for all). The authors concluded that plastic spinal closure for 1 to 4 level revision posterior thoraco-lumbar fusions had no advantage in reducing wound complications over spine surgeon closure but increased post-operative seroma formation. Level of Evidence = IV.

Hersh et al (2021) stated that SSI and dehiscence are devastating complications of surgery for spinal metastases. Wound closure involving plastic surgeons has been proposed as a strategy to lower post-operative complications. In a retrospective, single-center study, these investigators examined if plastic surgery closure is associated with lower rates of wound complications, wound infection, and wound re-operation compared to simple closure by spine surgeons. Patients surgically treated for metastatic tumors at a single comprehensive cancer center between April 2013 and 2020 were identified. Primary pathology, demographic information, clinical characteristics, pre-operative laboratory values, tumor location, operative characteristics, and post-operative outcomes were collected. Univariable analyses used student t-tests for continuous variables and $\chi 2$ tests for categorical variables. Multi-variable regressions were carried out to control for confounders. This trial included 317 patients, of which 56 underwent closure by plastic surgeons and 291 by neurosurgeons. Patients in the plastic surgery cohort were more likely to have received prior radiation to the surgical site, more often on long-term corticosteroid therapy, and more likely to have sacrococcygeal tumors. Operations involving plastic surgeons were more likely to be revision surgeries, corpectomies, and to involve a staged approach. Furthermore, patients in the plastic surgery cohort had longer incision lengths, longer surgeries, greater intra-operative blood loss (IOBL), were more likely to receive transfusions, and had longer hospitalizations. Local para-spinous advancement flaps were the most common complex wound closure technique. Plastic surgery closure was not significantly associated with a difference in rates of post-operative wound complications, wound infection, or wound-related re-operations compared to simple wound closure. The authors identified that patients undergoing plastic surgery wound closure had worse baseline risk, longer surgeries, greater IOBL, and longer hospitalizations compared to patients receiving simple closure. Despite their increased risk, complex wound closure did not significantly alter the rates of post-operative wound complications, wound infection, or wound-related re-operations. Consideration may be given to plastic surgery closure in patients at high-risk of wound complications or with extensive wound defects.

In a retrospective study, Ezeokoli et al (2023) compared wound complication rates between orthopedic closure (OC) and plastic multi-layered closure (PMC) in patients undergoing primary posterior spinal fusion for neuromuscular scoliosis (NMS). These investigators hypothesize that multi-layered closure will be associated with better post-operative outcomes. They collected data on pediatric patients diagnosed with NMS who underwent 1st-time spinal instrumentation between January 1, 2018 and May 31, 2021. Patient demographics, length of surgery, spinal levels fused and operative variables, wound complication rate, treatments, and need for wound wash-out were reviewed and recorded. A total of 86 patients were reviewed: 46 with OC and 40 with PMC. There was a significant increase in operating room (OR) time with PMC compared with OC (6.7 ± 1.2 hours versus 7.3 ± 1.3 hours, p = 0.016). There was no difference in complication rate, mean post-operative day of complication or unplanned return to the OR for OC and PMC, respectively. There was a slightly significant increase in the number of patients going home with a drain in the PMC cohort compared with the OC cohort (2.1 % versus 15 %, p = 0.046). The authors concluded that PMC showed longer OR times than OC, and did not show a statistically significant reduction in wound complications or unplanned returns to the OR. However, other studies have reported statistical and clinical significance with these variables. These researchers stated that surgical programs should review internal patient volumes and outcomes for spinal fusion in NMS patients, and consider if PMC after spinal fusions in pediatric patients with NMS or other scoliosis subtypes is an appropriate option in their institution to minimize post-operative wound complications.

The authors stated that drawbacks of this trial included those intrinsic to a retrospective design, including a lower level of evidence compared with prospective or randomized controlled trials, and selection bias. There may be variations in both PMC and OC that were surgeon-dependent, and this was a small volume of patients (40 in the PMC group). There may be other factors influencing wound healing and infection rates that would be apparent in a larger study. although there is a trend in complications favoring PMC, this difference was not statistically significant, and a larger cohort is needed to be statistically certain. This trial may be under-powered. During the PMC time frame, 2 pediatric orthopedic spine surgeons left the institution and 1 joined. Co-morbidities were not accounted for in the cohort. Revision surgeries were not included as a population for review.

Occipito-Cervical Fusions

Fisher et al (2001) provided an overview of atlanto-occipital dislocation (AOD) and associated occipital condyle fractures to alert physicians to this rare injury and potentially improve patient outcome. Data were obtained from a Medline search of the English literature from 1966 to 1999, and the experience of 4 spine surgeons at a quaternary care acute spinal cord injury (SCI) unit. Detailed anatomic and epidemiologically sound radiology studies were identified and analyzed. Only small retrospective studies or case series were available in the literature. Valid anatomic, biomechanical, and radiologic evaluations were extracted from studies. Clinical data came from limited studies and expert opinion. Early diagnosis is essential and is facilitated by a detailed clinical examination and strict adherence to an imaging algorithm that includes CT and MRI scanning. When the dislocation is identified, timely gentle reduction and prompt stabilization via non-operative or operative means is found to optimize patient outcome. The authors concluded that AOD should be suspected in any patient involved in a high-speed motor vehicle or pedestrian collision. Once suspected, proper imaging and appropriate management of these once fatal injuries can improve survival and neurologic outcome.

Dziurzynski et al (2005) noted that several methods are proposed for the diagnosis of AOD, including the Power's ratio, X-line method, basion-dens interval (BDI), condylar gap, and Harris method. No blinded comparison of the results of these methods has been compared to patient outcome, and there is no information available regarding the accuracy of these methods applied to CT scans. These investigators determined the best method to diagnose AOD. Plain lateral radiographs and CTs of the cervical spine were reviewed in 104 patients, including 6 with AOD. Images underwent a blinded review by a board certified neurosurgeon, orthopedist, radiologist, and emergency physician. Each diagnostic method for AOD was applied for determination of sensitivity, specificity, and positive and negative predictive values (PPV and NPV). The ability to identify relevant anatomic landmarks was also tabulated. Average values for sensitivities, specificities, PPV and NPV for each method applied to plain radiographs were: 0.4625 to 1.0, 0.8933 to 0.9725, 0.2775 to 0.45, and 0.975 to 1.0, respectively. These values for each method applied to CT scans are: 0.7075 to 1.0, 0.8725 to 0.9775, 0.3175 to 1, and 0.98 to 1.0, respectively. Identification of relevant anatomic landmarks occurred 99.75 % of the time when these methods were applied to CT scans compared to 39 % to 84 % of the time on plain radiographs. The authors concluded that sensitivity, specificity, PPV and NPV of these methods improved when applied to CT scans because of better visualization of anatomic landmarks. These researchers stated that these findings suggested CT scans of the cervical spine may be warranted in all trauma patients suspected of having cervical spine injury.

Horn et al (2007) stated that although rare, traumatic AOD injuries are associated with a high mortality rate. In a retrospective study, these researchers examined the imaging and clinical factors that determined treatment and were predictive of outcomes, respectively, in survivors of this injury. The medical records and imaging studies obtained in 33 patients with AOD were reviewed. Clinical factors that predicted outcomes, especially neurological injury at presentation and imaging findings, were evaluated. The most sensitive method for the diagnosis of AOD was the measurement of basion axial-basion dens interval on CT scanning. A total of 5 patients with severe traumatic brain injuries (TBIs) were not treated and subsequently died. Of the 28 patients in whom treatment was carried out, 23 underwent fusion and 5 were fitted with an external orthosis. Abnormal findings of the atlanto-occipital ligaments on MRI, associated with no or questionable abnormalities on CT scanning, provided the rationale for non-operative treatment. Of the 28 patients treated for their injuries, peri-operative death occurred in 5, 3 of whom had presented with severe neurological injuries. The mortality rate was highest in patients with a TBI at presentation. The mortality rate was lower in patients presenting with a SCI; however, in this group there was a higher rate of persistent neurological deficits. The authors concluded that the spines in patients with CT-documented AOD were most likely unstable and need surgical fixation. In patients for whom CT findings were normal and MR imaging findings suggested marginal abnormalities, non-operative treatment should be considered. The best predictors of outcome were severe brain or upper cervical injuries at initial presentation.

Mendenhall et al (2015) noted that only Level 3 evidence exists for the diagnosis and treatment of AOD with few studies examining mortality, neurologic improvement, and patient-reported outcomes (PROs). In a retrospective, cohort study, these researchers examined the incidence of AOD, 90-day surgical morbidity and mortality after AOD, patient factors that may be associated with delayed or missed diagnosis, as well as factors that were associated with mortality and neurologic improvement after AOD. In addition, these investigators quantified the pain, disability, and QOL experienced by patients surviving AOD. A total of 5,337 consecutive spine CT traumagrams from 1997 to 2012 were included. Outcome measures included mortality, neurologic improvement, complications, EQ-5D, NDI, Numeric Rating Scale (NRS)-neck, NRS-arm, and return-to-work. Patients were considered to have AOD if they met one of the following radiographic criteria: BDI greater than 10 mm; BAI: anterior displacement greater than 12 mm or posterior displacement greater than 4 mm between the basion and posterior C2 line; and condyle to C1 interval greater than 1.4 mm. Linear regression analysis was carried out to identify factors associated with 90-day mortality, neurologic improvement, and missed diagnosis. PROs were assessed via phone interview. A total of 31 patients met radiographic criteria for AOD; an incidence of 0.6 % over 15 years; and 21 (68 %) patients were treated with occipital cervical fusion. At 90 days post-operatively, there were no new neurologic deficits or re-operations. A total of 8 (26 %) patients died within 90 days. All patients who died had no documented AOD diagnosis and were not treated surgically. Missed AOD diagnosis was the strongest predictor of mortality. Younger age, lower Glasgow Coma Score (GCS), lower Injury Severity Score (ISS) score, and worse initial American Spinal Injury Association (ASIA) score were significantly associated with greater neurologic improvement. Higher ISS score and better ASIA score were significantly associated with missed AOD diagnosis. The average PROs metrics at time of telephone follow-up were as follows: EQ-5D = 0.73 ± 0.19, NDI = 30.89 ± 18.57, NRS-neck = 2.33 ± 2.21, NRS-arm = 2.00 ± 2.54. Of the patients with follow-up data, 4 were employed full-time, and 5 were receiving disability. The authors concluded that these findings suggested that failure to diagnose AOD was a powerful predictor of mortality. Higher ISS scores and better neurologic presentation were significantly associated with missed diagnosis. Cranio-cervical arthrodesis preserved neurologic function with low complication rate and unexpectedly high PROs and return-to-work. These investigators stated that these findings must be carefully interpreted because it was unclear if missed AOD diagnosis accompanied another death-causing injury (e.g., TBI) or if failure to treat AOD contributed to mortality in a multi-factorial manner.

Kasliwal et al (2016) stated that traumatic OCD results from ligamentous injury to the cranio-cervical junction and is associated with a high rate of mortality and significant neurologic morbidity. The diagnosis is often missed on initial lateral cervical spinal radiographs mainly due to inadequate visualization of radiological landmarks and low degree of suspicion. Widespread availability of multi-detector CT (MDCT) of the spine and development of better diagnostic radiological criteria has allowed timely diagnosis and good clinical outcome following posterior occipito-cervical fusion (OCF) and instrumentation for a pathology that was once considered uniformly fatal. The authors concluded that a high index of suspicion following high impact trauma with early recognition and prompt surgical fixation could result in good clinical outcome; posterior OCF and instrumentation remains the surgical treatment of choice.

Borgo et al (2018) noted that mucopolysaccharidoses (MPS) are a group of diseases characterized by abnormal accumulation of glycosaminoglycans (GAGs). Although there are differences among the various disease types, the osteoarticular system is always involved. These investigators established a framework for MPS-related orthopedic manifestations and for their treatment. These researchers, affiliated to 3 different Italian Orthopedic Centers, reviewed data in light of their accumulated professional experience. Bone alterations made up what is known as dysostosis multiplex, involving the trunk and limbs and with typical radiological findings. Joints are affected by pathological tissue infiltrations. The cervical spinal cord is involved, with stenosis and cervical and occipito-cervical instability. In MPS there is a much higher incidence of scoliosis compared with healthy subjects without any particular distinctive feature. Kyphosis of the spine is more frequent and also more severe because of its possible neurological complications, and it is localized at the thoraco-lumbar level with a malformed vertebra at the top of the deformity. Evolving forms, and those associated with neurological damage, require antero-posterior spine fusion. The hip is invariably involved, with dysplasia affecting the femoral neck (coxa valga), the femoral epiphysis (loss of sphericity, osteonecrosis), and the femoral acetabulum which is flared. All these features explain the tendency to progressive hip migration. Genu valgum is often found (a deviation of the physiological axis with an obtuse angle opening laterally). This deformity is often localized at the proximal tibial metaphysis; it causes functional limitations and results in an irregular erosion of the articular cartilage. In young patients who still have the growth plate, it is possible to execute a medial hemi-epiphysiodesis, a temporary inhibition of cartilage growth, with progressive axis correction. The authors concluded that MPS patients are at high risk of neurological, anesthesiologic, and general surgical complications.

Onodera et al (2023) stated that Down syndrome, also known as trisomy 21, is associated with congenital cervical spine abnormalities, including atlanto-axial instability with or without os odontoideum, atlanto-occipital instability, and hypoplasia of the atlas. These investigators reported a case of Down syndrome complicated by congenital AOD. The subject presented with severe cervical myelopathy at 13 years of age after a 10-year follow-up. Radiography and CT showed os odontoideum protruding into the foramen magnum and congenital anterior AOD. Furthermore, a bifurcated internal occipital crest with a thinned central portion of the occipital bone was noted. MRI showed kyphotic alignment of the spinal cord with severe compression at the foramen magnum level. As the neurological impairment was partially improved by halo vest immobilization, these researchers conducted in situ C0-C2 fusion with an iliac autograft and decompression of the foramen magnum and posterior arch of C1. An improvement was observed immediately after surgery. Two years after surgery, radiography and CT showed solid C0-C2 segment fusion. The authors stated that the accumulation of similar cases is essential for determining the prognosis or optimal treatment for this rare congenital condition.

In a retrospective study, Hong et al (2024) examined the impact of tumor extension into the occipital condyle (OC) in lower clival chordoma management and the need for occipito-cervical fusion (OCF). This trial was carried out on 35 patients with lower clival chordoma. The pre-operative area of the intact OCs, Hounsfield units, and the integrity of the apical ligament and the tectorial membrane were assessed using pre-operative imaging. A total of 7 (20 %) patients were in the OCF group. The OCF group exhibited a higher prevalence of pre-operative pain in the neck or head ($p = 0.006$), ligament absence ($p = 0.022$), and increased propensity for post-operative wound issues ($p = 0.022$) than the non-OCF group. The OCF group had less intact OCs ($p < 0.001$) and higher spinal instability neoplastic score ($p = 0.002$) than the non-OCF group. All patients with intact OCs of less than 60 % underwent OCF, and those with OCs 70 % or greater were treated without OCF. Those with OCs between 60 % and 69 % underwent OCF if the ligaments were eroded, and did not undergo OCF if the ligaments were intact. Treatment strategies varied, with endoscopic endonasal approach alone being common. Radiation therapy was administered to 89 % of patients. All 3 patients treated with OCF after tumor resection had wound issues; none treated with OCF before resection had wound issues. None developed atlanto-occipital instability. Survival rates did not significantly differ between the 2 groups. The authors concluded that in the absence of mobility-related neck pain, patients with lower clival chordoma and intact OC of 60 % or greater, intact apical ligament, and intact tectorial membrane, may not require OCF.

Use of Conservative Treatments Before Surgery for Lumbar Spinal Stenosis

Hilibrand and Rand (1999) noted that degenerative lumbar stenosis is a common cause of disabling back and lower extremity (LE) pain in the elderly. The process usually begins with degeneration of the intervertebral disks and facet joints, leading to narrowing of the spinal canal and neural foramina. Associated factors may include a developmentally narrow spinal canal and degenerative spinal instability. Non-operative management entails restriction of aggravating activities, PT, as well as anti-inflammatory medications. If non-operative treatment has failed, surgical treatment may be appropriate. Decompression should be carried out to address all clinically relevant neural elements while maintaining spinal stability. If instability is present, autogenous inter-transverse bone grafting is recommended. There may be an advantage to augmenting some of these procedures with internal fixation. Surgical success rates as high as 85 % have been reported, but may be compromised by inadequate decompression, inadequate stabilization, or medical co-morbidities. Short-term follow-up data showed that operative management provided more effective relief than non-operative treatment;

however, prospective studies comparing the effects of non-operative and operative interventions on the long-term natural history of lumbar spinal stenosis are needed.

Amundsen et al (2000) stated that surgical decompression has been considered the rational treatment for the treatment of lumbar spinal stenosis (LSS); however, clinical experience indicated that many patients do well with conservative treatment. In a prospective study, a cohort of 100 patients with symptomatic LSS were given surgical or conservative treatment and followed for 10 years. These researchers identified the short- and long-term results following surgical and conservative treatment, and examined if clinical or radiologic predictors for the treatment result could be defined. In this trial, a total of 19 patients with severe symptoms were selected for surgical treatment, and 50 patients with moderate symptoms for conservative treatment, whereas 31 patients were randomized between the conservative (n = 18) and surgical (n = 13) treatment groups. Pain was decisive for the choice of treatment group. All patients were observed for 10 years by clinical evaluation as well as questionnaires. The results, evaluated by patient and physician, were rated as excellent, fair, unchanged, or worse. After a period of 3 months, relief of pain had occurred in most patients. Some had relief earlier, whereas for others it took 1 year. After a period of 4 years, excellent or fair results were found in 50 % of the patients selected for conservative treatment, and in 80 % of the patients selected for surgery. Patients with an unsatisfactory result from conservative treatment were offered delayed surgery after 3 to 27 months (median of 3.5 months). The treatment result of delayed surgery was essentially similar to that of the initial group. The treatment result for the patients randomized for surgical treatment was considerably better than for the patients randomized for conservative treatment. Clinically significant deterioration of symptoms during the final 6 years of the follow-up period was not observed. Patients with multi-level afflictions, surgically treated or not, did not have a poorer outcome than those with single-level afflictions. Clinical or radiologic predictors for the final outcome were not found. There were no drop-outs, except for 14 deaths. The authors concluded that the outcome was most favorable for surgical treatment. Moreover,. these investigators stated that an initial conservative approach appeared advisable for many patients because those with an unsatisfactory result could be treated surgically later with a good outcome.

In a prospective study, Tadokoro et al (2005) identified outcomes of elderly LSS patients who were treated conservatively; and examined factors that control the prognosis. This trial included a total of 89 patients, 70 years of age or older, who underwent in-hospital conservative treatment. The Japanese Orthopedic Association's score (JOA score) and the disturbance level of activities of daily living (ADL) were used for evaluation. Nerve involvement was classified into radicular, cauda equina, and mixed type. Myelographic findings were classified into central defect with or without block and root defect. Associations between disturbance level of ADL, nerve involvement, as well as myelographic classifications were examined. The mean JOA score increased from 11.1 points at admission to 15.9 points at discharge, with 14.3 points maintained at the follow-up; 48.8 % of radicular type showed no obstacle in ADL at the follow-up compared with 33.3 % of the other types; 13.3 % of central defect with block showed no obstacle in ADL compared with 47.8 % of the other types with significant difference. The authors concluded that the prognosis of conservative treatment for elderly LSS patients was relatively good; and radicular type may be a candidate for conservative treatment. Moreover, these investigators stated that patients with complete block in the myelogram may not respond favorably to conservative treatment.

In a Cochrane review, Zaina et al (2016) examined the effectiveness of different types of surgery compared with different types of non-surgical interventions in adults with symptomatic LSS. Primary outcomes included QOL, disability, function and pain. In addition, these investigators examined complication rates and side effects, and assessed short-, intermediate- and long-term outcomes (6 months, 6 months to 2 years, 5 years or longer). They searched the Cochrane Central Register of Controlled Trials (CENTRAL), Medline, Embase, 5 other databases as well as 2 trials registries up to February 2015. These researchers also screened reference lists and conference proceedings related to treatment of the spine. RCTs comparing surgical versus non-operative treatments in participants with LSS confirmed by clinical and imaging findings were selected for analysis. From the 12,966 citations screened, these investigators examined 26 full-text articles and included 5 RCTs (643 participants). Low-quality evidence from the meta-analysis carried out on 2 studies using the ODI (pain-related disability) to compare direct decompression with or without fusion versus multi-modal non-operative care revealed no significant differences at 6 months (MD -3.66, 95 % CI: -10.12 to 2.80) and at 1 year (MD -6.18, 95 % CI: -15.03 to 2.66). At 24 months, significant differences favored decompression (MD -4.43, 95 % CI: -7.91 to -0.96). Low-quality evidence from 1 small study showed no difference in pain outcomes between decompression and usual conservative care (bracing and exercise) at 3 months (RR 1.38, 95 % CI: 0.22 to 8.59), 4 years (RR 7.50, 95 % CI: 1.00 to 56.48) and 10 years (RR 4.09, 95 % CI: 0.95 to 17.58). Low-quality evidence from 1 small study suggested no differences at 6 weeks in the ODI for patients treated with minimally invasive mild decompression versus those treated with epidural steroid injections (MD 5.70, 95 % CI: 0.57 to 10.83; 38 participants). Zurich Claudication Questionnaire (ZCQ) results were better for epidural injection at 6 weeks (MD -0.60, 95 % CI: -0.92 to -0.28), and VAS improvements were better in the mild decompression group (MD 2.40, 95 % CI: 1.92 to 2.88). At 12 weeks, many cross-overs prevented further analysis. Low-quality evidence from 1 study (n = 191 subjects) favored the inter-spinous spacer versus usual conservative treatment at 6 weeks, 6 months and 1 year for symptom severity and physical function. All remaining studies reported complications associated with surgery and conservative side effects of treatment: 2 studies reported no major complications in the surgical group, and the other study reported complications in 10 % and 24 % of subjects, including spinous process fracture, coronary ischemia, respiratory distress, hematoma, stroke, risk of re-operation and death due to pulmonary edema. The authors concluded that they had very little confidence to conclude if surgical treatment or a conservative approach was better for LSS, and they could provide no new recommendations to guide clinical practice. However, it should be noted that the rate of side effects ranged from 10 % to 24 % in surgical cases, and no side effects were reported for any conservative treatment. No clear benefits were observed with surgery versus non-surgical treatment. These investigators stated that these findings suggested that clinicians should be very careful in informing patients regarding possible therapeutic options, especially given that conservative treatments have resulted in no reported side effects. They stated that high-quality research is needed to compare surgical versus conservative care for individuals with LSS.

Ma et al (2017) stated that LSS is a common degenerative disease that affected the lumbar spine function and QOL, which can be treated by conservative treatment and surgical intervention. In a systematic review and meta-analysis, these investigators compared the effectiveness of conservative treatment versus surgical intervention for the management of patients with LSS. They searched PubMed as well as other databases on September 18, 2016; RCTs comparing conservative treatment versus surgical intervention for LSS patients were included for analysis. Outcomes and complications were collected with data selection criteria and analyzed with Review Manager Version 5.3. A total of 9 RCTs (14 articles) and 1,658 patients were included, and 3of them were high-quality studies. At first 6 months after treatment, there were no significant differences for ODI scores between 2 therapeutic groups ($p > 0.05$), however, surgery group demonstrated significant higher ODI scores at 1 year ($p < 0.05$) and 2 years ($p < 0.05$). Two studies reported no significant difference between laminectomy and conservative treatment for the SF-36 physical function scores at 3 months, 6 months, 12 months and 24 months ($p > 0.05$); and 2 studies reported patients were satisfied with X-STOP implanted at 6 weeks, 6 months, and 1 year. No statistical differences for adverse events (AEs) were observed intra-operatively or within 72 hours ($p > 0.05$) between surgery and non-surgery groups. Moreover, subgroup analysis showed there were no safety differences between laminectomy and conservative treatment, X-STOP and conservative treatment at early stage of duration. However, the surgical groups had higher complication rates than non-surgery groups throughout the follow-up duration. The authors concluded that surgery groups reported better late clinical outcomes after 1 year and higher complication rate throughout the follow-up duration, although it had no significant differences compared with conservative groups in the first 6 months post-treatment. However, there was no evidence that a definitive method could be firmly recommended to LSS patients. These researchers stated that further investigations are needed to achieve high quality and credible results.

In a systematic review, Peev et al (2024) formulated a list of recommendations for the role of spinal injections and surgery in the treatment of acute LBP. These investigators carried out a systematic literature search from 2012 to 2022 using PubMed, Medline, and Cochrane Central Register of Controlled Trials for studies examining the role of injections and surgery for the management of acute LBP. Inclusion criteria included RCTs, as well as prospective and retrospective studies reporting primary outcomes (pain improvement (VAS score) and back-specific functional status) and secondary outcomes (post-procedure complications). These data were reviewed, presented, and voted on by an expert panel consisting of 14 attending spine surgeons from 14 countries at the consensus meeting of the World Federation of Neurosurgical Societies (WFNS) Spine Committee. A 2-round consensus-based Delphi method was employed to generate consensus, and topics with greater than 66 % agreement were categorized as having reached consensus. A total of 100 studies met inclusion criteria. Of these, 20 were selected by the committee for full text review and presented at the consensus meeting. The committee voted on 8 statements and achieved consensus on the following 7 statements: First -- Epidural steroid injections (ESIs) showed significant benefit to discogenic back pain. Second -- A lateral approach is superior to a midline approach for ESIs. Third -- Short-term (less than 1 week) effect of ESIs was similar between steroids. Fourth -- ESIs exhibited a variety of potential complications. Fifth -- CT or fluoroscopy guidance can be used for lumbar medial branch blocks. Sixth -- Lumbar medial branch radiofrequency ablations (RFAs) can be carried out on patients with recurrent pain following a successful ESI. Seventh -- Acute LBP is usually self-limiting, resolves in less than 6 weeks, and does not require surgical intervention. The authors concluded that given significant treatment heterogeneity, these investigators provided the latest, evidence-based recommendations for management of acute LBP. They stated that ESIs are effective at short-term pain relief, and surgical intervention should be reserved for patients who fail conservative measures.

Computer-Assisted Surgical Navigation in Spinal Surgery

Malik et al (2021) examined medical and surgical complication rates between robotic-assisted versus conventional elective posterior lumbar fusions. The Symphony Integrated DataVerse was queried using International Classification of Diseases, 10th Edition, Clinical Modification procedure codes to identify patients undergoing elective posterior lumbar fusions for degenerative spine pathologies between 2015 and 2018. International Classification of Diseases, 10th Edition, Clinical

Modification procedure codes (8E0W4CZ, 8E0W0CZ, 8E0W3CZ) were used to identify patients undergoing a robotic-assisted spinal fusion. Outcome measures were 90-day medical and surgical complications, 1-year pseudarthrosis, and 1-year revision surgery rates. Multi-variate logistic regression analyses were employed to examine if undergoing a robotic-assisted fusion (versus conventional fusion) was associated with differences in wound complications, medical complications, pseudarthrosis, revision surgery, as well as re-admissions within 90 days of surgery. A total of 39,387 patients undergoing elective posterior lumbar fusions were included in the cohort-of whom 245 (0.62 %) patients underwent a robotic-assisted fusion. Multi-variate analysis revealed that robotic-assisted fusion (versus conventional fusion) was not associated with significant differences in 90-day rates of wound complications ($p = 0.299$), urinary tract infections ($p = 0.648$), acute myocardial infarctions ($p = 0.209$), acute renal failure ($p = 0.461$), pneumonia ($p = 0.214$), stroke ($p = 0.917$), deep venous thrombosis ($p = 0.562$), pulmonary embolism ($p = 0.401$), and re-admissions ($p = 0.985$). Furthermore, there were no significant differences in the 1-year rates of revision fusions ($p = 0.316$) and pseudarthrosis ($p = 0.695$). The authors concluded that patients who underwent a robotic-assisted fusion had similar rates of surgical and medical complications compared with those who underwent a conventional fusion. Moreover, these researchers stated that further investigations using more granular prospective data that collect information on patient-reported outcomes are needed to examine the future role of robots in spinal surgery.

Matur et al (2023) stated that navigated and robotic pedicle screw placement systems have been developed to improve the accuracy of screw placement; however, the literature comparing the safety and accuracy of robotic and navigated screw placement with fluoroscopic free-hand screw placement in thoraco-lumbar spinal fusion surgery has been limited. In a systematic review and meta-analysis, these researchers compared the accuracy and safety profiles of robotic and navigated pedicle screws with fluoroscopic free-hand pedicle screws. Only RCTs comparing robotic-assisted or navigated pedicle screws placement with free-hand pedicle screw placement in the thoraco-lumbar spine were included. Outcome measures included OR estimates for screw accuracy according to the Gertzbein-Robbins scale and RR for various surgical complications. These investigators systematically searched PubMed and Embase for English-language studies from inception through April 7, 2022, including references of eligible studies. The search was carried out according to PRISMA guidelines. Two reviewers performed a full abstraction of all data, and 1 reviewer verified accuracy. Information was extracted on study design, quality, bias, participants, and risk estimates. Data and estimates were pooled using the Mantel-Haenszel method for random-effects meta-analysis. A total of 14 studies entailing 12 RCTs were identified (n = 892 patients, 4,046 screws). The pooled analysis showed that robotic and navigated pedicle screw placement techniques were associated with higher odds of screw accuracy (OR 2.66, 95 % CI: 1.24 to 5.72, p = 0.01). Robotic and navigated screw placement was associated with a lower risk of facet joint violations (RR 0.09, 95 % CI: 0.02 to 0.38, p < 0.01) and major complications (RR 0.31, 95 % CI: 0.11 to 0.84, p = 0.02). There were no observed differences between groups in nerve root injury (RR 0.50, 95 % CI: 0.11 to 2.30, p = 0.37), or return to operating room for screw revision (RR 0.28, 9 5% CI: 0.07 to 1.13, p = 0.07). The authors concluded that these estimates suggested that robotic and navigated screw placement techniques were associated with higher odds of screw accuracy and superior safety profile compared with fluoroscopic free-hand techniques. Moreover, these researchers stated that ongoing prospective, multi-center registries are expected to provide additional relevant information on long-term clinical outcomes, and future RCTs are needed to further validate these findings.

The authors stated that this study had several drawbacks. First, although all included studies were RCTs, risks of bias were variable due to their surgical nature. Yet, the vast majority of included RCTs had low risk of bias, predisposing this meta-analysis to a low overall risk of bias. Owing to the recent introduction of multiple robotic and navigated systems across the literature, different protocols were included in this meta-analysis to collect the highest number of cases and provide the most in-depth evaluation of current technologies in spine fixation surgery compared with fluoroscopic-assisted free-hand screw placement techniques. However, this aggregation may have determined decreased effect sizes for each individual technique under analysis. Second, although these researchers did not identify any differences in procedure time between the 2 groups, the procedure time did not take into consideration the time required for the pre-operative CT scan and registration. The reduced screw insertion time associated with navigation may aid in off-setting this time, but these data were only available for navigation and suggested that this time may only be off-set to a significant degree in larger cases. Third, robotics and navigation may also suffer registration failures and may need to be abandoned for various reasons. These drawbacks could not be addressed in this study but should be considered when weighing the advantages and disadvantages of robotics and navigation. Fourth, as a consequence of the limited and heterogeneous data across included RCTs, patient-reported functional outcomes and radiation exposure could not be evaluated. Fifth, the lack of granular data prevented individual meta-analyses based on indications, different robotic and navigated systems, and free-hand techniques.

In a retrospective study, Li et al (2023) compared the effectiveness of percutaneous pedicle screw internal fixation with the aid of the TINAVI orthopedic surgery robot with that of traditional open surgery for Levine-Edward type IIA (post-reduction) hangman fractures and examined the safety and effectiveness of the TINAVI robot-assisted orthopedic surgery procedure. The medical records of 60 patients with Levine-Edward type IIA (post-reduction) hangman fractures treated surgically from June 2015 to February 2022 were analyzed. Among these patients, 25 were treated with percutaneous pedicle screw fixation under TINAVI (the robot group), and 35 were treated with pedicle screw implantation assisted by a conventional C-arm X-ray machine (the traditional operation group). The pedicle screw placement grade was assessed according to the Rampersaud scale. The correct rate of pedicle screw placement was calculated. The invasion of adjacent facet joints, VAS score, NDI score, SF-36 score, EQ-5D score, and operation-related data were recorded, and patients were followed-up. All subjects were followed-up for an average of 15.0 ± 3.4 months. The accuracy of screw placement in the robot group was higher than that in the traditional operation group, while the rates of intra-operative blood loss and invasion of the facet joint were lower and the incision length and hospital LOS were shorter. On the 3rd day after the operation, the VAS score in the robot group was significantly higher than that in the traditional operation group; however, there was no significant difference in the NDI score. On the 3rd day after the operation, the SF-36 and EQ-5 questionnaire scores of the robot group were better than those of the traditional operation group. No complications occurred in any of the patients. Post-operative cervical X-ray revealed that the cervical vertebra was stable, and there was no fracture, angle or displacement. Post-operative CT showed that all fractures healed, and the average healing time was 3.4 months. The authors concluded that the treatment of Levine-Edward IIA (post-repositioning) hangman fractures with percutaneous pedicle fixation assisted by the TINAVI orthopedic surgery robot could significantly improve screw placement accuracy with a low rate of invasion of the adjacent facet joint, a short operation time, a low bleeding rate, as well as high patient satisfaction. Although there were still many disadvantages, the TINAVI orthopedic surgery robot still has good prospects for application.

These researchers noted that this study had quite a few drawbacks. First, the machine is expensive, but some studies have concluded that it could shorten the total hospital LOS and could yield significant social benefits. Second, the size of the operating room is small, the robot is composed of more equipment with a larger size, the part in close contact with the patient requires strict aseptic protection, and the operator must plan the screw placement path several times during surgery. Third, the stability of the robot affects the accuracy of positioning. Fourth, screw planning was markedly influenced by subjective operator factors, and different surgeons with different surgical experience made large errors in planning, which affected the accuracy. Fifth, when the joint surface of the articular eminence was rough and the soft tissue pressure was high, the planning of the screw path and deviation from the planned screw path may occur when the articular surface of the articulation was rough, the angle between the planned screw path and the vertebral body bone was narrow, and the tip of the guide pin could slip. These researchers also lacked the ability to detect and correct slippage. To address these pain points, these investigators believed that a suitable incision type should be selected and that the soft tissues should be adequately released when placing the access and screw, then, the operator should select the guide pin, and the diameter of the guide pin should be closely matched to prevent excessive soft tissue tension or the embedded sleeve from affecting the tapping accuracy. Lastly, the actual screw path should be checked by lateral fluoroscopy during guide needle placement to see if it matches the planned screw path; the authors' institution adopted the spine stapling method to monitor accuracy, which meant that the screw entry point was placed at the vertex of the spine, and if it was not positioned correctly, the operator could distinguish it by visual inspection. If less than half of the guide-wires were found to be poorly positioned intra-operatively, the robot could be re-registered and operated gently; and if more than half of the guide-wires were poorly positioned, the robot could be checked for firm installation and obstruction of the optical tracking system and tracer. Sixth, the TINAVI robot has a long learning curve, and for the initial contact with the robot, familiarity with system placement, image acquisition and screwing tract planning would increase the surgical time. Long segmental spinal lesions require multiple data acquisitions, and the pre-operative planning of the surgical plan and “easy spine” free-hand screwing combined with “difficult spine” robot-assisted screwing could reduce the operative time. Seventh, the position of the vertebral body may change after unilateral screwing. The lack of intra-operative real-time detection may result in failure or poor screw placement on the opposite side, and excessive pressure of the guide on the underside should be avoided when the needle was drilled for placement, which may result in a change in the vertebral body position. These researchers often employed a Mayfield head-frame to fix the head prior to cervical spine surgery. Eighth, the O-arm increased the radiation dose to the patient. Ninth, previously built-in objects have had some effect on data acquisition, and the accuracy of revision surgery needs further study. Tenth, intra-operative positioning and execution depended on the tracer, which often led to additional trauma, and the authors attached it to the table-side column to reduce trauma and reduce the impact of respiratory mobility. These researchers stated that as they were still in the early stages of use, the sample size was insufficient, and the number of patients needs to be increased at a later stage to enhance the credibility of the study findings. The retrospective study design would inevitably lose some clinical data, and other clinical data need to be analyzed in further prospective studies with more comprehensive evaluation indices.

Papalia et al (2024) noted that in the last 10 years, intra-operative navigation has been introduced in spinal surgery to prevent risks and complications. In a systematic review and meta-analysis, these investigators compared pedicle screw accuracy, clinical outcomes, and complications between navigated and conventional techniques. They carried out a literature search using the Cochrane Central Library, PubMed, and Scopus data-bases on April 30, 2023. RCTs, prospective and retrospective studies that compared pedicle screw accuracy in the thoracic-lumbar-sacral segments, blood loss, operative time, hospital LOS, intra-operative and post-operative revision of screws, neurological and systemic complications, VAS, and ODI between navigated and free-hand or fluoroscopy-assisted techniques were included in this study. The meta-analysis was carried out using Review Manager software. Clinical outcomes were evaluated as continuous outcomes with MD, while pedicle screw accuracy and complications were assessed as dichotomous outcomes with OR, all with 95 % CIs. The statistical significance of the results was fixed at $p < 0.05$. This meta-analysis included 30 studies for a total of 17,911 patients and 24,600 pedicle screws. Statistically significant results in favor of the navigated technique were observed for the accuracy of pedicle screws ($p = 0.0001$), hospital LOS ($p = 0.0002$), blood loss ($p < 0.0001$), post-operative revision of pedicle screws ($p < 0.00001$), and systemic complications ($p = 0.0008$). In particular, the positioning of the screws was clinically acceptable in 96.2 % of the navigated group and 94.2 % with traditional techniques. No significant differences were found in VAS, ODI, and operative time between the 2 groups. The authors concluded that navigated pedicle screw fixation has been shown to be a safe and effective technique with high improvement in clinical outcomes and accuracy in patients undergoing spinal fusion compared with conventional techniques. Level of Evidence = III.

The authors stated that the key drawback of this study was the low level of evidence of the included studies. Due to the design of the procedure, most studies were retrospective, while few RCTs were identified via the literature search. Moreover, a high heterogeneity was reported among the studies for the surgical indication and the surgical approach; despite this, it was not possible to perform a subgroup analysis. In addition, the forest plots demonstrated heterogeneity of the data regarding the operative time and intra-operative revision outcomes. Several possible causes of heterogeneity may be suspected, such as differences in the surgeon’s expertise or learning curve. These factors may have primarily influenced studies with a low number of patients.

Aurouer et al (2024) stated that robotic spinal surgery may result in better pedicle screw placement accuracy, and reduction in radiation exposure and hospital LOS, compared to free-hand surgery. These researchers described the protocol of a RCT designed to compare screw placement accuracy of robot-assisted surgery with integrated 3D computer-assisted navigation versus free-hand surgery with 2D fluoroscopy for arthrodesis of the thoraco-lumbar spine. This is a evaluator-blinded, single-center RCT with a 1:1 allocation ratio. Participants (n = 300) will be randomized into 2 groups -- robot-assisted (Mazor X Stealth Edition) versus free-hand, after stratification based on the planned number of pedicle screws needed for surgery. The primary outcome is the proportion of pedicle screws placed with grade A accuracy (Gertzbein-Robbins classification) on post-operative CT images. The secondary outcomes are intervention time, operation room occupancy time, hospital LOS, estimated blood loss (EBL), surgeon's radiation exposure, screw fracture/loosening, superior-level facet joint violation, complication rate, re-operation rate on the same level or 1 level above, functional and clinical outcomes (ODI, pain, Hospital Anxiety and Depression Scale [HADS], sensory and motor status) as well as cost-utility analysis. The authors stated that this RCT will provide insight into whether robot-assisted surgery with the newest generation spinal robot will yield better pedicle screw placement accuracy than free-hand surgery. Potential benefits of robot-assisted surgery include lower complication and revision rates, shorter hospital LOS, lower radiation exposure, and reduction of economic cost of the overall care.

Head et al (2024) compared the 90-day surgical outcomes, as well as 1-year revision rate of robotic versus non-robotic lumbar fusion surgery. Patients greater than 18 years of age who underwent primary lumbar fusion surgery at the authors' institution were identified and propensity-matched in a 1:1 fashion based on robotic assistance during surgery. Patient demographics, surgical characteristics, and surgical outcomes, including 90-day surgical complications and 1-year revisions, were collected. Multi-variable regression analysis was carried out; significance was set to $p < 0.05$. A total of 415 patients were identified as having robotic lumbar fusion and were matched to a control group. Bi-variant analysis showed no significant difference in total 90-day surgical complications ($p = 0.193$) or 1-year revisions ($p = 0.178$). The operative duration was longer in robotic surgery (287 + 123 mins versus 205 + 88.3 mins, $p \leq 0.001$). Multi-variable analysis demonstrated that robotic fusion was not a significant predictor of 90-day surgical complications (OR = 0.76 [0.32 to 1.67], $p = 0.499$) or 1-year revisions (OR = 0.58 [0.28 to 1.18], $p = 0.142$). Other variables identified as the positive predictors of

1-year revisions included levels fused (OR = 1.26 [1.08 to 1.48], p = 0.004) and current smokers (OR = 3.51 [1.46 to 8.15], p = 0.004). The authors concluded that the findings of this study suggested that robotic-assisted and non-robotic-assisted lumbar fusions were associated with a similar risk of 90-day surgical complications and 1-year revision rates; however, robotic surgery increased time under anesthesia.

The authors stated that this study had several drawbacks, including those inherent to retrospective analysis. Furthermore, because this was not a randomized cohort, procedure selection and outcomes could not fully be unlinked. A surgeon’s choice of procedure may have been influenced by individual patient factors that impacted procedure outcomes. Additionally, these investigators could not assess screw placement directly as CT scans were not routinely ordered at their institution; thus, they could not comment on screw accuracy. Although this cohort was matched by age, sex, body mass index (BMI), Charlson Co-morbidity Index (CCI), and levels fused, other factors (such as history of revision, pre-operative diagnosis, and intra-operative durotomy) were not matched and therefore could have confounded the analysis; however, their statistical strategy aimed to reduce the impact of these factors. By conducting bi-variant logistic regression analysis before building their multi-variant logistic regression model, these researchers could independently examine each variable and its impact on the outcome of interest. By reporting on the multi-variable regression model developed from this list of variables, the authors were able to examine the impact of robots and variables of interest on the outcomes in the most accurate way possible through retrospective analysis.

In a systematic review and meta-analysis, Wu et al (2024) examined the accuracy and feasibility of robot-assisted cervical screw placement and factors that may affect the accuracy. These investigators carried out a comprehensive search on PubMed, Embase, Cochrane Library, Web of Science, CNKI, and Wanfang Med for the selection of potential eligible literature. The outcomes were examined in terms of the RR or SMD and corresponding 95 % CI. Subgroup analyses of the accuracy of screw placement at different cervical segments and with different screw placement approaches were conducted. A comparison was made between robotic navigation and conventional free-hand cervical screw placement. A total of 6 comparative cohort studies and 5 case-series studies with 337 patients and 1,342 cervical screws were included in this study. The perfect accuracy was 86 % (95 % CI: 82 % to 89 %) and the clinically acceptable rate was 98 % (95 % CI: 95 % to 99 %) in robot-assisted cervical screw placement. The perfect accuracy of robot-assisted C1 lateral mass screw placement was the highest (96 %), followed by C6 to C7 pedicle screw placement (93 %) and C2 pedicle screw placement (86 %), and the lowest was C3 to C5 pedicle screw placement (75 %). The open approach had a higher perfect accuracy than the percutaneous/intermuscular approach (91 % versus 83 %). Compared with conventional freehand cervical screw placement, robot-assisted cervical screw placement had a higher accuracy, a lower incidence of peri-operative complications, and less intra-operative blood loss. The authors concluded that with good collaboration between the operator and the robot, robot-assisted cervical screw placement was accurate and feasible. These researchers stated that robot-assisted cervical screw placement has a promising prospect.

Surgery for Bertolotti's Syndrome

Bertolotti’s syndrome is back pain with the presence of a fusion between the transverse process of the last lumbar vertebrae and the 1st sacral segment, also known as lumbo-sacral transitional vertebrae (LSTV). A recent review (Zhu et al, 2023) documented that treatment of only 419 patients with Bertolotti’s syndrome were described in the medical literature. There are no prospective or randomized studies regarding treatment, although the incidence of transitional vertebrae is quite high in the population overall (Crane et al, 2021). A retrospective study (Ashour et al, 2022) of 288 patients who underwent lumbar surgery for other causes were evaluated and the overall incidence of LSTV was 46.2 %. A prospective evaluation (Ravikanth, 2019) of 500 lumbo-sacral X rays from patients with LBP showed an incidence of 26.8 % of LSTV. A retrospective review (Vinha et al 2022) of consecutive CT scans (not performed for back pain) at a single institution found a 24.9 % incidence of LSVT. A population-based study (Tang et al, 2014) of 5,860 individuals found an incidence of LSTV of 15.8 %. This study suggested that certain types of abnormal articulations were associated with a higher incidence of back and gluteal pain. Medical management, PT, injections, and surgery including removal of the articulation and fusion have been attempted with no best practices or clear outcomes defined (Crane et al, 2021). In summary, there is a high incidence of lumbo-sacral transitional vertebrae in the population (both with and without back pain) and an absence of high quality studies showing that LSTV is the specific cause of back pain in individuals or that surgery for Bertolotti's is superior to medical/conservative management.

Avulsion of Spinal Nerves

Sanniec et al (2016) stated that surgical decompression of peripheral branches of the trigeminal and occipital nerves has been shown to alleviate migraine symptoms. Site II surgery entails decompression of the zygomaticotemporal branch of the trigeminal nerve (ZTBTN) by the technique developed by Guyuron. Failure of site II surgery may occur secondary to an inability to recognize a second temporal trigger: site V, the auriculotemporal nerve (ATN). A direct approach for site V has been used with no clear description in the literature. These investigators described a safe and efficient technique for decompression of the ATN that may be easily combined with ZTBTN decompression. Moreover, they stated that long-term follow-up is needed to compare this technique with the use of a separate incision for decompressing-avulsing the ATN at the lateral port level, as well as comparison with a combined decompression of the ATN with ligation of the superficial temporal artery (TA) at the proximal origin (5B) and 5A surgery. The outcome of the site V decompression will be analyzed in prospective and retrospective trials conducted in the future.

Kale et al (2022) noted that scapholunate advanced collapse (SLAC) and scaphoid nonunion advanced collapse (SNAC) are common patterns of wrist arthritis, and surgical interventions include partial and total wrist arthrodesis and wrist denervation, which maintains the current anatomy while relieving pain. These investigators examined current practices within the hand surgery community with respect to the use of anterior interosseous nerve/posterior interosseous nerve (AIN/PIN) denervation in the treatment of SLAC and SNAC wrists. An anonymous survey was distributed to 3,915 orthopedic surgeons via the American Society for Surgery of the Hand (ASSH) listserv. The survey collected information on conservative and operative management, indications, complications, diagnostic block, as well as coding of wrist denervation. A total of 298 answered the survey; 46.3 % (n = 138) of the respondents used denervation of AIN/PIN for every SNAC stage, and 47.7 % (n = 142) of the respondents used denervation of AIN/PIN for every SLAC wrist stage. AIN and PIN combined denervation was the most common stand-alone procedure (n = 185, 62.1 %). Surgeons were more likely to offer the procedure (n = 133, 55.4 %) if motion preservation had to be maximized (n = 154, 64.4 %). The majority of surgeons did not consider loss of proprioception (n = 224, 84.2 %) or diminished protective reflex (n = 246, 92.1 %) to be significant complications. A total of 33.5 % (90 respondents) reported never performing a diagnostic block before denervation. The authors concluded that both SLAC and SNAC patterns of wrist arthritis could result in debilitating wrist pain. There was a wide range of treatment for different stages of disease. Moreover, these researchers stated that further investigations are needed to identify ideal candidates and examine long-term outcomes.

The authors stated that sample size was an inherent drawback: the survey had 298 responses and a response rate of 7.61 %. Poor response rate was likely due to length of the survey. In addition, this study was limited to members of the ASSH. It was unclear if surgeons not in the ASSH would consider isolated denervation to treat SNAC or SLAC wrist. This survey also focused largely on the subjective views of the ASSH membership toward a set of procedures, rather than objective treatment outcomes. As such, these findings should not be taken as guidance for future therapeutic decisions, but rather should be viewed as the current state of attitudes toward denervation in the hand surgery community.

Excision of Spinal Osseous Neoplasms

Nguyen et al (2000) noted that spinal osseous neoplasms are often encountered and could be challenging when present as solitary lesions. Familiarity with the range of benign and malignant spinal pathology could aid radiologists in formulating a comprehensive differential diagnosis. These investigators discussed the spectrum of extradural spinal tumors, accounting for the majority of primary spinal tumors, by comparing the epidemiology, pathophysiology, clinical presentation, and characteristic imaging appearance of these lesions. The discussion entailed the commonly encountered benign lesions, such as vertebral venous vascular malformation and enostosis (a benign bone lesion), as well as malignant lesions including metastases and lymphoma. These investigators also discussed other less-encountered primary spinal tumors such as plasmacytoma, osteoid osteoma, osteoblastoma, giant cell tumor, eosinophilic granuloma, chordoma, chondrosarcoma, osteosarcoma, Ewing's sarcoma, and angiosarcoma. The authors concluded that familiarity with the characteristic imaging features could aid radiologists in making an accurate diagnosis, if not formulating a strong differential diagnosis, and guide clinical management.

Chalamgari et al (2023) stated that the assessment and treatment of vertebral primary bone lesions continue to pose a unique yet significant challenge. These investigators discussed the assessment of various vertebral primary bone lesions, outlining the relevant non-surgical and surgical interventional methods. They describe examples of primary benign and malignant tumors, comparing and contrasting their differences. In addition, these researchers highlighted emerging treatments and approaches for these tumors (e.g., cryoablation and stereotactic body radiation therapy [SBRT]). The authors stated that interventions for vertebral lesions are divided into surgical excision and non-surgical medical management. While primary benign vertebral lesions are often asymptomatic and undetected; however, when they progress and become symptomatic, surgical excision is needed. For patients with extremely poor prognoses, generally conservative therapies are recommended, while patients with good or acceptable prognoses are considered candidates for surgical excision and stabilization if needed. The standard treatment for complex benign and malignant vertebral primary bone lesions is the surgical intralesional excision/curettage, decompression, and stabilization of the affected area with subsequent bone grafting, namely with autogenous cancellous bone graft. Excision or curettage aimed toward removing the benign tumor is the most common surgical intervention. For lesions categorized as a benign aggressive or low-grade malignant bone tumor, cryotherapy with liquid nitrogen is as effective as wide excision.

Pelvic Fixation in Spinal Surgery

Jain et al (2015) stated that attaining solid osseous fusion across the lumbo-sacral junction has historically been, and continues to be, a challenge in spinal surgery. Robust pelvic fixation plays an important role in achieving this objective. These investigators described the history of and indications for spinopelvic fixation, examined conventional spinopelvic fixation techniques, and reviewed the newer S2-alar-iliac technique and its outcomes in adult as well as pediatric patients with spinal deformity. Since the introduction of Harrington rods in the 1960s, spinal instrumentation has evolved substantially. Indications for spinopelvic fixation as a means to achieve lumbo-sacral arthrodesis entail a long arthrodesis (5 or more vertebral levels) or the use of 3-column osteotomies in the lower thoracic or lumbar spine, surgical treatment of high-grade spondylolisthesis, and correction of lumbar deformity and pelvic obliquity. A variety of techniques have been described over the years, including Galveston iliac rods, Jackson intra-sacral rods, the Kostuik trans-iliac bar, iliac screws, and S2-alar-iliac screws. Modern iliac screws and S2-alar-iliac screws were associated with relatively low rates of pseudarthrosis. S2-alar-iliac screws had the advantages of less implant prominence and inline placement with proximal spinal anchors. The authors concluded that these techniques provided powerful methods for obtaining control of the pelvis in facilitating lumbo-sacral arthrodesis.

El Dafrawy e al (2019) noted that sacropelvic (SP) fixation is indicated in various clinical settings, most notably long spinal arthrodesis, reduction of high-grade spondylolisthesis, as well as complex sacral fractures. The sacropelvis is characterized by complex regional anatomy and poor bone quality. These factors make achieving solid fusion across the lumbo-sacral junction challenging; however, a better understanding of spinal biomechanics at that level has resulted in much higher fusion rates than those of the past. The newer fixation techniques are biomechanically superior to previous methods mainly because they achieve bony purchase anterior to the pivot point first described in 1994. Presently, the 2 most frequently employed fixation techniques are iliac screws and S2-alar-iliac screws. Although these techniques were associated with very high rates of fusion, instrumentation-related pain and re-operation remained problematic. The authors provided an overview of the regional anatomy and biomechanics at the lumbo-sacral junction, as well as a summary of fixation techniques with an emphasis on the most widely used techniques today.

Lee et al (2023) examined the incidence, mechanism, and potential protective strategies for pelvic fixation failure (PFF) within 2 years after adult spinal deformity (ASD) surgery. Data for ASD patients (aged 18 years or older, minimum of 6 instrumented levels) with pelvic fixation (S2-alar-iliac [S2AI] and/or iliac screws) with a minimum 2-year follow-up were consecutively collected (2015 to 2019). Patients with prior pelvic fixation were excluded. PFF was defined as any revision to pelvic screws, which may include broken rods across the lumbo-sacral junction requiring revision to pelvic screws, pseudarthrosis across the lumbo-sacral junction requiring revision to pelvic screws, a broken or loose pelvic screw, or sacral/iliac fracture. Patient information including demographic data as well as health history (age, sex, body mass index [BMI], smoking status, American Society of Anesthesiologists [ASA] score, osteoporosis), operative (total instrumented levels [TIL], 3-column osteotomy [3CO], interbody fusion), screw (iliac, S2AI, length, diameter), rod (diameter, kick-stand), rod pattern (number crossing lumbo-pelvic junction, lowest instrumented vertebra [LIV] of accessory rod[s], lateral connectors, dual-headed screws), and pre- and post-radiographic (lumbar lordosis, pelvic incidence, pelvic tilt, major Cobb angle, lumbo-sacral fractional curve, C7 coronal vertical axis [CVA], T1 pelvic angle, C7 sagittal vertical axis) parameters were collected. All rods across the lumbo-sacral junction were cobalt-chrome. All iliac and S2AI screws were closed-headed tulips. Both uni-variate and multi-variate analyses were carried out to determine risk factors for PFF. Of 253 patients (mean age of 58.9 years, mean TIL of 13.6, 3CO 15.8 %, L5 to S1 interbody 74.7 %, mean pelvic screw diameter/length of 8.6/87 mm), the 2-year failure rate was 4.3 % (n = 11). The mechanisms of failure included broken rods across the lumbo-sacral junction (n = 4), pseudarthrosis across the lumbo-sacral junction requiring revision to pelvic screws (n = 3), broken pelvic screw (n = 1), loose pelvic screw (n = 1), sacral/iliac fracture (n = 1), and painful/prominent pelvic screw (n = 1). A higher number of rods crossing the lumbo-pelvic junction (mean of 3.8 no failure versus 2.9 failure, p = 0.009) and accessory rod LIV to S2/ilium (no failure of 54.2 % versus failure 18.2 %, p = 0.003) were protective for failure. Multi-variate analysis showed that accessory rod LIV to S2/ilium versus S1 (OR 0.2, p = 0.004) and number of rods crossing the lumbar to pelvis (OR 0.15, p = 0.002) were protective, while worse post-operative CVA (OR 1.5, p = 0.028) was an independent risk factor for failure. The authors concluded that the 2-year PFF rate was low relative to what was reported in the literature, despite patients undergoing long fusion constructs for ASD. The number of rods crossing the lumbo-pelvic junction and accessory rod LIV to S2/ilium relative to S1 alone likely increased construct stiffness. Residual post-operative coronal mal-alignment should be avoided to reduce PFF.

Habib et al (2023) stated that SP fixation is the immobilization of the sacroiliac joint to attain lumbo-sacral fusion and prevent distal spinal junctional failure. SP fixation is indicated in many spinal conditions (e.g., scoliosis, multi-level spondylolisthesis, spinal/sacral trauma, tumors, or infections). Many SP fixation techniques have been described in the literature. Currently, the most commonly used surgical techniques for SP fixation are direct iliac screws and sacral-2-alar-iliac screws. There is currently no consensus in the literature on which technique carries more favorable clinical outcomes. The authors examined the available data on each technique and discussed their respective advantages and disadvantages. These investigators noted that with the available wide spectrum of surgical techniques used in SP fixation, the available data point toward the clinical advantage of S2AI screws on multiple fronts when compared with the conventional iliac screws (post-operative infection, symptomatic screw prominence, instrument failure, rate of revision surgery, etc.). Moreover, the newly emerging data reported the clinical outcomes of the sub-crestal iliac screw technique and how it required less complex instrumentation construct as these screws eliminated the need for an offset connector placement and resulted in lower rates of screw head prominence. In the authors’ experience, the sub-crestal iliac screw technique represented a safe, feasible, and reliable method of pelvic fixation.

Removal of the C1 Tubercle or Transverse Process or Cervical Laminectomy for Internal Jugular Vein Stenosis

Ding et al (2020) examined the clinical profiles of cervical spondylosis-related internal jugular vein stenosis (IJVS). A total of 46 patients, who were diagnosed as IJVS induced by cervical spondylotic compression were recruited. The clinical manifestations and imaging features of IJVS were presented. Vascular stenosis was present in 69 out of the 92 IJVs, in which, 50.7 % (35/69) of the stenotic vessels were compressed by the transverse process of C1, and 44.9 % (31/69) by the transverse process of C1 combined with the styloid process. The transverse process of C1 compression was more common in unilateral IJVS (69.6 % versus 41.3 %, p = 0.027) while the transverse process of C1 combined with the styloid process compression had a higher propensity to occur in bilateral IJVS (52.2 % versus 30.4 %, p = 0.087). A representative case underwent the resection of the elongated left lateral mass of C1 and styloid process. His symptoms were ameliorated at 6-month follow-up. The authors proposed cervical spondylotic IJV compression syndrome as a new cervical spondylosis subtype. A better understanding of the clinical presentations and imaging features of this type of cervical spondylosis can be of great relevance to clinicians in making a proper diagnosis. Compression from the transverse process of C1 alone was more common in unilateral IJVS, while dual compression from the transverse process combined with the styloid process was more likely in bilateral IJVS. Moreover, the lateral tubercle and styloid process resection may be a promising option to address cervical spondylotic IJVS-associated symptoms. Moreover, these researchers stated that well-designed studies with large sample size are needed to further investigate this disorder.

The authors stated that this study had several drawbacks. First, the small sample size (n = 46 patients) may bias the epidemiological results in this study. Second, the severity of some symptoms was not quantified. Given that, the exact difference between unilateral IJVS and bilateral IJVS could not be assessed accurately. Third, only 1 case underwent surgical intervention without complete follow-up data; thus, the safety and effectiveness of bone resection could not be concluded in this trial.

Fritch et al (2020) stated that idiopathic intracranial hypertension (IIH), also known as pseudotumor cerebri, often presents with severe headache and associated vision loss. Venous outflow obstruction has been noted as a prominent etiologic factor in many cases, and previous anatomic studies have revealed that the IJV at the base of the skull could be prone to compression by the neighboring bony structures. These investigators presented the case of 13-year-old boy with a multi-factorial intracranial hypertension including compression of the IJV by the transverse process of C1. Computerized tomography angiographic (CTA) imaging showed bilateral stenosis of the IJVs due to compression from the transverse processes of C1. Medical management and shunt were attempted without resolution of symptoms. A hemodynamically significant stenosis at the right IJV was confirmed with manometry; thus, the C1 transverse process was resected and a stent placed endovascularly with resolution of pressure gradient and clinical symptoms. The authors concluded that contribution of C1 compression to this patient's intracranial hypertension suggested that evaluation for IJV compression below the skull base may be needed to identify the underlying cause of intracranial hypertension in certain patients. In addition, surgical decompression of the IJV may be needed as part of the treatment strategy. If venous stenting is being considered, this decompressive step must be taken before stenting is carried out. The findings of this single-case study need to be validated by well-designed studies.

Scerrati et al (2021) noted that IJVS is associated with several central nervous system disorders such as Meniere disease or Alzheimer's disease. The extrinsic compression between the styloid process and the C1 transverse process, is an emerging biomarker related to several clinical manifestations. However, currently only a limited number of cases were reported, and few information are available regarding treatment, outcome and complications. These investigators examined clinical-radiological characteristics, diagnosis and treatment of the styloidogenic IJV compression. They carried out a comprehensive literature review. Studies reporting patients suffering from extracranial jugular stenosis were searched. These researchers reviewed demography, clinical and radiological characteristics, as well as outcome, type of treatment, and complications. A total of 13 studies reporting 149 patients were included. Clinical presentation was non-specific. Most frequent symptoms were headache (46.3 %), tinnitus (43.6 %), insomnia (39.6 %). The stenosis was unilateral in 51 patients (45.9 %) and bilateral in 60 (54.1 %). Anticoagulants were the most common prescribed drug (57.4 %). Endovascular treatment was carried out in 50 patients (33.6 %), surgery in 55 (36.9 %), combined in 28 (18.8 %). Improvement of general conditions was reported in 58/80 patients (72.5 %). Complications were reported in 23 % of cases. The authors concluded that jugular stenosis is a complex and often under-estimated disease. Conservative medical treatment usually fails while surgical, endovascular or a combined treatment improved general conditions in more than 70 % of patients.

The authors stated that this study had several drawbacks. First, most of the included studies were case-reports or case-series, with a poor quality of evidence. Second, timing, specific type and dosage of medical treatments were not reported in most of the studies. Third, clinical and radiological outcomes were reported in a small number of patients. Lack of manometry data was very concerning and may constitute a bias in the results evaluation. Fourth, long-term follow-up was missing for most of the cases. Fifth, statistics was not applied, because of the heterogeneity of the data. These researchers tried to obtain individual data from each selected study; however, in several studies this was not possible, and they collected only overall data of patients

Pang et al (2022) stated that small subset of patients with presumed IIH were found to have isolated IJVS. These investigators reviewed the current interventions used in patients who present with intracranial hypertension secondary to IJVS. They carried out a literature search in December 2020 on PubMed/Medline and Scopus data-bases for studies examining surgical and endovascular interventions used for the treatment of intracranial hypertension in the setting of IJVS. No date, patient population, or study type was excluded. All studies that included at least 1 case in which a surgical or endovascular intervention was employed for the treatment of IJVS were included. Selection criteria for patients varied, most commonly defined by identification of compression of the IJV. The 17 studies included in this review ranged from case-reports to large, single-center cohort studies. The most used surgical intervention was styloidectomy, which had an overall better outcome success rate (79 %) than angioplasty/stenting (66 %). No complications were recorded in any of the surgical cases analyzed. Outcome measures varied, but all studies recorded clinical symptoms of the patients. The authors concluded that few current large cohort studies examined surgical and endovascular interventions for patients with IJVS. Notably, the most common intervention was styloidectomy, followed by IJV stenting. Moreover, these researchers stated that y understanding the trends and experience of interventionalists and surgeons, more focused and larger studies can be carried out to determine effective strategies with the best clinical outcomes.

Physical Therapy for Cervical Deformity

Sharan et al (2012) noted that dropped head syndrome (DHS) is characterized by severe weakness of the cervical para-spinal muscles that results in the passively correctable chin-on-chest deformity. DHS is most commonly associated with neuromuscular disorders. However, it is not always accompanied by electromyographic (EMG) findings or noticeable changes on muscle biopsy. In such cases, the term isolated neck extensory myopathy (INEM) is used instead. The literature on the management of INEM is limited. Most reports suggest that non-surgical interventions aid in stabilizing the deformity. The literature on surgical management of INEM is limited and mixed, with outcomes ranging from poor to excellent. The prevalence of DHS likely will increase as life expectancy increases. Recent advances in the understanding of sagittal mal-alignment and surgical techniques have improved the ability to provide better QOL for patients with cervical deformity.

Brodell et al (2020) stated that DHS is a group of disorders with diverse etiologies involving different anatomical components of the neck, leading to a debilitating, flexible, anterior curvature of the cervical spine. Causes of DHS include myasthenia gravis, ALS, Parkinson disease (PD), radiation therapy, and cumulative age-related changes. Idiopathic cases have also been reported. Non-operative treatment of DHS entails orthotic bracing and PT; and surgical treatment consists of cervical spine fusion to correct the deformity. The limited data available examining the clinical and radiographic outcomes of surgical intervention indicated a higher rate of complications with the majority having favorable outcomes in the long-term.

Lee et al (2020) noted that neuromuscular disorders (NMDs) are diseases involving the upper and lower motor neurons and muscles. In patients with NMDs, cervical spinal deformities are a very common issue; however, unlike thoraco-lumbar spinal deformities, few studies have examined these disorders. Patients with NMDs exhibit irregular spinal curvature caused by poor balance and poor coordination of their head, neck, and trunk. Particularly, cervical deformity occurs at younger age, and is known to show more rigid and severe curvature at high cervical levels. Muscular physiologic dynamic characteristics such as spasticity or dystonia combined with static structural factors such as curvature flexibility can result in deformity and often lead to traumatic SCI. Furthermore, post-operative complication rate is higher due to abnormal involuntary movement and muscle tone; thus, it is important to control abnormal involuntary movement peri-operatively along with strong instrumentation for correction of deformity. Various methods such as botulinum toxin injection, PT, muscle division technique, or intra-thecal baclofen pump implant may help control abnormal involuntary movements and improve spinal stability. Surgical management for cervical deformities associated with NMDs requires a multi-disciplinary effort and a customized strategy.

Abadiyan et al (2021) examined the effect of adding a smartphone app to an 8-week global postural re-education (GPR) on neck pain, endurance, QOL, and forward head posture (FHP) in patients with chronic neck pain and FHP. A total of 60 male and female office workers (age of 38.5 ± 9.1 years) with chronic neck pain were randomly assigned into 3 groups: group 1 (GPR+ a smartphone app, n = 20), group 2 (GPR alone, n = 20), and group 3 (the control group, n = 20). The primary outcome was pain and the secondary outcomes were disability, QOL, endurance, and posture. Pain, disability, endurance, QOL, and posture were assessed using the VAS, NDI, progressive iso-inertial lifting evaluation (PILE) test, QOL questionnaire (SF-36), and photogrammetry, respectively, at pre-and post-8-week interventions. A 1-way analysis of co-variance (ANCOVA) was performed to analyze the data. The GPR+ a smartphone app had statistically significant improvements versus GPR alone in pain (MD, - 2.05 ± 0.65, ES (95 % CI: - 0.50 (- 1.04 to - 0.01), p = 0.04), disability (difference = 11.5 ± 1.2, ES (95 % CI: 0.31 (0.22 to 0.97), p = 0.033), FHP (difference = 1.6 ± 0.2, ES (95 % CI: 0.31 (0.09 to 0.92), p = 0.047), and endurance (difference = 2 ± 3.3, ES (95 % CI: 0.51 (0.02 to 1.03), p = 0.039). Both of the GPR+ a smartphone app and GPR alone groups had statistically significant differences versus the control group in all outcomes. The authors concluded that when a work-place assessment and management could not be as part of any intervention, adding a smartphone app to GPR for NP may be an appropriate tool to administer a home and work exercise program resulting in elevating pain and disability, as well as improving FHP and endurance.

Abd El-Azeim et al (2022) stated that one of the most overspread postural abnormalities is FHP and it is described as head projection anteriorly in relation to the trunk which appears mainly in sagittal plane. Scapular stabilization exercise (SSE) is capable of restoring each of thoracic cage and head neutral optimum position by neck and shoulder muscles interactions and via controlling scapular position and movement. In a RCT, these investigators examined the impact of adding SSE to postural correctional exercises (PCE) on individuals with symptomatic FHP. Participants were individuals with postural dysfunction in form of FHP admitted to the outer clinic of the Faculty of Physical Therapy. A total of 60 participants (age of 20 to 35 years) with symptomatic FHP were recruited for this trial. Subjects were allocated randomly by opaque sealed envelope to 2 groups who were referred from an orthopedist: Group "A" received SSE and postural correction exercises, whereas Group "B" received only postural correctional exercises. Treatments were carried out 3 times/week for 10 weeks. The cranio-vertebral angle, pressure pain threshold, cervical flexor and extensor muscles endurance, Arabic NDI, upper trapezius and sterno-cleidomastoid muscle root mean square during rest and activity were employed to evaluate the patients pre-treatment and post-treatment. Within group analysis for 60 participants reported statistically significant difference between baseline and post-treatment as p value of < 0.05 with more refinement in stabilization exercise group. The authors concluded that adding SSEs to PCEs was a more effective method than PCEs for the management of FHP patients.

Hemoglobin A1c and Spinal Surgery

The Endocrine Society Clinical Practice Guideline (2022) acknowledges the widely accepted notion that both preoperative and perioperative glycemic management influence surgical outcomes, yet there remains debate over whether specific preoperative HbA1c or blood glucose (BG) levels should be recommended prior to elective surgeries. Recommendation 5.2 suggests that for adult patients with diabetes undergoing elective procedures, if achieving a hemoglobin A1c (HbA1c) level below 8% (63.9 mmol/mol) is not possible, preoperative BG concentrations should be targeted between 100 and 180 mg/dL (5.6 to 10 mmol/L). A meta-analysis of 11 non-randomized controlled trials indicated that patients with a preoperative HbA1c of less than 7% experienced shorter hospital stays compared to those with HbA1c levels of 7% or higher, although the certainty of this finding was very low. Additionally, 10 non-RCTs suggested that postoperative infections were less frequent in patients with a preoperative HbA1c below 7%, with similar results observed in two studies comparing patients with HbA1c levels above and below 8%. One study identified a HbA1c threshold of 7.8% above which wound complications significantly increased, while other studies indicated that improved glycemic control, as reflected by various HbA1c cutoffs, may lower postoperative infection rates, despite six studies showing a higher need for reoperations in patients with HbA1c below 7%. Furthermore, comparisons of preoperative HbA1c levels suggested potential reductions in respiratory, neurologic, renal, and cardiac complications, although the certainty of these findings was also very low. The panel noted that preoperative BG levels and diabetes status significantly impacted hospital care costs, and while the effects of race and sex were minimally studied, socioeconomic factors could affect access to diabetes care and surgical interventions. Ultimately, the panel concluded that better preoperative glucose management is associated with improved outcomes, recommending a target HbA1c above 8% to identify patients at higher risk for complications, along with a preoperative BG target of 100 to 180 mg/dL. Proposed areas for future research include investigating the effects of preoperative fasting glucose versus HbA1c on outcomes, the relationship between preoperative time in range and postoperative results, and the impact of varying durations of strict glycemic control on patients with elevated HbA1c levels (Korytkowski et al., 2022).

In a systematic review and meta-analysis, Tao et al. (2023) evaluated the role of preoperative HbA1c in risk stratification for patients undergoing spinal procedures and to summarize consensus recommendations. The background highlights that diabetes mellitus and hyperglycemia are independent risk factors for increased surgical complications, with HbA1c serving as a key indicator of long-term glycemic control that can be optimized to enhance surgical outcomes and patient-reported results; however, comprehensive reviews on this topic in spine surgery have been scarce. The authors conducted a systematic search of PubMed, EMBASE, Scopus, and Web-of-Science for English-language studies up to April 5, 2022, adhering to PRISMA guidelines, and included studies that reported preoperative HbA1c values and postoperative outcomes in spine surgery patients. They identified 22 articles (18 retrospective cohort studies and 4 prospective observational studies) with a level of evidence of III or greater, finding that 17 studies indicated a correlation between elevated preoperative HbA1c and poorer outcomes or increased complication risks. The random-effects meta-analysis revealed that patients with preoperative HbA1c levels greater than 8.0% had a significantly higher risk of postoperative complications (RR: 1.85, 95% CI: [1.48, 2.31], P <0.01) and that those with surgical site infections had an average HbA1c increase of 1.49% compared to those without infections (Mean Difference: 1.49%, 95% CI: [0.11, 2.88], P =0.03). The study concludes that an HbA1c level above 8.0% is linked to a greater risk of complications, indicating that elevated HbA1c is associated with less favorable outcomes following spine surgery, with a level of evidence rated at IV.

Glossary of Terms

Term	**Definition**
Spinal stenosis	Central, lateral recess or foraminal stenosis
Non-union of prior fusion	Lack of bridging bone or abnormal motion at fused segment
Lumbar pseudarthrosis	Absence of bridging bone that connects the vertebrae

Appendix

Types of Spondylolisthesis Description

The following types of spondylolisthesis are based on etiology:

I. *Type 1*

The dysplastic (congenital) type represents a defect in the upper sacrum or arch of L5. A high rate of associated spina bifida occulta and a high rate of nerve root involvement exist.

II. *Type 2*

This results from a defect in pars inter-articularis, which permits forward slippage of the superior vertebra, usually L5.

The following 3 subcategories are recognized:

- Acutely fractured pars
- Elongated yet intact pars
- Lytic (i.e., spondylolysis) or stress fracture of the pars.

III. *Type 3*

The degenerative (late in life) type is an acquired condition resulting from chronic disc degeneration and facet incompetence, leading to long-standing segmental instability and gradual slippage, usually at L4-L5. Spondylosis is a general term reserved for acquired age-related degenerative changes of the spine (i.e., discopathy or facet arthropathy) that can lead to this type of spondylolisthesis.

IV. *Type 4*

The traumatic (any age) type results from fracture of any part of the neural arch or pars that leads to listhesis.

V. *Type 5*

The pathologic type results from a generalized bone disease, such as Paget disease or osteogenesis imperfecta.

The Myerding Grading System

The Myerding grading system measures the percentage of vertebral slip forward over the body beneath.

Table 1: Myerding Grading System Percentage of Vertebral Slip Forward

Grade	Percentage
Grade 1	25 % of vertebral body has slipped forward
Grade 2	25 % to 49 % of vertebral body has slipped forward
Grade 3	50 % to 74 % of vertebral body has slipped forward
Grade 4	75 % to 99 % of vertebral body has slipped forward
Grade 5	Vertebral body has completely fallen off (i.e., spondyloptosis)

Manual Muscle Testing Scale

Table 2: Manual Muscle Testing Scale

Grade	Description
Grade 5	Holds position against strong pressure
Grade 4+	Holds position against moderate to strong pressure
Grade 4	Holds position against moderate pressure
Grade 4-	Holds position against slight to moderate pressure
Grade 3+	Holds position against slight pressure
Grade 3	Holds position against gravity
Grades 0/1/2	Cannot hold position against gravity or worse

Adapted from Kendall, et al., 2011.

The following spine fracture types are not considered unstable (unless there is additional imaging demonstrating instability):

- Fractures of the lamina
- Isolated pedicle fractures.
- Lumbar pars defects/fractures
- Mild/moderate compression fractures without posterior element involvement
- Spinous process fractures
- Transverse process fractures.

Urinary Cotinine Levels (ng/ml) for Smokers and Non-Smokers

- Heavy smoker may be more than 500 ng/ml.
- Light smoker or someone exposed to second-hand smoke are 11 to 30 ng/ml.
- Non-smoker are generally less than 10 ng/ml.

Source: The University of Rochester Medical Center's Health Encyclopedia (Haldeman-Englert, et al., 2024)

References

The above policy is based on the following references:

1. Abadiyan F, Hadadnezhad M, Khosrokiani Z, et al. Adding a smartphone app to global postural re-education to improve neck pain, posture, quality of life, and endurance in people with nonspecific neck pain: A randomized controlled trial. Trials. 2021;22(1):274.
2. Abd El-Azeim AS, Mahmoud AG, Mohamed MT, El-Khateeb YS. Impact of adding scapular stabilization to postural correctional exercises on symptomatic forward head posture: A randomized controlled trial. Eur J Phys Rehabil Med. 2022;58(5):757-766.
3. Ammendolia C, Cote P, Southerst D, et al. Comprehensive nonsurgical treatment versus self-directed care to improve walking ability in lumbar spinal stenosis: A randomized trial. Arch Phys Med Rehabil. 2018;99(12):2408-2419.
4. Ammendolia C, Hofkirchner C, Plener J, et al. Non-operative treatment for lumbar spinal stenosis with neurogenic claudication: An updated systematic review. BMJ Open. 2022;12(1):e057724.
5. Ammendolia C, Stuber K, Tomkins-Lane C, et al. What interventions improve walking ability in neurogenic claudication with lumbar spinal stenosis? A systematic review. Eur Spine J. 2014;23(6):1282-1301.

6. American College of Occupational and Environmental Medicine (ACOEM). Low back complaints. Elk Grove Village, IL: ACOEM; 2004.
7. Amundsen T, Weber H, Nordal HJ, et al. Lumbar spinal stenosis: Conservative or surgical management?: A prospective 10-year study. Spine (Phila Pa 1976). 2000;25(11):1424-1435; discussion 1435-1436.
8. Anderson PA, Matz PG, Groff MW, et al; Joint Section on Disorders of the Spine and Peripheral Nerves of the American Association of Neurological Surgeons and Congress of Neurological Surgeons. Laminectomy and fusion for the treatment of cervical degenerative myelopathy. J Neurosurg Spine. 2009;11(2):150-156.
9. Ashour A, Hassan A, Aly M, Nafady MA. Prevalence of Bertolotti's syndrome in lumbosacral surgery procedures. Cureus. 2022;14(6):e26341.
10. Atlas SJ, Delitto A. Spinal stenosis: Surgical versus nonsurgical treatment. Clin Orthop Relat Res. 2006;443:198-207.
11. Aurouer N, Guerin P, Cogniet A, et al. Pedicle screw placement accuracy in robot-assisted versus image-guided freehand surgery of thoraco-lumbar spine (ROBARTHRODESE): Study protocol for a single-centre randomized controlled trial. Trials. 2024;25(1):106.
12. Bambakidis NC, Feiz-Erfan I, Klopfenstein JD, et al. Indications for surgical fusion of the cervical and lumbar motion segment. Spine. 2005;30(16 Suppl):S2-S6.
13. Bartels RH, van Tulder MW, Moojen WA, et al. Laminoplasty and laminectomy for cervical spondylotic myelopathy: A systematic review. Eur Spine J. 2015;24 Suppl 2:160-167.
14. Bigos S, Bowyer O, Braen G, et al. Acute low back problems in adults. Clinical Practice Guideline No. 14. AHCPR Publication No. 95-0642. Rockville, MD: Agency for Health Care Policy and Research (AHCPR); December 1994.
15. Blue Cross Blue Shield Association (BCBSA), Technology Evaluation Center (TEC). Artificial lumbar disc replacement. TEC Assessment Program. Chicago, IL: BCBSA; 2007;22(2).
16. Bohan JS. Surgery for Sciatica: Early symptom relief is the only real benefit. JWatch Emergency Med. July 13, 2007.
17. Bono CM, Ghiselli G, Gilbert TJ, et al. An evidence-based clinical guideline for the diagnosis and treatment of cervical

radiculopathy from degenerative disorders. Spine J. 2011;11(1):64-72.

18. Borgo A, Cossio A, Gallone D, et al. Orthopaedic challenges for mucopolysaccharidoses. Ital J Pediatr. 2018;44(Suppl 2):123.
19. Bridwell KH. Surgical treatment of high-grade spondylolisthesis. Neurosurg Clin N Am. 2006;17(3):331-338.
20. Brodell JD, Jr, Sulovari A, Bernstein DN, et al. Dropped head syndrome: An update on etiology and surgical management. JBJS Rev. 2020;8(1):e0068.
21. Brox JI, Nygaard OP, Holm I, et al. Four-year follow-up of surgical versus non-surgical therapy for chronic low back pain. Ann Rheum Dis. 2010;69(9):1643-1648.
22. Brox JI, Reikeras O, Nygaard O, et al. Lumbar instrumented fusion compared with cognitive intervention and exercises in patients with chronic back pain after previous surgery for disc herniation: A prospective randomized controlled study. Pain. 2006;122(1-2):145-155.
23. Brox JI, Sorensen R, Friis A, et al. Randomized clinical trial of lumbar instrumented fusion and cognitive intervention and exercises in patients with chronic low back pain and disc degeneration. Spine. 2003;28(17):1913-1921.
24. Carragee EJ, Han MY, Suen PW, et al. Clinical outcomes after lumbar discectomy for sciatica: The effects of fragment type and anular competence. J Bone Joint Surg Am. 2003;85-A(1):102-108.
25. Carreon LY, Glassman SD, Howard J. Fusion and nonsurgical treatment for symptomatic lumbar degenerative disease: A systematic review of Oswestry Disability Index and MOS Short Form-36 outcomes. Spine J. 2008;8(5):747-755.
26. Carroll S, Wiesel S. Neurologic complications and lumbar laminectomy. Clin Orthop Relat Res. 1992;284:14-23.
27. Chalamgari A, Valle D, Palau Villarreal X, et al. Vertebral primary bone lesions: Review of management options. Curr Oncol. 2023;30(3):3064-3078.
28. Chen TP, Qian LG, Jiao JB, et al. Anterior decompression and fusion versus laminoplasty for cervical myelopathy due to ossification of posterior longitudinal ligament: A meta-analysis. Medicine (Baltimore). 2019;98(1):e13382.
29. Chen Y-N, Yang S-H, Chou S-C, Kuo M-F. The role of sacral laminoplasty in the management of spina bifida and sacral cystic

lesions: Case series. Neurosurg Focus 2019;47(4):E20.
30. Chen Z, Liu B, Dong J, et al. Comparison of anterior corpectomy and fusion versus laminoplasty for the treatment of cervical ossification of posterior longitudinal ligament: A meta-analysis. Neurosurg Focus. 2016;40(6):E8.
31. Chi JH, Tay B, Stahl D, Lee R. Complex deformities of the cervical spine. Neurosurg Clin N Am. 2007;18(2):295-304.
32. Cho R, Fu R, Carrino J, et al. Imaging strategies for low-back pain: Systematic review and meta-analysis. Lancet. 2009;373:463-472. National Institute for Health and Clinical Excellence (NICE). Low back pain: Early management of persistent non-specific low back pain. NICE Clinical Guideline 88. London, UK: NICE; May 2009.
33. Cho SK, Kim JS, Overley SC, Merrill RK. Cervical laminoplasty: Indications, surgical considerations, and clinical outcomes. J Am Acad Orthop Surg. 2018;26(7):e142-e152.
34. Chou R, Baisden J, Carragee EJ, et al. Surgery for low back pain: A review of the evidence for an American Pain Society Clinical Practice Guideline. Spine. 2009;34(10):1094-1109.
35. Chou R, Huffman LH. Nonpharmacologic therapies for acute and chronic low back pain: A review of the evidence for an American Pain Society/American College of Physicians Clinical Practice Guideline. Ann Internal Med. 2007;147(7):492-504.
36. Chou R, Loeser JD, Owens DK, et al.; American Pain Society Low Back Pain Guideline Panel. Interventional therapies, surgery, and interdisciplinary rehabilitation for low back pain: An evidence-based clinical practice guideline from the American Pain Society. Spine. 2009;34(10):1066-1077.
37. Chou R, Qaseem A, Snow V, et al. Diagnosis and treatment of low back pain: A joint clinical practice guideline from the American College of Physicians and the American Pain Society. Ann Intern Med. 2007;147(7):478-491.
38. Chow NW, Southerst D, Wong JJ, et al. Clinical outcomes in neurogenic claudication using a multimodal program for lumbar spinal stenosis: A study of 49 patients with prospective long-term follow-up. J Manipulative Physiol Ther. 2019;42(3):203-209.
39. Chrobok J, Vrba I, Stetkárová I. Selection of surgical procedures for treatment of failed back surgery syndrome (FBSS). Chir Narzadow Ruchu Ortop Pol. 2005;70(2):147-153.

40. Comer C, Redmond AC, Bird HA, et al. A home exercise programme is no more beneficial than advice and education for people with neurogenic claudication: Results from a randomised controlled trial. PLoS One. 2013;8(9):e72878.
41. Connor M, Briggs RG, Bonney PA, et al. Tobacco use is associated with increased 90-day readmission among patients undergoing surgery for degenerative spine disease. Global Spine J. 2022;12(5):787-794.
42. Conti P, Pansini G, Mouchaty H, et al. Spinal neurinomas: Retrospective analysis and long-term outcome of 179 consecutively operated cases and review of the literature. Surg Neurol. 2004;61(1):34-43.
43. Corp N, Mansell G, Stynes S, et al. Evidence-based treatment recommendations for neck and low back pain across Europe: A systematic review of guidelines. Eur J Pain. 2021;25(2):275-295.
44. Crane J, Cragon R, O'Neill J, et al. A comprehensive update of the treatment and management of Bertolotti's syndrome: A best practices review. Orthop Rev (Pavia). 2021;13(2):24980.
45. Cunningham MR, Hershman S, Bendo J. Systematic review of cohort studies comparing surgical treatments for cervical spondylotic myelopathy. Spine (Phila Pa 1976). 2010;35(5):537-543.
46. Dawodu ST. Cauda equina and conus medullaris syndromes. eMedicine. Neuro Topic 667. Omaha, NE: eMedicine.com: updated October 5, 2005. Available at: http://www.emedicine.com/neuro/topic667.htm. Accessed April 14, 2006.
47. Debono B, Wainwright TW, Wang MY, et al. Consensus statement for perioperative care in lumbar spinal fusion: Enhanced Recovery After Surgery (ERAS®) Society recommendations. Spine J. 2021;21(5):729-752.
48. Del Castillo-Calcaneo JD, Navarro-Ramirez R, Nakhla J, et al. Minimally invasive treatment for a sacral Tarlov cyst through tubular retractors. World Neurosurg. 2017;108:993.e9-993.e11.
49. Delitto A, Piva SR, Moore CG, et al. Surgery versus nonsurgical treatment of lumbar spinal stenosis: A randomized trial. Ann Intern Med. 2015;162(7):465-473.
50. Dengler J, Duhon B, Whang P, et al. Predictors of outcome in conservative and minimally invasive surgical management of

pain originating from the sacroiliac joint: A pooled analysis. Spine (Phila Pa 1976). 2017;42(21):1664-1673.
51. Dettori JR, Skelly AC, Hashimoto RE, et al. Artificial disc replacement (ADR). Health Technology Assessment. Provided by Spectrum Research, Inc. for the Technology Assessment Program, Washington State Health Care Authority. Olympia, WA: Washington State Health Care Authority; September 19, 2008.
52. Devereaux MW. Neck pain. Prim Care. 2004;31(1):19-31.
53. Devulapalli C, Broyles JM, Bello R, et al. Soft-tissue reconstruction of large spinal defects: A 12-year institutional experience. Plast Reconstr Surg. 2017;140(4):806-814.
54. DeWald CJ, Vartabedian JE, Rodts MF, et al. Evaluation and management of high-grade spondylolisthesis in adults. Spine. 2005;30(6 Suppl):S49-S59.
55. Deyo RA, Gray DT, Kreuter W, et al. United States trends in lumbar fusion surgery for degenerative conditions. Spine. 2005;30(12):1441-1445.
56. Deyo RA, Nachemson A, Mirza SK. Spinal-fusion surgery - the case for restraint. N Engl J Med. 2004;350(7):722-726.
57. Ding JY, Zhou D, Pan LQ, et al. Cervical spondylotic internal jugular venous compression syndrome. CNS Neurosci Ther. 2020;26(1):47-54.
58. Diwan AD, Parvartaneni H, Cammisa F. Failed degenerative lumbar spine surgery. Orthop Clin North Am. 2003;34(2):309-324.
59. Dr. Robert Bree Collaborative, Accountable Payments Models Workgroup. Lumbar Fusion Surgical Bundle. Seattle, SA Bree Collaborative; September 2014.
60. Duggal N, Mendiondo I, Pares HR, et al. Anterior lumbar interbody fusion for treatment of failed back surgery syndrome: an outcome analysis. Neurosurgery. 2004;54(3):636-643.
61. Durbhakula MM, Ghiselli G. Cervical total disc replacement, part I: Rationale, biomechanics, and implant types. Orthop Clin North Am. 2005;36(3):349-354.
62. Dziurzynski K, Anderson PA, Bean DB, et al. A blinded assessment of radiographic criteria for atlanto-occipital dislocation. Spine (Phila Pa 1976). 2005;30(12):1427-1432.
63. ECRI Health Technology Assessment Group. Treatment of degenerative lumbar spinal stenosis. Volume 1: Evidence report. Volume 2: Evidence tables and bibliography. Evidence

Report/Technology Assessment No. 32. Rockville, MD: Agency for Healthcare Research and Quality (AHRQ); 2001.

64. Edwards CC 2nd, Riew KD, Anderson PA, et al. Cervical myelopathy. Current diagnostic and treatment strategies. Spine J. 2003;3(1):68-81.
65. El Dafrawy MH, Raad M, Okafor L, Kebaish KM. Sacropelvic fixation: A comprehensive review. Spine Deform. 2019;7(4):509-516.
66. Emery SE. Cervical spondylotic myelopathy: Diagnosis and treatment. J Am Acad Orthop Surg. 2001;9(6):376-388.
67. Engquist M, Löfgren H, Öberg B, et al. Factors affecting the outcome of surgical versus nonsurgical treatment of cervical radiculopathy. Spine. 2015;40(20):1553-1563.
68. Erkan S, Rivera Y, Wu C, et al. Biomechanical comparison of a two-level Maverick disc replacement with a hybrid one-level disc replacement and one-level anterior lumbar interbody fusion. Spine J. 2009;9(10):830-835.
69. Evans L, O'Donohoe T, Morokoff A, Drummond K. The role of spinal surgery in the treatment of low back pain. Med J Aust. 2023;218(1):40-45.
70. Ezeokoli EU, Tamma PL, Borici N, et al. Plastic multilayered closure versus orthopedic surgeon closure after spinal instrumentation in pediatric neuromuscular scoliosis. World J Pediatr Surg. 2023;6(1):e000485.
71. Fairbank J, Frost H, Wilson-MacDonald J, Yu LM, Barker K, Collins R; Spine Stabilisation Trial Group. Randomised controlled trial to compare surgical stabilisation of the lumbar spine with an intensive rehabilitation programme for patients with chronic low back pain: The MRC spine stabilisation trial. BMJ. 2005;330(7502):1233.
72. Fehlings MG, Santaguida C, Tetreault L, et al. Laminectomy and fusion versus laminoplasty for the treatment of degenerative cervical myelopathy: Results from the AOSpine North America and International prospective multicenter studies. Spine J. 2017;17(1):102-108.
73. Fiori R, Forcina M, Spiritigliozzi L, et al. Full percutaneous treatment of degenerative disc disease with intradiscal lumbar interbody fusion and posterior stabilization: Preliminary results. Cardiovasc Intervent Radiol. 2020;43(6):889-896.

74. Fischgrund JS, Mackay M, Herkowitz HN, et al. Degenerative lumbar spondylolisthesis with spinal stenosis: A prospective, randomized study comparing decompressive laminectomy and arthrodesis with and without spinal instrumentation. Spine. 1997;22(24):2807-2812.
75. Fisher CG, Sun JC, Dvorak M. Recognition and management of atlanto-occipital dislocation: improving survival from an often fatal condition. Can J Surg. 2001;44(6):412-20.
76. Fourney DR, Skelly AC, DeVine JG. Treatment of cervical adjacent segment pathology: A systematic review. Spine (Phila Pa 1976). 2012;37(22 Suppl):S113-S122.
77. Fouyas IP, Statham PFX, Sandercock PAG, Lynch C. Surgery for cervical radiculomyelopathy. Cochrane Database of Syst Rev. 2001;(3):CD001466.
78. Fritch C, Voronovich Z, Carlson AP. C1 transverse process resection for management of jugular stenosis. Oper Neurosurg (Hagerstown). 2020;19(2):E209-E213.
79. Fritsch EW, Heisel J, Rupp S. The failed back surgery syndrome: reasons, intraoperative findings, and long-term results: A report of 182 operative treatments. Spine. 1996;21(5):626-633.
80. Fritz JM, Lurie JD, Zhao W, et al. Associations between physical therapy and long-term outcomes for individuals with lumbar spinal stenosis in the SPORT study. Spine J. 2014;14(8):1611-1621.
81. Fritzell P, Hägg O, Wessberg P, et al. 2001 volvo award winner in clinical studies: Lumbar fusion versus nonsurgical treatment for chronic low back pain: A multicenter randomized controlled trial from the Swedish Lumbar Spine Study Group. Spine. 2001;26(23):2521-2532.
82. Froholdt A, Reikeraas O, Holm I, et al. No difference in 9-year outcome in CLBP patients randomized to lumbar fusion versus cognitive intervention and exercises. Eur Spine J. 2012;21(12):2531-2538.
83. Fu TS, Lai PL, Tsai TT, et al. Long-term results of disc excision for recurrent lumbar disc herniation with or without posterolateral fusion. Spine. 2005;30(24):2830-2834.
84. Garg B, Mehta N. Brothers-in-arms: Liaison between spine surgeons and plastic surgeons in wound repair after complex spine surgery. N Am Spine Soc J. 2020;4:100031.

85. Gatot C, Liow MHL, Goh GS, et al. Smoking is associated with lower satisfaction in nondiabetic patients undergoing minimally invasive single-level transforaminal lumbar interbody fusion. Clin Spine Surg. 2022;35(1):E19-E25.
86. Geck MJ, Eismont FJ. Surgical options for the treatment of cervical spondylotic myelopathy. Orthop Clin North Am. 2002;33(2):329-348.
87. Gibson JN, Grant IC, Waddell G. The cochrane review of surgery for lumbar disc prolapse and degenerative lumbar spondylosis. Spine. 1999;24:1820-1832.
88. Gibson JNA, Waddell G. Surgery for degenerative lumbar spondylosis. Cochrane Database Syst Rev. 2005;(4):CD001352.
89. Gibson JNA, Waddell G. Surgical interventions for lumbar disc prolapse. Cochrane Database of Systematic Reviews. 2007;(2):CD001350.
90. Greenfield K, Nelson RJ, Findlay GD, et al. Microdiscectomy and conservative treatment for lumbar disc herniation with back pain and sciatica: A randomised clinical trial. Proceedings of the International Society for the Study of Lumbar Spine (Abstract). 2003:245.
91. Guyer RD, Patterson M, Ohnmeiss DD. Failed back surgery syndrome: Diagnostic evaluation. J Am Acad Orthop Surg. 2006;14(9):534-543.
92. Habib A, Muthiah N, Alattar A, et al. Direct iliac screw vs sacral-2-alar-iliac screws technique for sacropelvic fixation: Technical nuances and a review of the literature. Int J Spine Surg. 2023;17(4):511-519.
93. Haldeman-Englert C, Turley RK, Turley R. Nicotine Cotinine (Urine). Health Encyclopedia. Rochester, NY: University of Rochester Medical Center; 2024. Available at: https://www.urmc.rochester.edu/encyclopedia.aspx. Accessed June 14, 2024.
94. Hale JJ, Gruson KI, Spivak JM. Laminoplasty: A review of its role in compressive cervical myelopathy. Spine J. 2006;6(6 Suppl):289S-298S.
95. Haouas MY, Khoulali M, En-Nhaili Z, et al. Surgical treatment of sacral Tarlov cysts: About 20 cases. Pan Afr Med J. 2019;33:98.
96. Hazard RG. Failed back surgery syndrome: Surgical and nonsurgical approaches. Clin Orthop Relat Res. 2006;443:228-

232.

97. Heard JC, Lee YA, D'Antonio ND, et al. The impact of robotic assistance for lumbar fusion surgery on 90-day surgical outcomes and 1-year revisions. J Craniovertebr Junction Spine. 2024;15(1):15-20.
98. Health Services Advisory Council (HSAC). Spinal fusion vs. conservative therapy. Evidence Summaries and Legislative Reports. St. Paul, MN: Minnesota Department of Human Services; September 10, 2007.
99. Health Services Advisory Council (HSAC). Spinal fusion. Evidence Summaries and Legislative Reports. St. Paul, MN: Minnesota Department of Human Services; April 12, 2007.
100. Heller JG, Raich AL, Dettori JR, Riew KD. Comparative effectiveness of different types of cervical laminoplasty. Evid Based Spine Care J. 2013;4(2):105-115.
101. Herron L. Recurrent lumbar disc herniation: results of repeat laminectomy and discectomy. J Spinal Disord. 1994 Apr;7(2):161-166.
102. Hersh AM, Pennington Z, Schilling AT, et al. Plastic surgery wound closure following resection of spinal metastases. Clin Neurol Neurosurg. 2021;207:106800.
103. Hilibrand AS, Rand N. Degenerative lumbar stenosis: Diagnosis and management. J Am Acad Orthop Surg. 1999;7(4):239-249.
104. Hipp A, Sinert R. Evidence-based emergency medicine/rational clinical examination abstract. Clinical assessment of low back pain. Ann Emerg Med. 2006;47(3):283-285.
105. Hong S, Shinya Y, Lakomkin N, et al. Predicting the need for occipitocervical fusion for patients with lower clival chordoma: A single-center retrospective study. World Neurosurg. 2024;187:e321-e330.
106. Horn EM, Feiz-Erfan I, Lekovic GP, et al. Survivors of occipitoatlantal dislocation injuries: Imaging and clinical correlates. J Neurosurg Spine. 2007;6(2):113-120.
107. Huang D, Du X, Liang H, et al. Anterior corpectomy versus posterior laminoplasty for the treatment of multilevel cervical myelopathy: A meta-analysis. Int J Surg. 2016;35:21-27.
108. Humadi A, Chao T, Dawood S, et al. A meta-analysis of cervical laminoplasty techniques: Are mini-plates superior? Global Spine J. 2017;7(4):373-381.

109. Inglesby DC, Young ZT, Alshareef M, et al. Paraspinous muscle flaps for the treatment of complex spinal wounds. Spine (Phila Pa 1976). 2020;45(9):599-604.
110. Ishii M, Wada E, Ishii T, et al. Laminoplasty for patients aged 75 years or older with cervical myelopathy. J Orthop Surg (Hong Kong). 2008;16(2):211-214.
111. Iwasaki M, Okuda S, Miyauchi A, et al. Surgical strategy for cervical myelopathy due to ossification of the posterior longitudinal ligament: Part 1: Clinical results and limitations of laminoplasty. Spine. 2007;32(6):647-653.
112. Iwasaki M, Okuda S, Miyauchi A, et al. Surgical strategy for cervical myelopathy due to ossification of the posterior longitudinal ligament: Part 2: Advantages of anterior decompression and fusion over laminoplasty. Spine. 2007;32(6):654-660.
113. Jacobi S, Beynon A, Dombrowski SU, et al. Effectiveness of conservative nonpharmacologic therapies for pain, disability, physical capacity, and physical activity behavior in patients with degenerative lumbar spinal stenosis: A systematic review and meta-analysis. Arch Phys Med Rehabil. 2021;102(11):2247-2260.
114. Jacobs WCH, Anderson PG, van Limbeek J, et al. Single or double-level anterior interbody fusion techniques for cervical degenerative disc disease. Cochrane Database of Syst Rev. 2004;(4):CD004958.
115. Jain A, Hassanzadeh H, Strike SA, et al. Pelvic fixation in adult and pediatric spine surgery: Historical perspective, indications, and techniques: AAOS exhibit selection. J Bone Joint Surg Am. 2015;97(18):1521-1528.
116. Jang JS, Lee SH, Min JH, et al. Surgical treatment of failed back surgery syndrome due to sagittal imbalance. Spine. 2007;32(26):3081-7.
117. Jarvik JG, Deyo RA. Diagnostic evaluation of low back pain with emphasis on imaging. Ann Intern Med. 2002;137(7):586-597.
118. Jiang L, Tan M, Dong L, et al. Comparison of anterior decompression and fusion with posterior laminoplasty for multilevel cervical compressive myelopathy: A systematic review and meta-analysis. J Spinal Disord Tech. 2015;28(8):282-290.
119. Jiang YQ, Li XL, Zhou XG, et al. A prospective randomized trial comparing anterior cervical discectomy and fusion versus plate-

only open-door laminoplasty for the treatment of spinal stenosis in degenerative diseases. Eur Spine J. 2017;26(4):1162-1172.

120. Kahwati L, Weber RP, Lim M, et al. Surgery for lumbar radiculopathy/ sciatica. Final Evidence Report. Prepared by the RTI International–University of North Carolina Evidence-based Practice Center for the Health Technology Assessment Program (HTA). Olympia, WA: Washington State Health Care Authority; April 13, 2018.
121. Kale NN, Foote J, Medvedev G. Use of wrist denervation in the treatment of SLAC and SNAC wrist by ASSH members. J Wrist Surg. 2022;12(3):280-286.
122. Kang Y, Lee JW, Koh YH. New MRI grading system for the cervical canal stenosis. Am J Roentgenol. 2011;197: W134-W140.
123. Karaikovic EE, Pacheco HO. Treatment options for thoracolumbar spine fractures. Bosn J Basic Med Sci. 2005;5(2):20-26.
124. Kasliwal MK, Fontes RB, Traynelis VC. Occipitocervical dissociation-incidence, evaluation, and treatment. Curr Rev Musculoskelet Med. 2016;9(3):247-254.
125. Kelly DM, McCarthy RE, McCullough FL, Kelly HR. Long-term outcomes of anterior spinal fusion with instrumentation for thoracolumbar and lumbar curves in adolescent idiopathic scoliosis. Spine. 2010;35(2):194-198.
126. Kendall FP, McCreary EK, Provance PG, et al. Muscles: Testing and Function with Posture and Pain. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2011.
127. Khalid SI, Thomson KB, Chilakapati S, et al. The impact of smoking cessation therapy on lumbar fusion outcomes. World Neurosurg. 2022;164:e119-e126.
128. Khoury C. Closed spinal dysraphism: Clinical manifestations, diagnosis, and management. UpToDate [online serial]. Waltham, MA: UpToDate; reviewed November 2020.
129. Klimo P Jr, Dailey AT, Fessler RG. Posterior surgical approaches and outcomes in metastatic spine-disease. Neurosurg Clin N Am. 2004;15(4):425-435.
130. Koakutsu T, Morozumi N, Ishii Y, et al. Anterior decompression and fusion versus laminoplasty for cervical myelopathy caused by soft disc herniation: A prospective multicenter study. J Orthop Sci. 2010;15(1):71-78.

131. Koda M, Mochizuki M, Konishi H, et al. Comparison of clinical outcomes between laminoplasty, posterior decompression with instrumented fusion, and anterior decompression with fusion for K-line (-) cervical ossification of the posterior longitudinal ligament. Eur Spine J. 2016;25(7):2294-2301.
132. Korytkowski MT, Muniyappa R, Antinori-Lent K, et al. Management of hyperglycemia in hospitalized adult patients in non-critical care settings: An Endocrine Society clinical practice guideline. J Clin Endocrinol Metab. 2022;107(8):2101-2128.
133. Kreiner DS, Matz P, Bono CM, et al. Guideline summary review: An evidence-based clinical guideline for the diagnosis and treatment of low back pain. Spine J. 2020;20(7):998-1024.
134. Kuijper B, Tans JT, Beelen A, Nollet F, de Visser M. Cervical collar or physiotherapy versus wait and see policy for recent onset cervical radiculopathy: Randomised trial. BMJ. 2009;339:b3883.
135. Kwon BK, Hilibrand AS, Malloy K, et al. A critical analysis of the literature regarding surgical approach and outcome for adult low-grade isthmic spondylolisthesis. J Spinal Disord Tech. 2005;18 Suppl:S30-S40.
136. Kwon BK, Vaccaro AR, Grauer JN, et al. Indications, techniques, and outcomes of posterior surgery for chronic low back pain. Orthop Clin North Am. 2003;34(2):297-308.
137. Laminaplasty. In: Spine-Health.com. PF Ullrich, ed. Evanston, IL: Spine-Health.com; 1999-2001. Available at: http://www.spine-health.com/dir/laminaplasty.html. Accessed July 31, 2001.
138. Laminaplasty. In: Wheeless' Textbook of Orthopaedics. CR Wheeless III, ed. European ed. CR Wheeless; 1996. Available at: http://wheeless.belgianorthoweb.be/o11/1180.htm. Accessed July 31, 2001.
139. Lao L, Zhong G, Li X, et al. Laminoplasty versus laminectomy for multi-level cervical spondylotic myelopathy: A systematic review of the literature. J Orthop Surg Res. 2013;8:45.
140. Lau D, Winkler EA, Than KD, et al. Laminoplasty versus laminectomy with posterior spinal fusion for multilevel cervical spondylotic myelopathy: Influence of cervical alignment on outcomes. J Neurosurg Spine. 2017;27(5):508-517.
141. Lee CH, Jahng TA, Hyun SJ, et al. Expansive laminoplasty versus laminectomy alone versus laminectomy and fusion for cervical ossification of the posterior longitudinal ligament: Is there a

difference in the clinical outcome and sagittal alignment? Clin Spine Surg. 2016;29(1):E9-15.

142. Lee CH, Lee J, Kang JD, et al. Laminoplasty versus laminectomy and fusion for multilevel cervical myelopathy: A meta-analysis of clinical and radiological outcomes. J Neurosurg Spine. 2015;22(6):589-595.
143. Lee JJ, Oh SH, Jeong YH, et al. Surgical strategies for cervical deformities associated with neuromuscular disorders. Neurospine. 2020;17(3):513-524.
144. Lee NJ, Marciano G, Puvanesarajah V, et al. Incidence, mechanism, and protective strategies for 2-year pelvic fixation failure after adult spinal deformity surgery with a minimum six-level fusion. J Neurosurg Spine. 2022;38(2):208-216.
145. Lehman RA Jr, Lenke LG. Long-segment fusion of the thoracolumbar spine in conjunction with a motion-preserving artificial disc replacement: Case report and review of the literature. Spine (Phila Pa 1976). 2007;32(7):E240-E245.
146. Lehrich JR, Katz, JN, Sheon, RP. Approach to the diagnosis and evaluation of low back pain in adults. UpToDate [online serial]. Waltham, MA: UpToDate; 2007.
147. Lehrich JR, Sheon, RP. Treatment of subacute and chronic low back pain. UpToDate [online serial]. Waltham, MA: UpToDate; 2007.
148. Li S, Du J, Huang Y, et al. Comparison of the efficacies of TINAVI robot-assisted surgery and conventional open surgery for Levine-Edward type IIA (postreduction) hangman fractures. Sci Rep. 2023;13(1):15934.
149. Liang L, Feng M, Cui X, et al. The effect of exercise on cervical radiculopathy: A systematic review and meta-analysis. Medicine (Baltimore). 2019;98(45):e17733.
150. Liew SM, Simmons ED. Cervical deformity. Rationale for selecting the appropriate fusion technique. Orthoped Clin North Am. 1998;29(4):779-786.
151. Lin X, Cai J, Qin C, et al. Comparison of clinical outcomes and safety between laminectomy with instrumented fusion versus laminoplasty for the treatment of multilevel cervical spondylotic myelopathy. Medicine (Baltimore). 2019;98(8):e14651.
152. Liu FY, Ma L, Huo LS, et al. Mini-plate fixation versus suture suspensory fixation in cervical laminoplasty: A meta-analysis.

Medicine (Baltimore). 2017;96(5):e6026.

153. Liu FY, Yang SD, Huo LS, et al. Laminoplasty versus laminectomy and fusion for multilevel cervical compressive myelopathy: A meta-analysis. Medicine (Baltimore). 2016;95(23):e3588.
154. Liu X, Wang H, Zhou Z, Jin A. Anterior decompression and fusion versus posterior laminoplasty for multilevel cervical compressive myelopathy. Orthopedics. 2014;37(2):e117-e122.
155. Lown Institute. Unnecessary back surgery. Older Americans put at risk while billions in Medicare funds wasted. Lown Institute Hospitals Index. Needham, MA: Lown Institute; 2024.
156. Luo W, Li Y, Xu Q, et al. Cervical spondylotic amyotrophy: A systematic review. Eur Spine J. 2019 Oct;28(10):2293-2301.
157. Ma X-L, Zhao X-W, Ma J-X, et al. Effectiveness of surgery versus conservative treatment for lumbar spinal stenosis: A system review and meta-analysis of randomized controlled trials. Int J Surg. 2017;329-338.
158. Machino M, Yukawa Y, Imagama S, et al. Surgical treatment assessment of cervical laminoplasty using quantitative performance evaluation in elderly patients: A prospective comparative study in 505 patients with cervical spondylotic myelopathy. Spine (Phila Pa 1976). 2016;41(9):757-763.
159. Maghout Juratli S, Franklin GM, Mirza SK, et al. Lumbar fusion outcomes in Washington State workers' compensation. Spine. 2006;31(23):2715-2723.
160. Magnaes B. Surgical treatment of low back pain. Tidsskr Nor Laegeforen. 1999;119(12):1773-1777.
161. Malik AT, Drain JP, Jones J, et al. Robotic-assisted versus conventional posterior lumbar fusion -- An analysis of 90-day complications and readmissions. World Neurosurg. 2021;152:e168-e174.
162. Marchand AA, Suitner M, O'Shaughnessy J, et al. Feasibility of conducting an active exercise prehabilitation program in patients awaiting spinal stenosis surgery: A randomized pilot study. Sci Rep. 2019;9(1):12257.
163. Marchetti PG, Binazzi R, Briccoli A, et al. The surgical treatment of spondylolisthesis. Chir Organi Mov. 1994;79(1):85-91.
164. Martin CR, Gruszczynski AT, Braunsfurth HA, et al. The surgical management of degenerative lumbar spondylolisthesis: A systematic review. Spine. 2007;32(16):1791-1798.

165. Martinez Ferez IM, Molina Linde JM, Villegas Portero R. Estandares de uso adecuado de la artrodesis vertebral. [Appropriateness criteria for vertebral arthrodesis] [summary]. Informes de Evaluacion de Tecnologias Sanitarias AETSA 2007/06. Seville, Spain: Andalusian Agency for Health Technology Assessment (AETSA); 2009.
166. Masaki Y, Yamazaki M, Okawa A, et al. An analysis of factors causing poor surgical outcome in patients with cervical myelopathy due to ossification of the posterior longitudinal ligament: Anterior decompression with spinal fusion versus laminoplasty. J Spinal Disord Tech. 2007;20(1):7-13.
167. Matsudaira K, Yamazaki T, Seichi A, et al. Spinal stenosis in grade I degenerative lumbar spondylolisthesis: A comparative study of outcomes following laminoplasty and laminectomy with instrumented spinal fusion. J Orthop Sci. 2005;10(3):270-6.
168. Matur AV, Palmisciano P, Duah HO, et al. Robotic and navigated pedicle screws are safer and more accurate than fluoroscopic freehand screws: A systematic review and meta-analysis. Spine J. 2023;23(2):197-208.
169. Matz PG, Anderson PA, Groff MW, et al; Joint Section on Disorders of the Spine and Peripheral Nerves of the American Association of Neurological Surgeons and Congress of Neurological Surgeons. Cervical laminoplasty for the treatment of cervical degenerative myelopathy. J Neurosurg Spine. 2009;11(2):157-169.
170. Mavrocordatos P, Cahana A. Minimally invasive procedures for the treatment of failed back surgery syndrome. Adv Tech Stand Neurosurg. 2006;31:221-252.
171. McCrory DC, Turner DA, Patwardhan MB, et al. Spinal fusion for treatment of degenerative disease affecting the lumbar spine. Technology Assessment [draft]. Prepared for the Agency for Healthcare Research and Quality (AHRQ) by the Duke Evidence-based Practice Center. Rockville, MD: AHRQ; November 1, 2006.
172. McCullen G, Vaccaro AR, Garfin SR. Thoracic and lumbar trauma: Rationale for selecting the appropriate fusion technnique. Orthop Clin North Am. 1998;29(4):813-828.
173. Mendenhall SK, Sivaganesan A, Mistry A, et al. Traumatic atlantooccipital dislocation: Comprehensive assessment of mortality, neurologic improvement, and patient-reported

outcomes at a Level 1 trauma center over 15 years. Spine J. 2015;15(11):2385-2395.

174. Mikhael MM, Wolf CF, Wang JC. Cervical spine surgery: Cervical laminaplasty. Instr Course Lect. 2012;61:461-468.
175. Minetama M, Kawakami M, Nakagawa M, et al. A comparative study of 2-year follow-up outcomes in lumbar spinal stenosis patients treated with physical therapy alone and those with surgical intervention after less successful physical therapy. J Orthop Sci. 2018;23(3):470-476.
176. Minetama M, Kawakami M, Teraguchi M, et al. Supervised physical therapy vs. home exercise for patients with lumbar spinal stenosis: A randomized controlled trial. Spine J. 2019;19(8):1310-1318.
177. Minetama M, Kawakami M, Teraguchi M, et al. Supervised physical therapy versus unsupervised exercise for patients with lumbar spinal stenosis: 1-year follow-up of a randomized controlled trial. Clin Rehabil. 2021;35(7):964-975.
178. Minetama M, Kawakami M, Teraguchi M, et al. Supervised physical therapy versus surgery for patients with lumbar spinal stenosis: A propensity score-matched analysis. BMC Musculoskelet Disord. 2022;23(1):658.
179. Mirza SK, Deyo RA. Systematic review of randomized trials comparing lumbar fusion surgery to nonoperative care for treatment of chronic back pain. Spine. 2007;32(7):816-823.
180. Mummaneni PV, Kaiser MG, Matz PG, et al; Joint Section on Disorders of the Spine and Peripheral Nerves of the American Association of Neurological Surgeons and Congress of Neurological Surgeons. Cervical surgical techniques for the treatment of cervical spondylotic myelopathy. J Neurosurg Spine. 2009;11(2):130-141.
181. Nachemson A, Carlsson C-A, Englund L, et al. Back and neck pain [summary]. SBU Report No.145. Stockholm, Sweden: Swedish Council on Technology Assessment in Health Care (SBU); 2000.
182. Nachemson AL. Instability of the lumbar spine. Pathology, treatment, and clinical evaluation. Neurosurg Clin N Am. 1991;2(4):785-790.
183. Nakashima H, Kato F, Yukawa Y, et al. Comparative effectiveness of open-door laminoplasty versus French-door laminoplasty in

cervical compressive myelopathy. Spine (Phila Pa 1976). 2014;39(8):642-647.

184. Narayan P, Haid RW. Treatment of degenerative cervical disc disease. Neurol Clin. 2001;19(1):217-229.
185. Nguyen TT, Thelen JC, Bhatt AA. Bone up on spinal osseous lesions: A case review series. Insights Imaging. 2020;11(1):80.
186. Nikolaidis I, Fouyas IP, Sandercock PA, Statham PF. Surgery for cervical radiculopathy or myelopathy. Cochrane Database Syst Rev. 2010;2010(1):CD001466.
187. North American Spine Society (NASS). Adult isthmic spondylolisthesis. North American Spine Society Public Education Series. LaGrange, IL: NASS; 2006.
188. North American Spine Society (NASS). Diagnosis and treatment of degenerative lumbar spinal stenosis. Burr Ridge, IL: North American Spine Society (NASS); 2011.
189. Norwegian Knowledge Centre for the Health Services (NOKC). Treatment of lumbar disc herniation. SMM-Report 1/2001. Oslo, Norway: NOKC; 2001.
190. Nunna RS, Ostrov PB, Ansari D, et al. The risk of nonunion in smokers revisited: A systematic review and meta-analysis. Global Spine J. 2022;12(3):526-539.
191. Ohnari H, Sasai K, Akagi S, et al. Investigation of axial symptoms after cervical laminoplasty, using questionnaire survey. Spine J. 2006;6(3):221-227.
192. Onesti ST. Failed back syndrome. Neurologist. 2004;10(5):259-264.
193. Onodera R, Sakamoto R, Taniguchi Y, et al. Congenital atlanto-occipital dislocation in a patient with Down syndrome: A case report. Skeletal Radiol. 2023;52(9):1785-1789.
194. Osman NS, Cheung ZB, Hussain AK, et al. Outcomes and complications following laminectomy alone for thoracic myelopathy due to ossified ligamentum flavum: A systematic review and meta-analysis. Spine (Phila Pa 1976). 2018;43(14):E842-E848.
195. Pang S, Kolarich AR, Brinjikji W, et al. Interventional and surgical management of internal jugular venous stenosis: A narrative review. J Neurointerv Surg. 2022;14(5):neurintsurg-2021-017937.
196. Papadopoulos EC, Girardi FP, Sandhu HS, et al. Outcome of revision discectomies following recurrent lumbar disc herniation.

Spine. 2006;31(13):1473-1476.

197. Papalia GF, Vadala G, Russo F, et al. Higher accuracy and better clinical outcomes in navigated thoraco-lumbar pedicle screw fixation versus conventional techniques: A systematic review and meta-analysis. Spine (Phila Pa 1976). 2024;49(19):1370-1380.
198. Park HJ, Kim SS, Lee SY, et al. A practical MRI grading system for cervical foraminal stenosis based on oblique sagittal images. Br J Radiol. 2013;86(1025):20120515.
199. Patel, RK. Lumbar degenerative disk disease. eMedicine Physical Medicine and Rehabilitation. Topic 67. Omaha, NE: eMedicine.com; updated January 18, 2007. Available at: http://www.emedicine.com/pmr/topic67.htm. Accessed May 21, 2003.
200. Pearson RG, Clement RGE, Edwards KL, Scammell BE. Do smokers have greater risk of delayed and non-union after fracture, osteotomy and arthrodesis? A systematic review with meta-analysis. BMJ Open. 2016;6(11):e010303.
201. Peev N, Zygourakis C, Sippl C, et al. Acute back pain - Role of injection techniques and surgery: WFNS spine committee recommendations. World Neurosurg X. 2024;22:100315.
202. Perron AD, Huff JS. Spinal Cord Disorders. In: Marx, JA, ed, Rosen's Emergency Medicine: Concepts and Clinical Practice. 5th ed. St. Louis, MO: Mosby;2002:1499.
203. Peul WC, van Houwelingen HC, van den Hout WB, et al. Leiden-The Hague Spine Intervention Prognostic Study Group. Surgery versus prolonged conservative treatment for sciatica. NEJM. 2007;356(22):2245-2256.
204. Phan K, Fadhil M, Chang N, et al. Effect of smoking status on successful arthrodesis, clinical outcome, and complications after anterior lumbar interbody fusion (ALIF). World Neurosurg. 2018;110:e998-e1003.
205. Phan K, Scherman DB, Xu J, et al. Laminectomy and fusion vs laminoplasty for multi-level cervical myelopathy: A systematic review and meta-analysis. Eur Spine J. 2017;26(1):94-103.
206. Phillips FM, Cunningham B. Managing chronic pain of spinal origin after lumbar surgery: the role of decompressive surgery. Spine. 2002;27(22):2547-2553.
207. Pietrantonio A, Trungu S, Fama I, et al. Long-term clinical outcomes after bilateral laminotomy or total laminectomy for

lumbar spinal stenosis: A single-institution experience. Neurosurg Focus. 2019;46(5):E2.

208. Plener J, Csiernik B, To D, et al. Conservative management of cervical radiculopathy: A systematic review. Clin J Pain. 2023;39(3):138-146.
209. Postacchini F. Results of surgery compared with conservative management for lumbar disc herniations. Spine. 1996;21(11):1383-1387.
210. Qi Q, Chen Y, Ling Z, et al. Modified laminoplasty preserving the posterior deep extensor insertion into C2 improves clinical and radiologic results compared with conventional laminoplasty: A meta-analysis. World Neurosurg. 2018;111:157-165.
211. Qi Q, Li L, Luo J, et al. Is mini-plate fixation superior to suture suspensory fixation in cervical laminoplasty? A meta-analysis. World Neurosurg. 2016;93:144-153.
212. Qian S, Wang Z, Jiang G, et al. Efficacy of laminoplasty in patients with cervical kyphosis. Med Sci Monit. 2018;24:1188-1195.
213. Qin R, Chen X, Zhou P, et al. Anterior cervical corpectomy and fusion versus posterior laminoplasty for the treatment of oppressive myelopathy owing to cervical ossification of posterior longitudinal ligament: A meta-analysis. Eur Spine J. 2018;27(6):1375-1387.
214. Ravikanth R, Majumdar P. Bertolotti's syndrome in low-backache population: Classification and imaging findings. Tzu Chi Med J. 2019;31(2):90-95.
215. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 1: Introduction and methodology.J Neurosurg Spine. 2005;2(6):637-638.
216. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 2: Assessment of functional outcome. J Neurosurg Spine. 2005;2(6):639-646.
217. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for

degenerative disease of the lumbar spine. Part 3: Assessment of economic outcome. J Neurosurg Spine. 2005;2(6):647-652.

218. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 4: Radiographic assessment of fusion. J Neurosurg Spine. 2005;2(6):653-657.
219. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 5: Correlation between radiographic and functional outcome. J Neurosurg Spine. 2005;2(6):658-661.
220. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 6: Magnetic resonance imaging and discography for patient selection for lumbar fusion. J Neurosurg Spine. 2005;2(6):662-669.
221. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 7: Intractable low-back pain without stenosis or spondylolisthesis. J Neurosurg Spine. 2005;2(6):670-672.
222. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 8: Lumbar fusion for disc herniation and radiculopathy. J Neurosurg Spine. 2005;2(6):673-678.
223. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 9: Fusion in patients with stenosis and spondylolisthesis. J Neurosurg Spine. 2005;(6):679-685.
224. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons.

Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 10: Fusion following decompression in patients with stenosis without spondylolisthesis. J Neurosurg Spine. 2005;2(6):686-691.

225. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 11: Interbody techniques for lumbar fusion. J Neurosurg Spine. 2005;2(6):692-699.
226. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 12: Pedicle screw fixation as an adjunct to posterolateral fusion for low-back pain. J Neurosurg Spine. 2005;2(6):700-706.
227. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 13: Injection therapies, low-back pain, and lumbar fusion. J Neurosurg Spine. 2005;2(6):707-715.
228. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 14: Brace therapy as an adjunct to or substitute for lumbar fusion. J Neurosurg Spine. 2005;2(6):716-724.
229. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 15: Electrophysiological monitoring and lumbar fusion. J Neurosurg Spine. 2005;2(6):725-732.
230. Resnick DK, Choudhri TF, Dailey AT, et al; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for

degenerative disease of the lumbar spine. Part 16: Bone graft extenders and substitutes. J Neurosurg Spine. 2005;2(6):733-736.
231. Resnick DK, Choudhri TF, Dailey AT, et al.; American Association of Neurological Surgeons/Congress of Neurological Surgeons. Guidelines for the performance of fusion procedures for degenerative disease of the lumbar spine. Part 17: Bone growth stimulators and lumbar fusion. J Neurosurg Spine. 2005;2(6):737-740.
232. Rhee JM, Basra S. Posterior surgery for cervical myelopathy: Laminectomy, laminectomy with fusion, and laminoplasty. Asian Spine J. 2008;2(2):114-126.
233. Ricciardi L, Stifano V, Sturiale CL, et al. Minimally invasive decompression with posterior elements preservation versus laminectomy and fusion for lumbar degenerative spondylolisthesis: A systematic review and meta-analysis of surgical, clinical and radiological outcomes. Surg Technol Int. 2020;36:457-463.
234. Robertson PA, Jackson SA. Prospective assessment of outcomes improvement following fusion for low back pain. J Spinal Disord Tech. 2004;17(3):183-188.
235. Ronthal M. Surgery for herniated lumbar disc and for spondylolisthesis. JWatch Neurol. July 17, 2007.
236. Ryken TC, Heary RF, Matz PG, et al; Joint Section on Disorders of the Spine and Peripheral Nerves of the American Association of Neurological Surgeons and Congress of Neurological Surgeons. Cervical laminectomy for the treatment of cervical degenerative myelopathy. J Neurosurg Spine. 2009;11(2):142-149.
237. Sakaura H, Hosono N, Mukai Y, et al. Long-term outcome of laminoplasty for cervical myelopathy due to disc herniation: A comparative study of laminoplasty and anterior spinal fusion. Spine. 2005;30(7):756-759.
238. Sanniec K, Borsting E, Amirlak B. Decompression-avulsion of the auriculotemporal nerve for treatment of migraines and chronic headaches. Plast Reconstr Surg Glob Open. 2016;4(4):e678.
239. Scerrati A, Norri N, Mongardi L, et al. Styloidogenic-cervical spondylotic internal jugular venous compression, a vascular disease related to several clinical neurological manifestations: Diagnosis and treatment -- a comprehensive literature review. Ann Transl Med. 2021;9(8):718.

240. Schaller B. Failed back surgery syndrome: the role of symptomatic segmental single-level instability after lumbar microdiscectomy. Eur Spine J. 2004;13(3):193-198.
241. Schimmel J, George N, Schwarz J, et al. Carboxyhemoglobin levels induced by cigarette smoking outdoors in smokers. J Med Toxicol. 2018;14(1):68-73.
242. Schneider MJ, Ammendolia C, Murphy DR, et al. Comparative clinical effectiveness of nonsurgical treatment methods in patients with lumbar Spinal stenosis: A randomized clinical trial. JAMA Netw Open. 2019;2(1):e186828.
243. Schoelles K, Reston J, Treadwell J, et al. Spinal fusion and discography in chronic uncomplicated lumbar degenerative disc disease. Health Technology Assessment. Prepared by the ECRI Institute for the Technology Assessment Program, Washington State Health Care Authority. Contract No. 6020-0017, Task Order No. 2. Olympia, WA: Washington State Health Care Authority; October 19, 2007.
244. Seo D-H, Yoon K-W, Lee SK, Kim Y-J. Microsurgical excision of symptomatic sacral perineurial cyst with sacral recapping laminectomy: A case report in technical aspects. J Korean Neurosurg Soc 2014;55(2):110-113.
245. Sharan AD, Kaye D, Charles Malveaux WM, Riew KD. Dropped head syndrome: Etiology and management. J Am Acad Orthop Surg. 2012;20(12):766-774.
246. Shigematsu H, Ueda Y, Koizumi M, et al. Does developmental canal stenosis influence surgical results of bilateral open-door laminoplasty for cervical spondylotic myelopathy? J Neurosurg Spine. 2008;9(4):358-362.
247. Skaf G, Bouclaous C, Alaraj A, et al. Clinical outcome of surgical treatment of failed back surgery syndrome. Surg Neurol. 2005;64(6):483-488.
248. Slater J, Kolber MJ, Schellhase KC, et al. The influence of exercise on perceived pain and disability in patients with lumbar spinal stenosis: A Systematic review of randomized controlled trials. Am J Lifestyle Med. 2015;10(2):136-147.
249. Small SA, Perron AD, Brady WJ. Orthopedic pitfalls: Cauda equina syndrome. Am J Emerg Med. 2005;23(2):159-163.
250. Stam J. Consensus on diagnosis and treatment of the lumbrosacral radicular syndrome. Ned Tijdschr Geneeskd.

1996;140:2621-2627.

251. Steinmetz MP, Resnick DK. Cervical laminoplasty. Spine J. 2006;6(6 Suppl):274S-281S.
252. Strayer A. Lumbar spine: Common pathology and interventions. J Neurosci Nurs. 2005;37(4):181-193.
253. Sun S, Li Y, Wang X, et al. Safety and efficacy of laminoplasty versus laminectomy in the treatment of spinal cord tumors: A systematic review and meta-analysis. World Neurosurg. 2019;125:136-145.
254. Sun Y, Li L, Zhao J, Gu R. Comparison between anterior approaches and posterior approaches for the treatment of multilevel cervical spondylotic myelopathy: A meta-analysis. Clin Neurol Neurosurg. 2015;134:28-36.
255. Tadokoro K, Miyamoto H, Sumi M, Shimomura T. The prognosis of conservative treatments for lumbar spinal stenosis: Analysis of patients over 70 years of age. Spine (Phila Pa 1976). 2005;30(21):2458-2463.
256. Takeshita K, Seichi A, Akune T, et al. Can laminoplasty maintain the cervical alignment even when the C2 lamina is contained? Spine. 2005;30(11):1294-1298.
257. Tamai K, Terai H, Suzuki A, et al. Anterior cervical discectomy and fusion provides better surgical outcomes than posterior laminoplasty in elderly patients with C3-4 level myelopathy. Spine (Phila Pa 1976). 2017;42(8):548-555.
258. Tang M, Yang XF, Yang SW, et al. Lumbosacral transitional vertebra in a population-based study of 5860 individuals: Prevalence and relationship to low back pain. Eur J Radiol. 2014;83(9):1679-1682.
259. Tao X, Matur AV, Palmisciano P, et al. Preoperative HbA1c and postoperative outcomes in spine surgery: A systematic review and meta-analysis. Spine (Phila Pa 1976). 2023;48(16):1155-1165.
260. Tis JE, O'Brien MF, Newton PO, et al. Adolescent idiopathic scoliosis treated with open instrumented anterior spinal fusion: Five-year follow-up. Spine. 2010;35(1):64-70.
261. Tsuji H. Laminoplasty for patients with compressive myelopathy due to so-called spinal canal stenosis in cervical and thoracic regions. Spine. 1982;7(1):28-34.
262. Tsutsumimoto T, Shimogata M, Yoshimura Y, Misawa H. Union versus nonunion after posterolateral lumbar fusion: a

comparison of long-term surgical outcomes in patients with degenerative lumbar spondylolisthesis. 1: Eur Spine J. 2008;17(8):1107-1112.
263. Turner JA, Ersek M, Herron L, et al. Surgery for lumbar spinal stenosis. Attempted meta-analysis of the literature. Spine. 1992;17(1):1-8.
264. Umeta RS, Avanzi O. Techniques of lumbar-sacral spine fusion in spondylosis: Systematic literature review and meta-analysis of randomized clinical trials. Spine J. 2011;11(7):668-676.
265. Vinha A, Bartolo J, Lemos C, et al. Lumbosacral transitional vertebrae: Prevalence in a Southern European population and its association with low back pain. Eur Spine J. 2022;31(12):3647-3653.
266. Vokshoor A. Spondylolisthesis, spondylolysis, and spondylosis. eMedicine. Orthopedic Topic 560. Omaha, NE:eMedicine.com: Updated June 30, 2004. Available at: http://www.emedicine.com/orthoped/topic560.htm. Accessed July 2, 2007.
267. Walker MJ, Boyles RE, Young BA, et al. The effectiveness of manual physical therapy and exercise for mechanical neck pain: A randomized clinical trial. Spine. 2008;33(22):2371-2378.
268. Wang MY, Green BA. Open-door cervical expansile laminoplasty. Neurosurgery. 2004;54(1):119-123; discussion 123-124.
269. Washington State Department of Labor and Industries. Guidelines for lumbar fusion (arthrodesis). Medical Treatment Guidelines. Olympia, WA: Washington State Department of Labor and Industries; 2002.
270. Weber H. Lumbar disc herniation. A controlled, prospective study with ten years of observation. Spine. 1983;8(2):131-140.
271. Weinstein JN, Lurie JD, Tosteson TD, et al. Surgical versus nonsurgical treatment for lumbar degenerative spondylolisthesis. NEJM. 2007;356(22):2257-2270.
272. Weinstein JN, Lurie JD, Tosteson TD, et al. Surgical vs nonoperative treatment for lumbar disk herniation: The Spine Patient Outcomes Research Trial (SPORT) observational cohort. JAMA. 2006;296(20):2451-2459.
273. Weinstein JN, Tosteson TD, Lurie JD, et al. Surgical vs nonoperative treatment for lumbar disk herniation: The Spine

Patient Outcomes Research Trial (SPORT): A randomized trial. JAMA. 2006;296(20):2441-2450.

274. Weinstein JN, Tosteson TD, Lurie JD, et al; SPORT Investigators. Surgical versus nonsurgical therapy for lumbar spinal stenosis. N Engl J Med. 2008;358(8):794-810.
275. Weissler EH, Jenkins AL, 3rd, Hecht AC, Taub PJ. Plastic surgeon closure of index spinal cases: A single-institution review of 928 cases. Clin Spine Surg. 2019;32(9):E397-E402.
276. Wheeless CR, Nunley JA, Urbaniak JR, eds. Wheeless' Textbook of Orthopaedics. WheelessOnline.com. Brooklandville, MD: Data Trace Internet Publishing LLC; 2008. Available at: http://www.wheelessonline.com/. Accessed May 15, 2008.
277. Windsor RE. Cervical disc injuries. eMedicine. Spine Topic 18. Omaha, NE:eMedicine.com: updated April 6, 2006. Available at: http://www.emedicine.com/sports/fulltopic/topic18.htm. Accessed April 14, 2006.
278. Winters ME, Kluetz P, Zilberstein J. Back pain emergencies. Med Clin North Am. 2006;90(3):505-523.
279. Work Loss Data Institute. Neck and upper back (acute & chronic). Encinitas, CA: Work Loss Data Institute; 2011.
280. Wright MA, Weinstein AL, Bernstein JL, et al. Muscle flap closure following complex spine surgery: A decade of experience. Plast Reconstr Surg. 2020;146(5):642e-650e.
281. Wu J, Fan M, He D, et al. Accuracy and clinical characteristics of robot-assisted cervical spine surgery: A systematic review and meta-analysis. Int Orthop. 2024;48(7):1903-1914.
282. Xia Y, Wan J, Yu B, et al. Treatment of fluorosis cervical canal stenosis by open-door cervical expansive laminoplasty using anchor fixation. Zhongguo Xiu Fu Chong Jian Wai Ke Za Zhi. 2009;23(10):1204-1208.
283. Xu P, Zhuang JS, Huang YS, et al. Is anterior decompression and fusion superior to laminoplasty for cervical myelopathy due to ossification of posterior longitudinal ligament? A systematic review and meta-analysis. J Spinal Cord Med. 2021;44(2):169-183.
284. Yang S, Yang D, Ma L, et al. Clinical efficacy of laminectomy with instrumented fixation in treatment of adjacent segmental disease following ACCF surgery: A retrospective observational study of 48 patients. Sci Rep. 2019;9(1):6551.

285. Yang SC, Niu CC, Chen WJ, et al. Open-door laminoplasty for multilevel cervical spondylotic myelopathy: Good outcome in 12 patients using suture anchor fixation. Acta Orthop. 2008;79(1):62-66.
286. Yavin D, Casha S, Wiebe S, et al. Lumbar fusion for degenerative disease: A systematic review and meta-analysis. Neurosurgery. 2017;80(5):701-715.
287. Yoon ST, Hashimoto RE, Raich A, et al. Outcomes after laminoplasty compared with laminectomy and fusion in patients with cervical myelopathy: A systematic review. Spine (Phila Pa 1976). 2013;38(22 Suppl 1):S183-S194.
288. Yoshii T, Egawa S, Chikuda H, et al. Comparison of anterior decompression with fusion and posterior decompression with fusion for cervical spondylotic myelopathy --A systematic review and meta-analysis. J Orthop Sci. 2020;25(6):938-945.
289. Yu H, Li X, Chen S, et al. Comparative effectiveness and safety of anterior cervical corpectomy with fusion, laminoplasty, and laminectomy and instrumented fusion for ossification of the posterior longitudinal ligament: A systematic review and network meta-analysis. J Invest Surg. 2022;35(3):667-676.
290. Yuan PS, Booth RE Jr, Albert TJ. Nonsurgical and surgical management of lumbar spinal stenosis. Instr Course Lect. 2005;54:303-312.
291. Yuan W, Zhu Y, Liu X, et al. Laminoplasty versus skip laminectomy for the treatment of multilevel cervical spondylotic myelopathy: A systematic review. Arch Orthop Trauma Surg. 2014;134(1):1-7.
292. Yuan X, Wei C, Xu W, et al. Comparison of laminectomy and fusion vs laminoplasty in the treatment of multilevel cervical spondylotic myelopathy: A meta-analysis. Medicine (Baltimore). 2019;98(13):e14971.
293. Yukawa Y, Kato F, Ito K, et al. Laminoplasty and skip laminectomy for cervical compressive myelopathy: Range of motion, postoperative neck pain, and surgical outcomes in a randomized prospective study. Spine. 2007;32(18):1980-1985.
294. Zaina F, Tomkins-Lane C, Carragee E, Negrini S. Surgical versus non-surgical treatment for lumbar spinal stenosis. Cochrane Database Syst Rev. 2016;2016(1):CD010264.
295. Zhong J, Balouch E, O'Malley N, et al. Comparison of plastic surgeon and spine surgeon closure in revision short segment

thoracolumbar spinal fusions. Spine (Phila Pa 1976). 2021;46(18):1279-1286.

296. Zhu B, Xu Y, Liu X, et al. Anterior approach versus posterior approach for the treatment of multilevel cervical spondylotic myelopathy: A systemic review and meta-analysis. Eur Spine J. 2013;22(7):1583-1593.
297. Zhu W, Ding X, Zheng J, et al. A systematic review and bibliometric study of Bertolotti's syndrome: Clinical characteristics and global trends. Int J Surg. 2023;109(10):3159-3168.
298. Zhuang T, Ku S, Shapiro LM, et al. A cost-effectiveness analysis of smoking-cessation interventions prior to posterolateral lumbar fusion. J Bone Joint Surg Am. 2020;102(23):2032-2042.